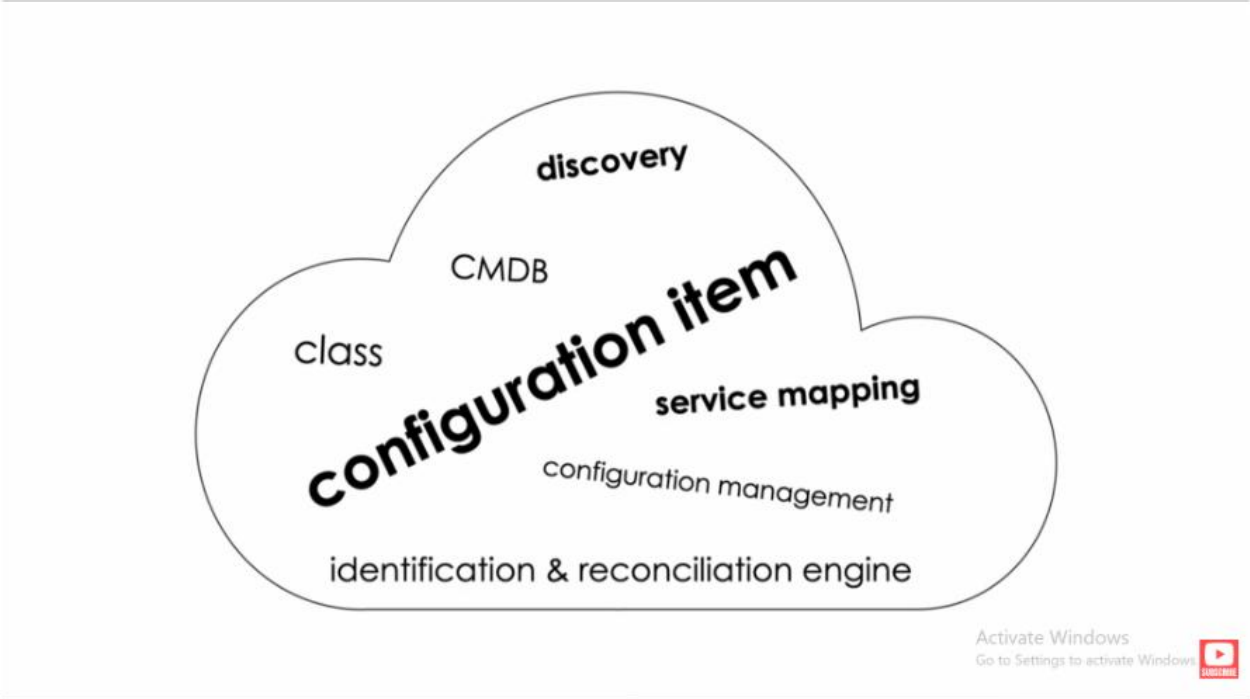
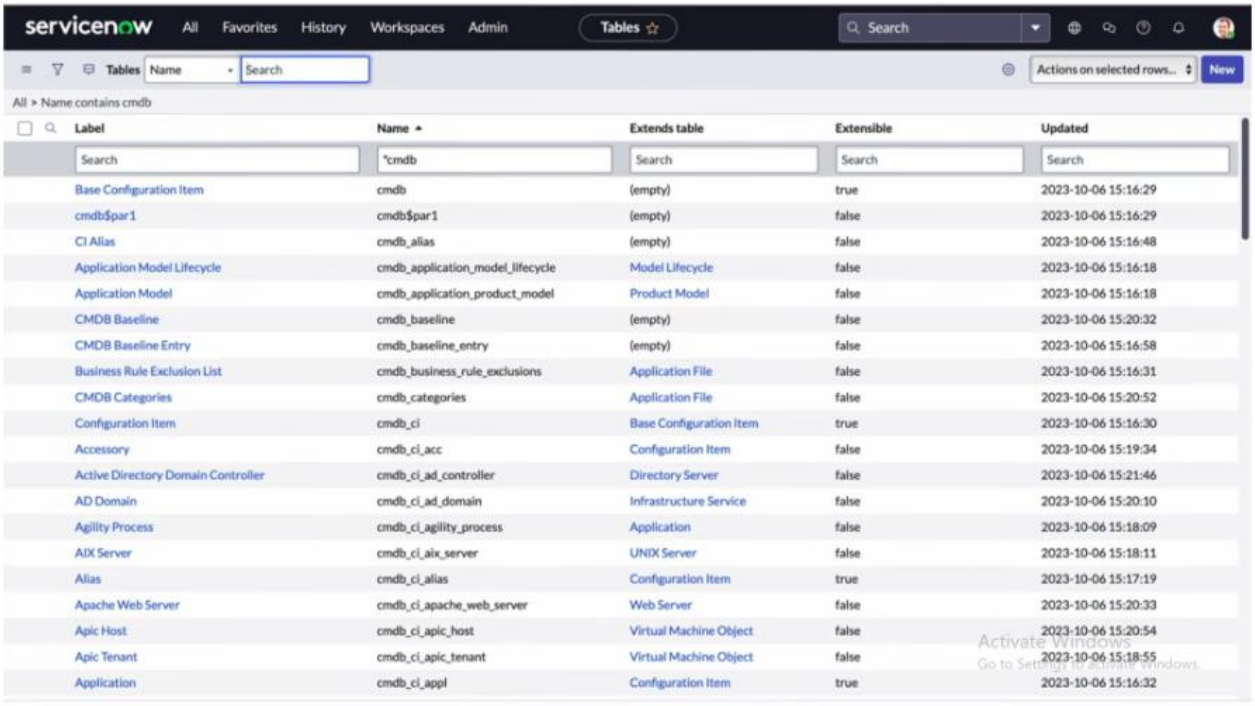
CMDB MANAGEMENT BASICS







Configuration Management is a process that enables a company to understand from an operational perspective how IT components are configured and connected to provide value in the form of IT Services.

Activate Windows
Go to Settings to activate Windows.

A **CMDB** is a set of database tables that store a logical model of a company's IT components, the services they provide, and the relationships between them.

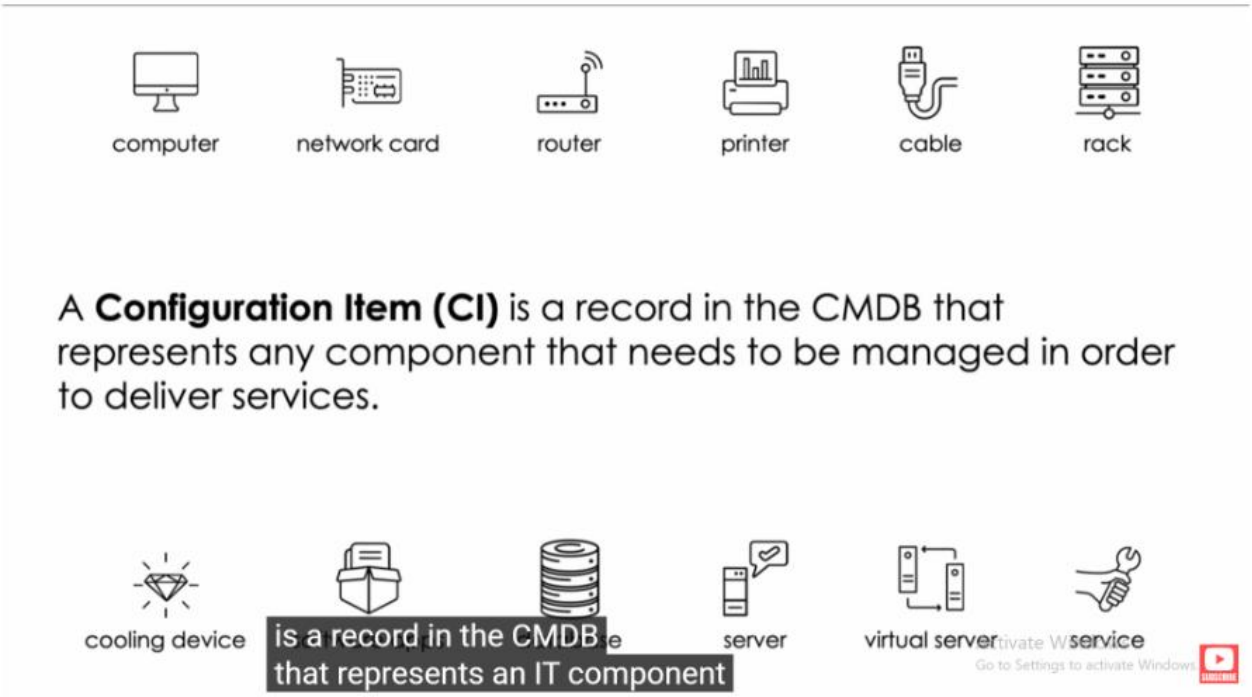
Activate Windows
Go to Settings to activate Windows.



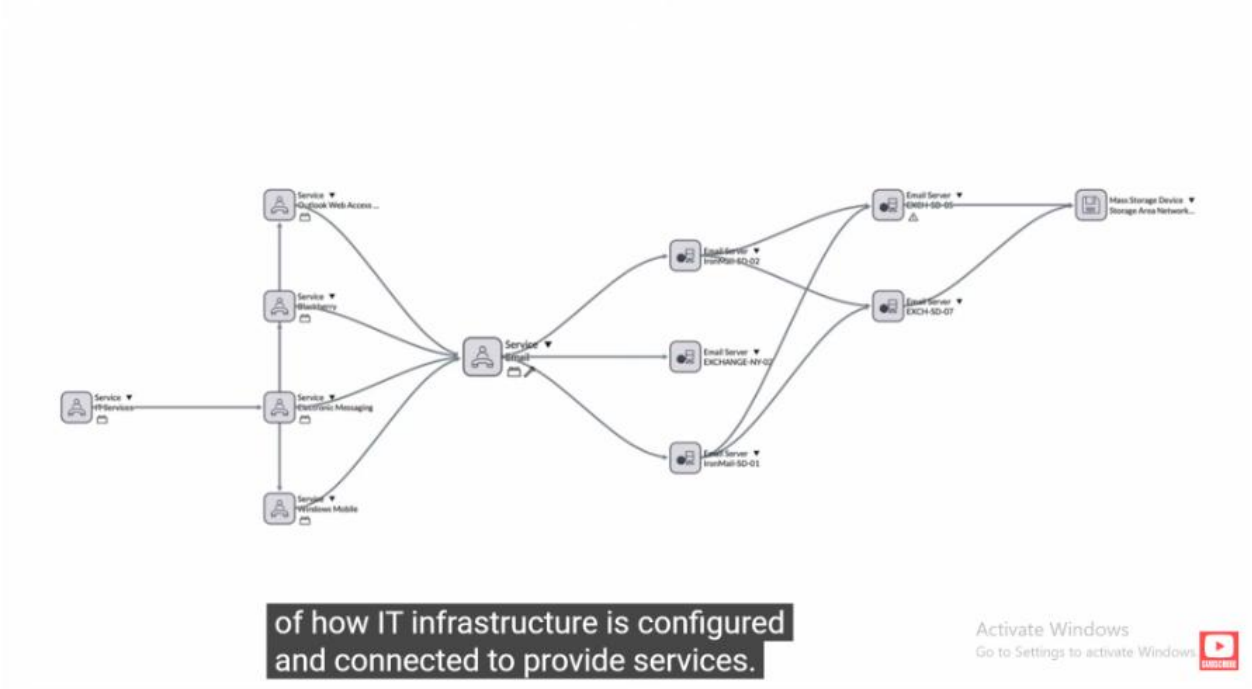


Activate Windows
Go to Settings to activate Windows.

Each record in the cmdb represents one of these IT components and gets stored in a table that's designed to store the attributes of that item.
Each record in cmdb is called CI



NOTE: CIs can be tangible like hardware and software, or they can be intangible like services.



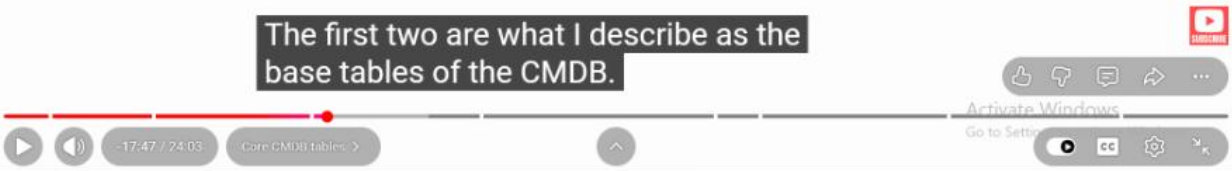
HOW THESE RELATIONSHIP HELPFUL ?

This makes possible to quickly diagnose problems, predict potential issues, and understand the impact of making changes in future.

ServiceNow CMDB Explained: What It Is, Why It Matters, and How It Works

Four Important Tables

The first two are what I describe as the base tables of the CMDB.



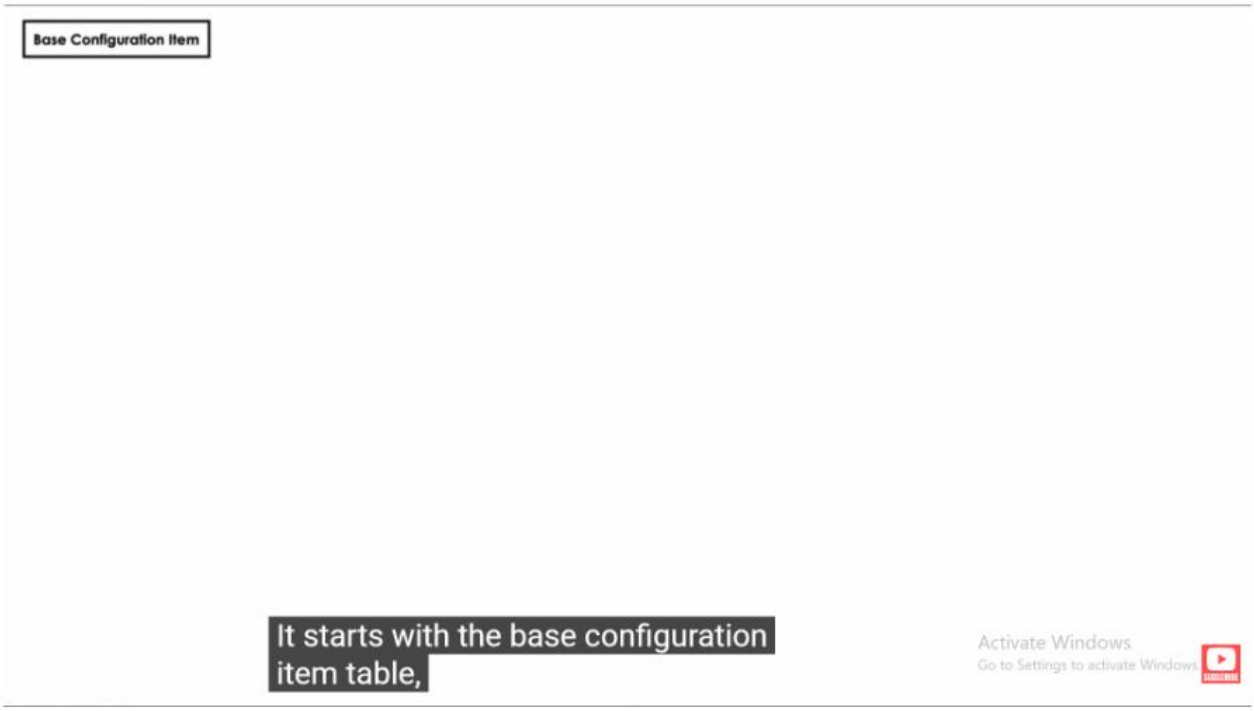
ServiceNow CMDB Explained: What It Is, Why It Matters, and How It Works

Base Configuration Item
[cmdb]

Configuration Item
[cmdb_ci]

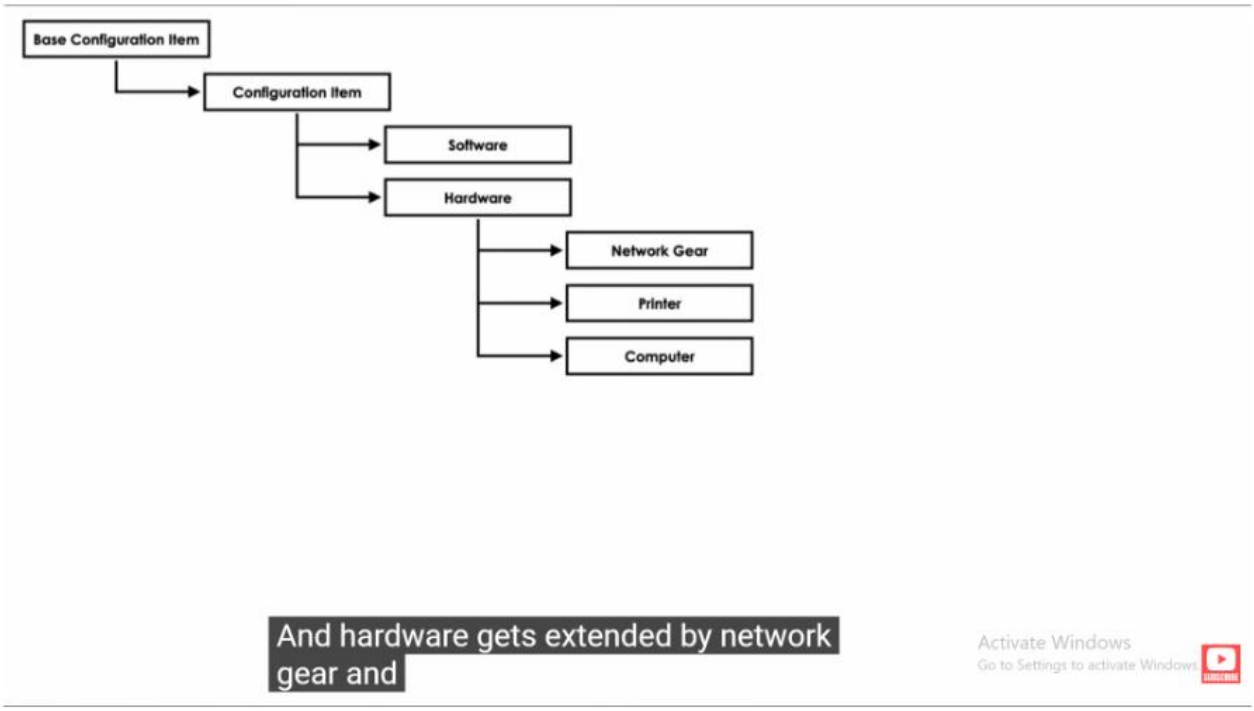
and the configuration item table CMDB_CI.

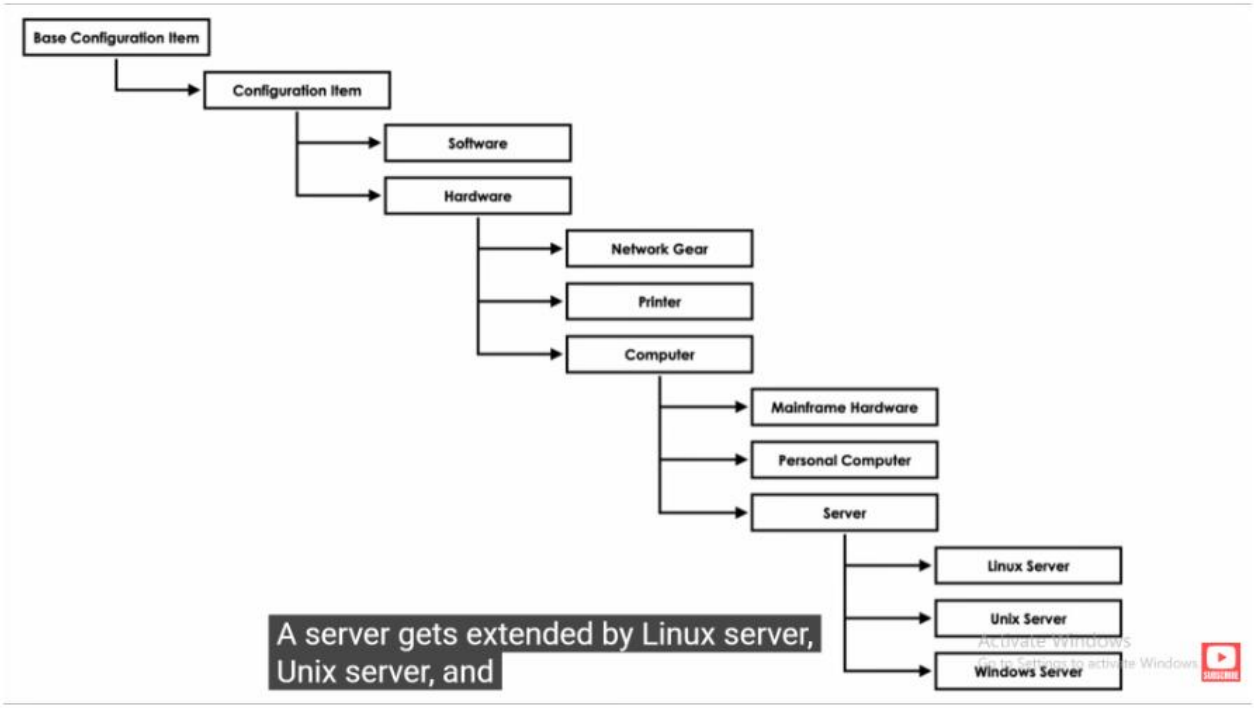




NOTE : This Base configuration item table (cmdb) defines 47 fields. And this cmdb table is extended by cmdb_ci table. Which additionally have 33 fields.







And other 2 main tables are :



and the CI relationship type table,

Activate Windows
Go to Settings to activate Windows.

These 2 tables are used to store relationships that exist b/w the CIs.

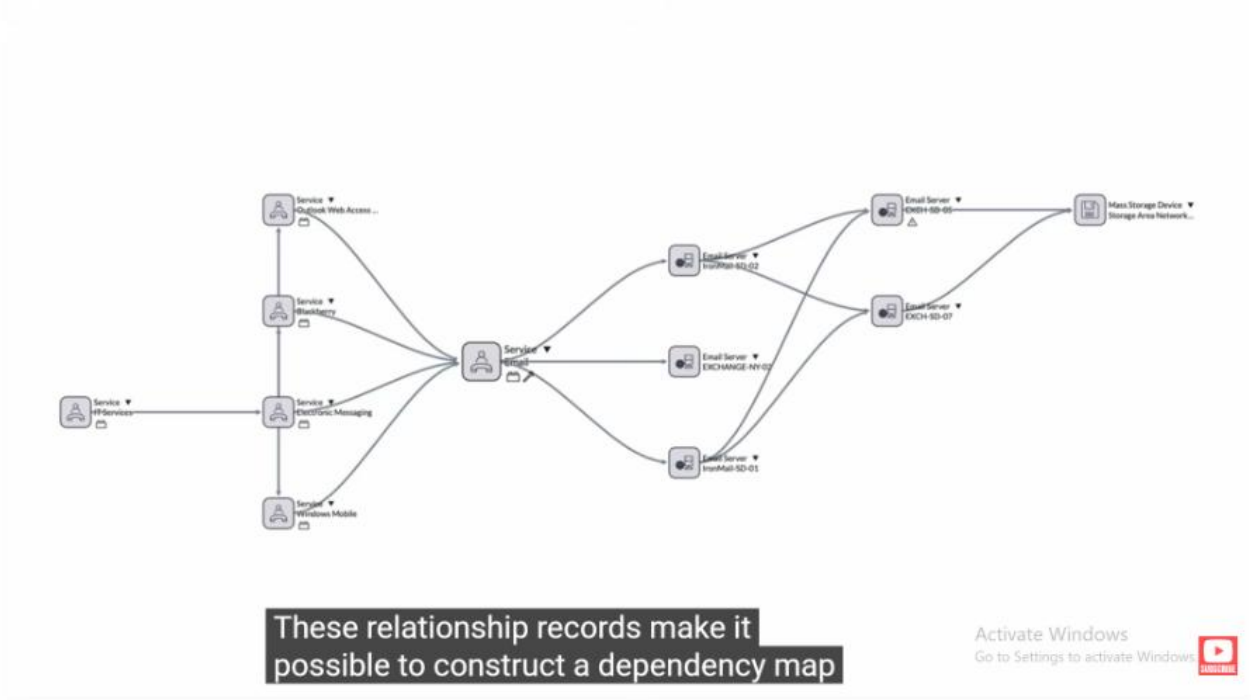


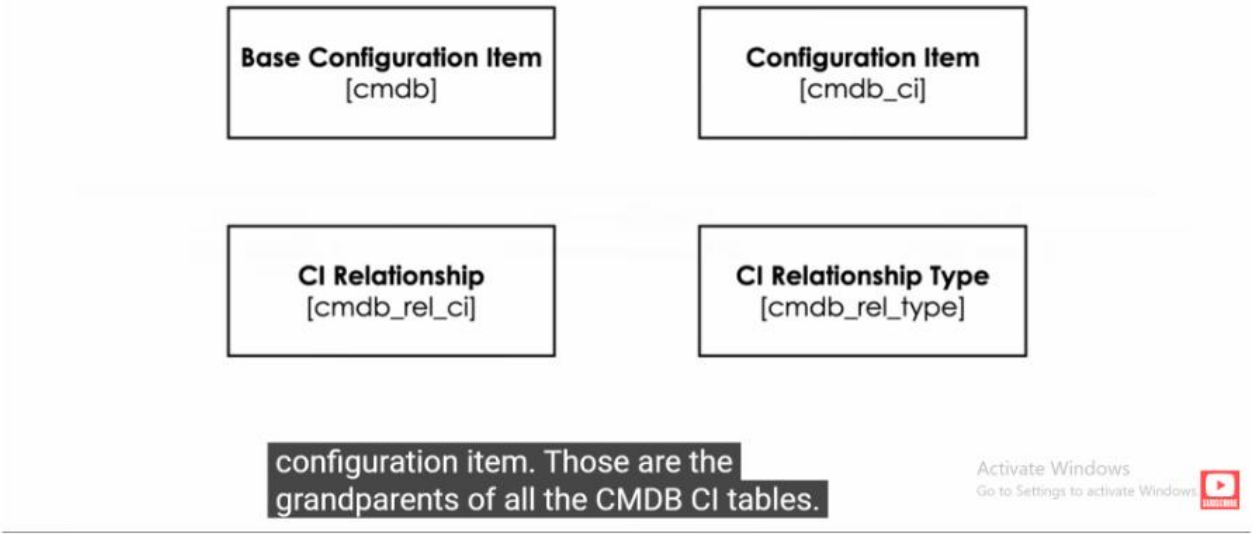
Parent CI	Relationship Type	Child CI
Email Service A	Depends on :: Used by	Email Server B

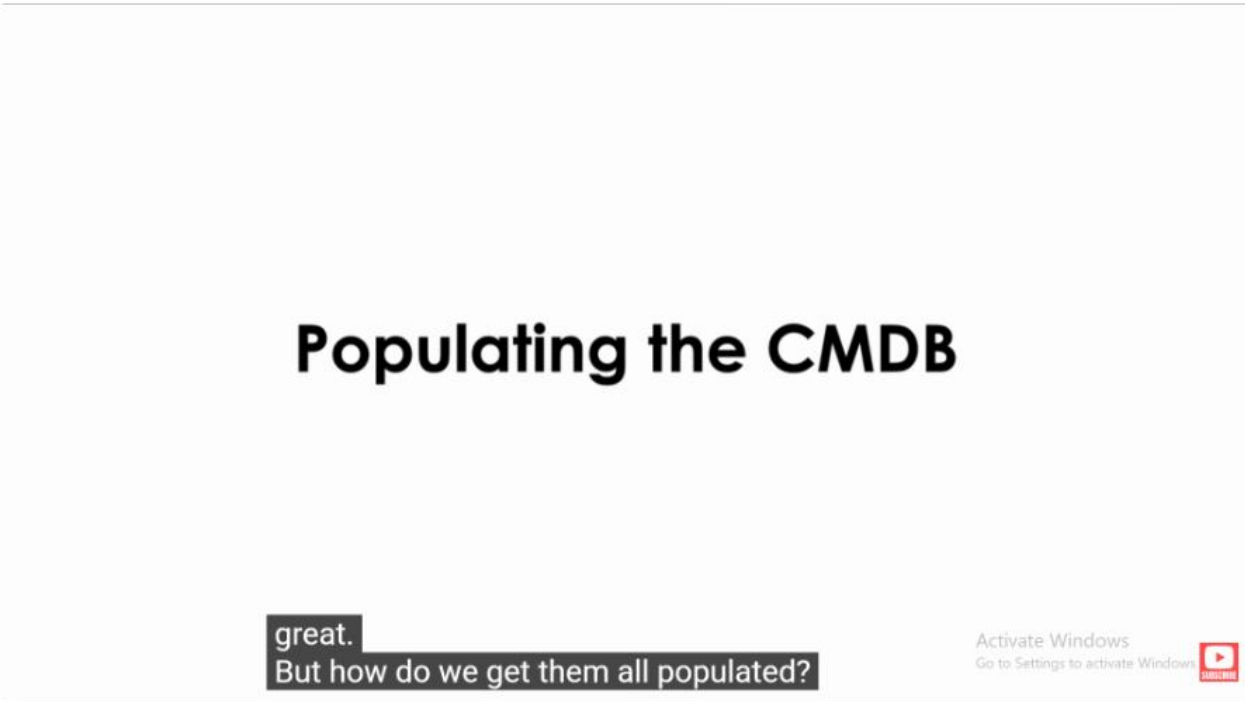
parent CI, a child CI, and a relationship type.

Activate Windows
Go to Settings to activate Windows.

And these relationship records makes it possible to construct a dependency map.







Now, how can we put all the configuration details or data in this ci table records ?
First we can do it by manual entry.

Manual Entry



Manually populate and submit forms

A form exists for every type of CI

Time-consuming and error-prone

I'm currently working with a client that has many millions of CIs

Activate Windows
Go to Settings to activate Windows.



Manual Entry



Manually populate and submit forms

A form exists for every type of CI

Time-consuming and error-prone

Integrations



Import from legacy inventory systems

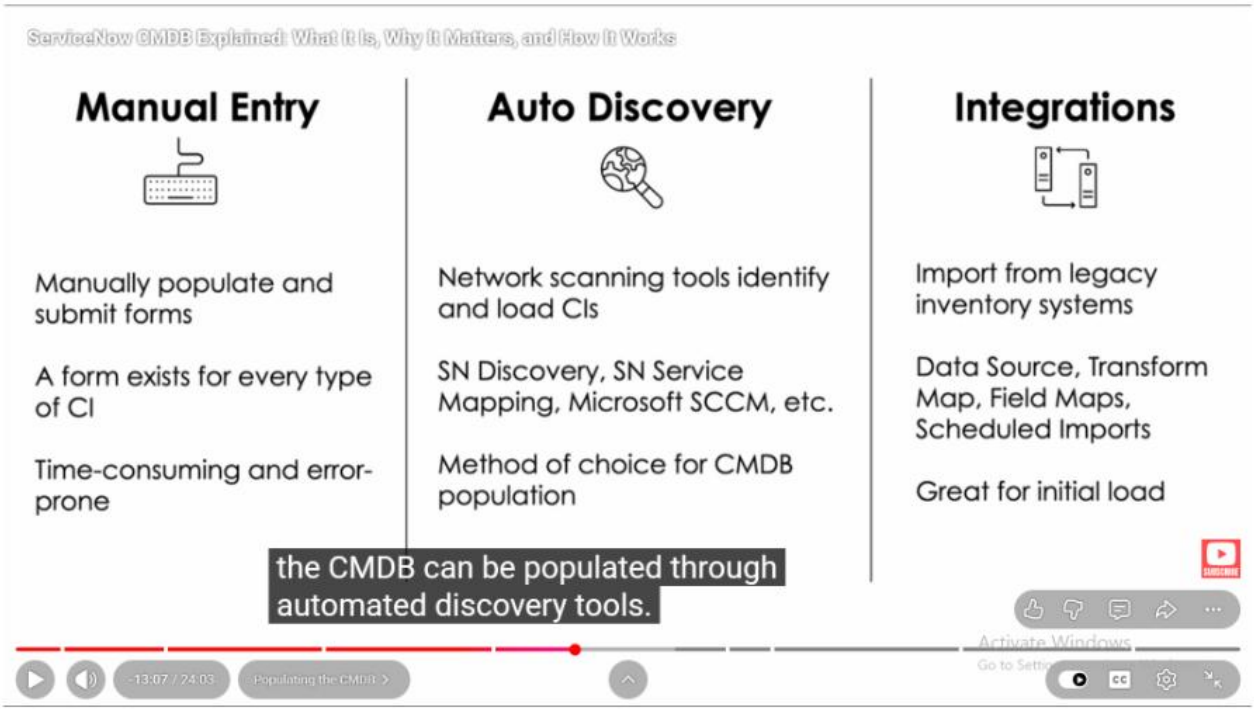
Data Source, Transform Map, Field Maps, Scheduled Imports

Great for initial load

Another option is to build imports and integrations to load the data.

Activate Windows
Go to Settings to activate Windows.



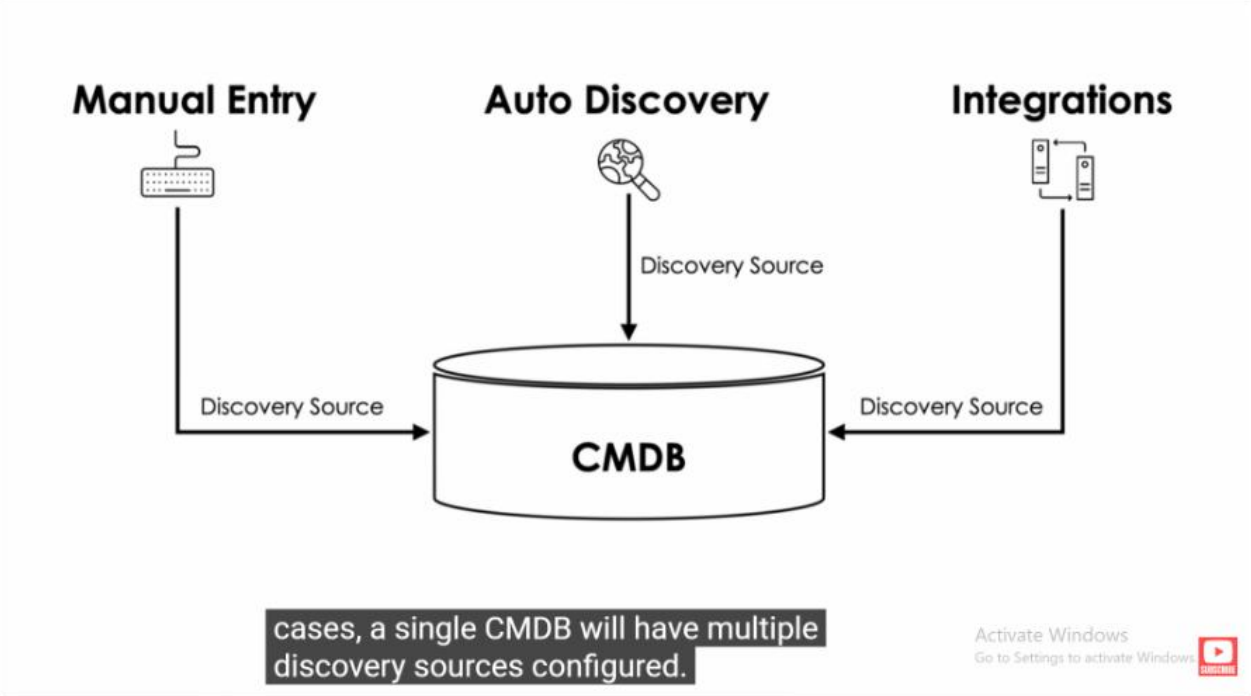


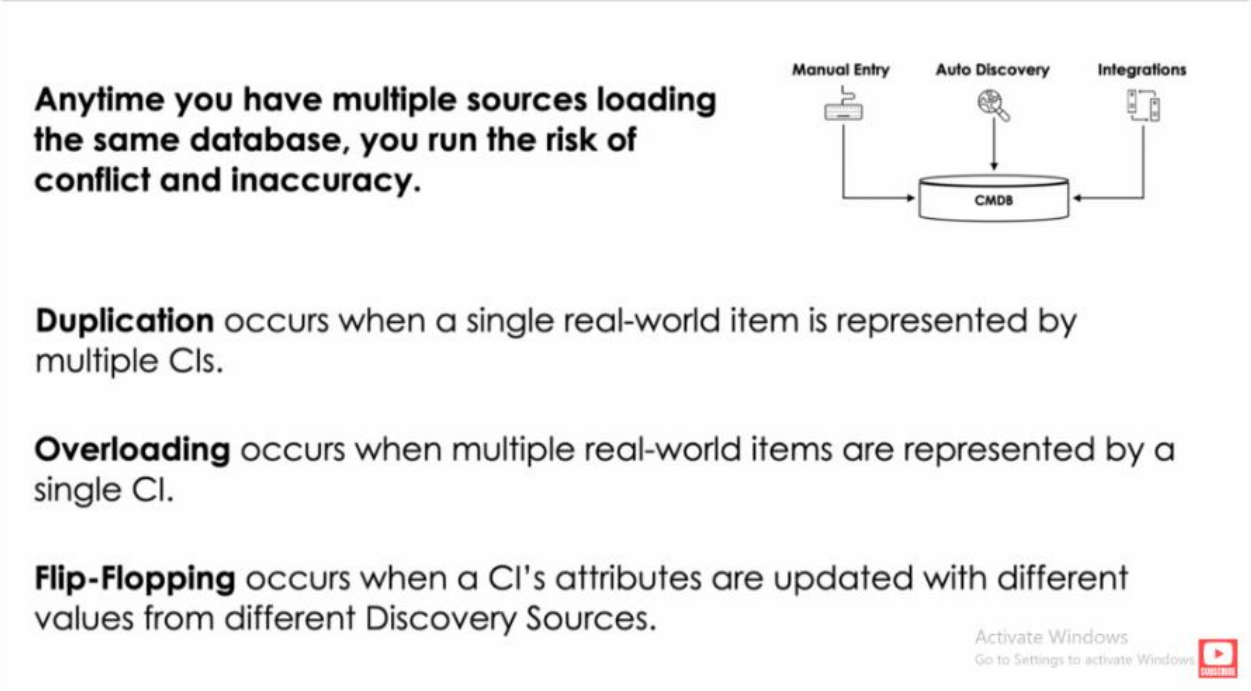
NOTE : Auto discovery is the mostly used method to populate CIs. In most of the cases this is the most accurate and preferred method of CI ingestion.

SNOW offers two of their own automated discovery tools named ; discovery, service mapping. The discovery tool identifies devices and software running on them. They call that horizontal discovery.
The service mapping tool performs vertical discovery. Which identifies relationships b/w CIs and maps how they work together to provide services.

It's also possible to use third party discovery tools like microsoft SCCM.

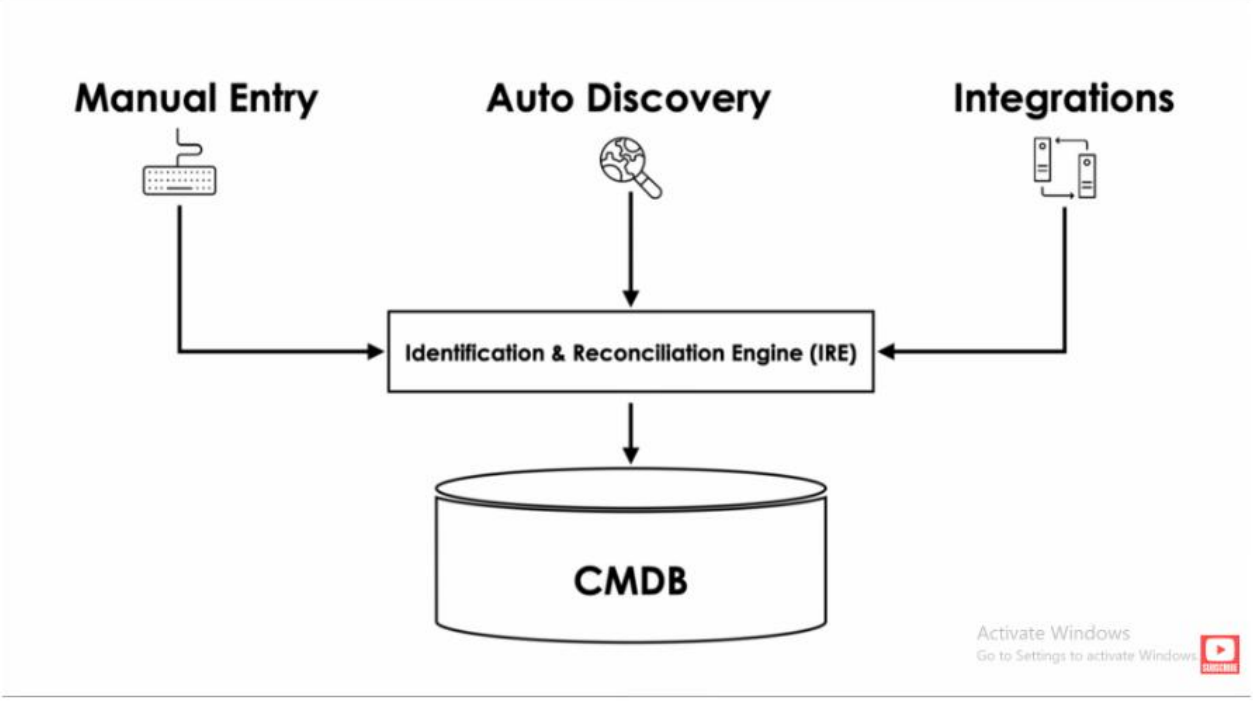
Each source of data entering the CMDB is called discovery source. And in most of the cases a single cmdb will have multiple discovery source configured.

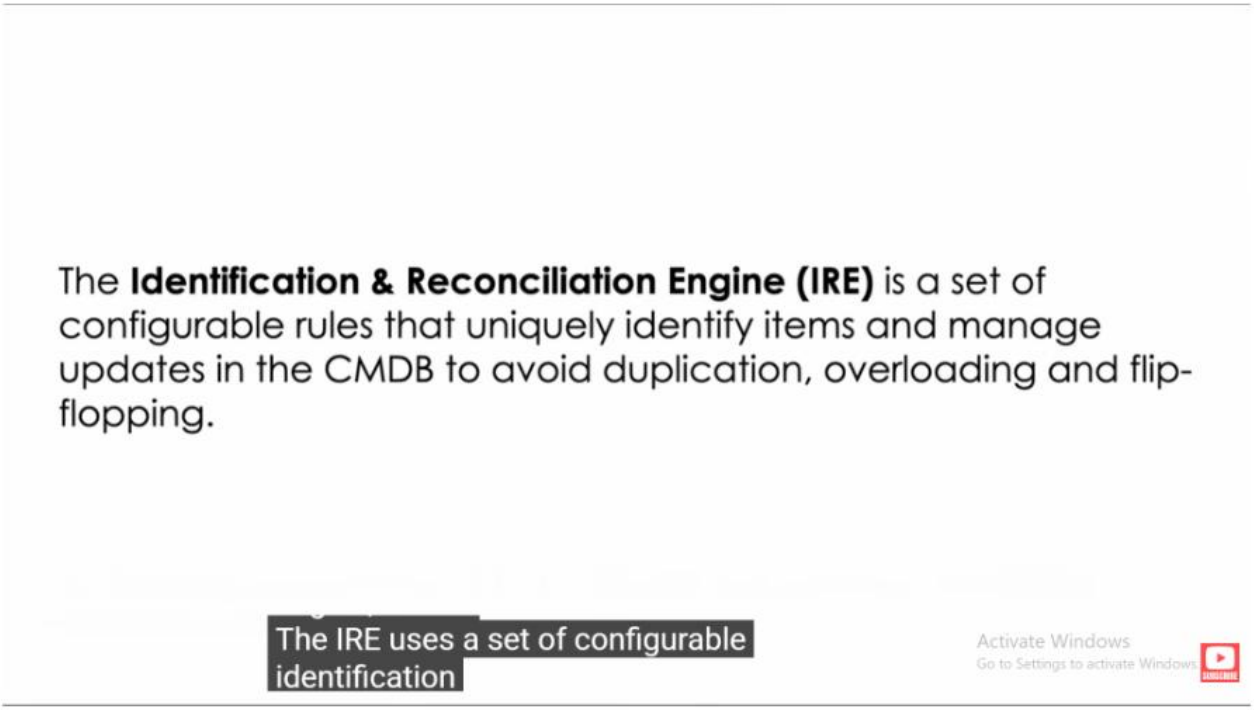




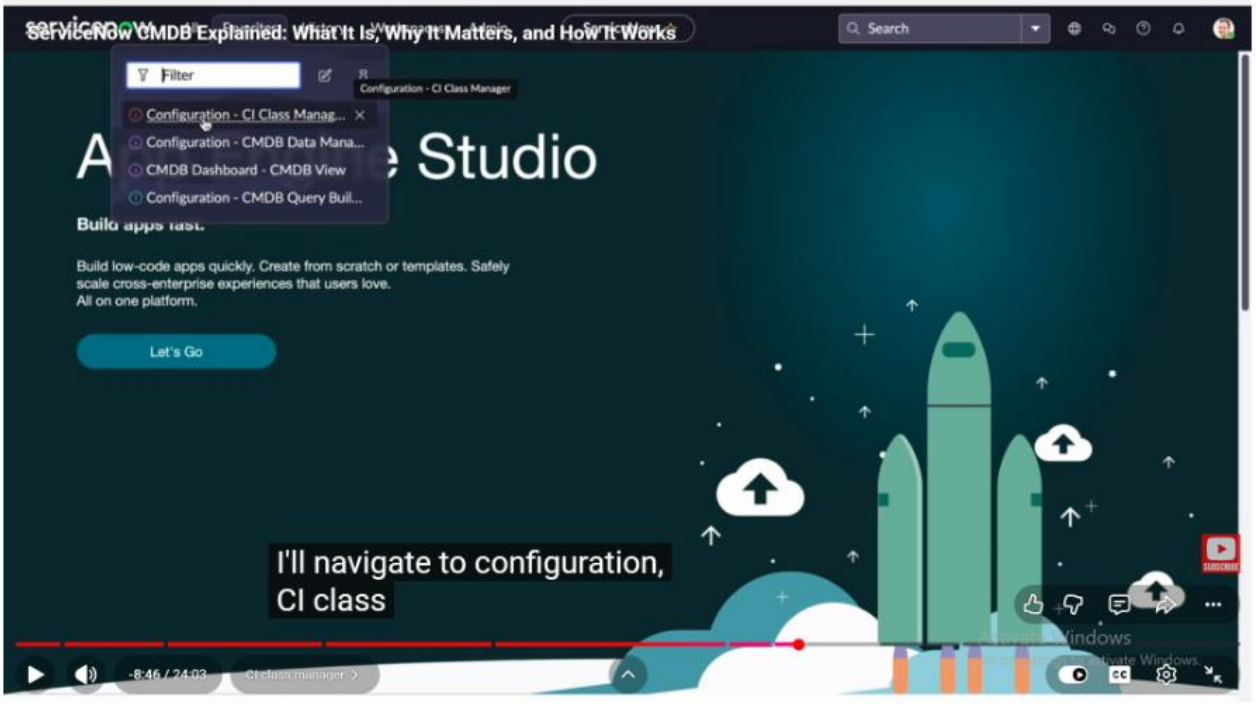
When multiple discovery sources load data for the same CI, but with different attribute values, the values gets updated repeatedly which is called flip flopping. And left by whichever source runs last.

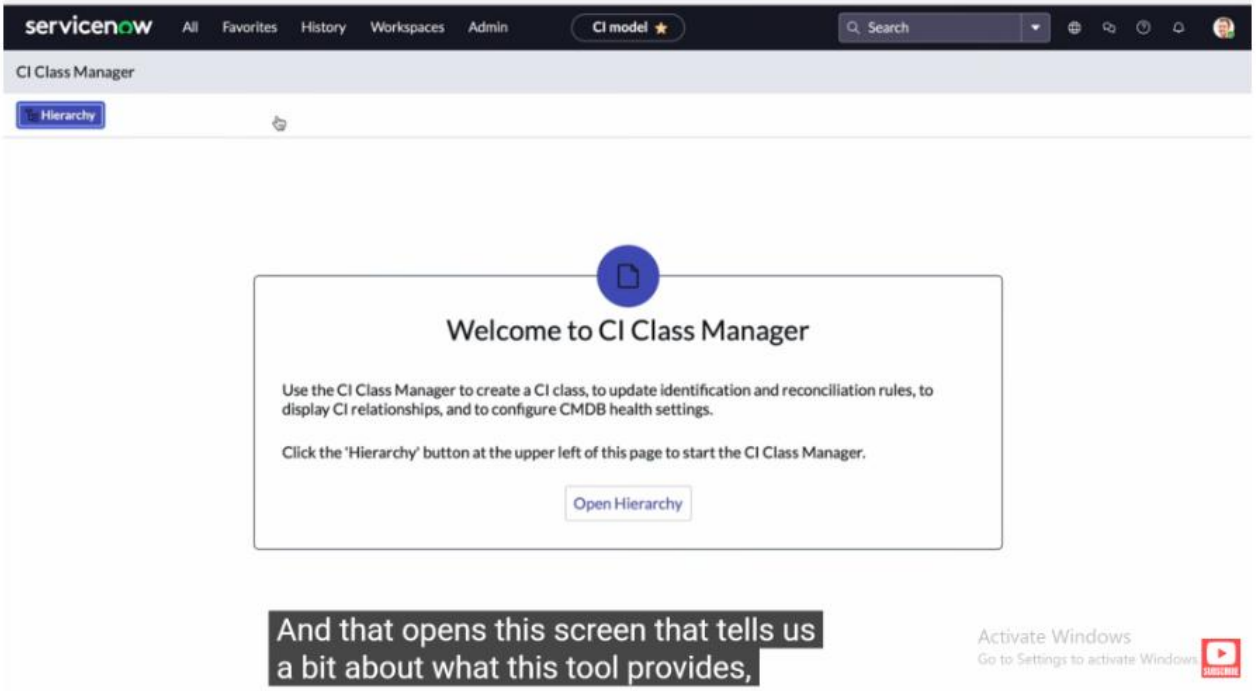
To solve all these problems, all data entering the cmdb must first pass through, what snow calls, the identification and reconciliation engine (IRE).



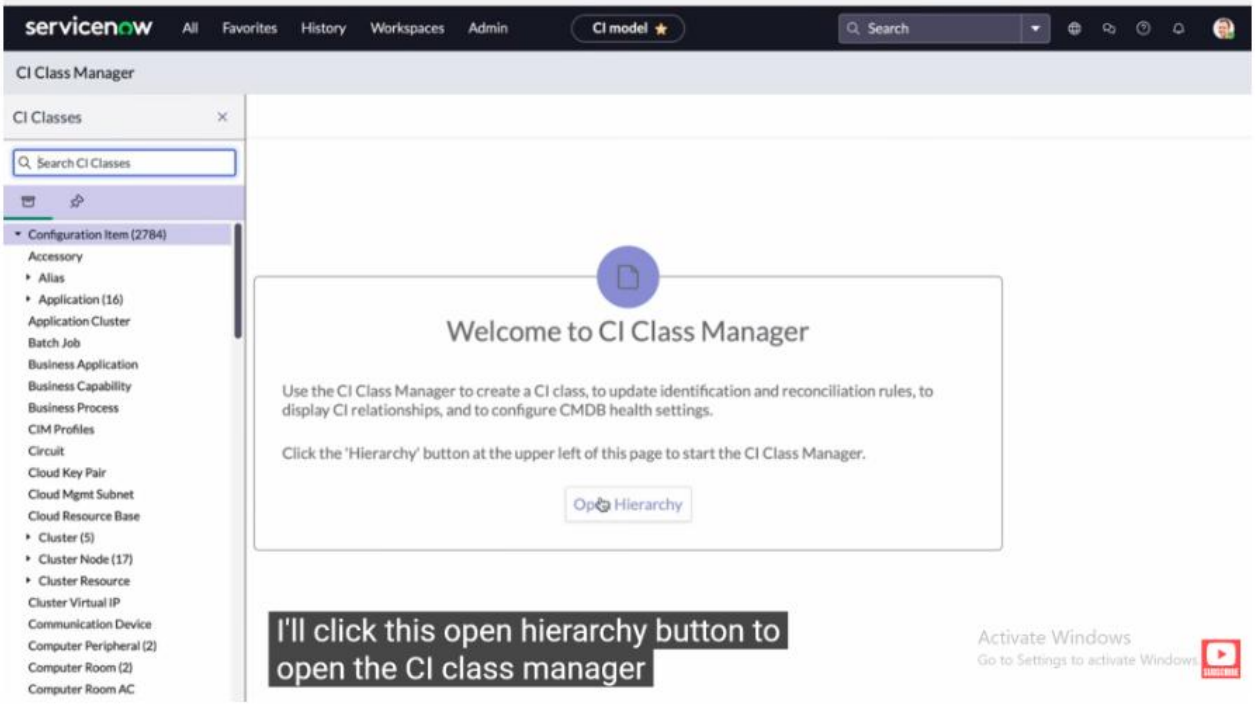


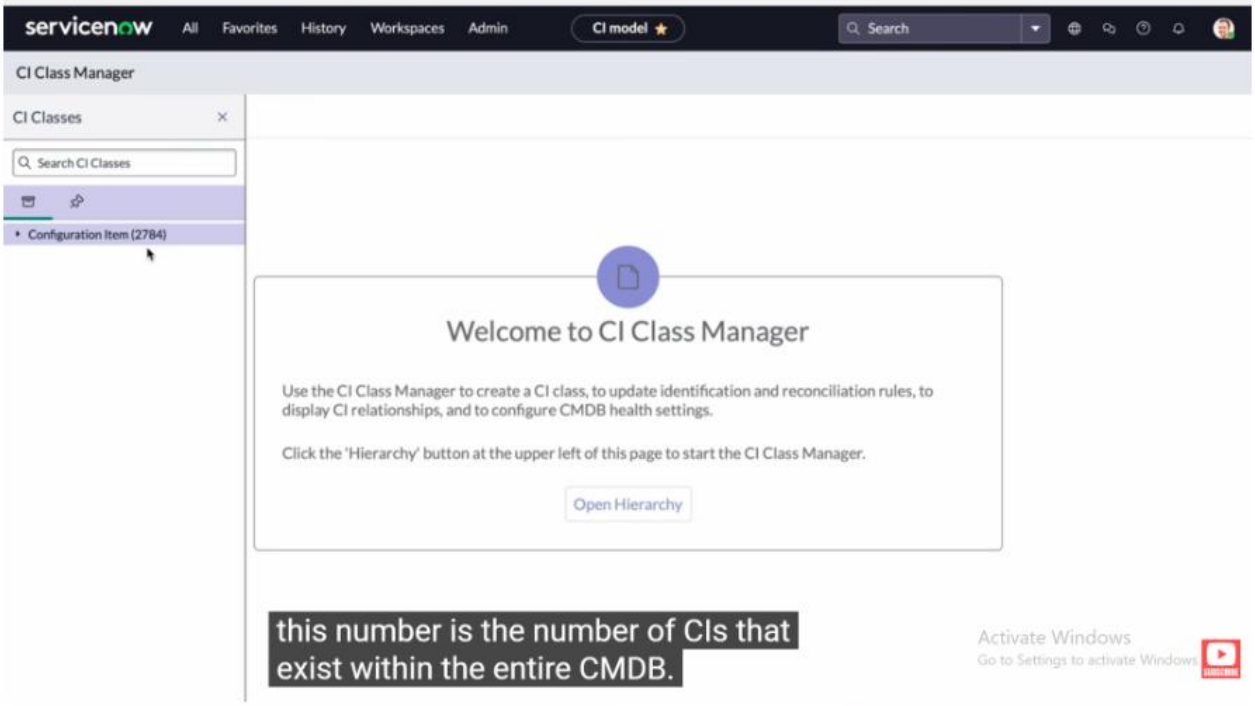
Now, let's move on the one of the important configuration module that is "CI Class manager". It is like the master front end of the CMDB and IRE.

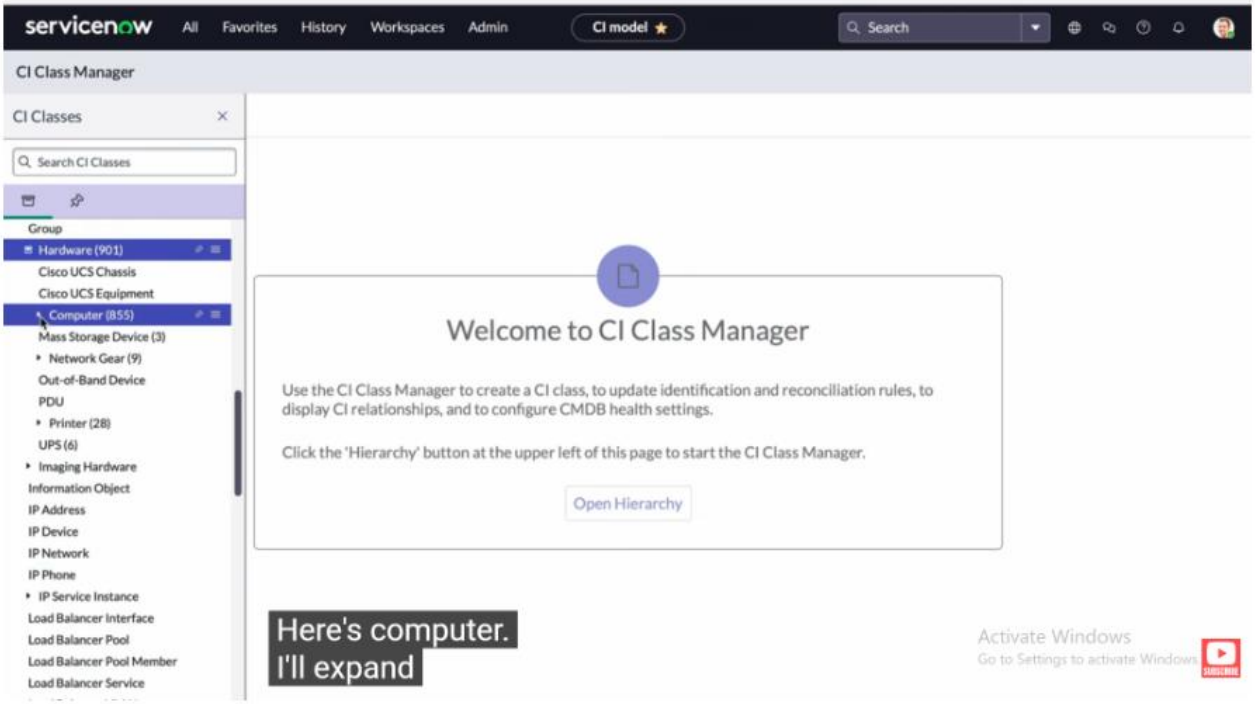


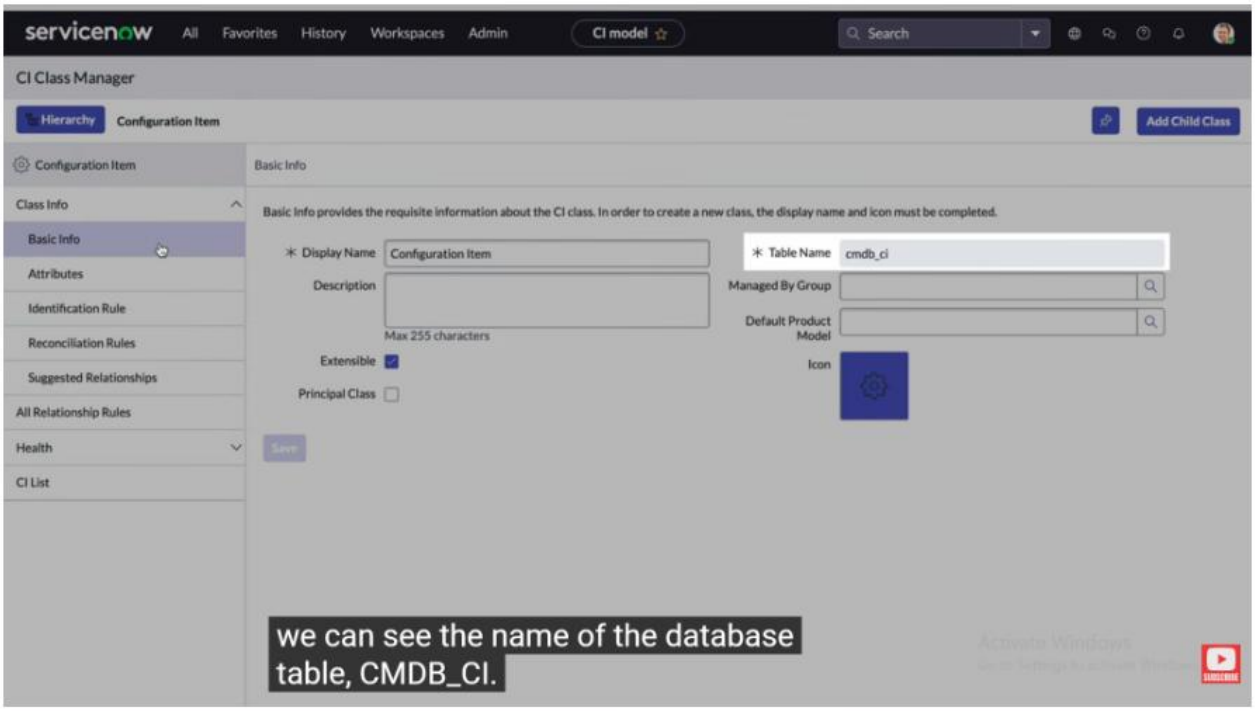


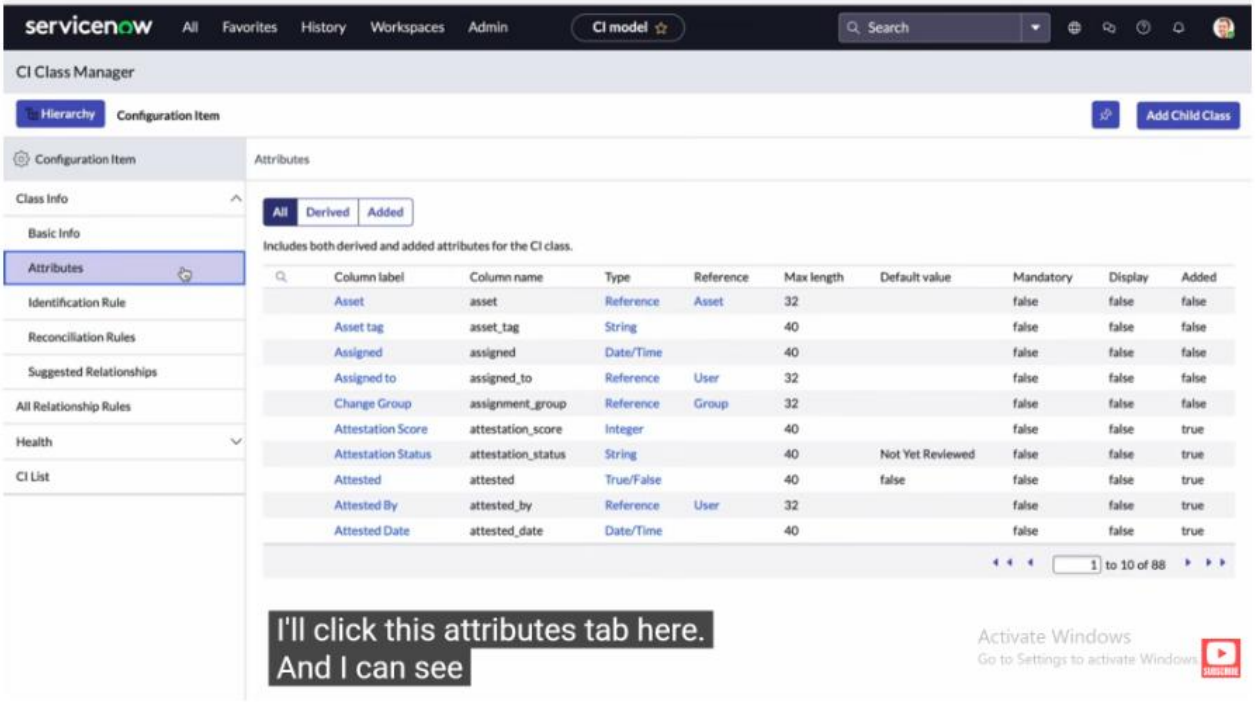
It provides you the ability to create new classes, update IRE rules, manage how relationships are configured and setup health settings.
For each class, there's a corresponding database table in CMDB

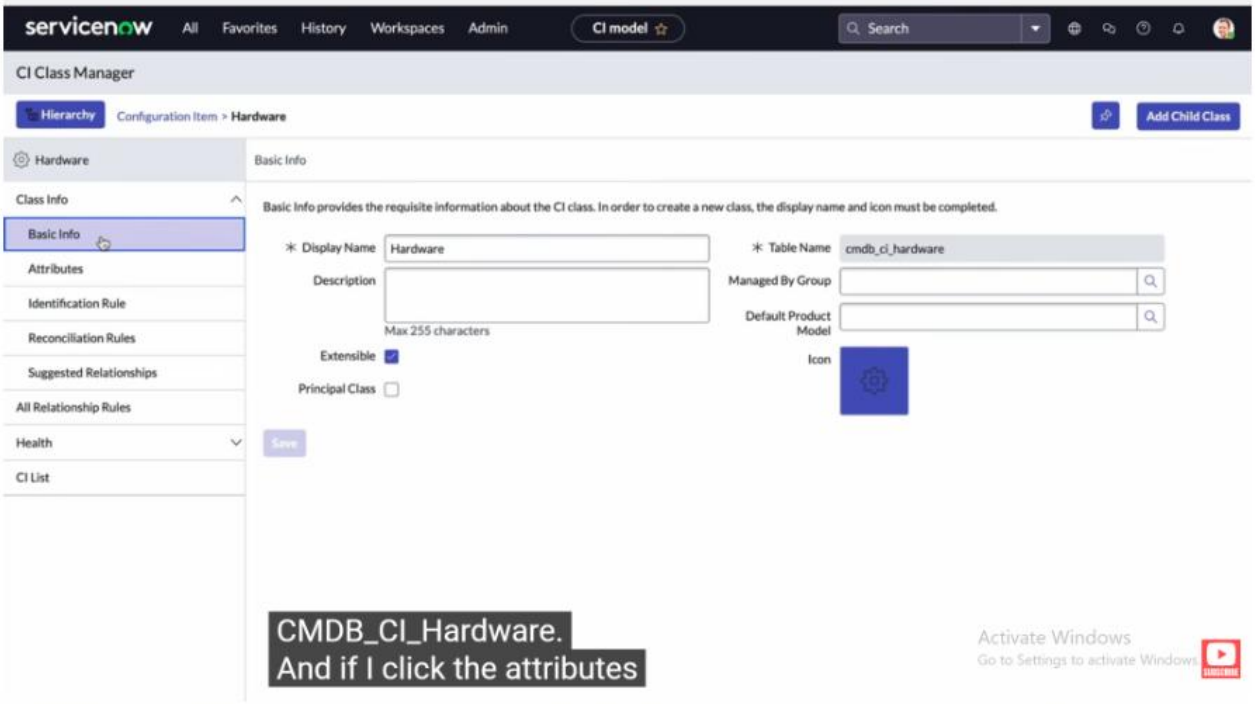

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
CI Classes
Search CI Classes
Configuration Item (2784)
Accessory
Alias
Application (16)
Application Cluster
Batch Job
Business Application
Business Capability
Business Process
CIM Profiles
Circuit
Cloud Key Pair
Cloud Mgmt Subnet
Cloud Resource Base
Cluster (5)
Cluster Node (17)
Cluster Resource
Cluster Virtual IP
Communication Device
Computer Peripheral (2)
Computer Room (2)
Computer Room AC
Welcome to CI Class Manager
Use the CI Class Manager to create a CI class, to update identification and reconciliation rules, to display CI relationships, and to configure CMDB health settings.
Click the 'Hierarchy' button at the upper left of this page to start the CI Class Manager.
Open Hierarchy
I'll click this open hierarchy button to open the CI class manager
Activate Windows
Go to Settings to activate Windows.

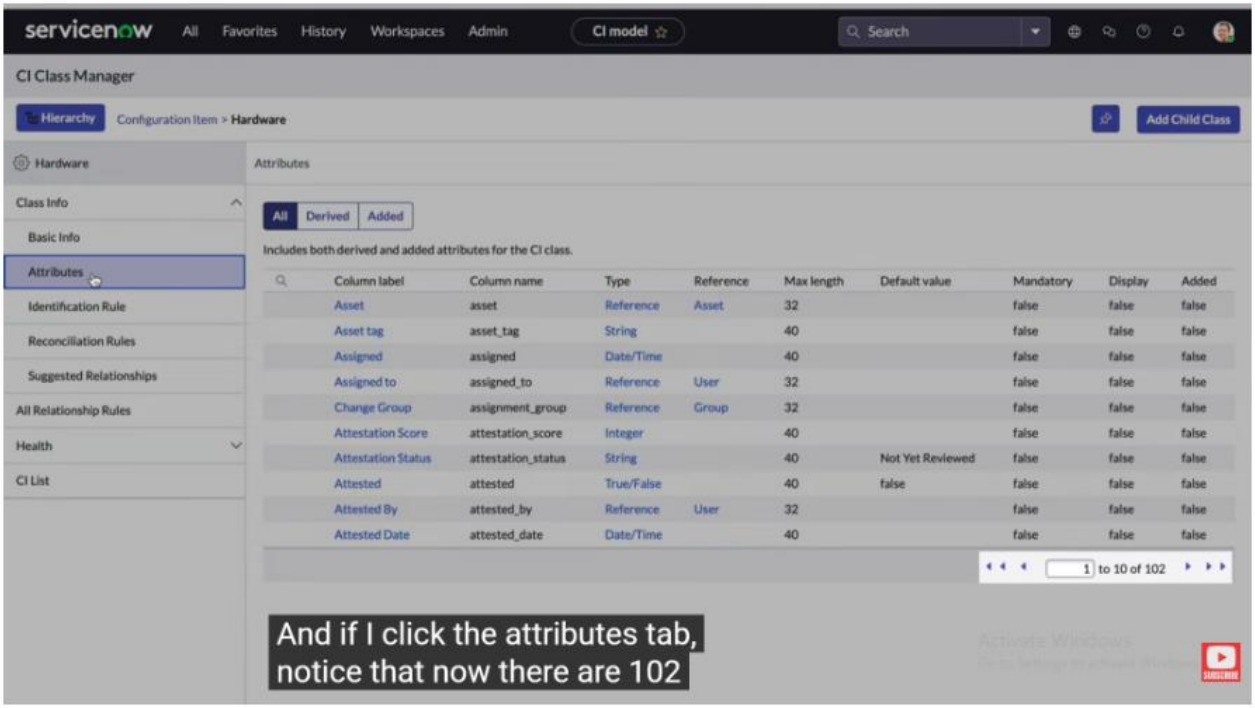

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
CI Classes
Search CI Classes
Configuration Item (2784)
Welcome to CI Class Manager
Use the CI Class Manager to create a CI class, to update identification and reconciliation rules, to display CI relationships, and to configure CMDB health settings.
Click the 'Hierarchy' button at the upper left of this page to start the CI Class Manager.
Open Hierarchy
this number is the number of CIs that exist within the entire CMDB.
Activate Windows
Go to Settings to activate Windows.

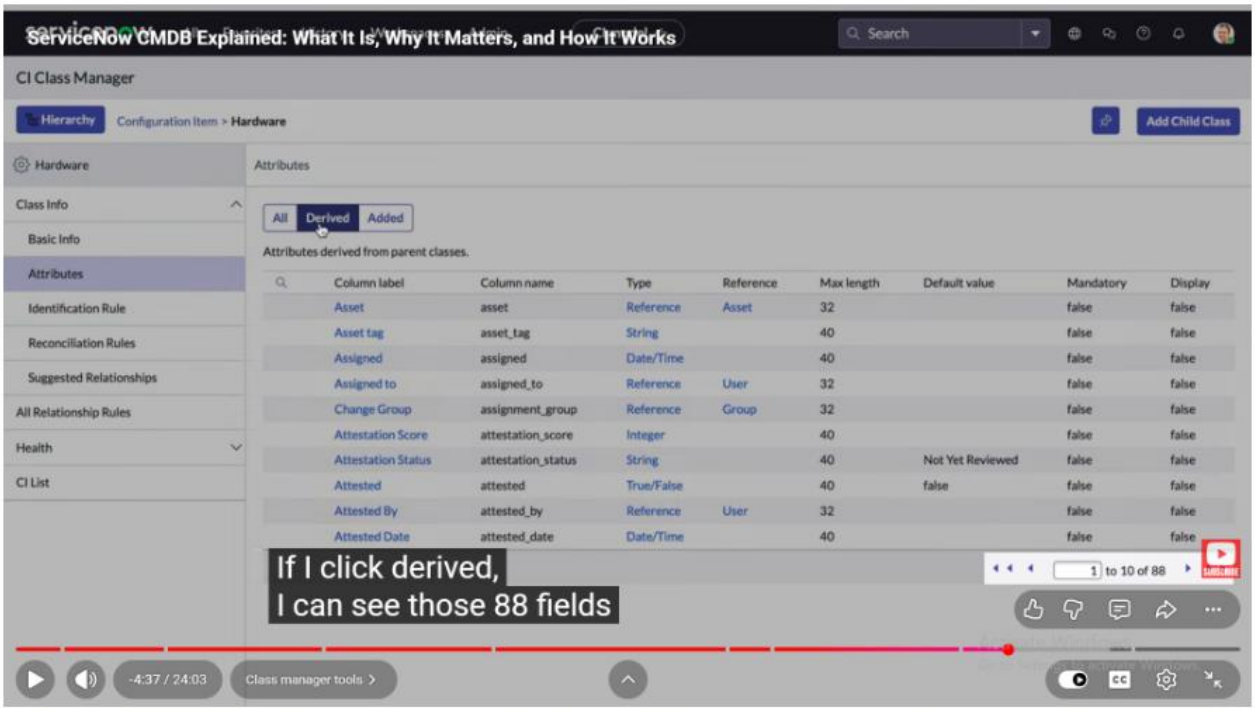

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
CI Classes
Search CI Classes
Group
Hardware (901)
Cisco UCS Chassis
Cisco UCS Equipment
Computer (855)
Mass Storage Device (3)
Network Gear (9)
Out-of-Band Device
PDU
Printer (28)
UPS (6)
Imaging Hardware
Information Object
IP Address
IP Device
IP Network
IP Phone
IP Service Instance
Load Balancer Interface
Load Balancer Pool
Load Balancer Pool Member
Load Balancer Service
Welcome to CI Class Manager
Use the CI Class Manager to create a CI class, to update identification and reconciliation rules, to display CI relationships, and to configure CMDB health settings.
Click the 'Hierarchy' button at the upper left of this page to start the CI Class Manager.
Open Hierarchy
Here's computer.
I'll expand
Activate Windows
Go to Settings to activate Windows.

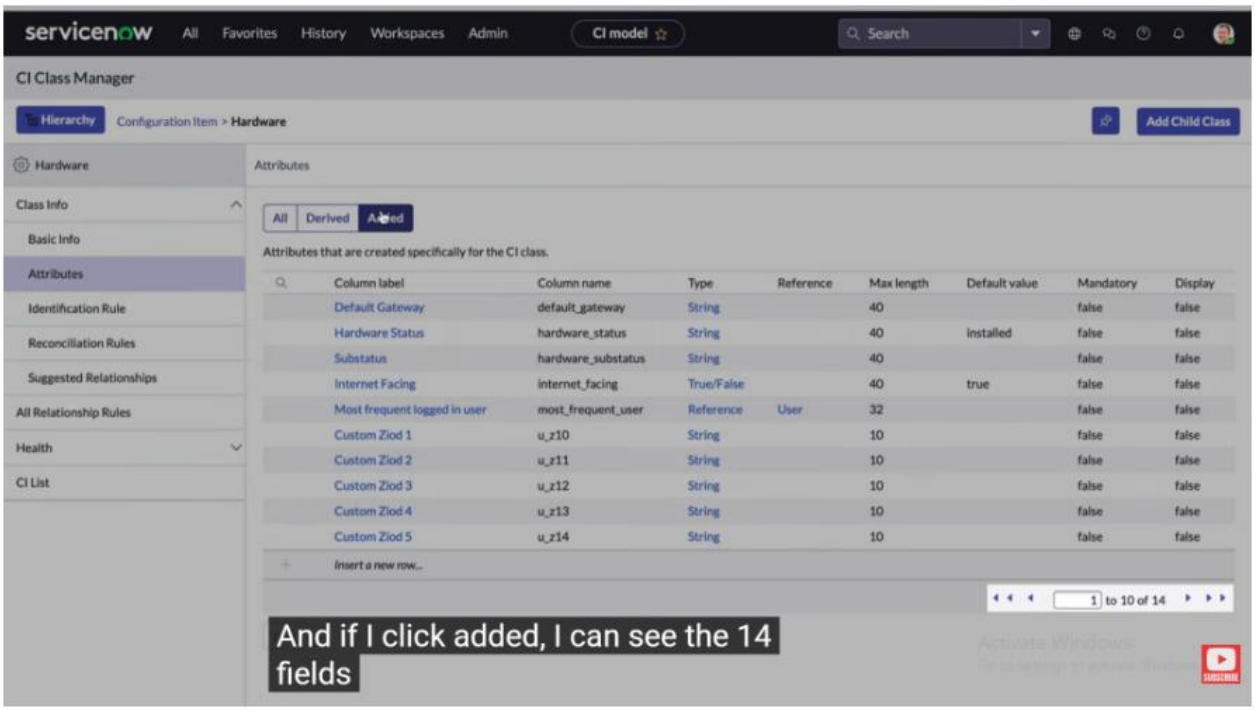

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item
Add Child Class
Configuration Item
Basic Info
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Basic Info provides the requisite information about the CI class. In order to create a new class, the display name and icon must be completed.
Display Name
Configuration Item
Description
Max 255 characters
Extensible
Principal Class
Table Name
cmdb_ci
Managed By Group
Default Product Model
Icon
Save
we can see the name of the database table, CMDB_CI.


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item
Add Child Class
Configuration Item
Attributes
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
All
Derived
Added
Includes both derived and added attributes for the CI class.
Column label
Column name
Type
Reference
Max length
Default value
Mandatory
Display
Added
Asset asset Reference Asset 32 false false false
Asset tag asset_tag String 40 false false false
Assigned assigned Date/Time 40 false false false
Assigned to assigned_to Reference User 32 false false false
Change Group assignment_group Reference Group 32 false false false
Attestation Score attestation_score Integer 40 false false true
Attestation Status attestation_status String 40 Not Yet Reviewed false false true
Attested attested True/False 40 false false false true
Attested By attested_by Reference User 32 false false true
Attested Date attested_date Date/Time 40 false false true
1 to 10 of 88
I'll click this attributes tab here.
And I can see

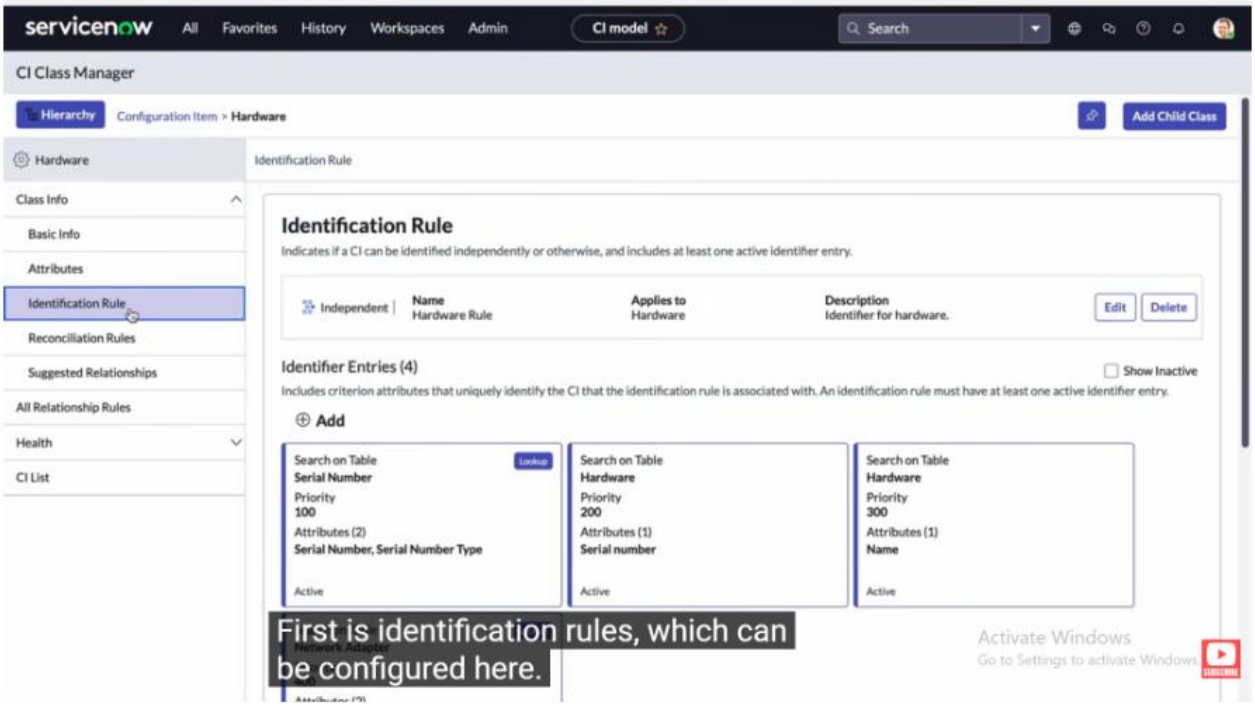

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item > Hardware
Add Child Class
Hardware
Basic Info
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Basic Info provides the requisite information about the CI class. In order to create a new class, the display name and icon must be completed.
Display Name
Hardware
Table Name
cmdb_ci_hardware
Description
Max 255 characters
Managed By Group
Default Product Model
Extensible
Principal Class
Icon
Save
CMDB_CI_Hardware.
And if I click the attributes
Activate Windows
Go to Settings to activate Windows.


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item > Hardware
Add Child Class
Hardware
Attributes
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
All
Derived
Added
Includes both derived and added attributes for the CI class.
Column label
Column name
Type
Reference
Max length
Default value
Mandatory
Display
Added
Asset asset Reference Asset 32 false false false
Asset tag asset_tag String 40 false false false
Assigned assigned Date/Time 40 false false false
Assigned to assigned_to Reference User 32 false false false
Change Group assignment_group Reference Group 32 false false false
Attestation Score attestation_score Integer 40 false false false
Attestation Status attestation_status String 40 Not Yet Reviewed false false false
Attested attested True/False 40 false false false false
Attested By attested_by Reference User 32 false false false
Attested Date attested_date Date/Time 40 false false false
1 to 10 of 102
And if I click the attributes tab,
notice that now there are 102

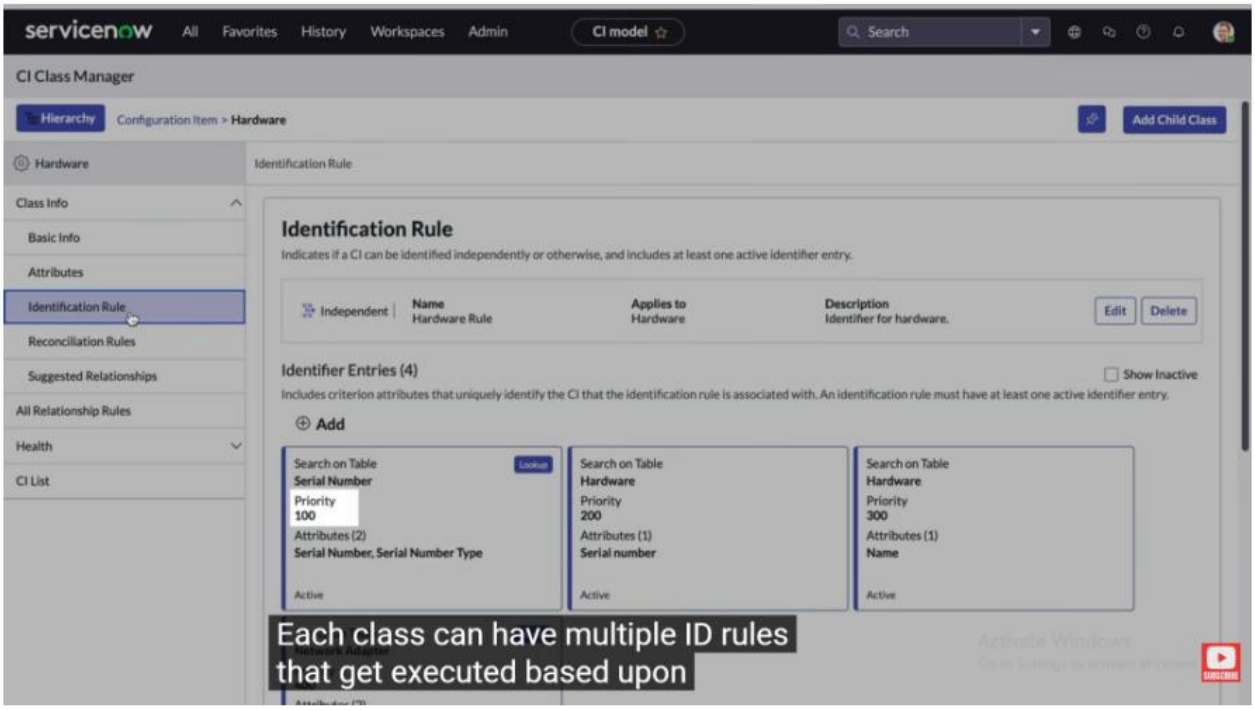




There are a lot of things, i can show some of the important things. Like this identification rule :

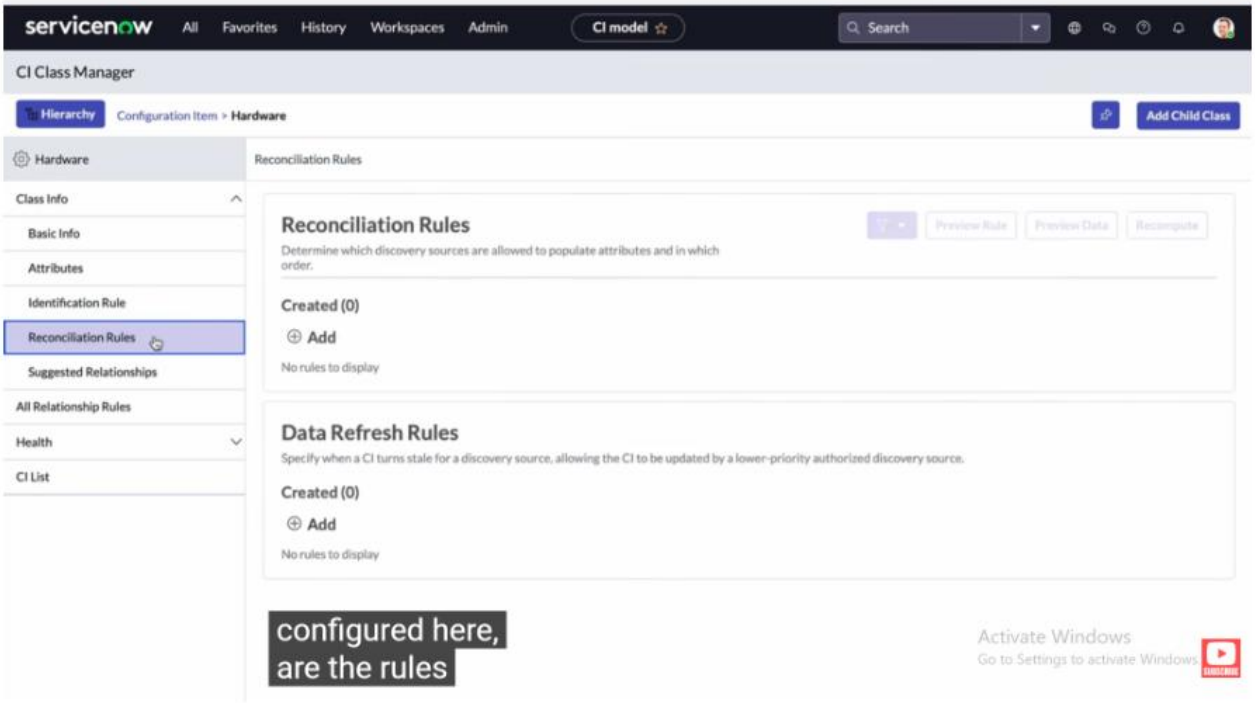


Here we see the working of IRE.
An identification rule is a rule that tells the IRE which attributes or fields to use to uniquely identify an item in that CMDB table.

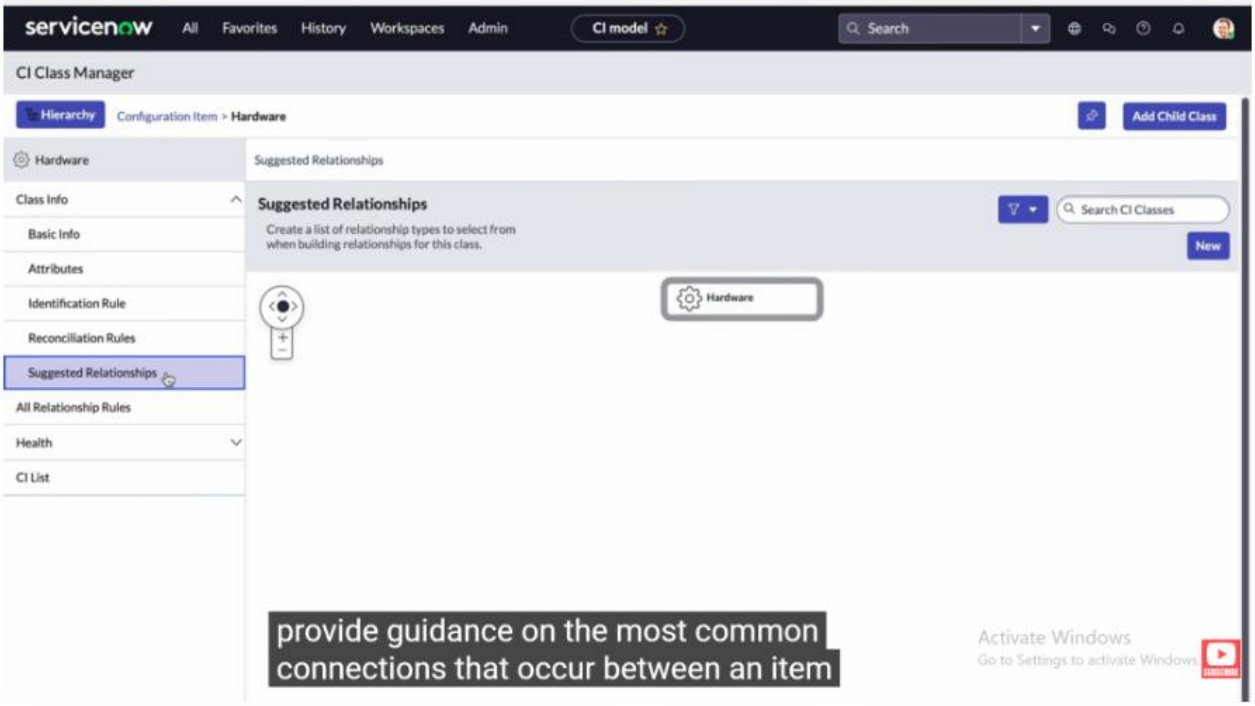


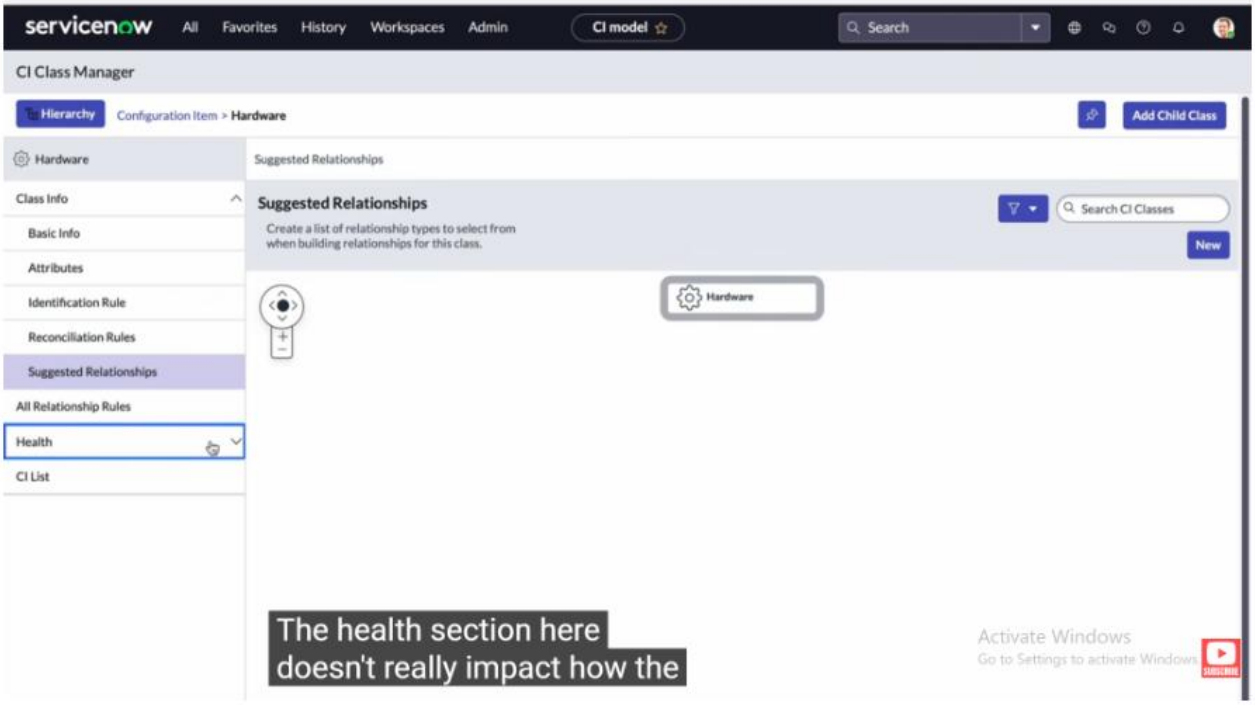
So, when a discovery source presents an item for entry into the cmdb, the identification rules that are set here are executed by the IRE to check and see if a CI is already exists for that item.

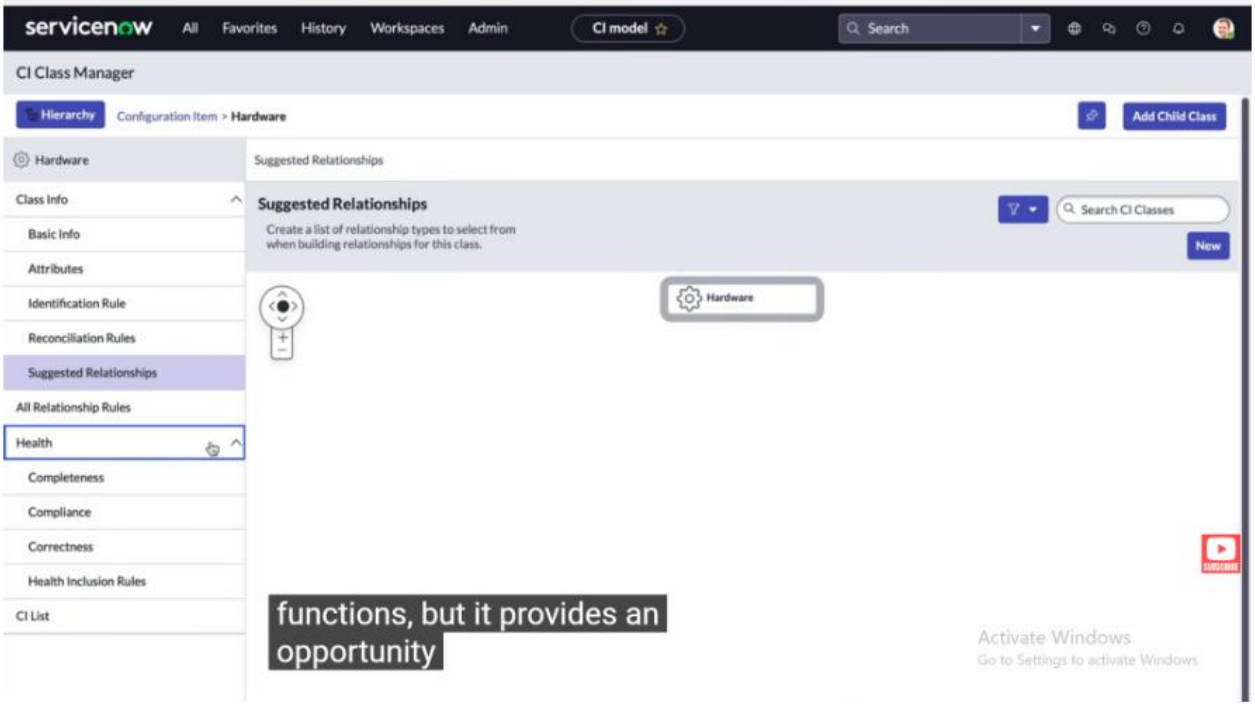
And if a match is found that existing ci gets updated with any changes. And if match is not found a new record is created in the corresponding table.

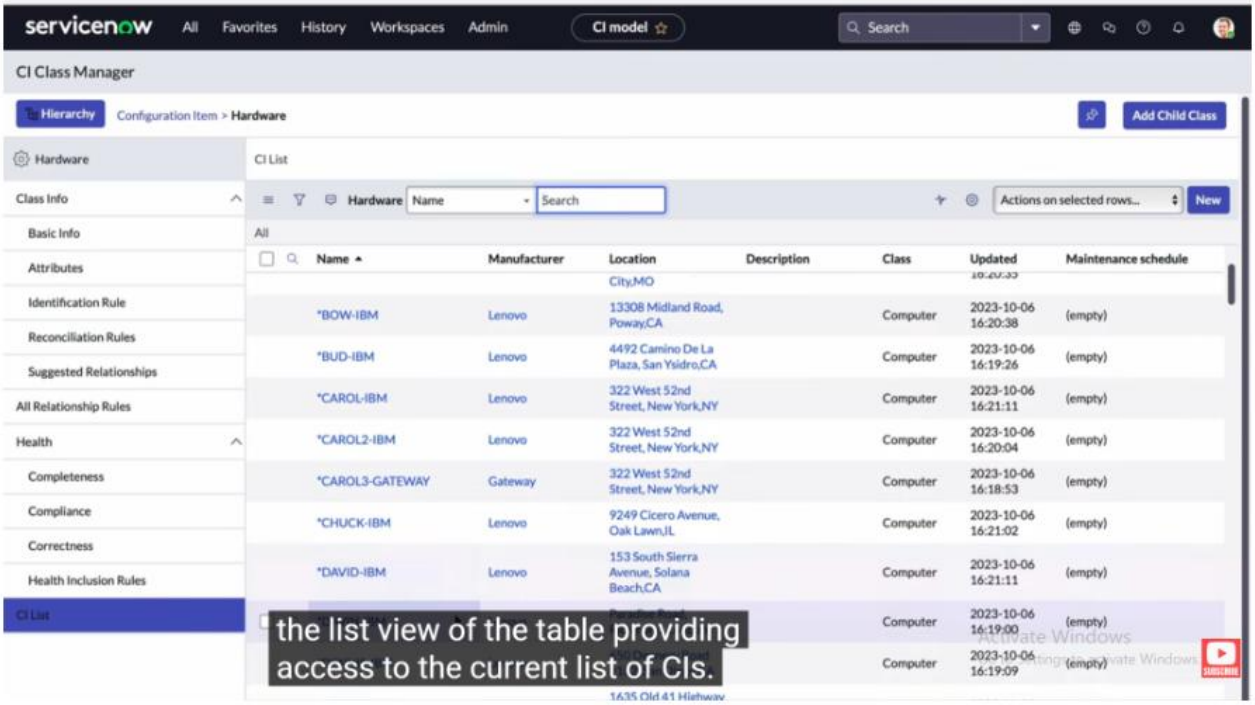


Reconciliation rules which are configured here, tells the IRE how to proceed when making updates to an existing CI. These are all about managing flip flopping that i mentioned earlier. When we have two or more discovery sources telling us different things about a CI, which one should we trust the most? That's the question that reconciliation rules answer. The rules can be applied at the record level or even at the individual field level.


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item > Hardware
Add Child Class
Hardware
Suggested Relationships
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Suggested Relationships
Create a list of relationship types to select from when building relationships for this class.
Search CI Classes
New
Hardware
provide guidance on the most common connections that occur between an item
Activate Windows
Go to Settings to activate Windows.

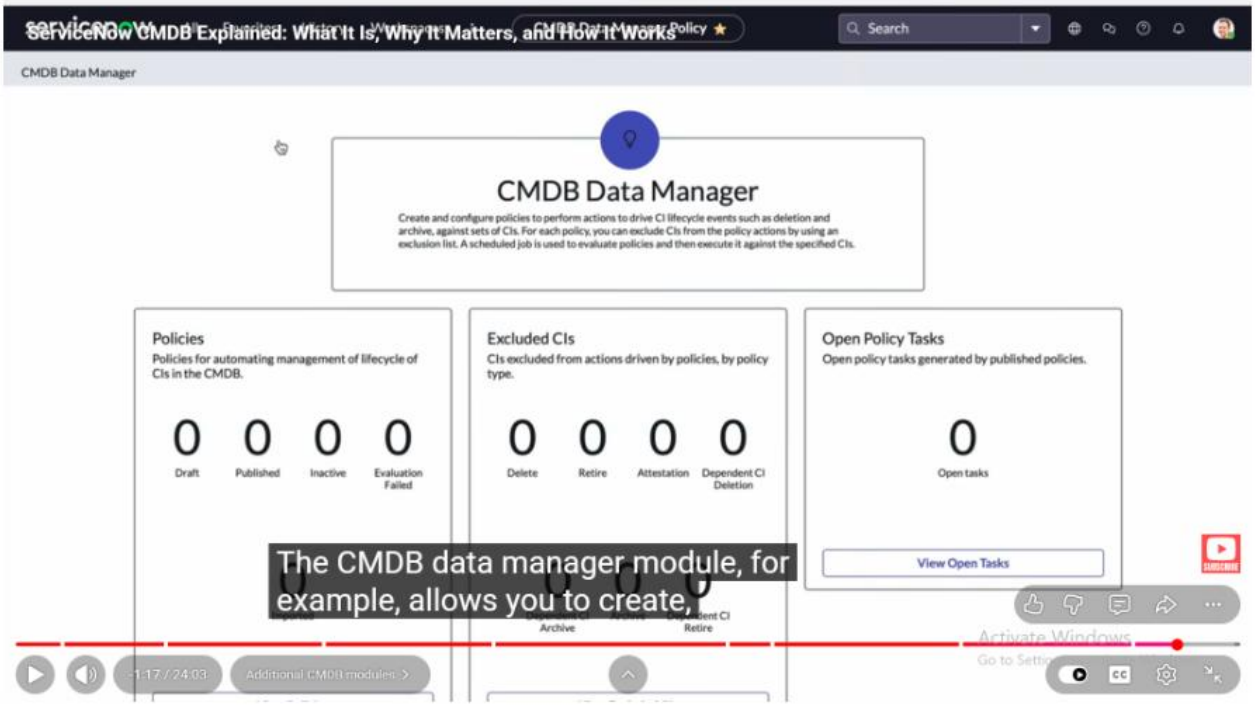

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Search
CI Class Manager
Hierarchy
Configuration Item > Hardware
Add Child Class
Hardware
Suggested Relationships
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Suggested Relationships
Create a list of relationship types to select from when building relationships for this class.
Search CI Classes
New
Hardware
The health section here doesn't really impact how the
Activate Windows
Go to Settings to activate Windows.



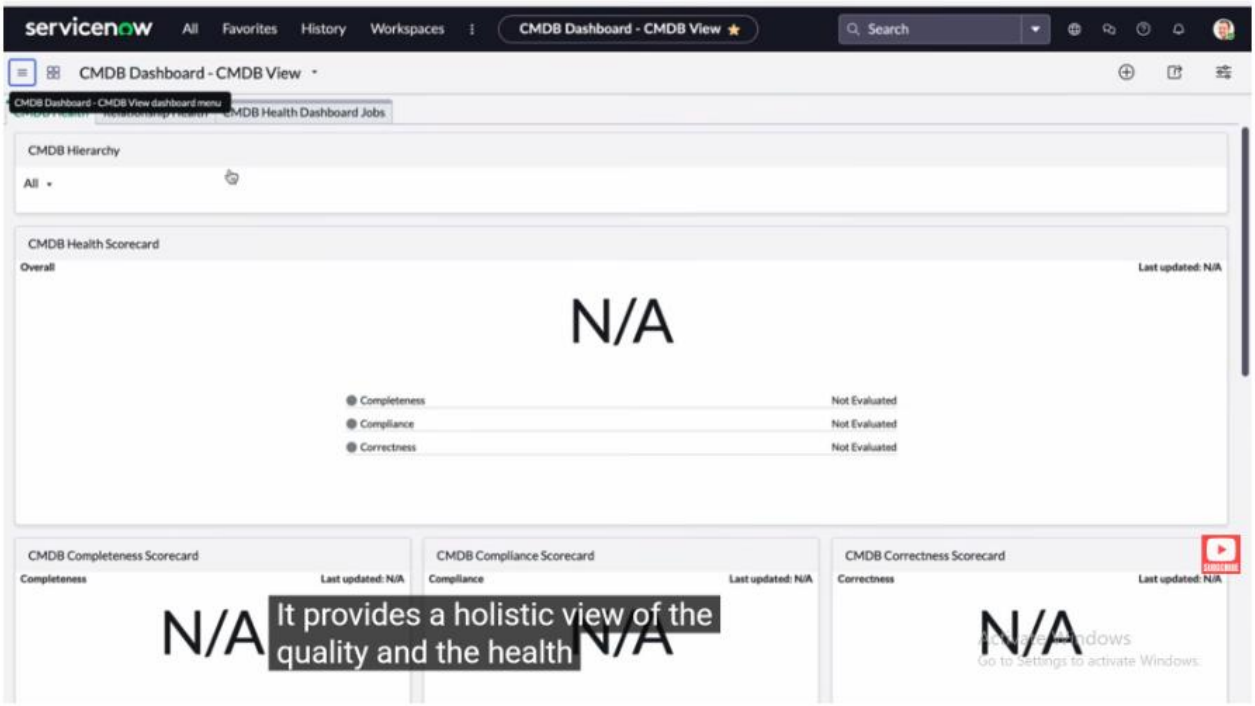


NOTE : this is one of over 50 modules

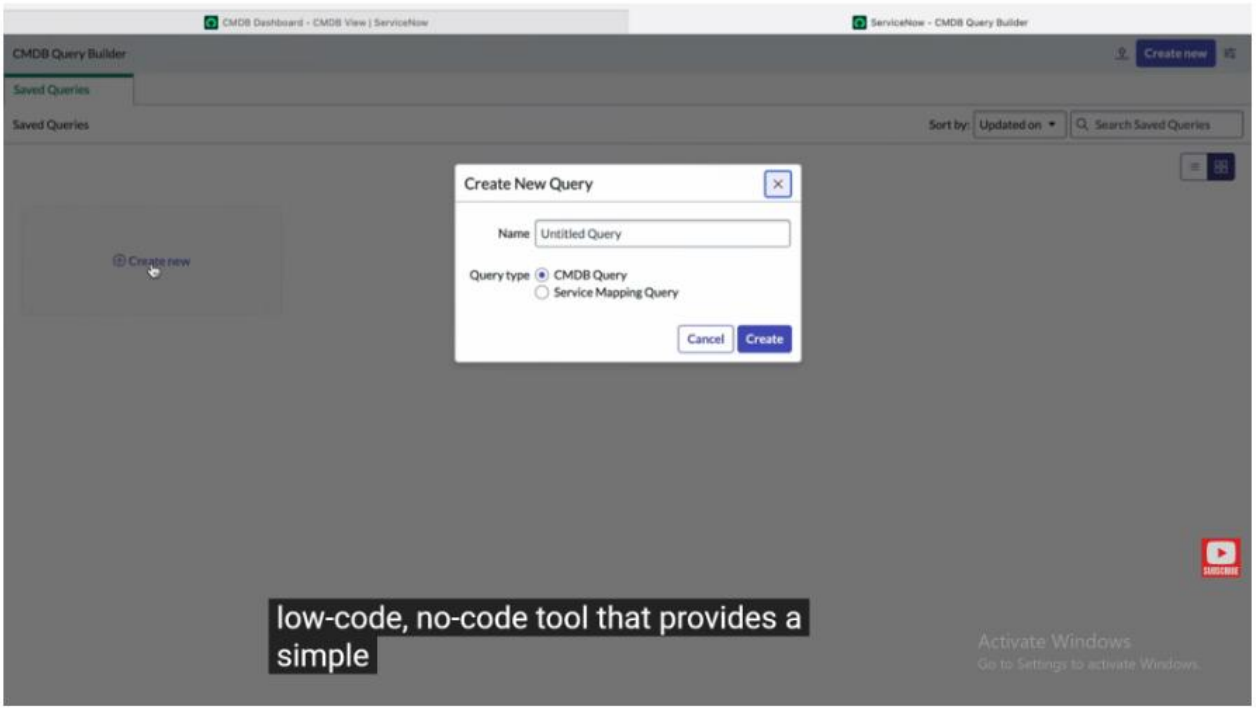
For eg. another module is cmdb data manager that allows you to create , publish, and manage ci policies to help maintain the cmdb as it grows

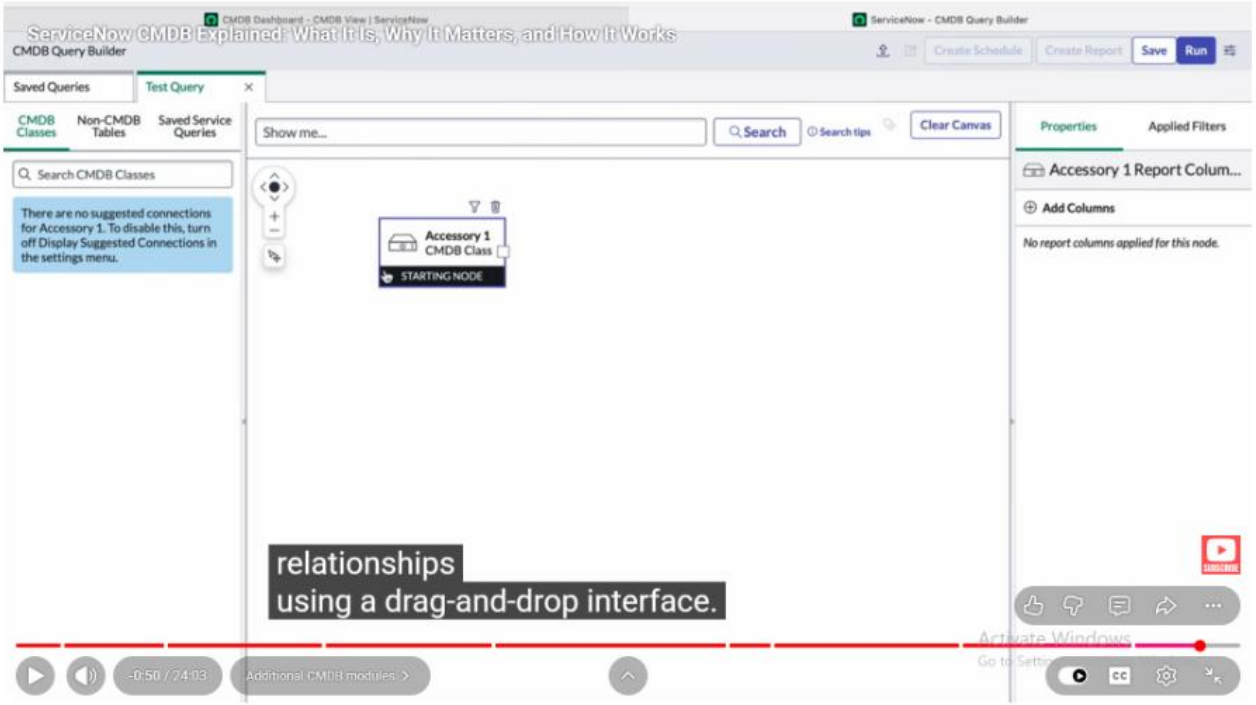


Another module is cmdb view :

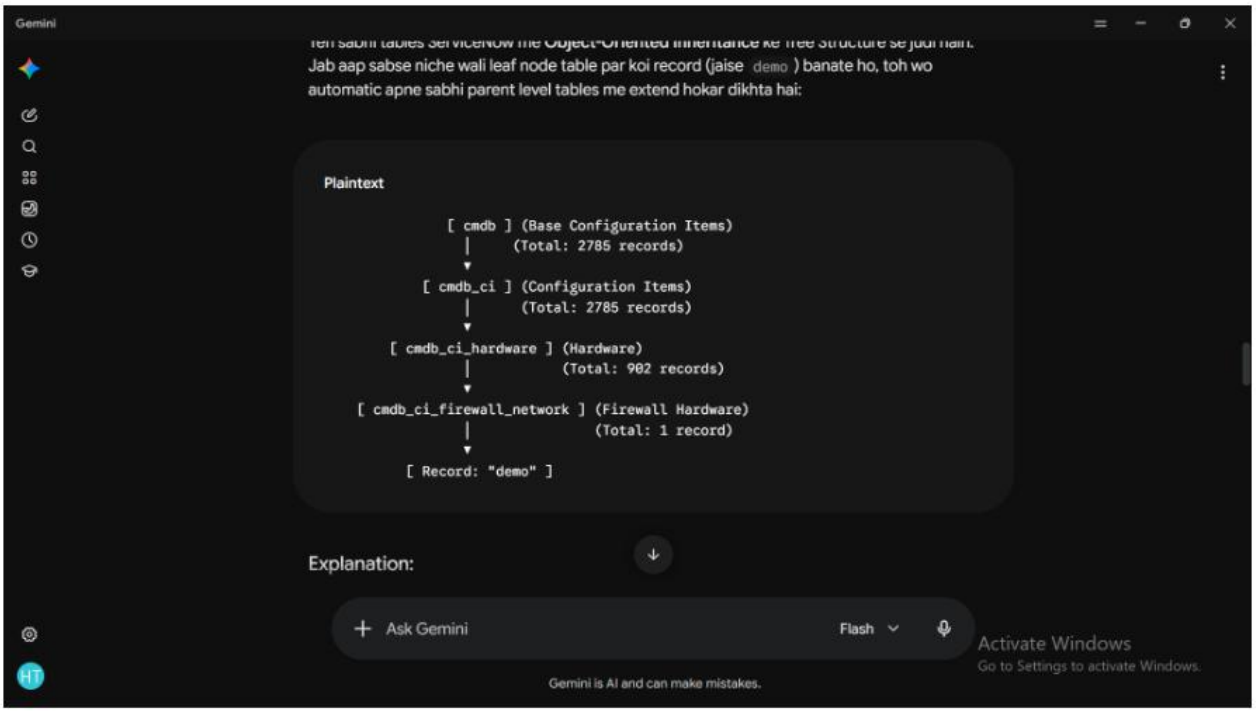


Another module is cmdb query builder :





HOW CMDB, CMDB_CI, HARDWARE, AND OTHER ITS CLASSES CONNECTED ?



Question : cmdb and cmdb_ci dono same hi hai lgta hai, kyuki dono me same number of records hi hai . yaa koi orr bhi difference hai ? mujhe lgta hai ki cmdb me kuch predefined fields hai orr cmdb ci unn fields ko to inherit kar hi rha hai or iski khud ki bhi kuch fields hai and aage iske child bhi yhi kar rhe hai ? aisa hai kya ? in veyr short

Answer:
Searched for ServiceNow CMDB base table hierarchy non cmdb_ci tables cmdb_rel cmdb_baseline to provide an accurate breakdown of tables under cmdb.

Aapka doubt bilkul valid hai.

cmdb base table me cmdb_ci ke alawa aur kya hota hai?

cmdb base root table se direct 2 major branches nikalti hain:

1. **IT Configuration Items (cmdb_ci)**: Isme saare IT assets (hardware, software, servers, IP switches, databases) aate hain.
2.
3. **Non-IT / Non-Standard Base Classes**: Jo core IT assets ka part nahi hain, wo direct cmdb se extend hotey hain.
4.

Non-CI Base Classes aur Metadata Tables kon-kon si hain?

1. **Non-IT Asset Classes**:

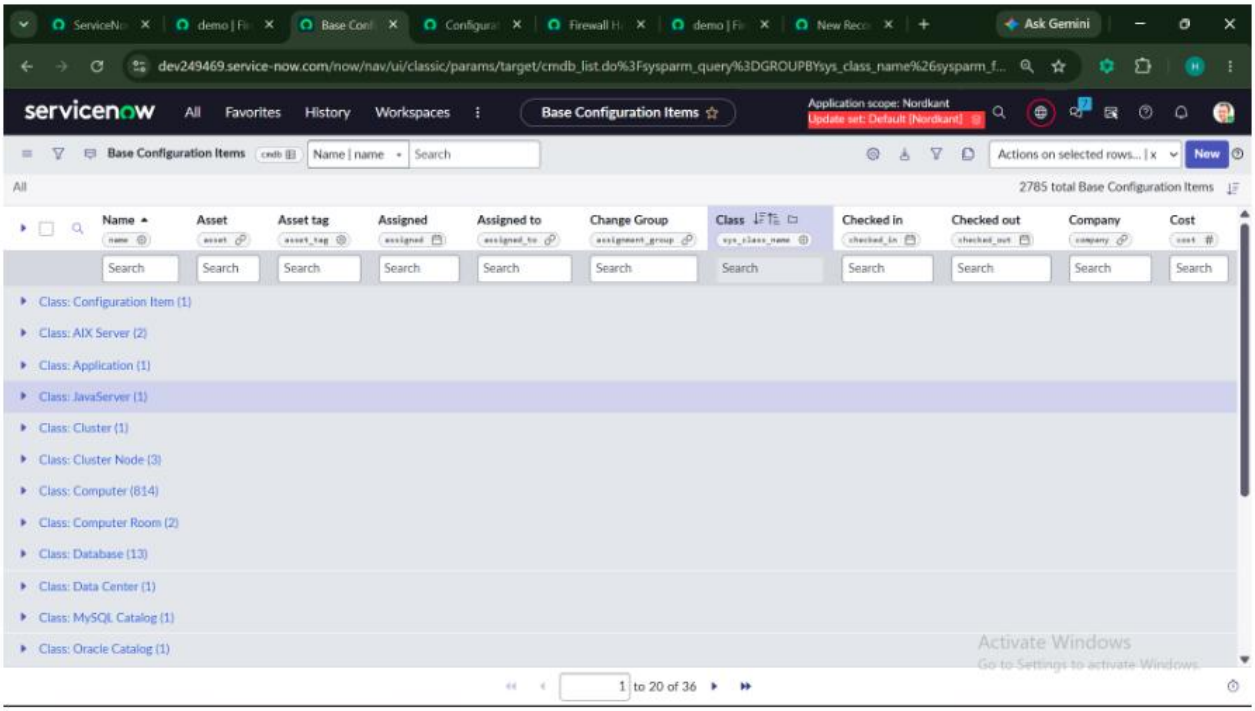
2.

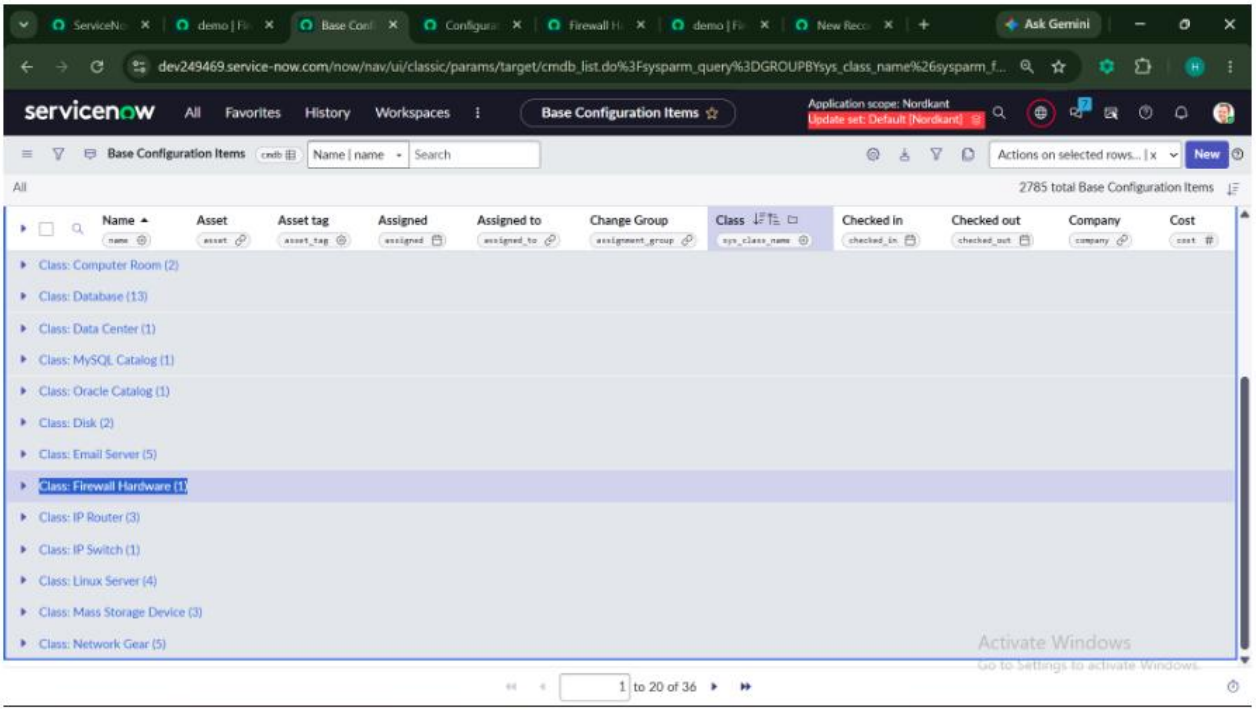
   - **Facility / Physical Location Items**: Physical infrastructure items (jaise AC units, generators, physical lockers) jo direct cmdb level parent se belong karti hain.
   -
   - **Operational Data**: Non-technical configuration entities jo specific IT CIs nahi hain.
   -

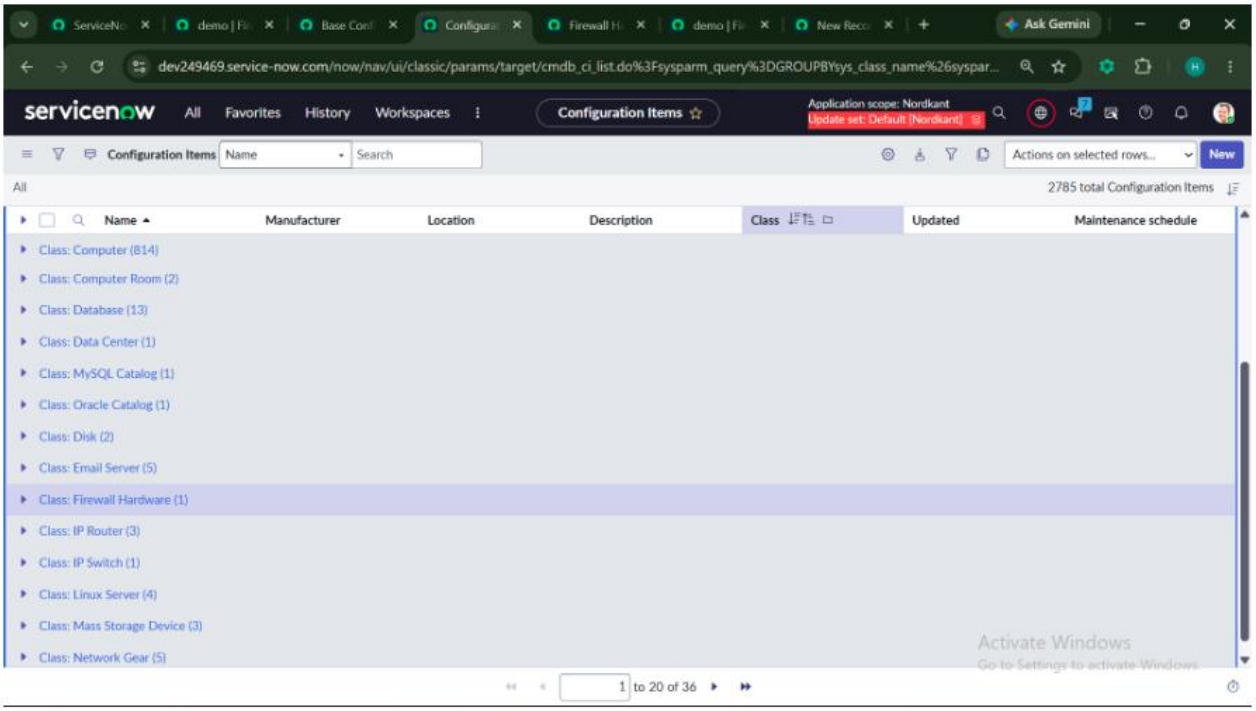
3. **CMDB Metadata & Relationship Tables (Base CMDB Framework)**:
4.

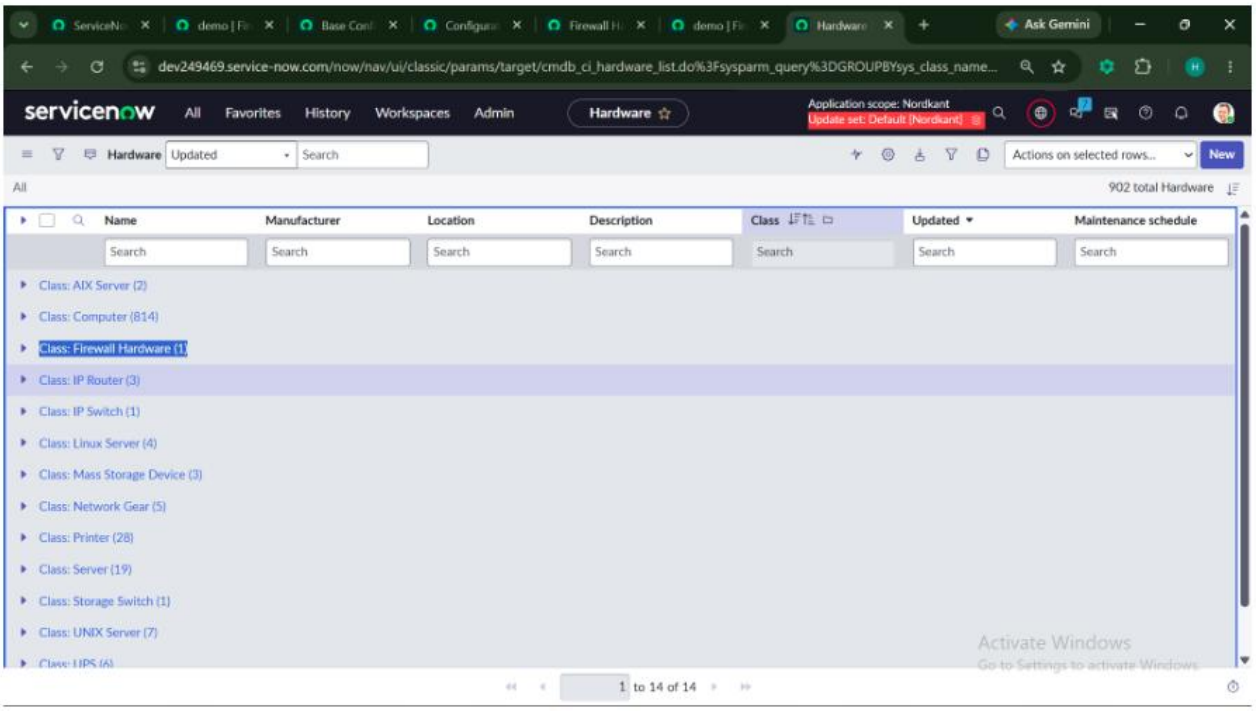
   - **CI Relationships (cmdb_rel_ci)**: CIs ke aapas ke connections aur dependency mappings (jaise *Server hosting Database*).
   -
   - **CMDB Baselines (cmdb_baseline)**: CIs ke purane vs naye configuration state ko track karne ke snapshots.
   -
   - **CI Audits & Health (cmdb_health_metric / cmdb_audit)**: CMDB data ki quality, duplicate identification, aur staleness check karne ki tables.
   -

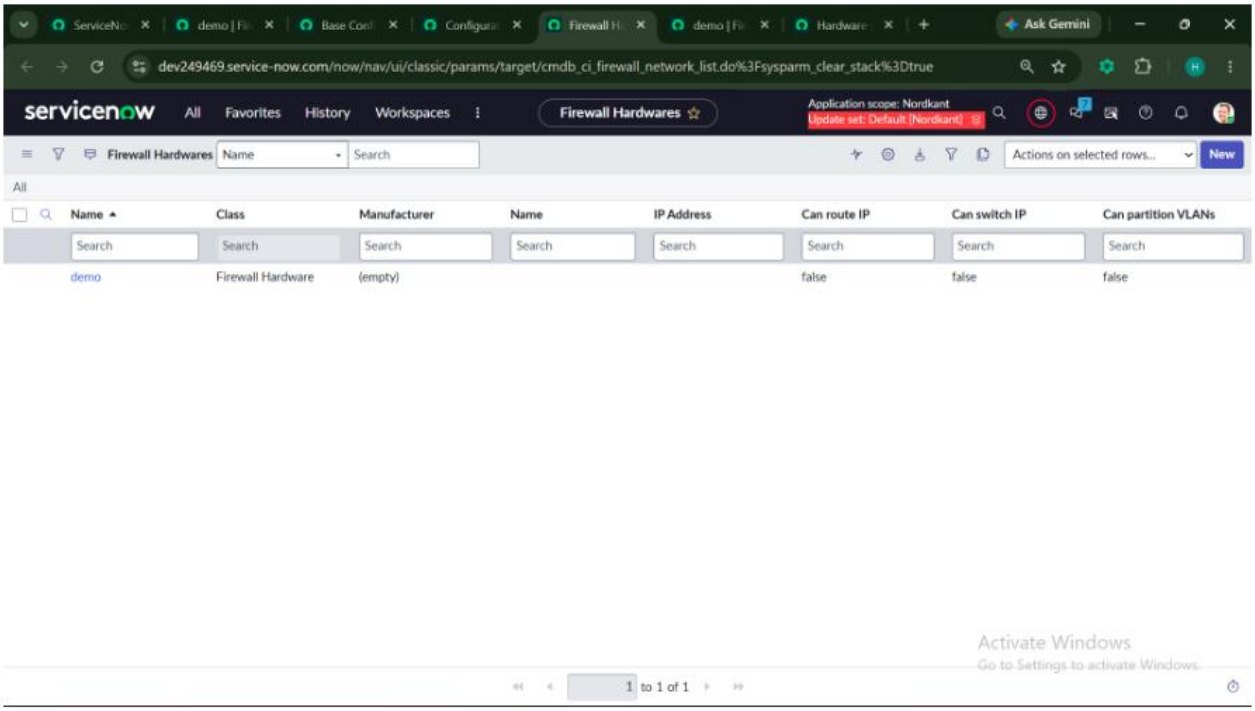
Summary: cmdb_ci sabhi **IT Assets** ke liye standard entry point hai, jabki cmdb poore ServiceNow Configuration system ka foundation structure rakhta hai (including non-IT items and relationship metadata).




Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/cmdb_list.do%3Fsysparm_query%3DGROUPBYsys_class_name%26sysparm_f...
servicenow
All
Favorites
History
Workspaces
Base Configuration Items
Application scope: Nordkant
Update set: Default [Nordkant]
Base Configuration Items
Name | name
Search
Actions on selected rows... | x
New
All
2785 total Base Configuration Items
Name
Asset
Asset tag
Assigned
Assigned to
Change Group
Class
Checked in
Checked out
Company
Cost
Class: Computer Room (2)
Class: Database (13)
Class: Data Center (1)
Class: MySQL Catalog (1)
Class: Oracle Catalog (1)
Class: Disk (2)
Class: Email Server (5)
Class: Firewall Hardware (1)
Class: IP Router (3)
Class: IP Switch (1)
Class: Linux Server (4)
Class: Mass Storage Device (3)
Class: Network Gear (5)
Activate Windows
Go to Settings to activate Windows.
1 to 20 of 36


Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/cmdb_ci_list.do%3Fsysparm_query%3DGROUPBYsys_class_name%26syspar...
servicenow
All
Favorites
History
Workspaces
Configuration Items
Application scope: Nordkant
Update set: Default [Nordkant]
Configuration Items
Name
Search
Actions on selected rows...
New
All
2785 total Configuration Items
Name
Manufacturer
Location
Description
Class
Updated
Maintenance schedule
Class: Computer (814)
Class: Computer Room (2)
Class: Database (13)
Class: Data Center (1)
Class: MySQL Catalog (1)
Class: Oracle Catalog (1)
Class: Disk (2)
Class: Email Server (5)
Class: Firewall Hardware (1)
Class: IP Router (3)
Class: IP Switch (1)
Class: Linux Server (4)
Class: Mass Storage Device (3)
Class: Network Gear (5)
Activate Windows
Go to Settings to activate Windows.
1 to 20 of 36


Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/cmdb_ci_hardware_list.do%3Fsysparm_query%3DGROUPBYsys_class_name...
servicenow
All
Favorites
History
Workspaces
Admin
Hardware
Application scope: Nordkant
Update set: Default [Nordkant]
Hardware
Updated
Search
Actions on selected rows...
New
All
902 total Hardware
Name
Manufacturer
Location
Description
Class
Updated
Maintenance schedule
Class: AIX Server (2)
Class: Computer (814)
Class: Firewall Hardware (1)
Class: IP Router (3)
Class: IP Switch (1)
Class: Linux Server (4)
Class: Mass Storage Device (3)
Class: Network Gear (5)
Class: Printer (28)
Class: Server (19)
Class: Storage Switch (1)
Class: UNIX Server (7)
Activate Windows
Go to Settings to activate Windows.
1 to 14 of 14


Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/cmdb_ci_firewall_network_list.do%3Fsysparm_clear_stack%3Dtrue
servicenow
All
Favorites
History
Workspaces
Firewall Hardwares
Application scope: Nordkant
Update set: Default [Nordkant]
Firewall Hardwares
Name
Search
Actions on selected rows...
New
All
Name
Class
Manufacturer
Name
IP Address
Can route IP
Can switch IP
Can partition VLANs
Search
demo
Firewall Hardware
(empty)
false
false
false
Activate Windows
Go to Settings to activate Windows.
1 to 1 of 1

Gemini



configuration tab me jo ye sab hai kya ye bhi cmdb se hi inherited chije hai jaise hardware , software hai ? in very short

Configuration tab me **2 alag-alag tarah ki cheezein** hain:

1. **Specific CI Modules (Derived Tables):** Jo categories infrastructure components dikhati hain (jaise *Servers*, *Application Servers*, *Database Servers*, *Clusters*, *Storage*, *Base Items*) — **haan, ye sab** `cmdb` / `cmdb_ci` **se hi inherited tables hain.** JPG + 1
2. **CMDB Admin & Governance Tools:** Jo management modules hain (jaise *CI Class Manager*, *CMDB Health Dashboard*, *Identification/Reconciliation*, *CMDB Properties*, *Query Builder*) — **ye CI tables nahi hain**, balki CMDB ko manage aur configure karne ke system modules/tools hain. JPG + 2

Ask Gemini Flash

Gemini is AI and can make mistakes.

Activate Windows
Go to Settings to activate Windows.

Gemini

ci class kya hoti hai ? in very short

ServiceNow me **CI Class (Configuration Item Class)** ek **table Blueprint / Template** hoti hai jo kisi specific category ke IT asset (jaise Server, Database, Router, Firewall) ke attributes aur properties ko define karti hai. PNG + 2

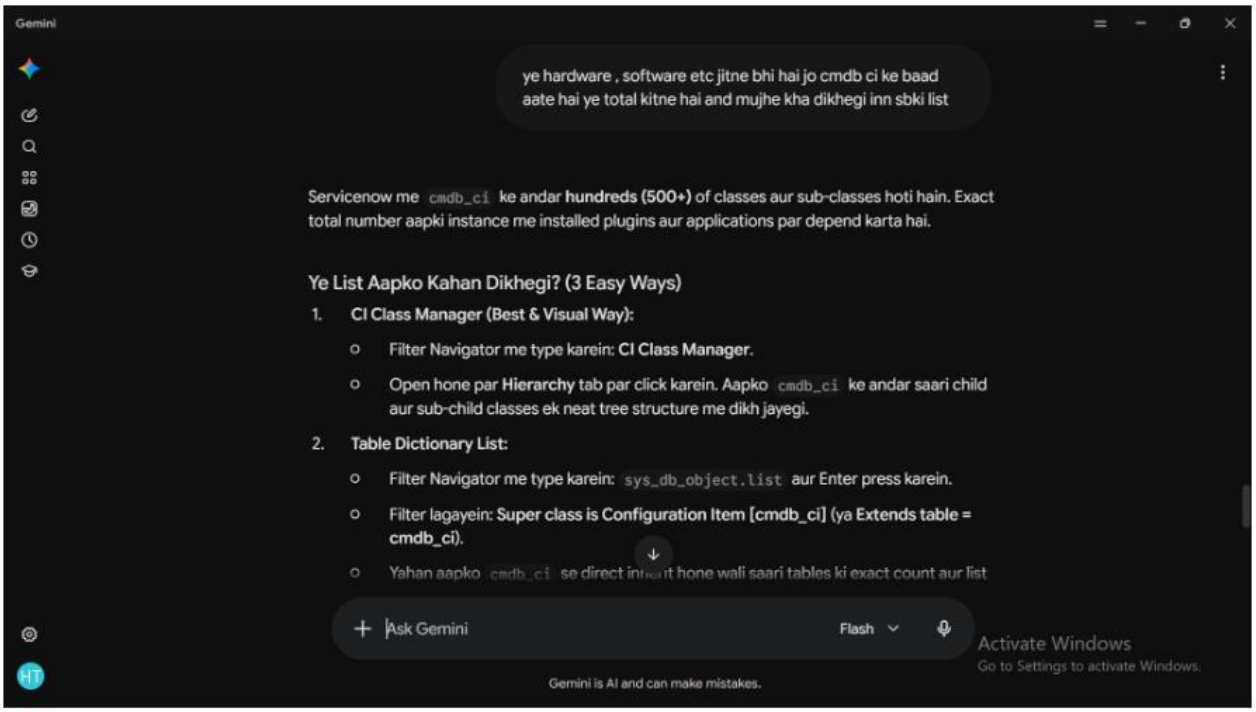
Simple word me:

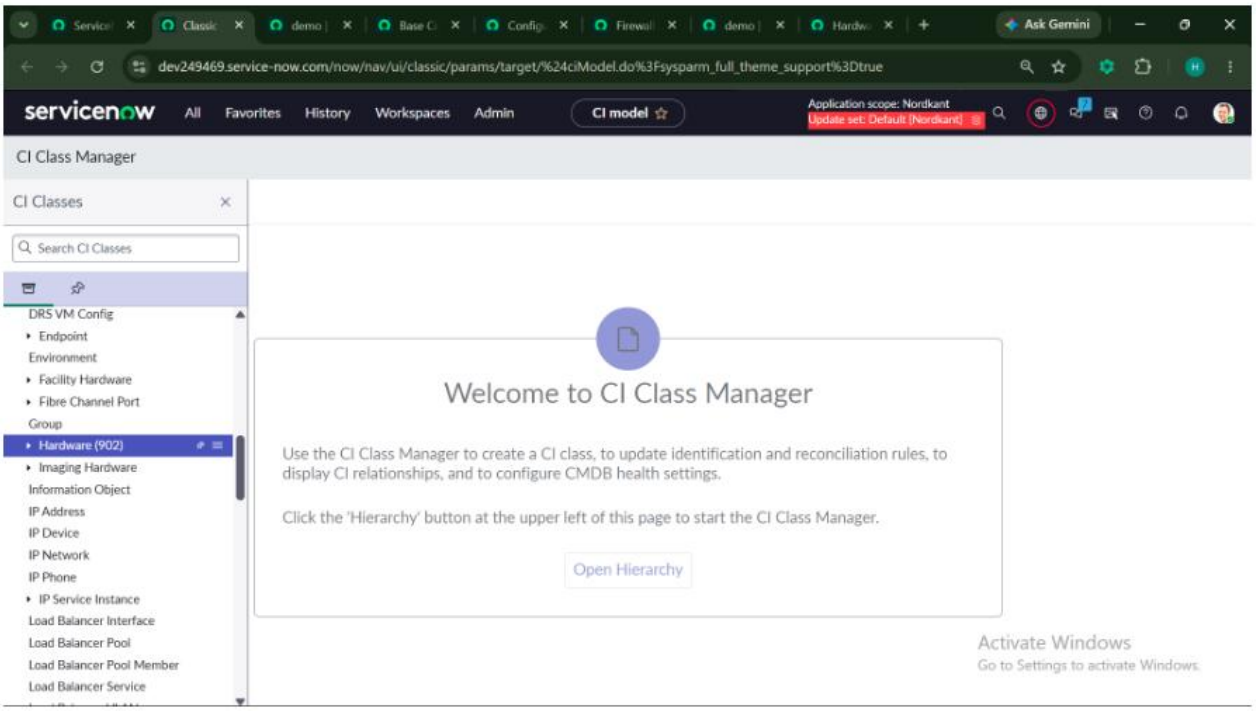
- **CI Class** = Blueprint / Category (e.g., `Firewall Hardware`) PNG + 1
- **CI Record** = Actual Device (e.g., your `demo` firewall) PNG

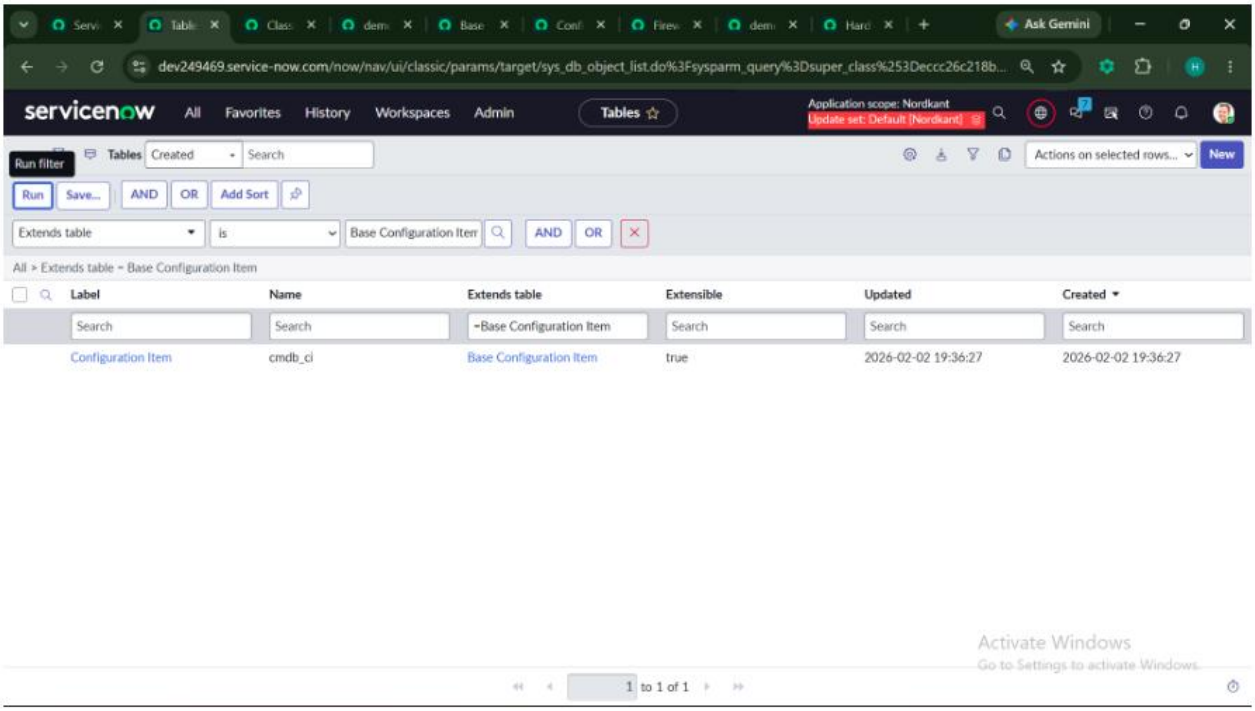
Ask Gemini Flash

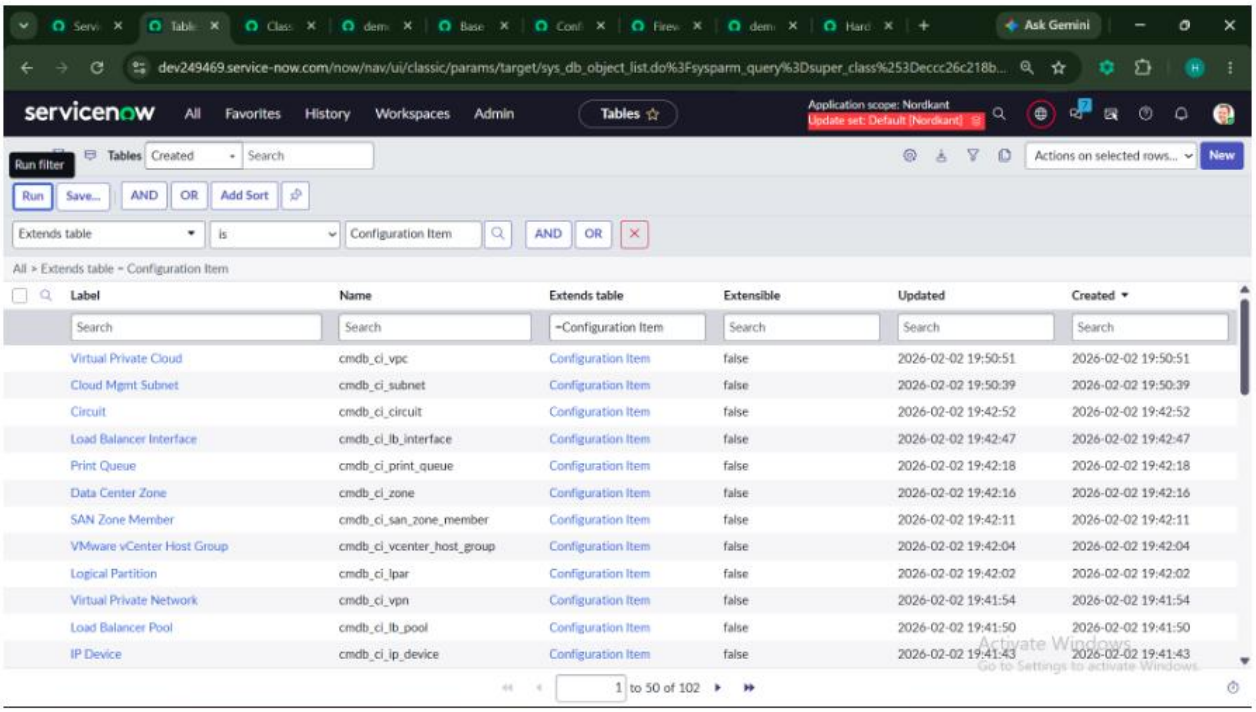
Gemini is AI and can make mistakes.

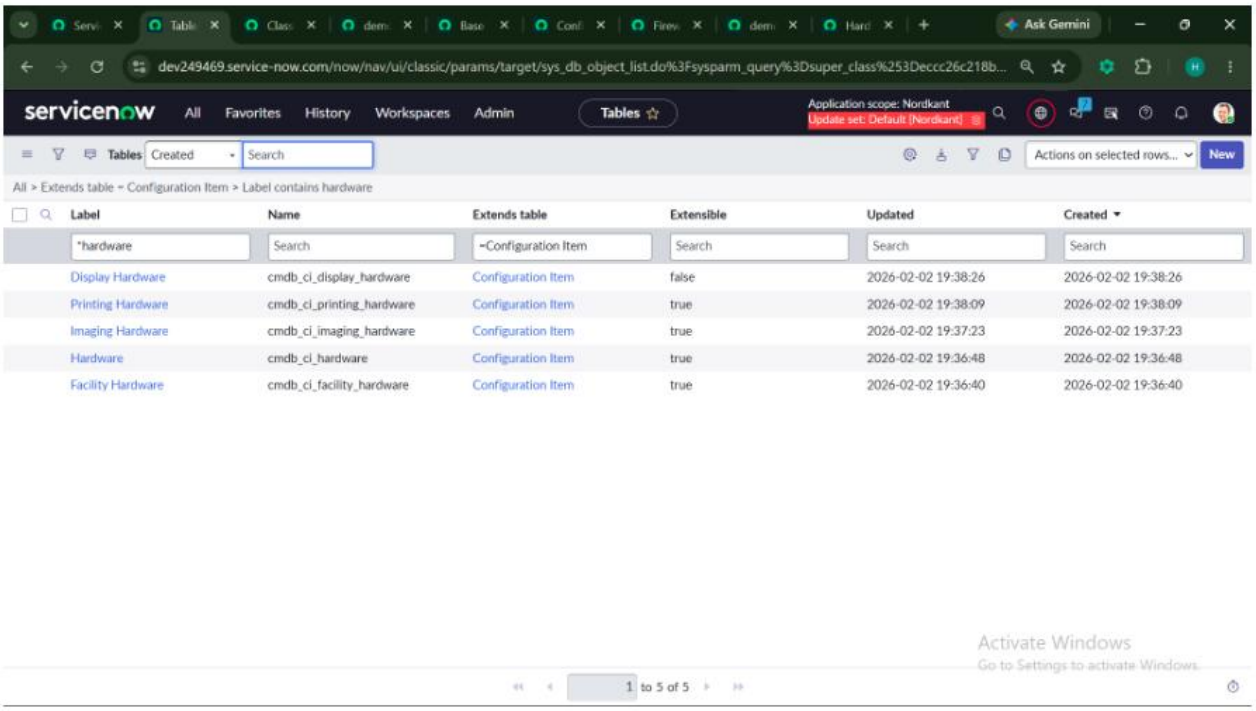
Activate Windows
Go to Settings to activate Windows.


Gemini
ye hardware , software etc jitne bhi hai jo cmdb ci ke baad aate hai ye total kitne hai and mujhe kha dikhegi inn sbki list
Servicenow me cmdb_ci ke andar hundreds (500+) of classes aur sub-classes hoti hain. Exact total number aapki instance me installed plugins aur applications par depend karta hai.
Ye List Aapko Kahan Dikhegi? (3 Easy Ways)
1. CI Class Manager (Best & Visual Way):
Filter Navigator me type karein: CI Class Manager.
Open hone par Hierarchy tab par click karein. Aapko cmdb_ci ke andar saari child aur sub-child classes ek neat tree structure me dikh jayegi.
2. Table Dictionary List:
Filter Navigator me type karein: sys_db_object.list aur Enter press karein.
Filter lagayein: Super class is Configuration Item [cmdb_ci] (ya Extends table = cmdb_ci).
Yahan aapko cmdb_ci se direct inherit hone wali saari tables ki exact count aur list
Ask Gemini
Flash
Gemini is AI and can make mistakes.
Activate Windows
Go to Settings to activate Windows.

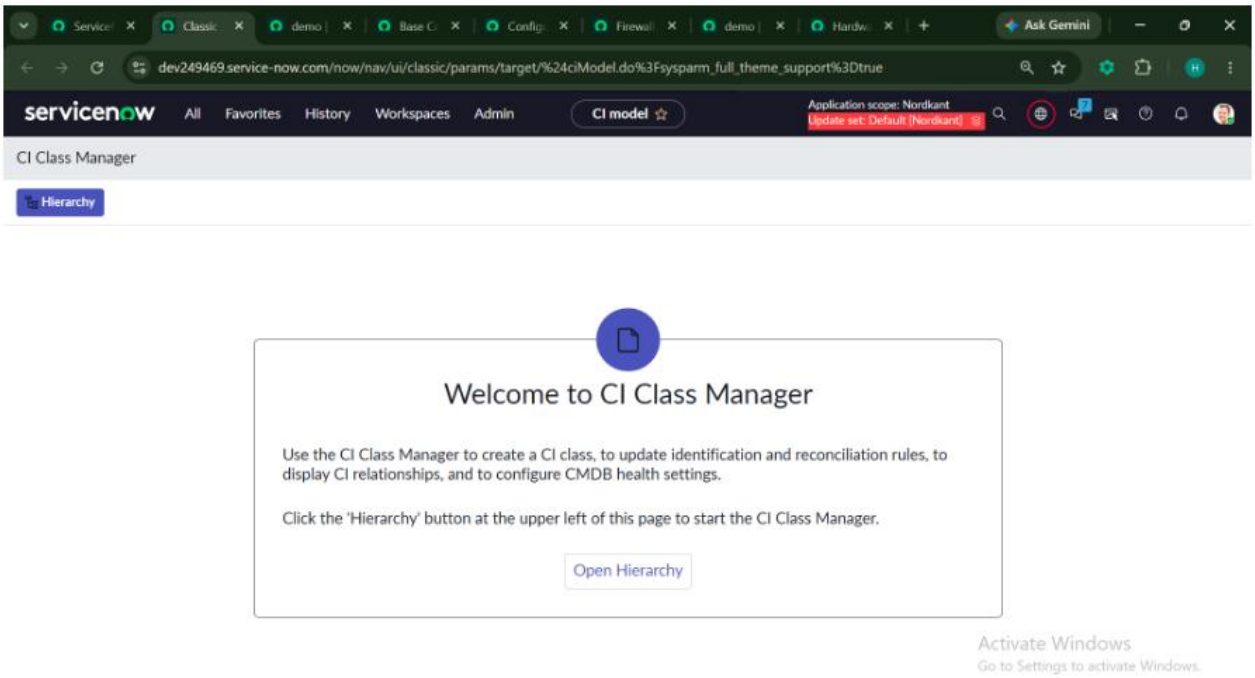

Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/%24ciModel.do%3Fsysparm_full_theme_support%3Dtrue
servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
CI Classes
Search CI Classes
DRS VM Config
Endpoint
Environment
Facility Hardware
Fibre Channel Port
Group
Hardware (902)
Imaging Hardware
Information Object
IP Address
IP Device
IP Network
IP Phone
IP Service Instance
Load Balancer Interface
Load Balancer Pool
Load Balancer Pool Member
Load Balancer Service
Welcome to CI Class Manager
Use the CI Class Manager to create a CI class, to update identification and reconciliation rules, to display CI relationships, and to configure CMDB health settings.
Click the 'Hierarchy' button at the upper left of this page to start the CI Class Manager.
Open Hierarchy
Activate Windows
Go to Settings to activate Windows.


Ask Gemini
dev249469.service-now.com/now/nav/ui/classic/params/target/sys_db_object_list.do%3Fsysparm_query%3Dsuper_class%253Deccc26c218b...
servicenow
All
Favorites
History
Workspaces
Admin
Tables
Application scope: Nordkant
Update set: Default [Nordkant]
Run filter
Tables
Created
Search
Actions on selected rows...
New
Run
Save...
AND
OR
Add Sort
Extends table
is
Base Configuration Item
AND
OR
All > Extends table = Base Configuration Item
Label
Name
Extends table
Extensible
Updated
Created
Search
Search
=Base Configuration Item
Search
Search
Search
Configuration Item
cmdb_ci
Base Configuration Item
true
2026-02-02 19:36:27
2026-02-02 19:36:27
Activate Windows
Go to Settings to activate Windows.
1 to 1 of 1





CMDB CI CLASS MANAGER :

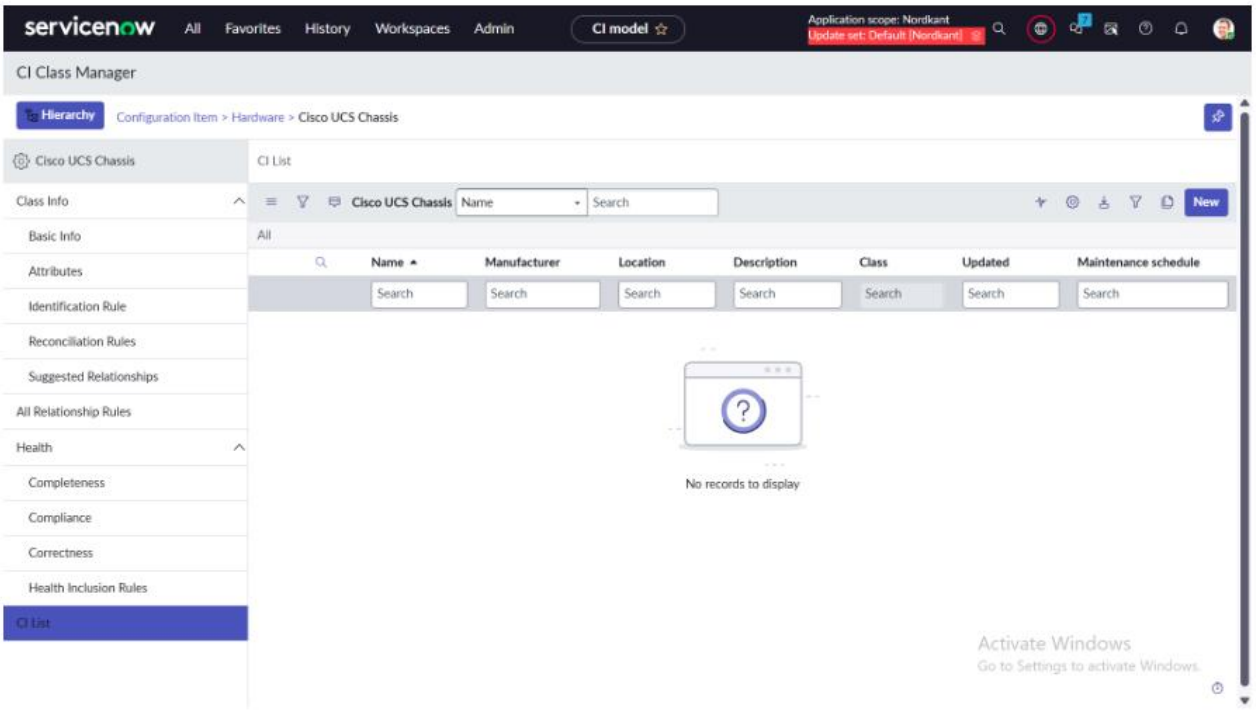


Yeh **CI Class Manager** tool hai.

Is centralized tool se aap:

- CMDB ke poore **Table Hierarchy Tree** ko dekhte aur manage karte hain.
-
- Nayi **CI Classes create** ya existing classes ko edit karte hain.
-
- **Identification & Reconciliation Rules** aur **CMDB Health settings** configure karte hain.

Question : what all are these ?



Answer :
Yeh UI **CI Class Manager** ka Control Panel hai jahan aap kisi specific CI Class (jaise image me **Cisco UCS Chassis** selected hai) ke saare rules, structure, aur settings configuration control karte ho.

Ek easy **Aadhaar / ID Card Analogy** se samjho:

1. Hierarchy Path (Configuration Item > Hardware > Cisco UCS Chassis)

- **Analogy:** Location Address (India > State > District).
-
- **Meaning:** Yeh dikhata hai ki yeh class kitni deep hierarchy me hai.
-

2. Class Info (Basic Details)

- **Basic Info:** Class ka Label, Technical Name aur Table Type.
-
- **Attributes:** Forms par kaun-kaun se fields/columns honge (jaise Name, IP Address, Serial Number).
-
    - *Analogy:* Passport ya Form me kya-kya details bharni hain.

3. Identification & Reconciliation Rules (Very Important!)

- **Identification Rule:** System duplicate device ko kaise pehchane (e.g., matching Serial Number + IP Address).
-
    - *Analogy:* Aadhaar Card Identification — Bio-metric fingerprint aur Fingerprints se exact unique banda match karna.
    -
- **Reconciliation Rules:** Agar 2 alag sources (jaise Discovery tool aur SCCM) same device ka data update karein, toh kiska data powerful/authoritative mana jaye.
-
    - *Analogy:* Document validation preference (Passport update > Voter ID update).
    -

4. Relationships (Connections)

- **Suggested / All Relationship Rules:** Is Class ka baaki items ke saath kya connection ho sakta hai (e.g., *Cisco UCS Chassis* **Runs On** / **Hosts** *Virtual Servers*).
-
    - *Analogy:* Family Tree (Kaun kiska parent/child hai).
    -

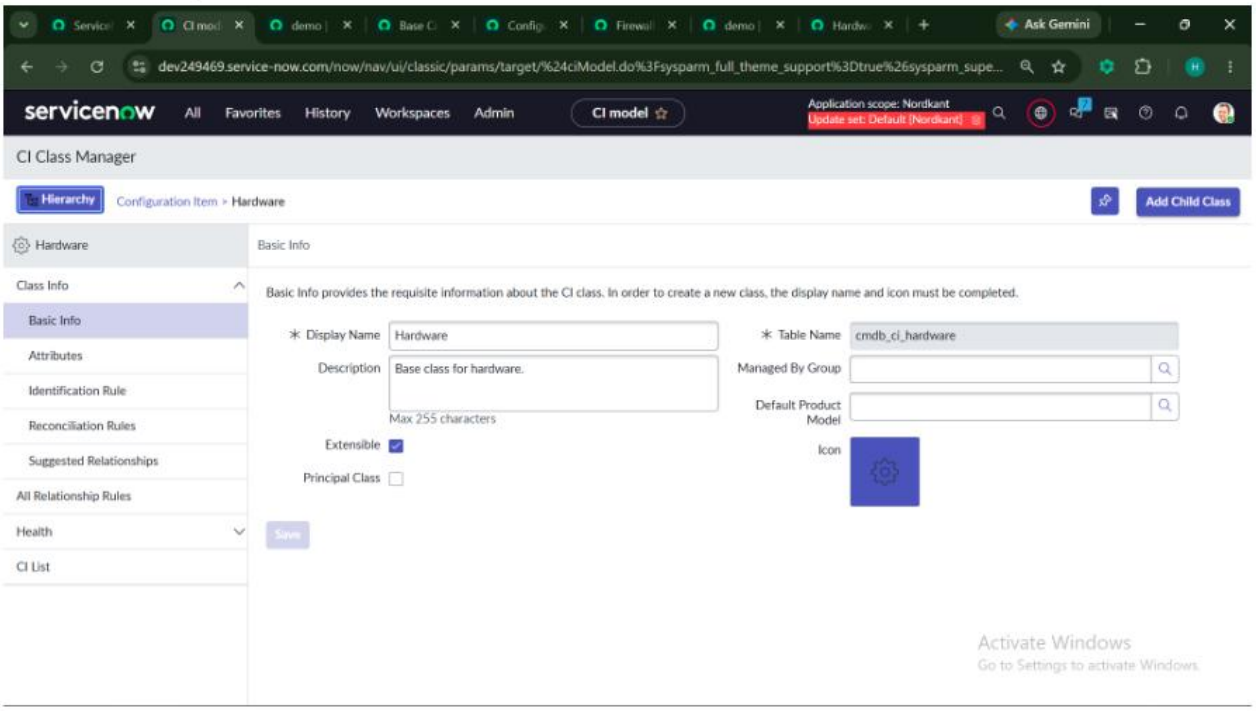
5. Health (Data Quality Checks)

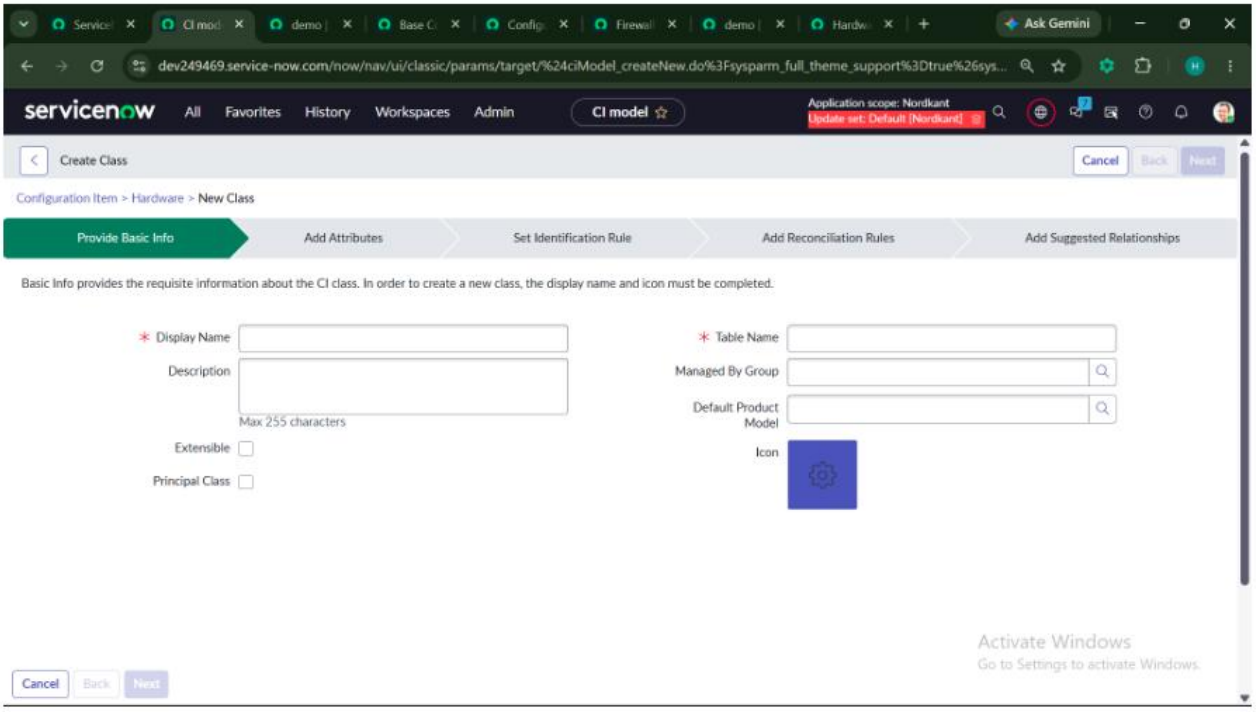
- **Completeness:** Kya saare important fields bhare hue hain? (e.g., Missing Serial Number?)
-
- **Compliance:** Kya device Security Audit pass kar raha hai?
-
- **Correctness:** Kya koi duplicate record ya orphan (unlinked) record toh nahi hai?
-
- **Health Inclusion Rules:** Kine devices par health check apply hoga.
-
    - *Analogy:* School Report Card (Data kitna clean aur complete hai).
    -

6. CI List (Main Screen)

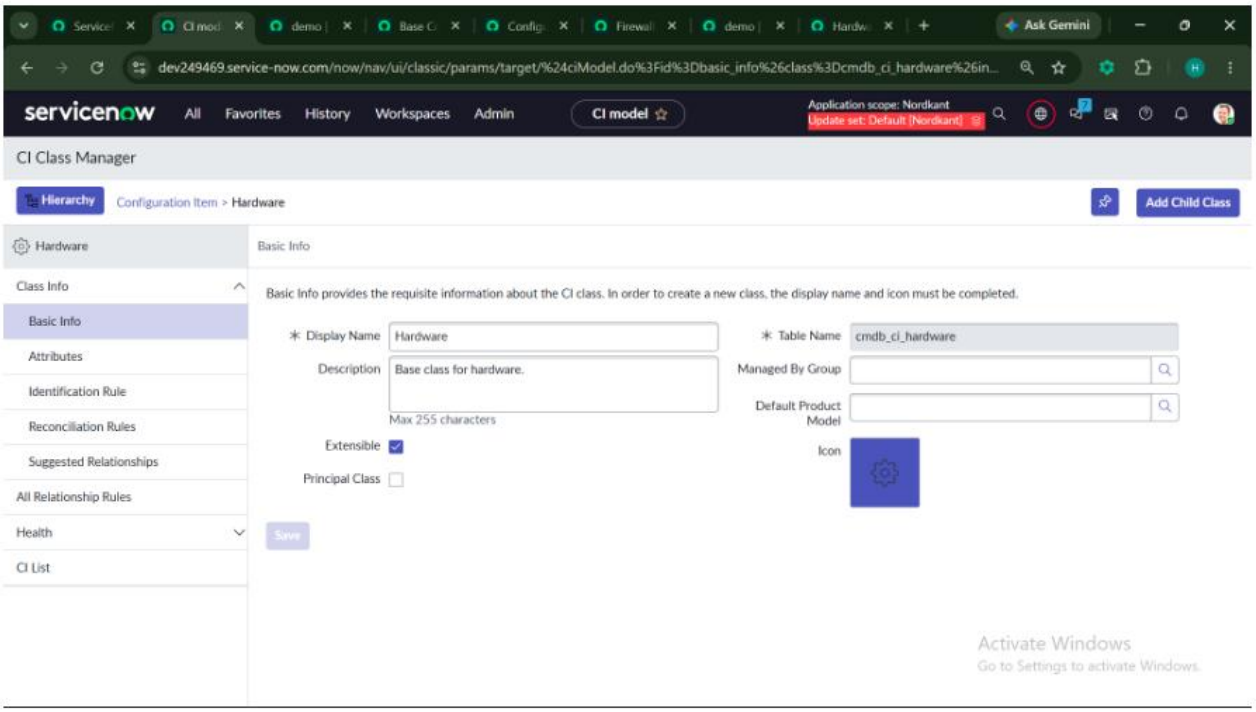
- **Screen View:** Is class me abhi jitne actual records exist karte hain unki list dikhana (filhaal No records to display dikh raha hai).
-
    - *Analogy:* Registered Candidate List.

Question : hardware me ek new class add karni ho to ?
Answer :
Hardware ke upar ek child class create ka buttton click kro



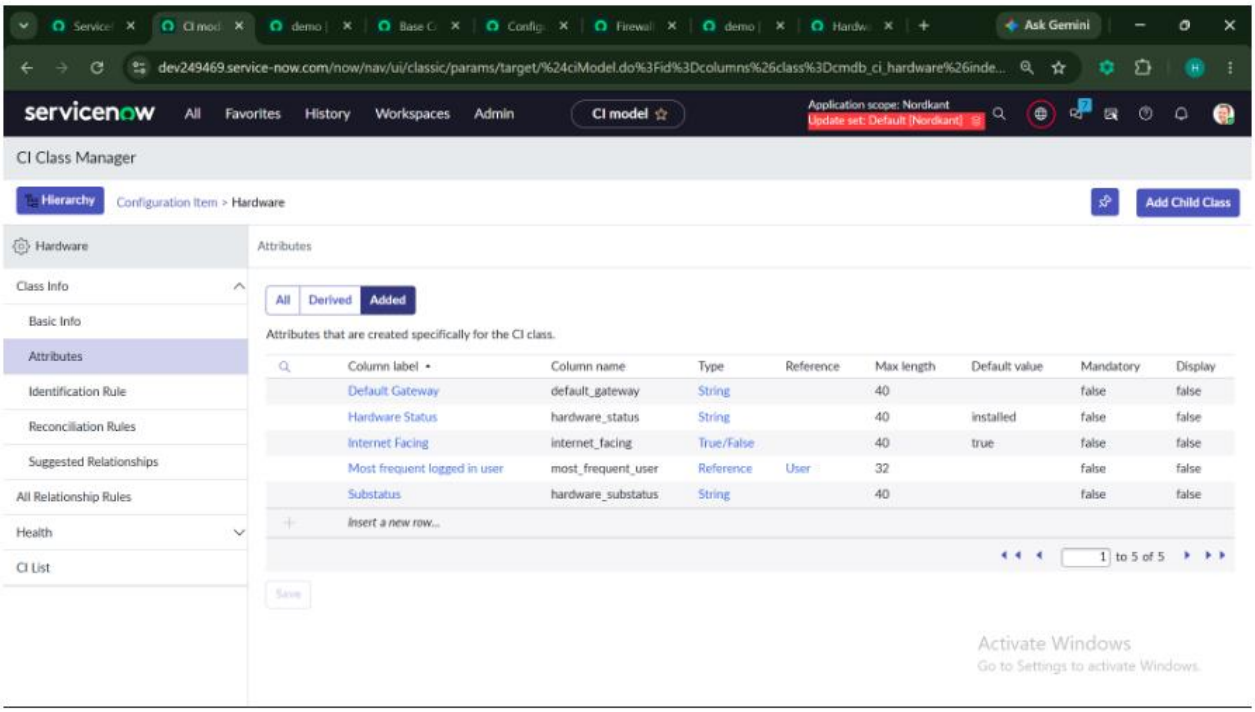


IN DETAIL :

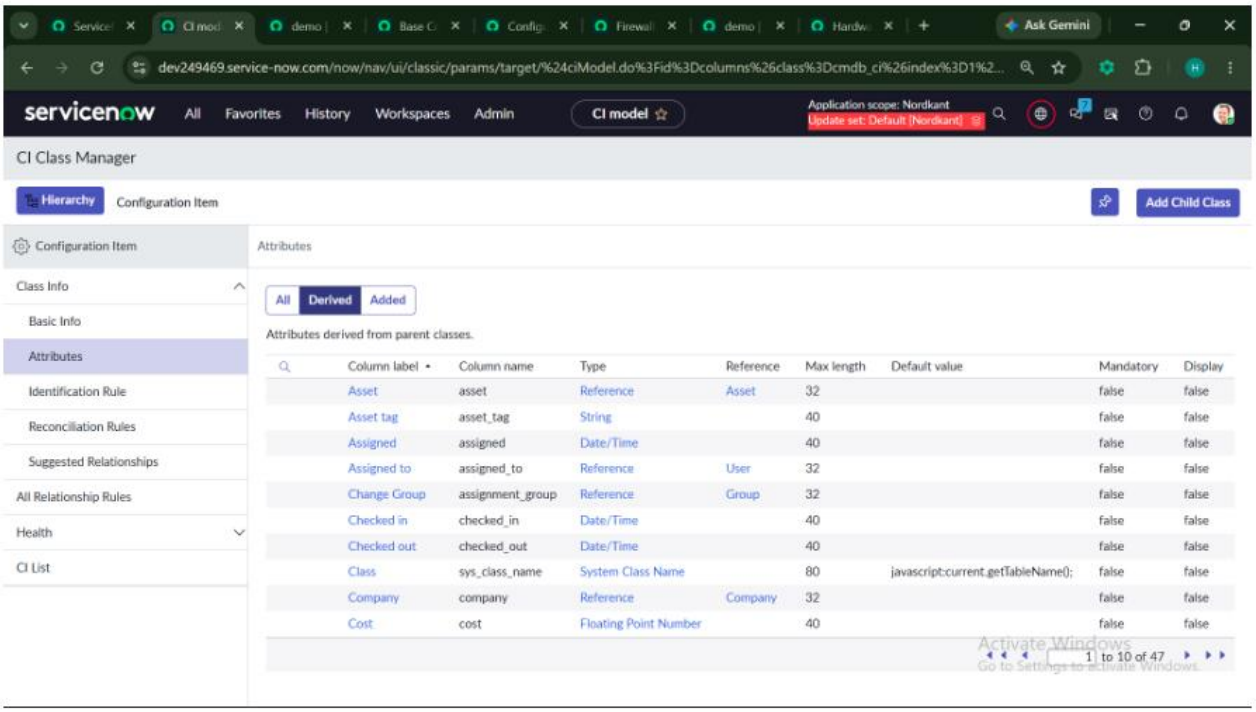


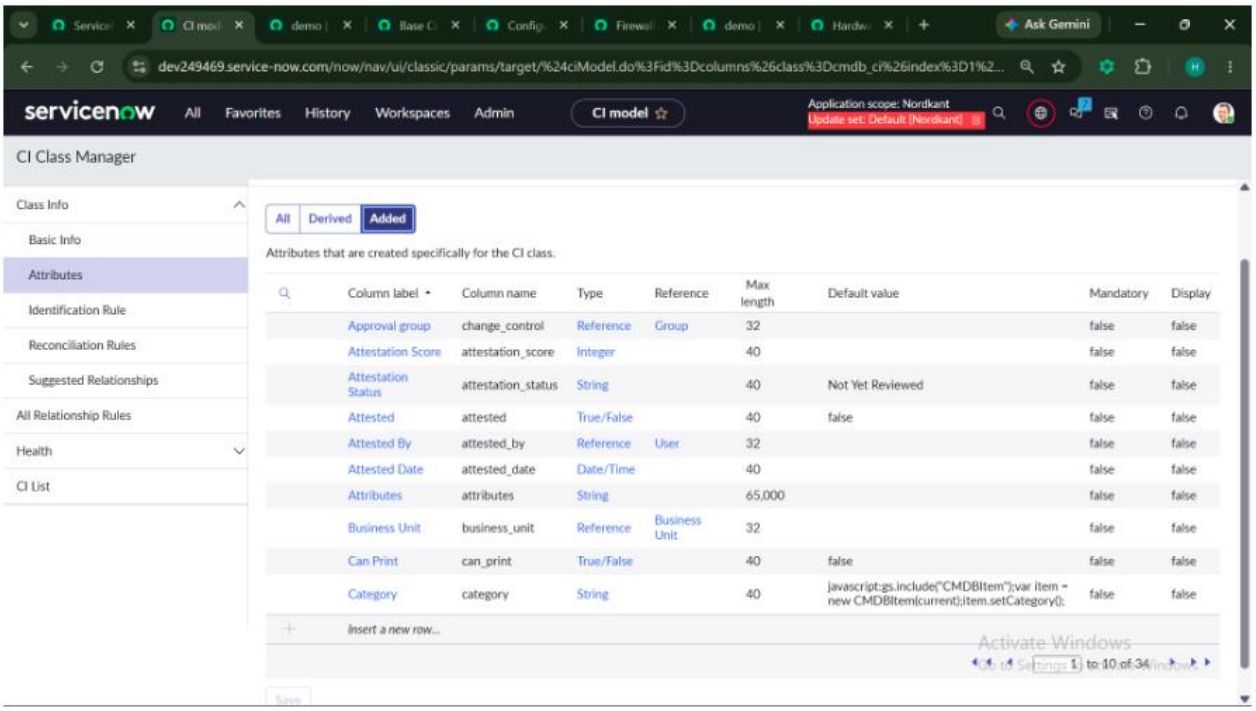
CI Class Manager ke **Basic Info** page ki har field ka matlab short me:

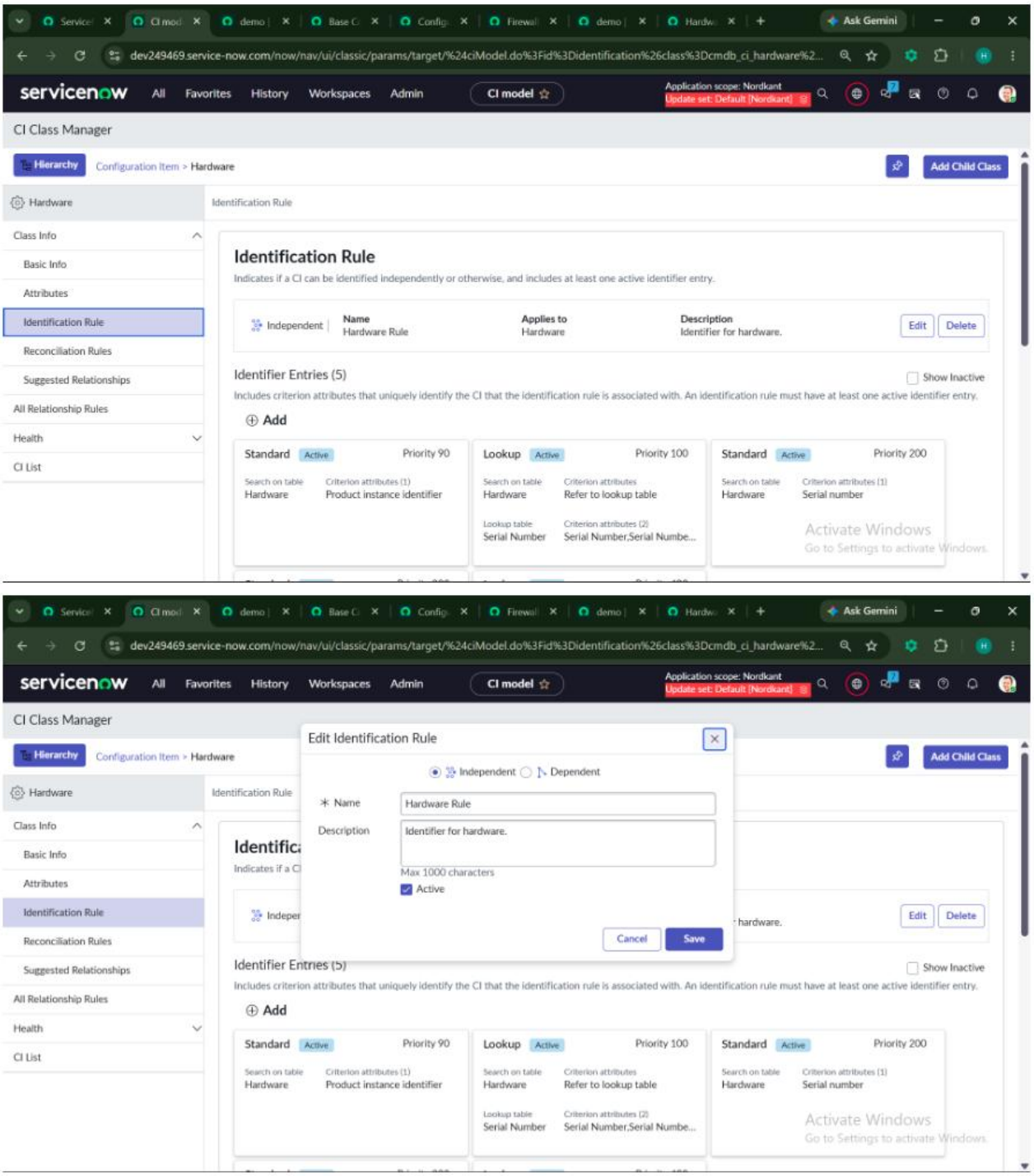
- **Display Name**: UI par class ka jo naam user ko dikhta hai (e.g., Hardware).
- **Table Name**: Database me table ka actual technical backend name (e.g., cmdb_ci_hardware).
- **Description**: Class ki brief details ya purpose (e.g., Base class for hardware.).
- **Extensible (Checkbox)**: Agar checked hai, toh is class se aage nayi child classes extend/derive ki ja sakti hain.
- **Principal Class (Checkbox)**: ServiceNow CMDB me hazaron classes hoti hain. Principal Class mark karne se system ko pata chalta hai ki yeh ek **important / key class** hai (jaise Servers, Laptops), jisse Discovery tools, Health Dashboards, aur Reporting me unnecessary extra classes filter out ho jati hain.
- **Managed By Group**: Is class ke CIs aur unke data ko maintain karne wali responsible team/group.
- **Default Product Model**: Asset Management me har CI kisi specific Hardware/Software catalog item (model) se juda hota hai. Jab bhi is class ka naya record banta hai, toh yeh field automatic batati hai ki is item ka default **Product Model / Specification** kaun sa hoga (e.g., *Dell PowerEdge R740*).
- **Icon**: Visual symbol jo class ko UI par represent karta hai.



NOTE : CMDB CI ki derived fields(yani cmdb ki fields) and iski khudki added fields kitni hai and total fields kitni hai :
Derived fields : 47
Added fields : 34
Total fields : 81


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Hierarchy
Configuration Item
Add Child Class
Configuration Item
Attributes
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
All
Derived
Added
Attributes derived from parent classes.
Column label
Column name
Type
Reference
Max length
Default value
Mandatory
Display
Asset asset Reference Asset 32 false false
Asset tag asset_tag String 40 false false
Assigned assigned Date/Time 40 false false
Assigned to assigned_to Reference User 32 false false
Change Group assignment_group Reference Group 32 false false
Checked in checked_in Date/Time 40 false false
Checked out checked_out Date/Time 40 false false
Class sys_class_name System Class Name 80 javascript:current.getTableName(); false false
Company company Reference Company 32 false false
Cost cost Floating Point Number 40 false false
1 to 10 of 47

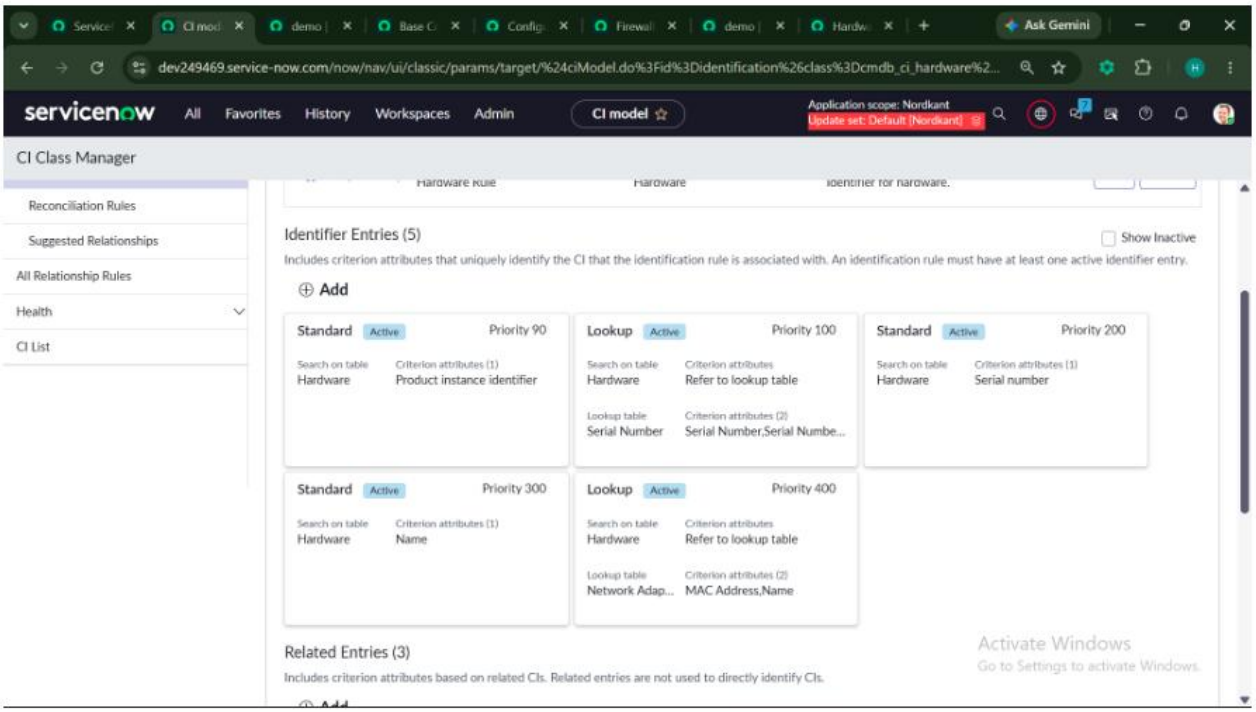

servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
All
Derived
Added
Attributes that are created specifically for the CI class.
Column label
Column name
Type
Reference
Max length
Default value
Mandatory
Display
Approval group change_control Reference Group 32 false false
Attestation Score attestation_score Integer 40 false false
Attestation Status attestation_status String 40 Not Yet Reviewed false false
Attested attested True/False 40 false false false
Attested By attested_by Reference User 32 false false
Attested Date attested_date Date/Time 40 false false
Attributes attributes String 65,000 false false
Business Unit business_unit Reference Business Unit 32 false false
Can Print can_print True/False 40 false false false
Category category String 40 javascript:gs.include("CMDBItem");var item = new CMDBItem(current);item.setCategory(); false false
Insert a new row...
1 to 10 of 34
Save

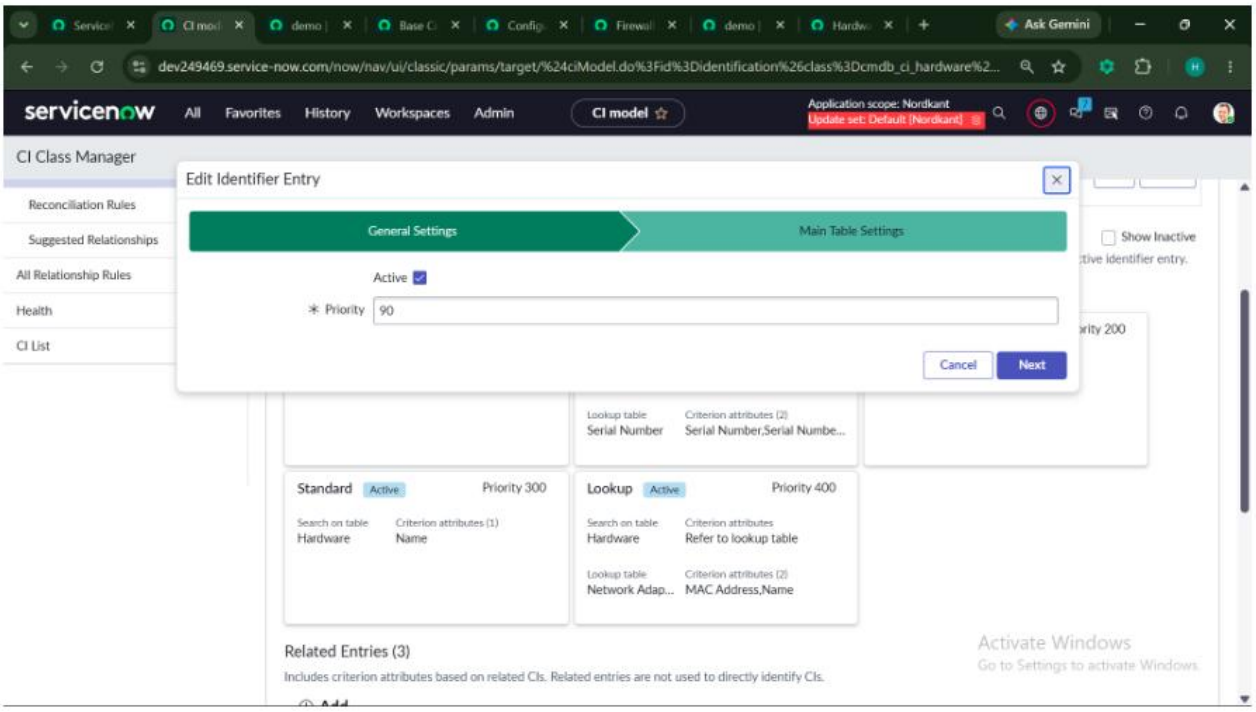


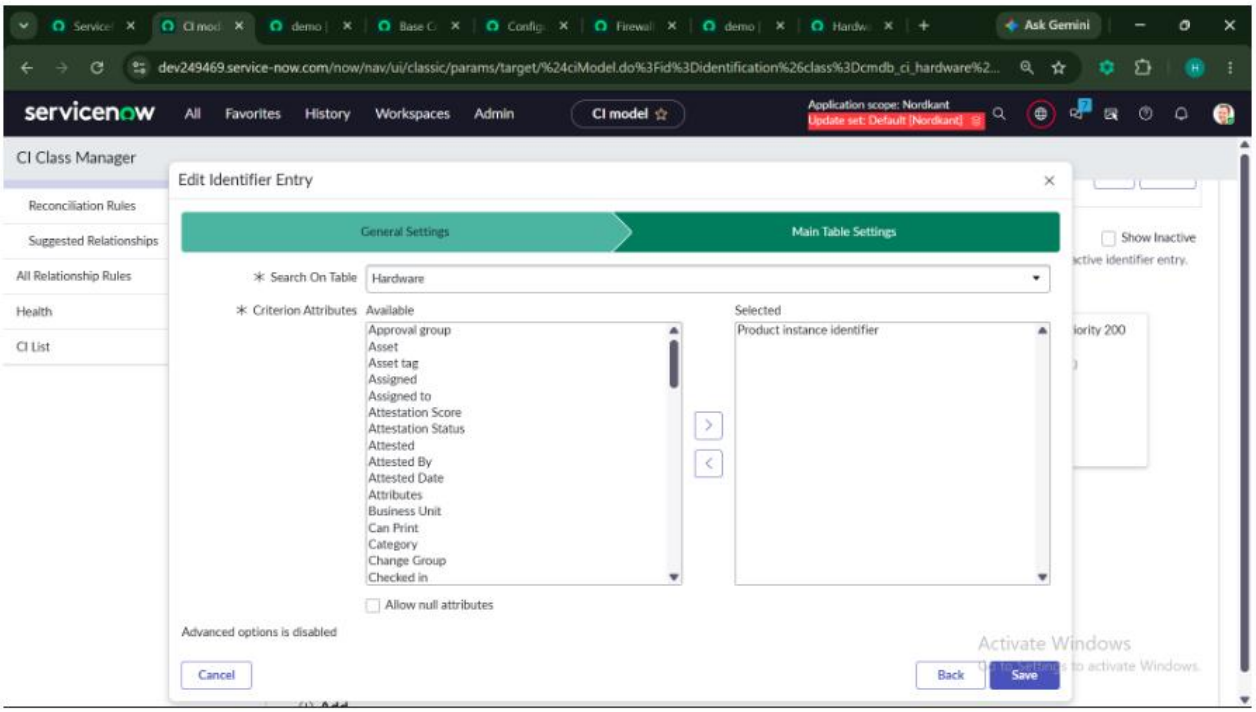
Identification Rules me **Independent** aur **Dependent** ka simple matlab yeh hai:

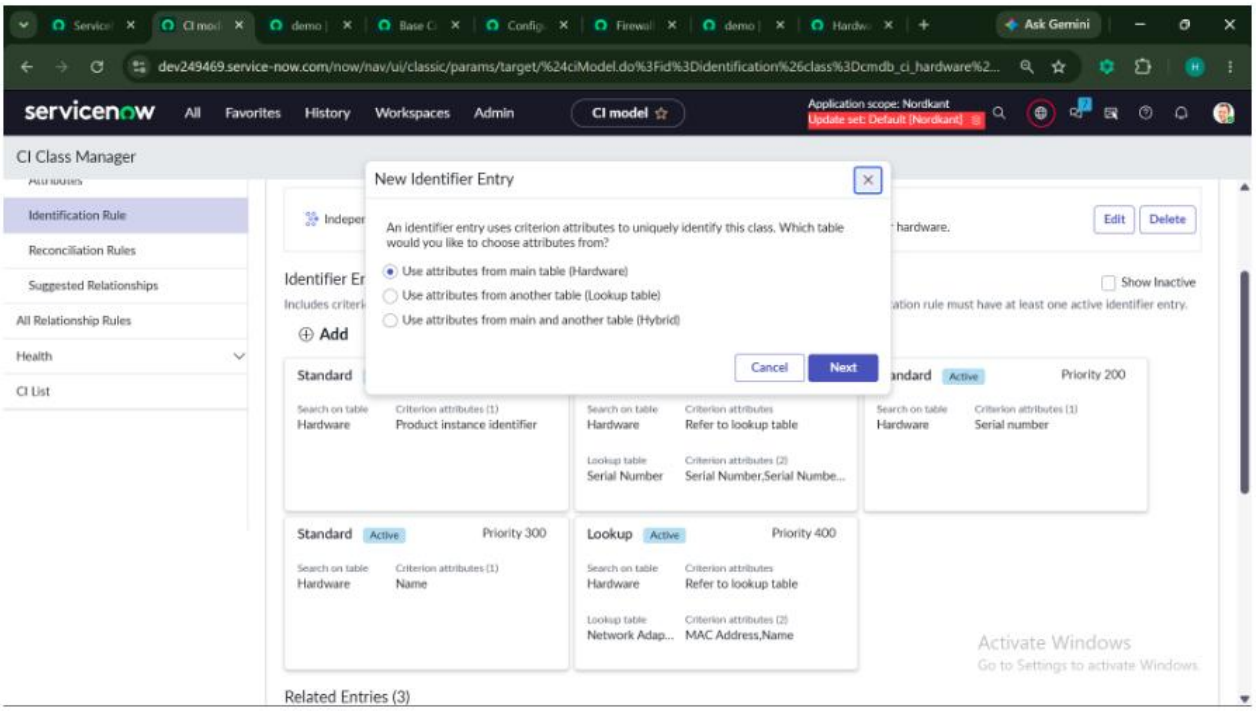
- **Independent CI (Standalone Device)**:
-
    - Yeh device akela uniquely identify ho sakta hai.
    -

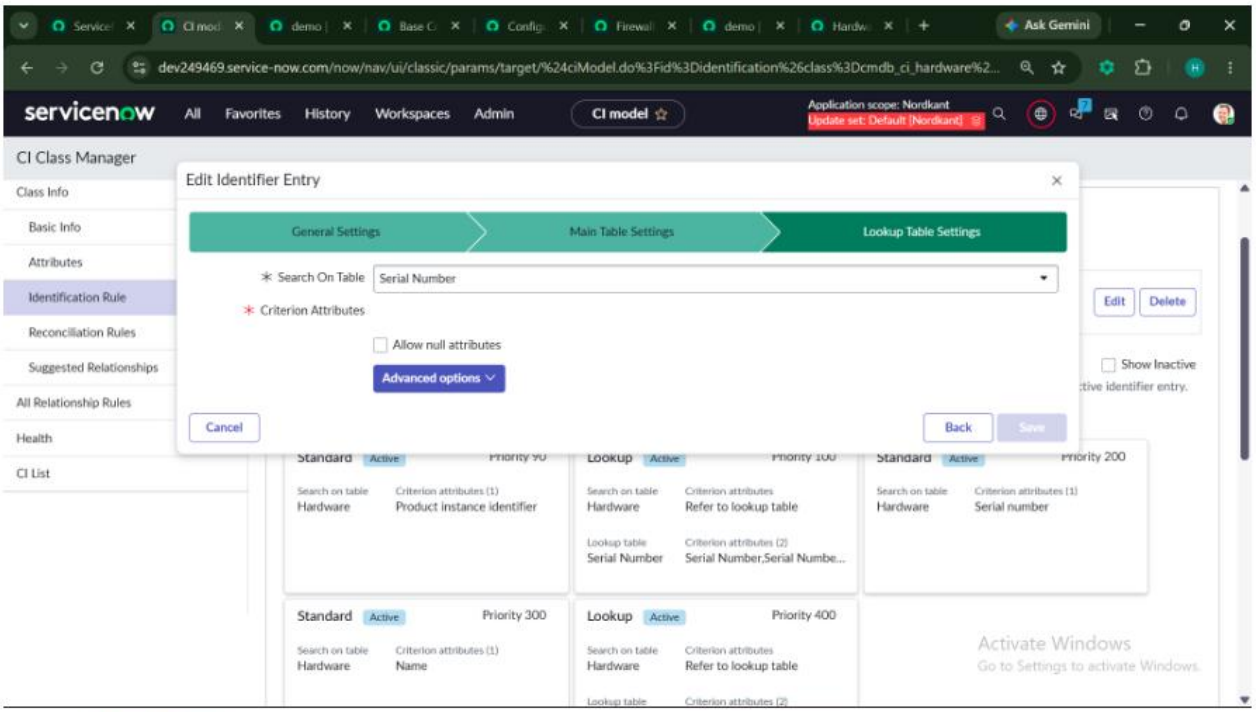
    - Isko identify hone ke liye kisi dusre CI par depend hone ki zarurat nahi hoti.
    -
    - *Examples:* Server, Laptop, Router (inke paas apna unique Serial Number ya MAC Address hota hai).
    -
- **Dependent CI (Sub-component / Attached Device)**:
-
    - Yeh CI akela uniquely identify **nahi** ho sakta. Isko identify karne ke liye iske **Parent CI** ki zarurat hoti hai.
    -
    - *Examples:*
    -
        - **Network Interface Card (NIC)** ya **Disk Drive**: Drive C: bina Server ke unique nahi hai, wo Server ke sath match hokar hi identify hoga.
        -
        - **Database Instance**: Database app hamesha us Host Server ke sath connect hokar hi identify hoti hai jis par wo run kar rahi hai.


dev249469.service-now.com/now/nav/ui/classic/params/target/%24ciModel.do%3Fid%3Didentification%26class%3Dcmdb_ci_hardware%2...
servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Identifier Entries (5)
Show Inactive
Includes criterion attributes that uniquely identify the CI that the identification rule is associated with. An identification rule must have at least one active identifier entry.
Add
Standard Active Priority 90
Search on table Hardware
Criterion attributes (1) Product instance identifier
Lookup Active Priority 100
Search on table Hardware
Criterion attributes Refer to lookup table
Lookup table Serial Number
Criterion attributes (2) Serial Number,Serial Numbe...
Standard Active Priority 200
Search on table Hardware
Criterion attributes (1) Serial number
Standard Active Priority 300
Search on table Hardware
Criterion attributes (1) Name
Lookup Active Priority 400
Search on table Hardware
Criterion attributes Refer to lookup table
Lookup table Network Adap...
Criterion attributes (2) MAC Address,Name
Related Entries (3)
Includes criterion attributes based on related CIs. Related entries are not used to directly identify CIs.
Activate Windows
Go to Settings to activate Windows.


dev249469.service-now.com/now/nav/ui/classic/params/target/%24ciModel.do%3Fid%3Didentification%26class%3Dcmdb_ci_hardware%2...
servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Edit Identifier Entry
General Settings
Main Table Settings
Active
Priority 90
Cancel
Next
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Show Inactive
Lookup table Serial Number
Criterion attributes (2) Serial Number,Serial Numbe...
Standard Active Priority 300
Search on table Hardware
Criterion attributes (1) Name
Lookup Active Priority 400
Search on table Hardware
Criterion attributes Refer to lookup table
Lookup table Network Adap...
Criterion attributes (2) MAC Address,Name
Related Entries (3)
Includes criterion attributes based on related CIs. Related entries are not used to directly identify CIs.
Activate Windows
Go to Settings to activate Windows.


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Edit Identifier Entry
General Settings
Main Table Settings
Search On Table
Hardware
Criterion Attributes
Available
Selected
Approval group
Asset
Asset tag
Assigned
Assigned to
Attestation Score
Attestation Status
Attested
Attested By
Attested Date
Attributes
Business Unit
Can Print
Category
Change Group
Checked in
Product instance identifier
Allow null attributes
Advanced options is disabled
Cancel
Back
Save
Show Inactive
Activate Windows


servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Nordkant
Update set: Default [Nordkant]
CI Class Manager
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
New Identifier Entry
An identifier entry uses criterion attributes to uniquely identify this class. Which table would you like to choose attributes from?
Use attributes from main table (Hardware)
Use attributes from another table (Lookup table)
Use attributes from main and another table (Hybrid)
Cancel
Next
Edit
Delete
Show Inactive
Add
Standard
Active
Priority 200
Search on table
Hardware
Criterion attributes (1)
Product instance identifier
Lookup
Refer to lookup table
Serial Number
Criterion attributes (2)
Serial Number,Serial Numbe...
Serial number
Priority 300
Name
Priority 400
Network Adap...
MAC Address,Name
Related Entries (3)
Activate Windows



Aapko Hardware class ke example se ek real-life scenario se samjhata hoon:

Maan lo aapke paas ek **Dell Server (Hardware)** hai jisko aap ServiceNow CMDB me register/discover kar rahe ho. System ke paas 5 rules hain **Priority order** me:

Step-by-Step Priority Matching:

1. **Priority 90 (Product Instance Identifier)**:
2.
    - System sabse pehle dekhta hai: *"Kya is Server ke paas specific Product Instance ID hai?"*
    -
    - Agar mil gayi ➔ Match done! (Baaki rules check hi nahi hone).
    -
    - Agar nahi mili ➔ System next step par jayega.
    -
3. **Priority 100 & 200 (Lookup / Standard Serial Number)**:
4.
    - System check karta hai: *"Kya iska **Serial Number** (e.g., DELL-998877) pehle se database me majood kisi Server se match hota hai?"*
    -
    - Agar match ho gaya ➔ System naya record banane ki bajaye purane record ko hi update karega.
    -

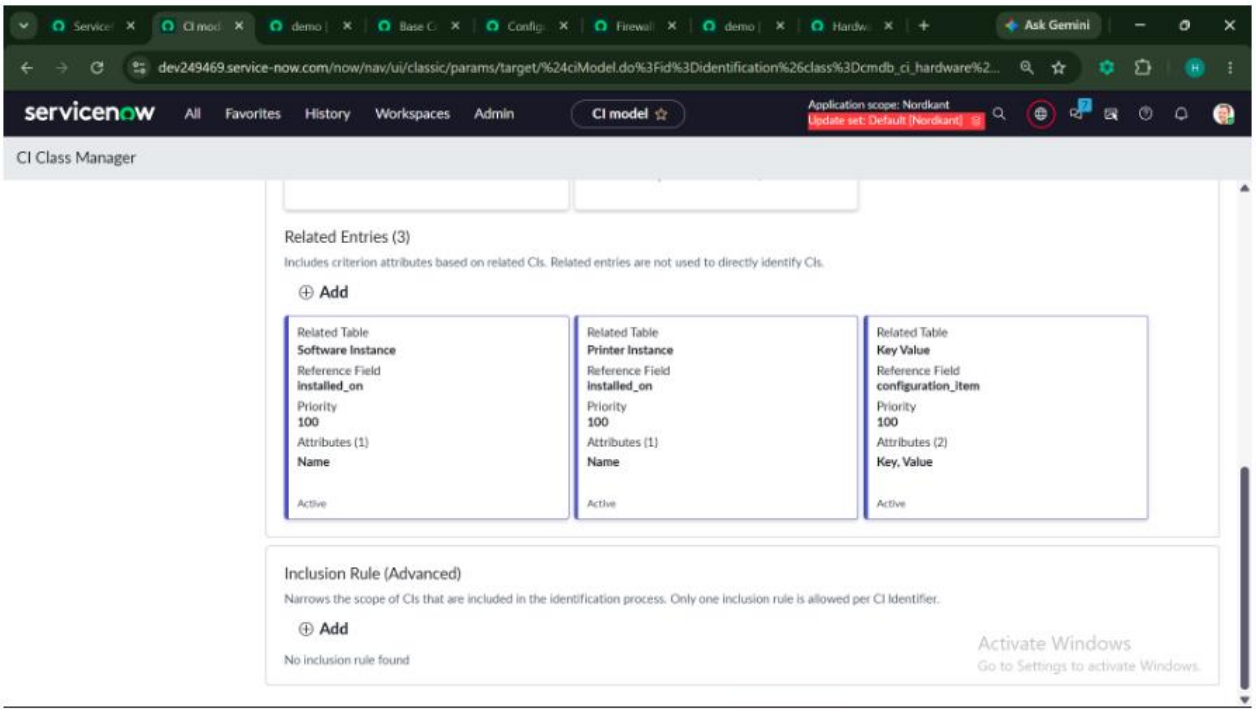
    - Agar match nahi hua ➔ System agle rule par badhega.
    -
5. **Priority 300 (Standard - Name)**:
6.
    - System check karta hai: *"Kya is name ka (Dell-Server-01) koi hardware pehle se register hai?"*
    -
7. **Priority 400 (Lookup - MAC Address)**:
8.
    - Sabse last me system Network Card ka **MAC Address** match karke identification verify karta hai.
    -

Summary:

- **Standard**: Direct Hardware table par majood fields (jaise Serial Number, Name) check karna.
-
- **Lookup**: Dedicated sub-tables (jaise Serial Number list ya Network Adapter table) me jakar check karna.
-
- **Priority**: Jo rule pehle (sabse low priority number) match ho gaya, system wahi ruk kar device ko identify kar leta hai.

NOTE : Agar koi bhi rule match nahi hota, toh **IRE (Identification Engine)** ise ek **brand-new device** maantaa hai aur database me **ek naya CI record create kar deta hai**.

Orr Agar koi bhi rule match ho jaata hai, toh **IRE** us existing record ko find kar leta hai aur naye data se us **purane record ko update** kar deta hai (naya duplicate record nahi banata).



Ek real-life example se poora flow samjho:

Maan lo aapke paas ek **Server (Hardware)** hai jiska naam **"Server-01"** hai. Is Server ke andar ek **Oracle Software** installed hai.

Jab ServiceNow Discovery is Server ko scan karta hai, toh ek saath 2 data points milte hain:

1. **Main Item:** Server-01 (Hardware)
2.
3. **Child Item:** Oracle (Software Instance)
4.

Phase 1: Main CI Matching (Identifier Entries)

System sabse pehle **Identifier Entry** chalata hai taaki pehchan sake ki yeh Server kaun sa hai:

- Server ka **Serial Number** check kiya ➔ Database me **Server-01** match ho gaya.
-

Phase 2: Sub-component Matching (Related Entries)

Ab system dekhta hai ki Server-01 ke andar installed **Oracle Software** ko kis record ke saath connect karna hai. Yahan **Related Entries** kaam aati hain:

- **Related Table:** Software Instance
-
- **Reference Field:** installed_on (jo batata hai ki yeh kis server par hai)
-
- **Attribute:** Name (Software ka naam, i.e., "Oracle")

-

Result: System samajh jaata hai: *"Acha, yeh Oracle software wahi wala hai jo pehle se Server-01 par* installed_on *field se linked tha."* Toh system ek naya software record banane ki bajaye, **Server-01 ke existing Oracle Software record ko update kar deta hai**.

matlab ek hardware ke andar multiple components orr connect hote hai, to jab vo hardware ci aata hai to uske ye multiple components bhi aate hai and unn multiple components ki duplicate copy naa bne issliye ye related entries hai taaki ye inke rules ke according phle check kre ki ye entry phle se to nhi stored hai ? agar h to update kro nhi hai to insert kro ? right ?

Bilkul correct! Aapne exact sahi samjha hai.

Jab koi Hardware CI aata hai, toh uske sath aane wale saare child/sub-components (jaise Software, Network Adapters, Disk Partitions) ke liye **Related Entries** kaam karti hain.

Flow bilkul yahi hota hai:

1. **Rule Match Ho Gaya** ➔ Existing sub-component ko **update** kar do.

2. **Rule Match Nahi Hua** ➔ Naya sub-component record **insert** kar do.

Isi mechanism ki wajah se CMDB me sub-components ke duplicates nahi bante aur clean hierarchy bani rehti hai.

More easy eg :
Ek simple **Phone aur Apps** ke real-life example se samjho:

Maan lo aapke paas ek **iPhone** hai aur usme **WhatsApp** installed hai.

Jab ServiceNow scan (Discovery) karta hai, toh usko 2 cheezein milti hain:

1. **Main Item:** iPhone (Hardware)
2.
3. **Child Item:** WhatsApp (Software)
4.

Step 1: Phone ko Pehchano (Identifier Entries)

System sabse pehle dekhta hai: *"Yeh kaun sa phone hai?"*

- System iPhone ka **IMEI / Serial Number** check karta hai.

-
- Database me match mil gaya ➔ *"Haan, yeh wohi Harsh ka iPhone hai!"*
-

Step 2: App ko Pehchano (Related Entries)

Ab system phone ke andar ki apps dekhta hai: *"Is iPhone me WhatsApp pehle se hai ya naya install hua hai?"*

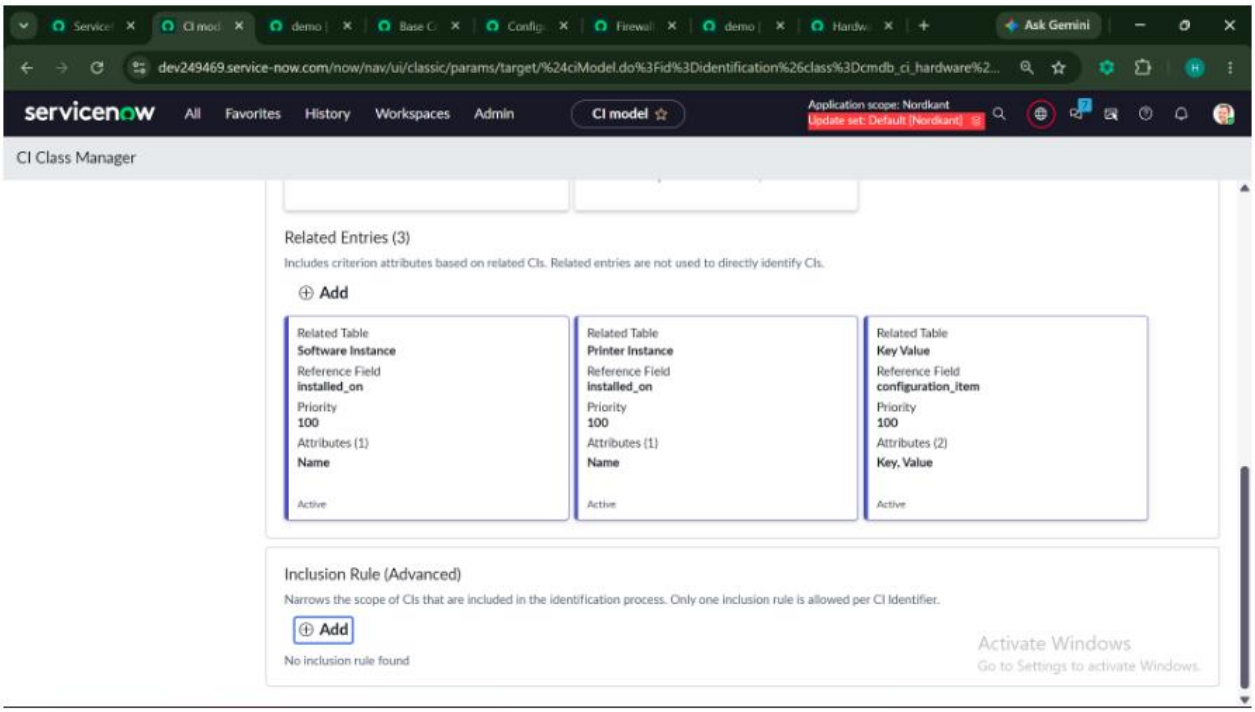
- System dekhta hai:
-
    - **App Name:** WhatsApp
    -
    - **Linked Phone (**installed_on**):** Harsh ka iPhone
    -

Result:

- **Agar WhatsApp pehle se installed tha:** System use pehchan kar bus update kar dega (naya duplicate WhatsApp icon nahi banayega).
-
- **Agar pehli baar mila:** System iPhone ke andar naya WhatsApp record entry kar dega.
-

Simple Line Me:

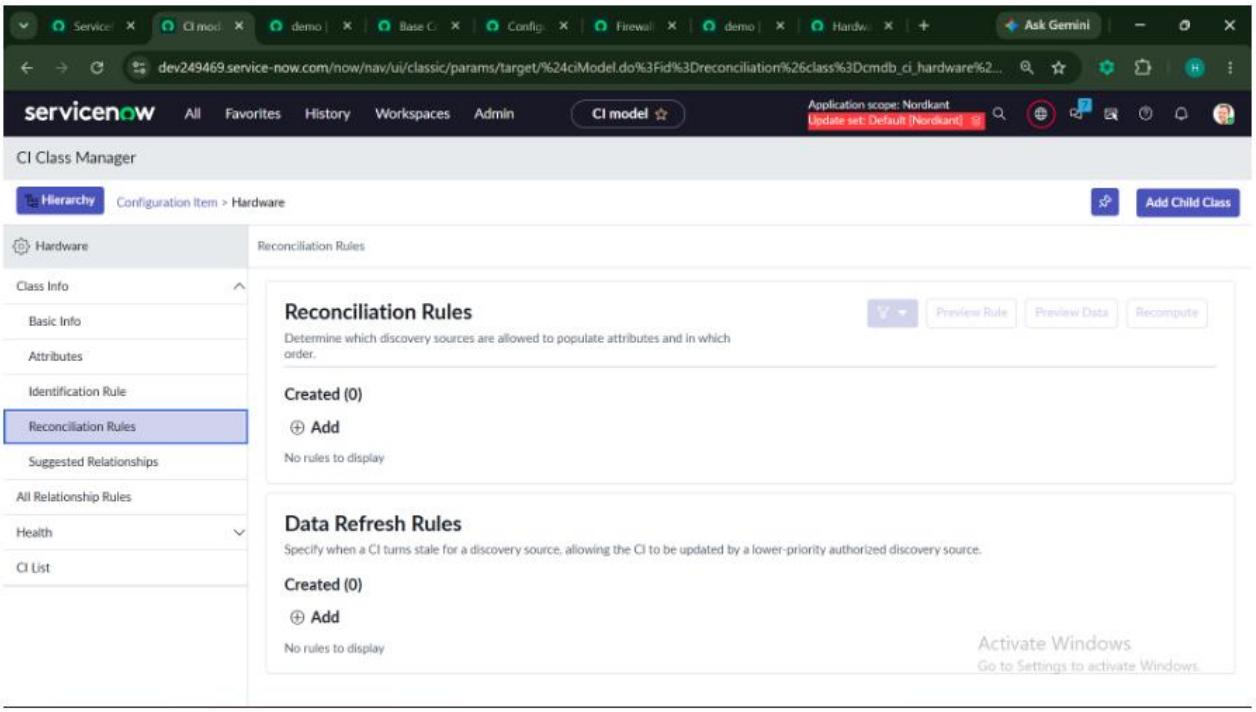
- **Identifier Entries ➔** Phone (Main Hardware) ki identity verify karti hain.
-
- **Related Entries ➔** Phone ke andar ki Apps (Sub-components) ki identity check karti hain taaki duplicate apps na banein!



Inclusion Rule (Advanced) ek filter condition hai jo yeh decide karti hai ki kaun-kaun se CIs identification process me aane chahiye.

- **Kaam:** Scope ko narrow karna (sirf specific condition match karne wale records ko hi identification rules me evaluate hone dena).
-
- **Example:** Agar aap lagao Install Status = Installed, toh system sirf active/installed status wale CIs par hi identification check karega, retired/in-trash devices ko process nahi karega.
-

In short: Yeh ek Entry Gate Guard ki tarah hai jo decide karta hai ki identification rules kis par chalenge aur kis par nahi.



Yeh **Reconciliation Rules** screen hai. Identification Rule ne CI toh pehchan liya, lekin jab **multiple sources** (jaise ServiceNow Discovery, SCCM, ya Manual CSV Import) same CI ka data bhejte hain, toh **kiska data maana jaaye aur kiska reject ho**, yeh yahan decide hota hai.

1. Reconciliation Rules (Data Priority Rule)

Yeh rules decide karte hain ki kaun sa **Data Source** kis field/attribute ko update kar sakta hai.

Real-life Example: Maan lo ek Server ki RAM size batani hai:

- **Source A (ServiceNow Discovery):** Auto-scan tool hai, yeh continuous live network scan karta hai. Iski priority **100 (High)** set kar di.
-
- **Source B (Excel Upload / Manual Import):** Human entry hai. Iski priority **200 (Low)** set kar di.
-

Scenario:

1. **Discovery** ne scan kiya aur Server RAM **64 GB** update kar di.
2.
3. Agle din kisi ne **Excel Upload** karke uski RAM **32 GB** likh di.
4.
5. **Result:** ServiceNow Excel wale update को **reject** kar dega kyunki **Discovery (Priority 100)** ka data **Excel (Priority 200)** se zyada trustworthy hai.
6.

2. Data Refresh Rules (Stale Data Rule)

Yeh rules handle karte hain ki agar High Priority source quiet/inactive ho jaaye, toh Low Priority source kab update kar sakta hai.

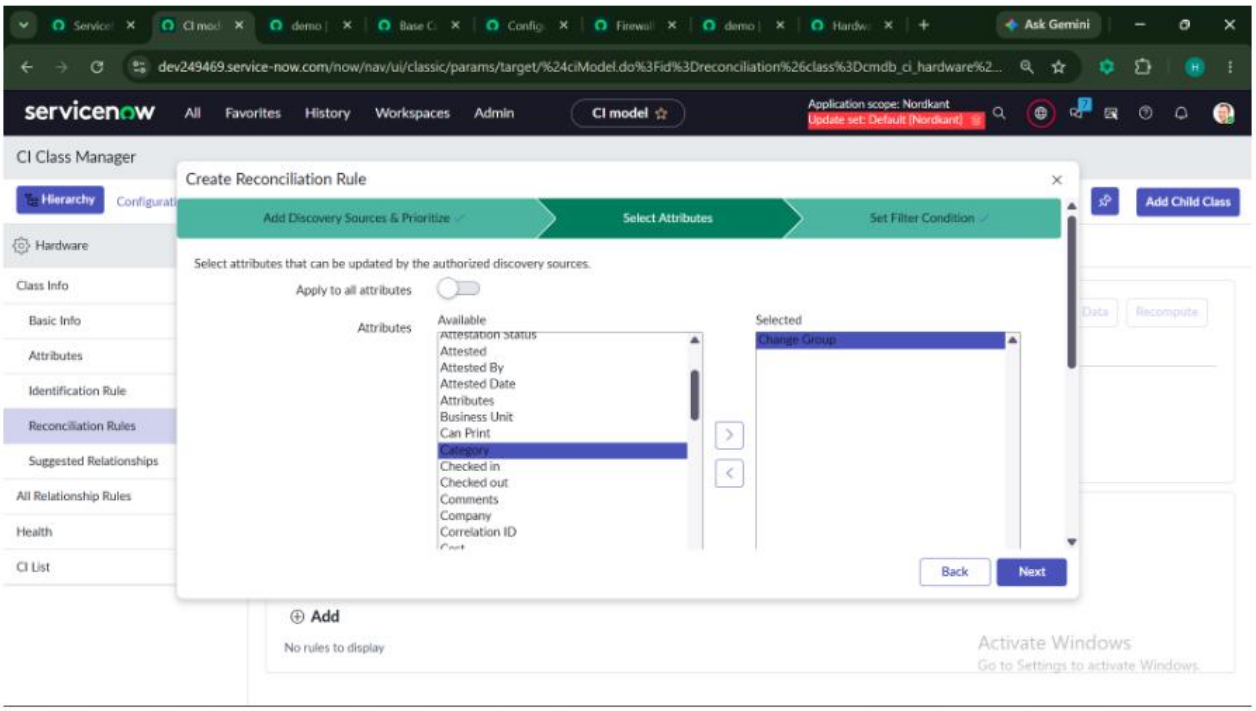
Real-life Example:

- **Discovery (Priority 100)** 30 dino se down hai aur koi naya data nahi bhej raha. Server ab **Stale (Purana)** ho gaya hai.
-
- **Rule Setup:** *"Agar Priority 100 source 30 dino tak data update na kare, toh Priority 200 (Excel Upload) ko update karne ki permission de do."*
-

Scenario: 30 din baad jab kisi ne Manual/Excel se update kiya, toh system abuse block karne ki bajaye permission **allow** kar dega kyunki pehla source stale ho chuka hai.

Quick Summary

- **Reconciliation Rules:** Decide karta hai ki overlapping data milne par **kaun si Discovery Source dominate/win karegi**.
-
- **Data Refresh Rules:** Decide karta hai ki agar top source band/stale ho jaaye, toh **lower source kab takeover kar sakta hai**.



Yeh **Reconciliation Rule create karne ka Wizard** ka Step 2 (Select Attributes) hai.

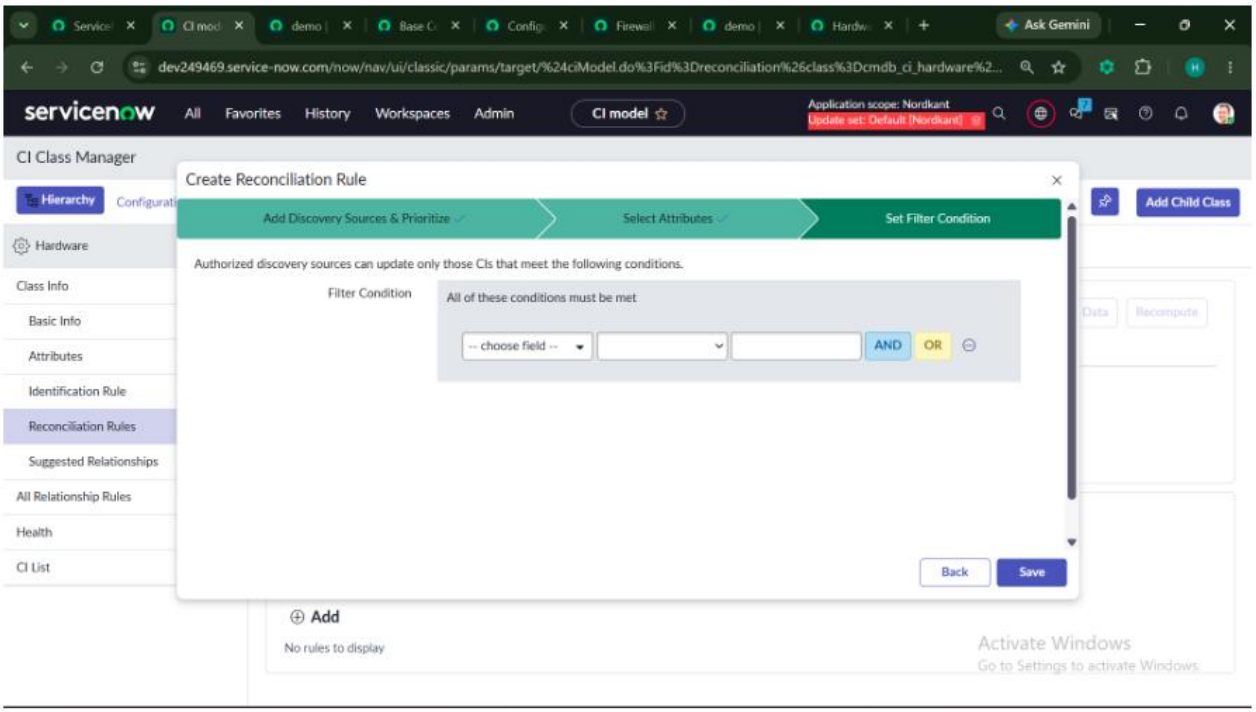
Iska kaam yeh select karna hai ki specific **Discovery Source** ko table ke **kaun-kaun se exact fields (attributes)** update karne ki permission deni hai.

Easy Example se samjho:

Maan lo aap chahte ho ki **SCCM (Discovery Source)** sirf Server ka Change Group aur Category field hi update kar sake, baki fields ko na chede.

1. **Apply to all attributes (Toggle OFF)**: Iska matlab hai ki rule saari fields par nahi, sirf chuni hui specific fields par chalega.
2.
3. **Available -> Selected**: Aapne Change Group ko right side move kar diya.
4.

Result: Ab SCCM ke paas permission hogi ki jab bhi data aaye, wo Hardware CI ke sirf Change Group field ko update karega. Baaki fields (jaise RAM, IP Address, Serial Number) ko wo touch nahi kar payega.



Yeh Reconciliation Rule creation ka Step 3 (**Set Filter Condition**) hai.

Iska kaam yeh restrict karna hai ki yeh Reconciliation Rule **sirf specific conditions match karne wale CIs par hi apply ho**.

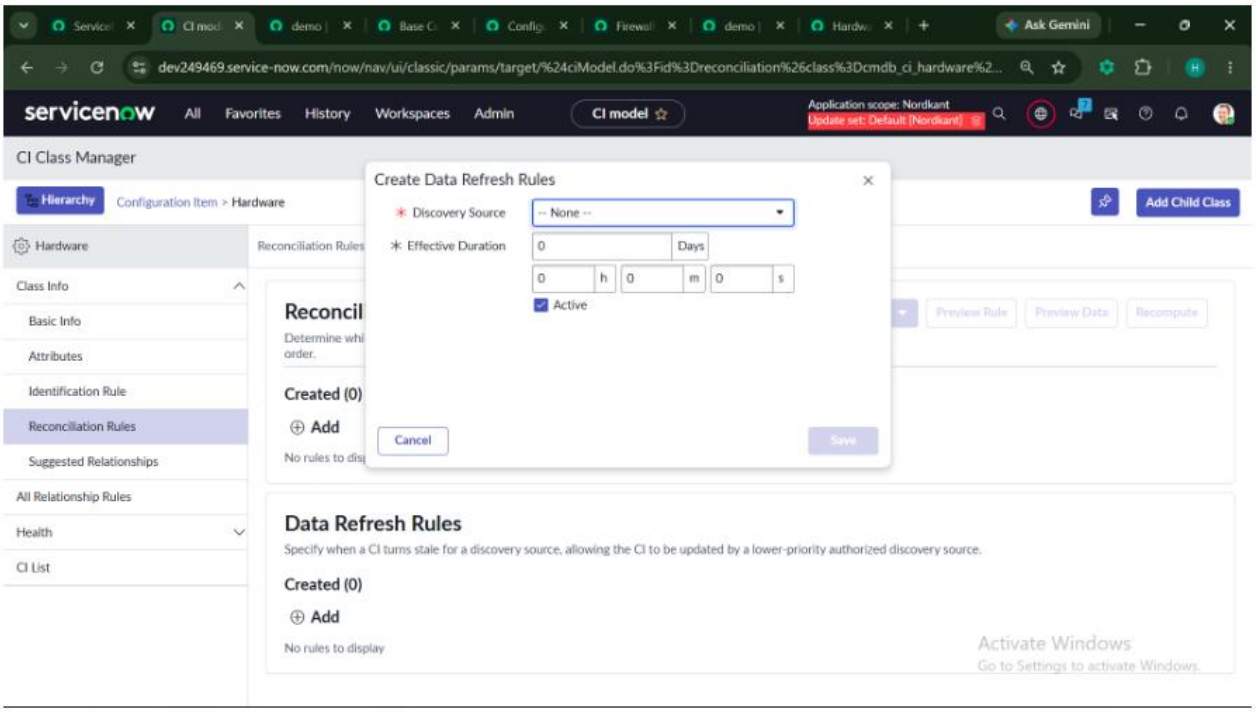
Easy Example:

Maan lo aap chaos nahi chahte aur condition lagate ho:

- Location is Delhi ya Environment is Production

Result: Discovery Source sirf unhi Hardware CIs ko update kar payegi jo Delhi me hain ya Production environment ke hain. Baaki sabhi CIs par yeh rule ignore ho jayega.

In short: Yeh ek Extra Security Gate/Filter hai jo decide karta hai ki Data Source **kaha aur kis record par** apna change apply kar sakta hai.

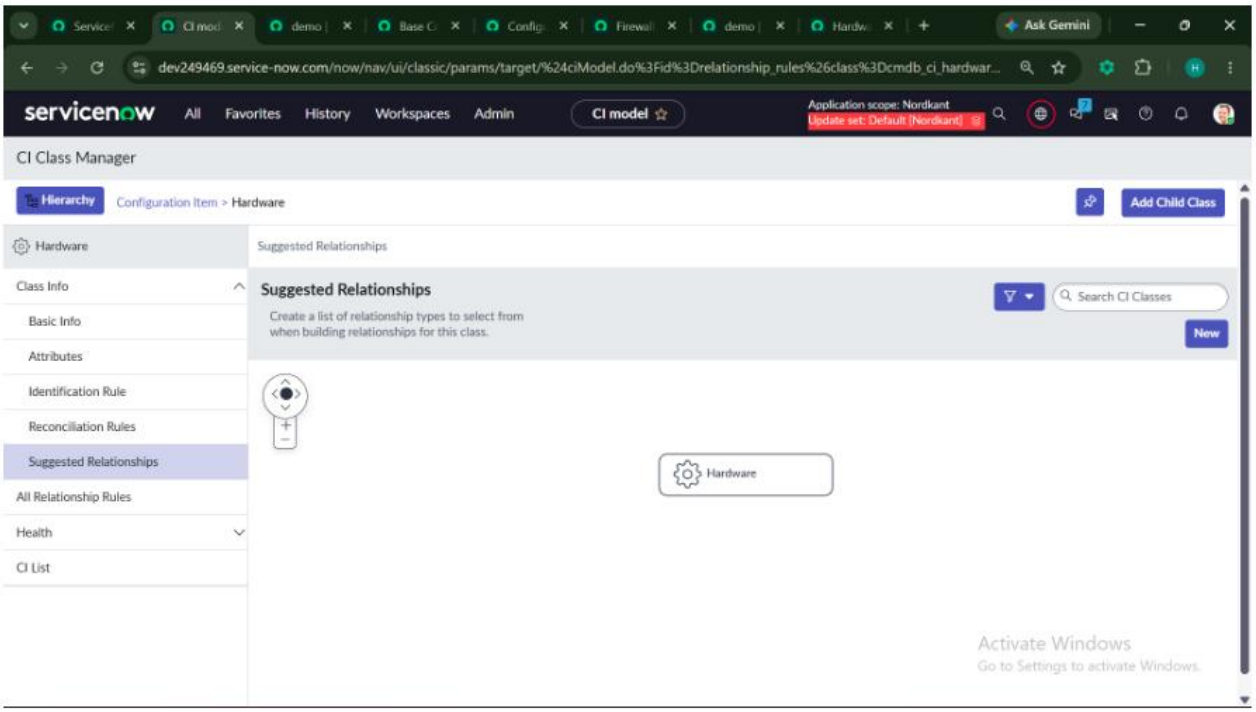


Yeh **Data Refresh Rule** create karne ka pop-up hai.

Iska kaam yeh Time Duration set karna hai ki high-priority source kitne din/ghante tak active na rahe toh **data ko "Stale" (purana) maan liya jaaye**.

- **Discovery Source**: Woh main tool select karo (e.g., *ServiceNow Discovery*).
- **Effective Duration**: Time Limit set karo (e.g., *30 Days*).

Result: Agar chosen Discovery Source ne 30 din tak kisi CI ko update nahi kiya, toh system lower-priority sources (jaise Manual Upload) ko us CI ko update karne ki **permission de dega**.



Yeh **Suggested Relationships** ki screen hai.

Iska main kaam yeh define karna hai ki is class (e.g., Hardware) ka dusri classes ke saath **kaun sa logical/valid connection ya relation** ho sakta hai. Yeh dropdown menu me sirf valid options dikhane ke liye use hota hai taaki users galat relationships na bana dein.

Easy Example se samjho:

Maan lo aapke paas **Server (Hardware)** aur **MySQL (Database Software)** hai.

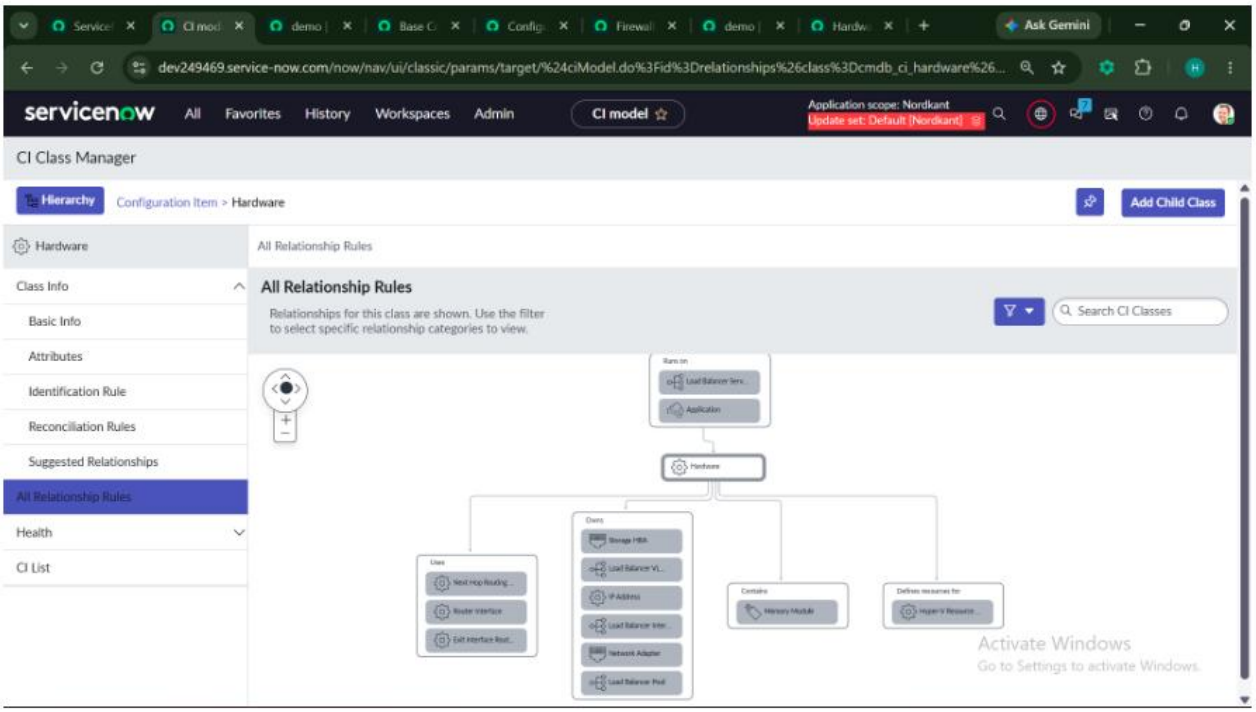
- **Valid Relation:** Server Runs on::Runs MySQL Database.
- **Invalid Relation:** Server Is Powered By MySQL Database (yeh logic wise galat hai).

Suggested Relationships me hum pehle se fix kar dete hain ki **Hardware** class kis-kis ke saath connect ho sakti hai:

1. Hardware ➔ **Runs on::Hosts** ➔ Software
2. Hardware ➔ **Connects to::Connected by** ➔ Network Switch

Key Benefits:

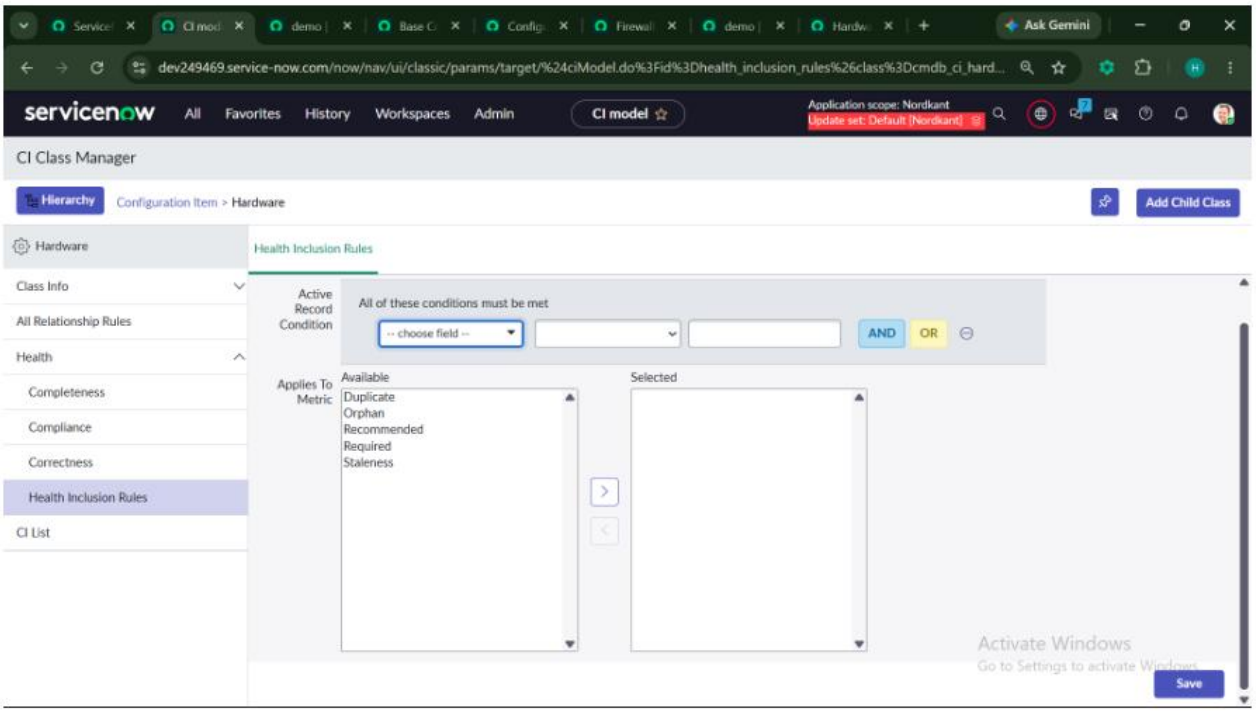
- **Dropdown Suggestions:** Jab koi admin ya discovery tool manual relationship banata hai, toh unhe 100 random options ke bajaye sirf sensible options hi dikhte hain.
- **Data Consistency:** CMDB clean rehta hai aur galat relationships nahi banti.



Agar aapko kisi ko bilkul simple words me samjhana ho:

- **Suggested Relationships**: Yeh hamari banayi hui **Approved Checklist** hai jo batati hai ki is Class ke liye kaun-se relationships valid hain, taaki user manual entry karte waqt koi galat connection na bana de.
-
- **All Relationship Rules**: Yeh **Full Complete Tree** hai jo is Class ke apne rules + Parent Class se mile hue saare inherited rules ko ek saath visual map me dikhata hai.
-

One-Liner: *Suggested* shortlist karta hai allowed options ko, jabki *All Relationship Rules* poori hierarchy ka total system view dikhata hai.



CMDB **Health** section ka main kaam yeh measure karna hai ki aapke CMDB data ki quality kitni achhi aur reliable hai. Isme 4 main modules hote hain:

1. Completeness (Pura Data Hai Ya Nahi?)

Check karta hai ki CIs me saare zaruri details bhare hain ya mandatory fields khaali (blank) chhodi gayi hain.

- **Mandatory Fields**: Batata hai ki is Class ke CI me kaun si fields hona compulsory hai (e.g., *Serial Number*).
-
- **Recommended Fields**: Batata hai ki achhi quality ke liye kaun si fields hona advisable hai (e.g., *Assigned to*, *Asset Tag*).
-

2. Compliance (Company Standards Follow Ho Rahe Hain?)

Check karta hai ki CIs company ke predefined policies aur audit standards ko follow kar rahe hain ya nahi.

- **Example**: Check karna ki kya saare Production Servers standard OS version par chal rahe hain ya koi un-authorized changes hain.
-

3. Correctness (Data Sahi Aur Accurate Hai?)

Check karta hai ki CMDB me koi junk, outdated, ya duplicate data toh nahi hai. Yeh 3 cheezon par focus karta hai:

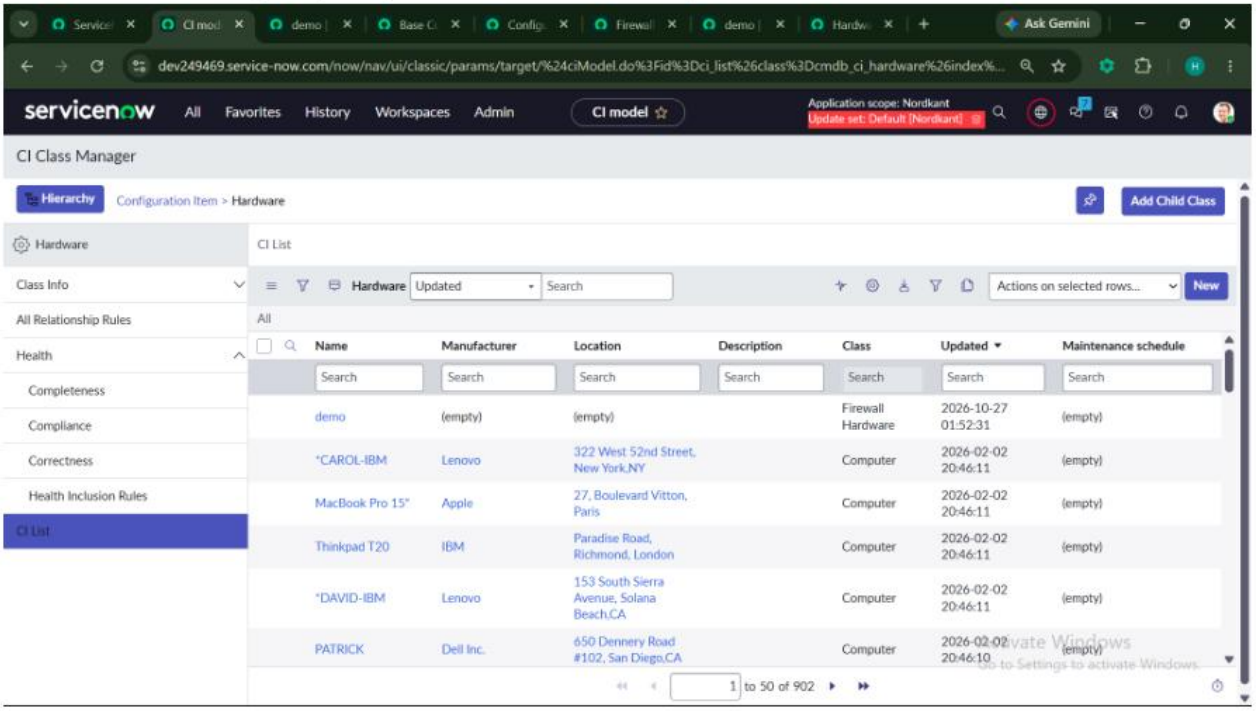
- **Duplicates**: Kya ek hi Hardware 2 baar create ho gaya hai?

- **Orphan Records**: Kya koi aisa child record hai jiska parent CI hi delete ho chuka hai?
- **Stale CIs**: Kya koi aisa CI hai jiska data months se scan/update nahi hua?

4. Health Inclusion Rules (Kiske Upar Health Check Chalana Hai?)

Yeh ek gatekeeping filter hai jo decide karta hai ki Health KPIs/Metrics (Completeness, Compliance, Correctness) **kaun-kaun se CIs par calculate honge**.

- **Example**: Health calculation me sirf Install Status = Installed wale live devices ko include karo, aur junk ya retired devices ko report se exclude kar do.



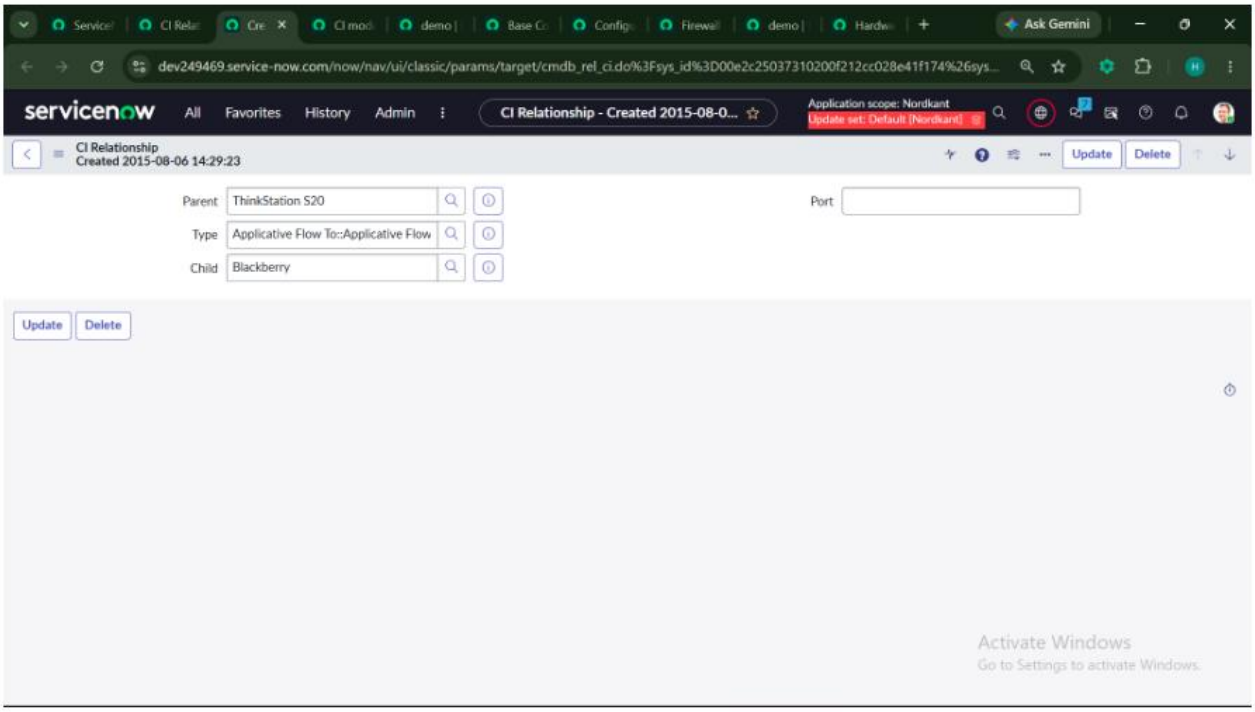
Question : Relationship table and concepts ?
Answer :
Connection Strength (cmdb_rel_ci table me) batati hai ki do CIs ke beech ka connection **kitna permanent ya frequent** hai.

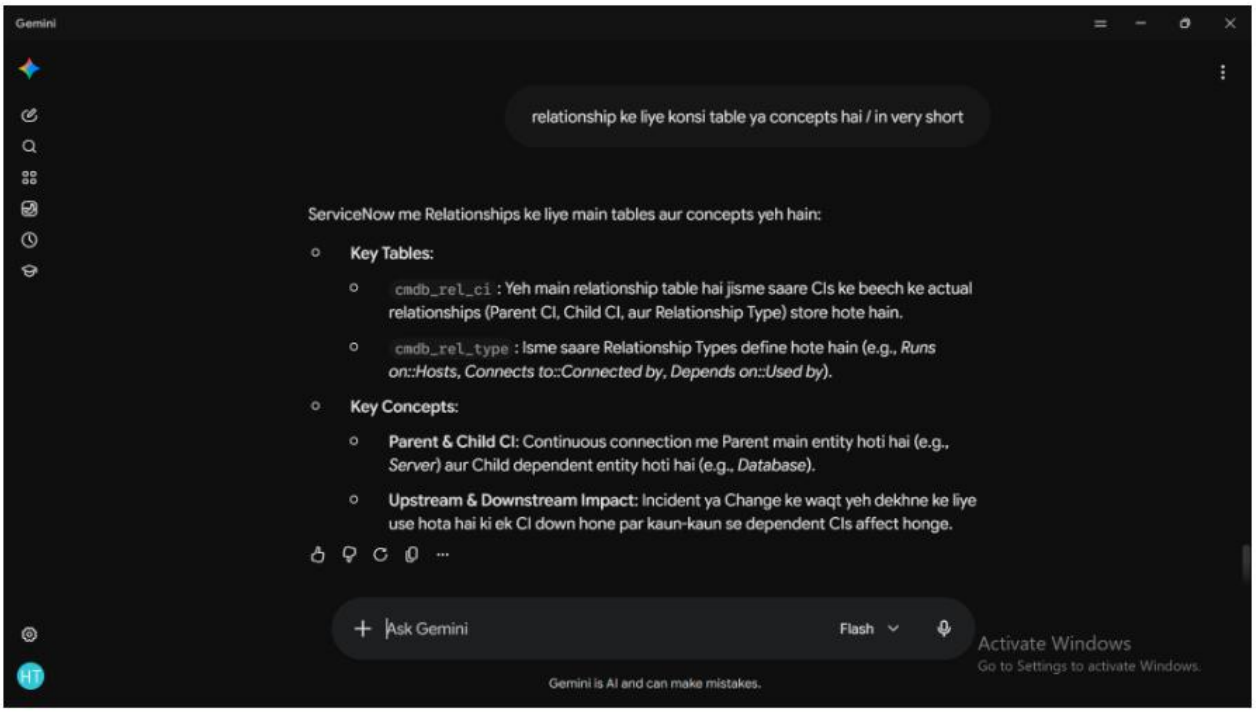
- **Values**: Isme options hote hain jaise Always (hamesha connected), Occasional, Infrequent, ya Cluster.
- **Purpose**: Yeh batata hai ki relationship continuously active hai (jaise ek Physical Server par App ka chalna = *Always*) ya fir intermittent/temporary hai.

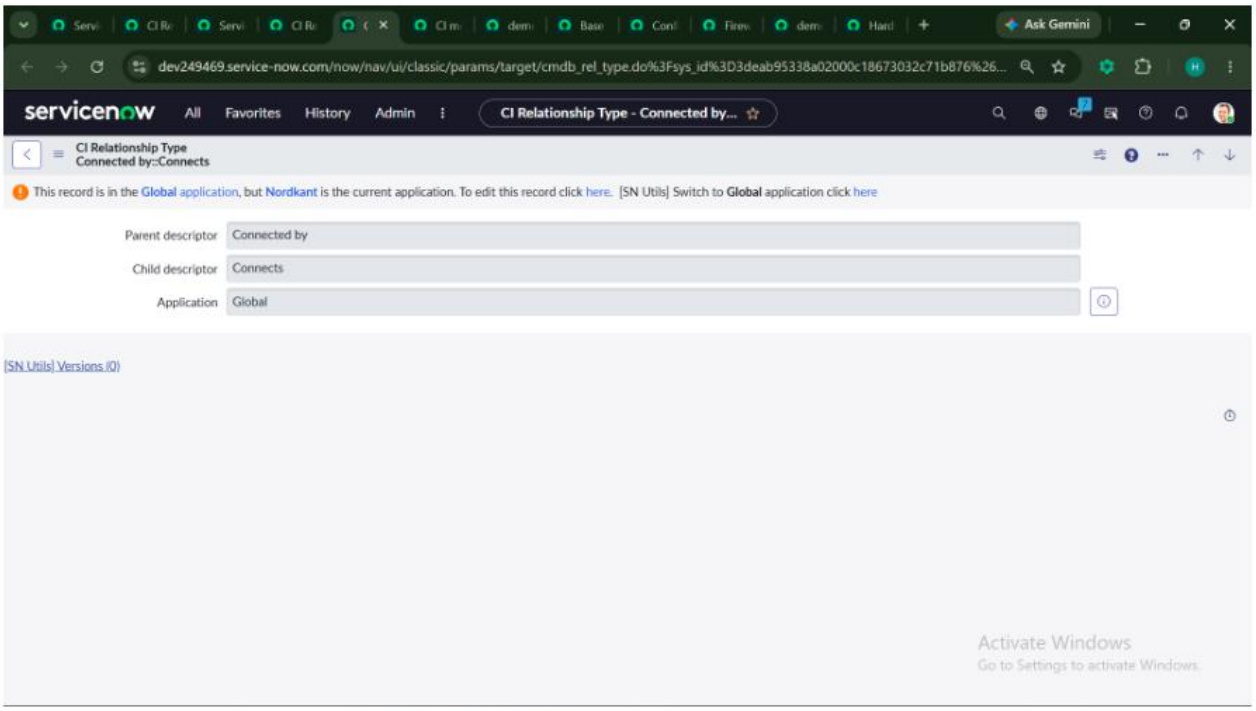
Note: Yeh ServiceNow ka ek **legacy/older field** hai jo built-in features me heavy use nahi hota, isiliye default entry aksar **"Always"** rehti hai.

cmdb_rel_ci table me **Port** field specific network ya application connections ke **Port Number** ko store karne ke liye hoti hai.

- **Kaam:** Ye batata hai ki Parent CI aur Child CI kis network port ke zariye communicate kar rahe hain (e.g., 80 HTTP ke liye, 443 HTTPS ke liye, 3306 MySQL ke liye).
-
- **Usage:** Special relationship types (jaise *Applicative Flow To* ya network mapping) me connection ka exact endpoint Specify karne ke liye optional detail ki tarah use hoti hai.







Yeh **CI Relationship Type** (cmdb_rel_type table) ka record definition form hai.

Iska kaam Parent aur Child CIs ke beech ki **2-way (directional) relationship labels** ko define karna hai:

- **Parent descriptor (**Connected by**)**: Batata hai ki Parent CI ki nazar se Child CI kya hai (e.g., *Server A is Connected by Switch B*).

-
- **Child descriptor (**Connects**)**: Batata hai ki Child CI ki nazar se Parent CI kya hai (e.g., *Switch B Connects to Server A*).
-

In Short: Yeh batata hai ki jab do CIs aapas me link honge, toh Parent ki taraf se kya naam dikhega aur Child ki taraf se kya naam dikhega.

NOTE :
Aap interview ya deep technical discussion ke liye ye high-impact doubts pooch sakte ho:

1. **IRE vs Normal Script:** *"Jab custom Background Script se* gr.insert() *karte hain, toh IRE execution skip kyun ho jata hai aur duplicate records kyun ban jate hain?"*
2.
3. **Dependent CI Cascading:** *"Agar Parent Hardware CI delete ho jaye, toh unke Child CIs (e.g., Software Instances) ke Related Entries ka kya hota hai?"*
4.
5. **Reconciliation Precedence:** *"Agar two Discovery Sources ki priority SAME set kar dein, toh ServiceNow konse source ke data ko accept karega?"*

Yeh rahe aapke teeno questions ke precise aur direct answers:

1. IRE vs Normal Script

- ***Kyun skip hota hai:*** gr.insert() *direct database level operation karta hai, jo ServiceNow ke standard Identification Rules, Reconciliation Rules, aur CI Class Manager logic ko completely bypassed/bypass kar deta hai.*
-
- ***Duplicate kyun bante hain:*** *Script database me bina identity matching rules run kiye raw row insert kar deti hai.*
-
- ***Solution:*** *Duplicate roknay ke liye background script me* gr.insert() *ki jagah* IdentificationEngineScriptable ***API*** *(*createOrUpdateCI()*) ka use kiya jata hai taaki IRE execute ho.*
-

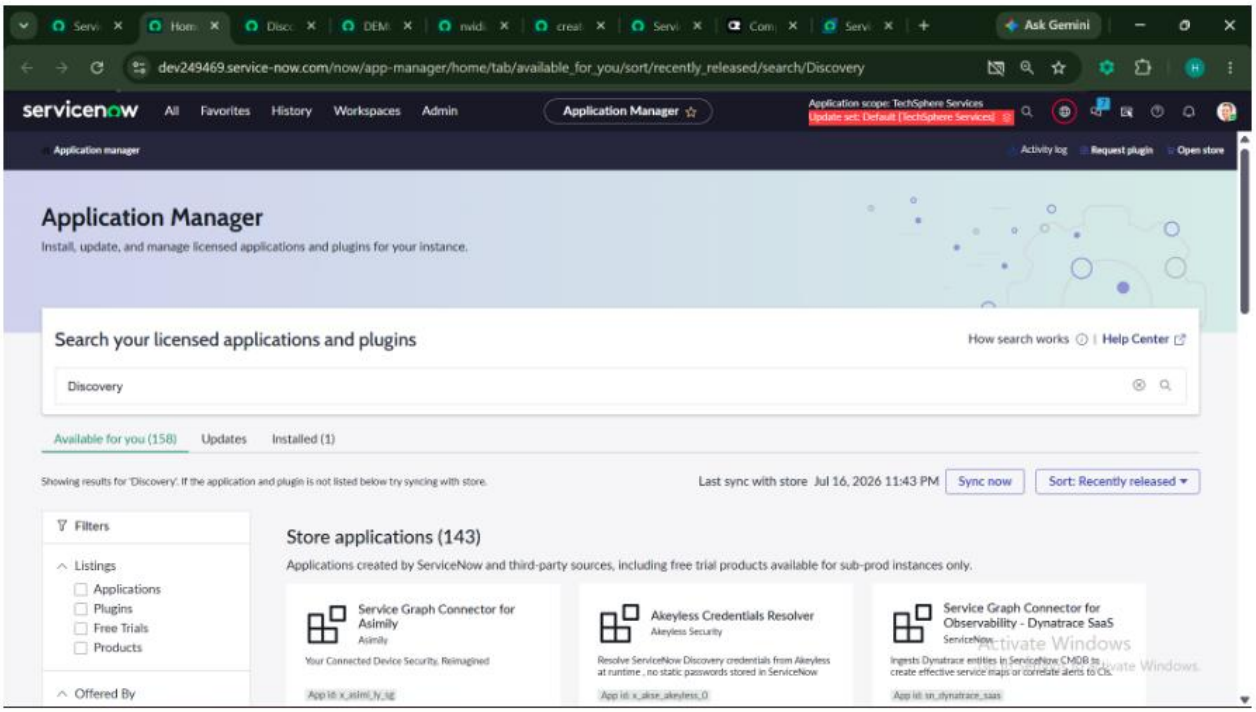
2. Dependent CI Cascading

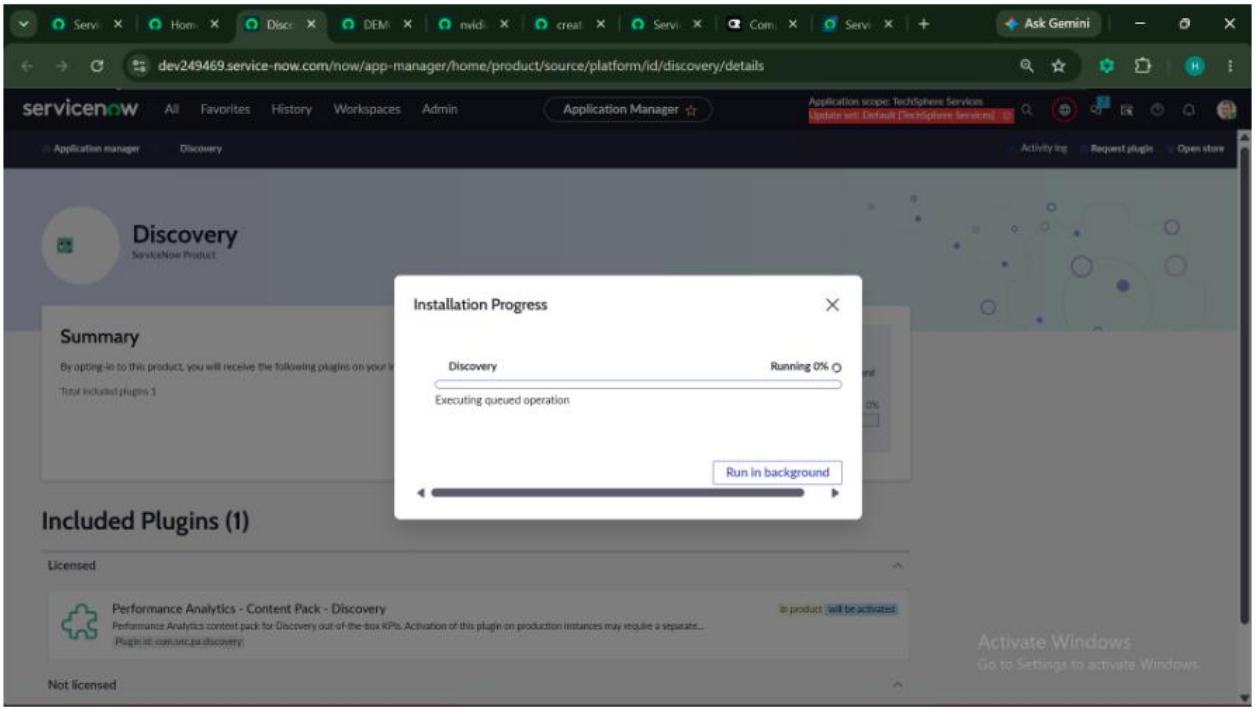
- ***Related Entries ka kya hota hai:*** *Jab Parent Hardware CI delete hota hai, toh uspar dependent child records (jaise Software Instances, Disk Partitions)* ***Orphan Records*** *ban jate hain.*
-
- ***System Behavior:*** *Related Entries script/rule unhe automatic cascade-delete nahi karti (jab dekha jaye standard CMDB settings me). Yeh child records reference field (*installed_on*) empty ya orphaned state me CMDB me bina parent ke pade rehte hain aur CMDB Health me Orphan/Correctness issue flag karte hain.*
-

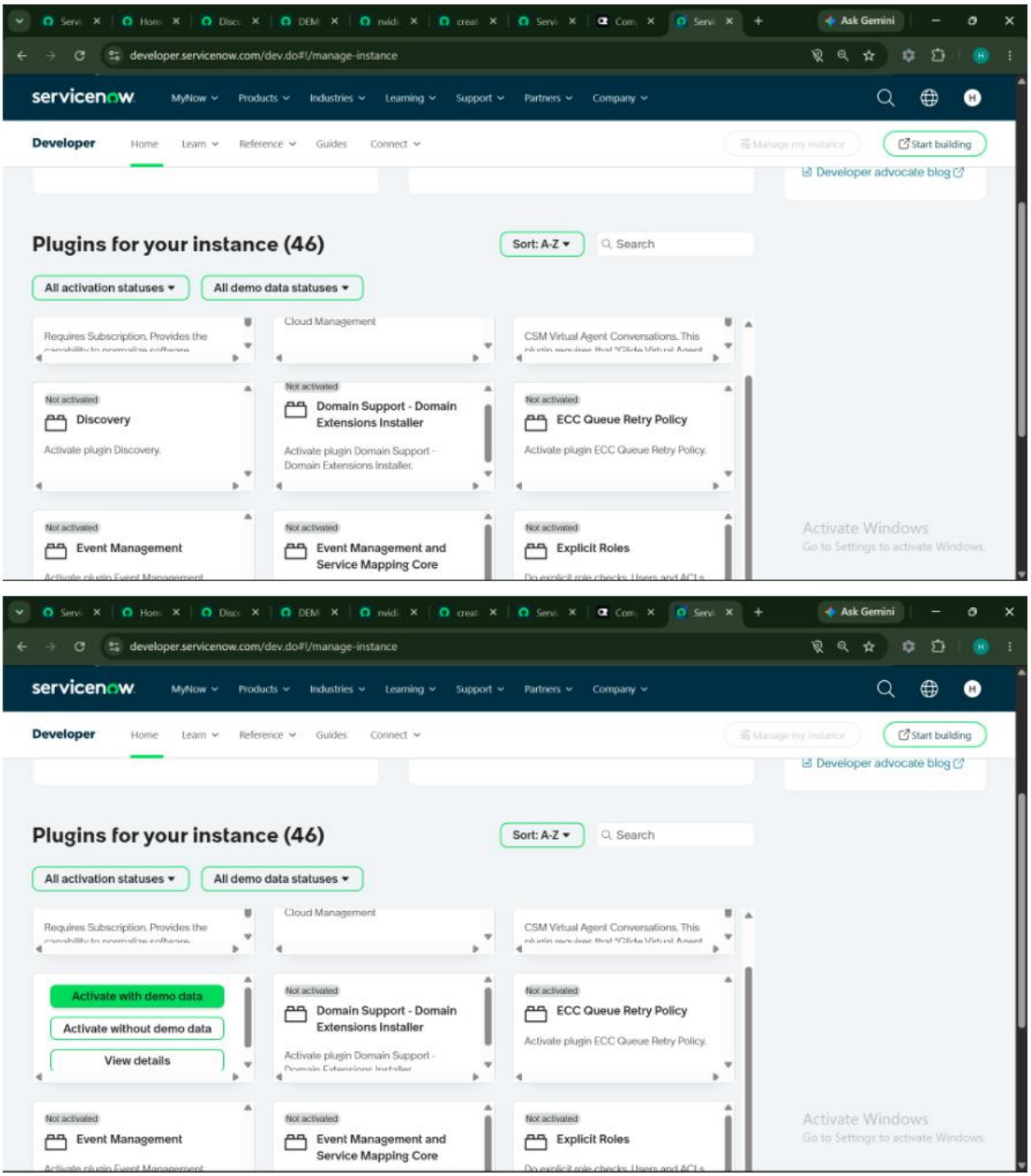
3. Reconciliation Precedence (Same Priority Tie)

- ***Konsa source win karta hai:*** *Agar do Discovery Sources (e.g., Source A aur Source B) ki priority score* ***exact SAME*** *hai (e.g., dono 100 hain), toh ServiceNow* ***"Most Recent Update" (Latest Timestamp)*** *ko accept karta hai.*
-
- ***Tie-Breaker Rule:*** *Jo Data Source sabse nayi/latest timestamp ke sath payload bhejegi, usko preference di jayegi aur existing value overwrite ho jayegi.*

QUERY BUILDER:


dev249469.service-now.com/now/app-manager/home/tab/available_for_you/sort/recently_released/search/Discovery
servicenow
All
Favorites
History
Workspaces
Admin
Application Manager
Application manager
Activity log
Request plugin
Open store
Application Manager
Install, update, and manage licensed applications and plugins for your instance.
Search your licensed applications and plugins
How search works
Help Center
Discovery
Available for you (158)
Updates
Installed (1)
Showing results for 'Discovery'. If the application and plugin is not listed below try syncing with store.
Last sync with store Jul 16, 2026 11:43 PM
Sync now
Sort: Recently released
Filters
Listings
Applications
Plugins
Free Trials
Products
Offered By
Store applications (143)
Applications created by ServiceNow and third-party sources, including free trial products available for sub-prod instances only.
Service Graph Connector for Aslmily
Aslmily
Your Connected Device Security, Reimagined
Akeyless Credentials Resolver
Akeyless Security
Resolve ServiceNow Discovery credentials from Akeyless at runtime , no static passwords stored in ServiceNow
Service Graph Connector for Observability - Dynatrace SaaS
ServiceNow
Ingests Dynatrace entities in ServiceNow CMDB to create effective service maps or correlate alerts to CIs.
Activate Windows


dev249469.service-now.com/now/app-manager/home/product/source/platform/id/discovery/details
servicenow
All
Favorites
History
Workspaces
Admin
Application Manager
Application manager
Discovery
Activity log
Request plugin
Open store
Discovery
ServiceNow Product
Summary
Installation Progress
Discovery
Running 0%
Executing queued operation
Run in background
Included Plugins (1)
Licensed
Performance Analytics - Content Pack - Discovery
Performance Analytics content pack for Discovery out-of-the-box KPIs. Activation of this plugin on production instances may require a separate...
Not licensed
Activate Windows
Go to Settings to activate Windows.



CSDM (common service data model)

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Common Service Data Model Overview

Briefing

Mark Bodman
Sr. Product Manager



Agenda

Strategic Context

3 pillars of a successful data foundation

CSDM Overview

Features supporting CSDM

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The CMDB/Service Graph *is* your digital foundation

It is critical to driving outcomes for your organization and ServiceNow



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3 Pillars for a successful Data Foundation (CMDB+)



01

Ingest

- Continuously bring in data, merge, update, delete
- Automate Physical CI's and Cloud resources / structure
- Merge separate sources into one trusted model
- Connect the discoverable to non-discoverable
- Ensure managed ownership



02

Govern

- Governance team & roles
- Ensure health metrics improve
- Establish lifecycle processes
- Manage & communicate structure and definitions
- Support use and customization / extension needs



03

Insight

- Establish analysis guidance
- Provide operational support for incident, change mgmt.
- Use in reporting and metrics
- Leverage for planning
- Support consuming products, roles, personas, processes



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a single trusted repository we want to connect the discoverable to the

The 3 Pillars – CMDB Features

01

Ingest

- Ingestion Pipeline
- IntegrationHub ETL
- Service Graph Connectors
- ITOM Visibility (Discovery, SM)
- Agent Client Collector
- IRE
- Multisource CMDB

02

Govern

- CMDB Data Manager
- CI Class Manager
- CMDB 360
- CSDM docs, guides, videos etc.
- Dynamic CI Groups
- Data Foundations Dashboards
- CMDB Health Dashboards

03

Insight

- CMDB Workspace
- CMDB Query Builder
- Service Graph Data Visualizer
- CSDM Product Views
- CMDB Search (NLQ)
- Support Products; ITSM, ITAM, ITOM Health, SPM, etc.

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the way we come to Market with around these three pillars so we can

Strong Data Foundation = value across the enterprise

Critical for managing digital products and services

Product family	ITSM	ITOM	ITAM	SPM	SecOps	DevOps	CSM	IRM (GRC)
Value of CMDB	Critical	Very Critical	Very Critical	High	Critical	High	Critical	High
Benefits of a mature CMDB & CSDM Framework	• Change mgmt. • True service level reporting • Remove semantic disparity	• Visibility across environment • System health and availability • Map issues to business processes for prioritization	• Accurate license positions • Visibility across operational and financial data	• Mapping of resources to business processes • Service types based on business context • Planning scope & impact of investments / Divestments	• Prioritization of vulnerabilities based on business criticality • Faster incident isolation and response • End-to-end risk visibility	• Automated DevOps change management • Link between configuration changes and impacted configuration items (CIs)	• Visibility across customer environment • Accurate response to incidents	• Ensures quality data used for audit • Healthy, trusted data is critical for audit scoping & activities
Key metrics	• Reduction in tickets • Reduction in MTTR	• Elimination of service outages • Reduction in MTTR	• Reduction in software license fines that can run in the millions • Reduction in failed audits	• Reduced business risk and outages from outdated technologies	• Reduction in time to remediate priority 1 vulnerabilities	• Automated change	• Customer satisfaction	• Accurately identify audited



the value of the data across many different areas and the criticality of

New Products and Features Requiring / Supporting CSDM

Every release more products, features, and functions use CSDM guidance

New product features that use CSDM

- CMDB Data Manager – implement CI policies using CSDM Lifecycle Stage, Lifecycle status
- Dynamic CI Group - New in Paris, part of Event alignment to CSDM
- Business Applications per Incident – C I-> Application Service -> Business Application
- CSDM in Change – Using Dynamic CI Groups for change processes
- Application Service Wizard and API's – Under CSDM nav, integrate with CI/CD tools
- Data Synchronization - Technical Service -> (Technical) Offering -> Dynamic CI Group -> CIs

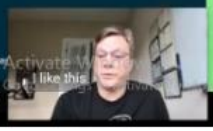
New products that use CSDM

- Site Reliability Operations
- Service Builder - Creates Service -> Offering -> Application Service & Catalog Item
- Digital Portfolio Management
- DevOps Config – New SDLC Component in Build domain (CSDM V4)

11 Product Views in DOC site explains how our products use CSDM, contribute to and uses the CSDM model.

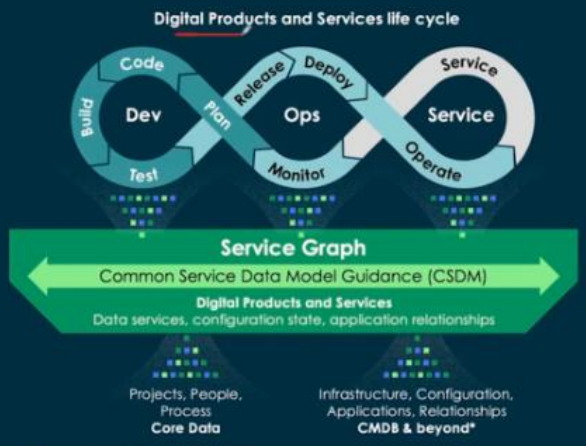
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there's a lot of features that require the csdm structure csdm was introduced a

Service Graph: next generation system of record for digital products and services



Service Graph Overview

Connected
Service Graph connects all functions necessary to manage Digital Products at every layer of the modern IT stack

Integrated
Planning, Architecture, DevOps, Traditional or Cloud—all guided by a single CSDM Framework

Comprehensive
Service Graph is broader than traditional CMDBs which were limited in scope and capability

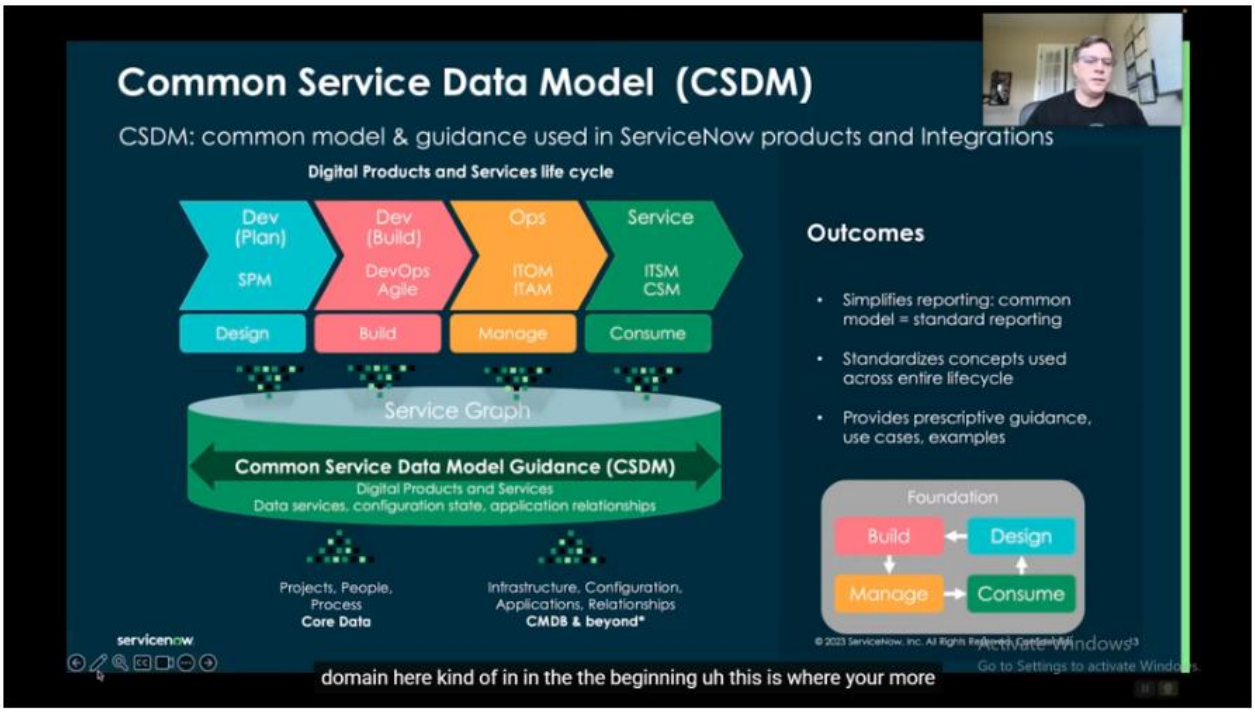
servicenow

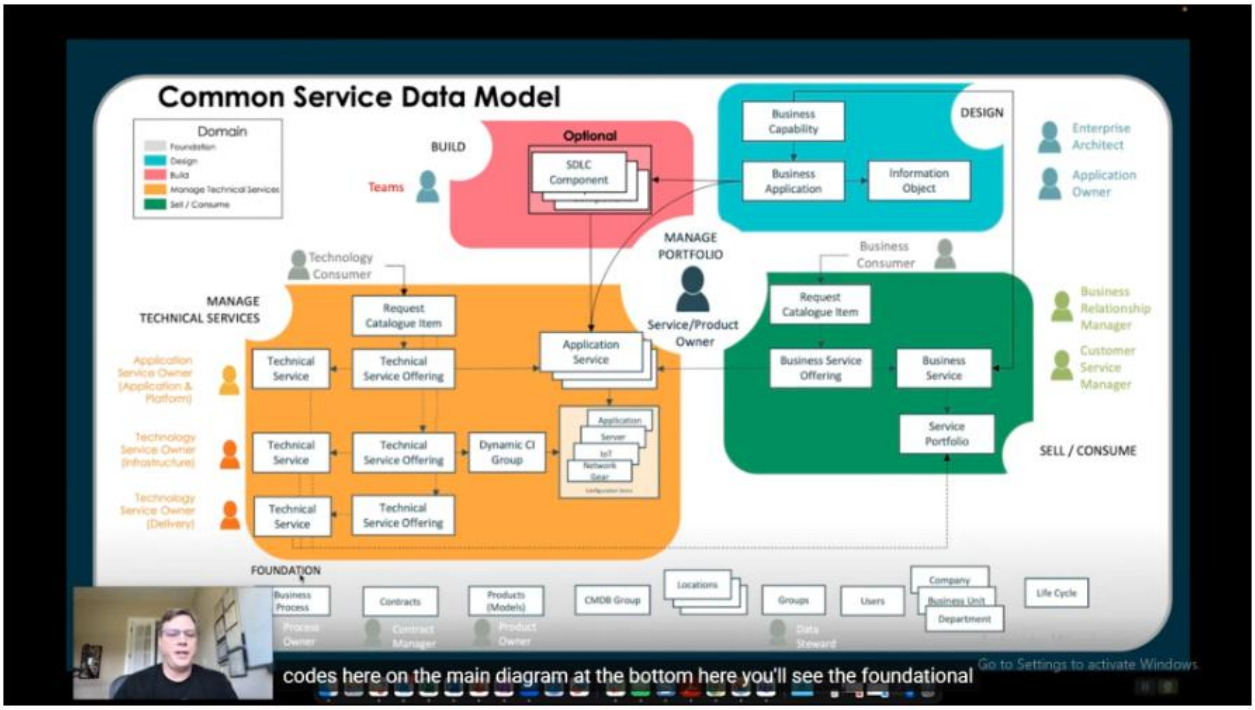
*rolling out over next several releases

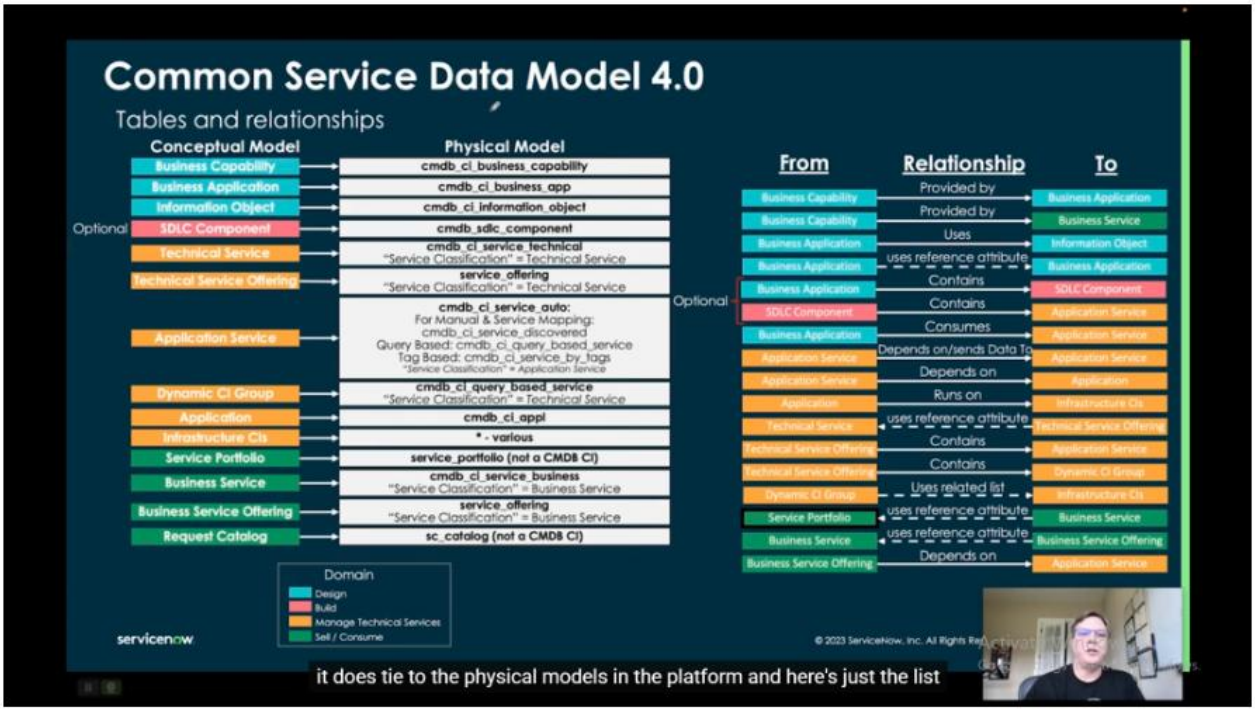
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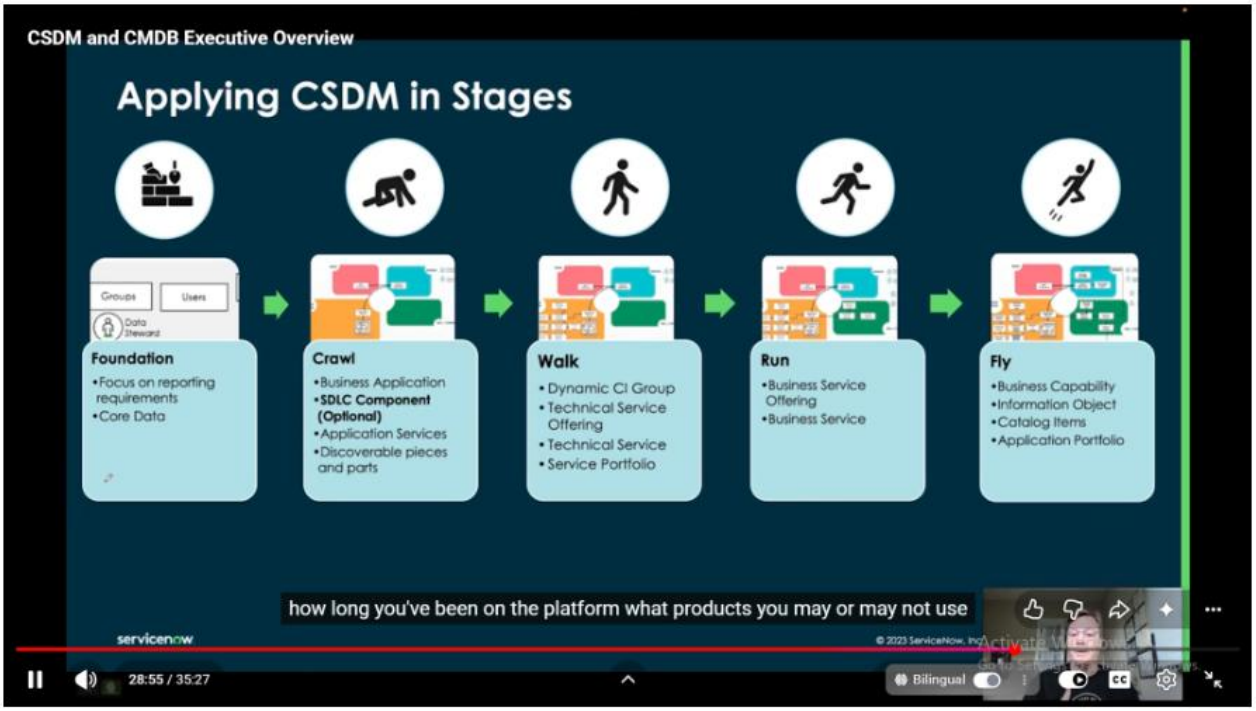


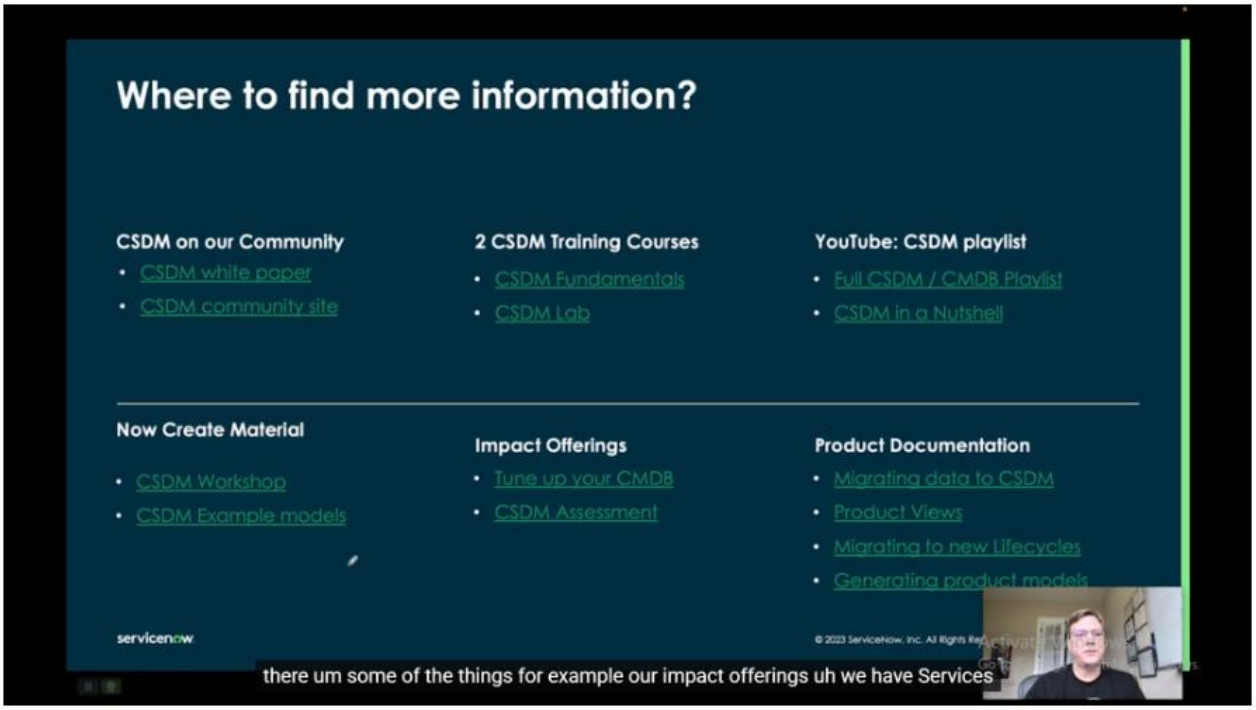
cstm is in the context of that full what I call digital products and services

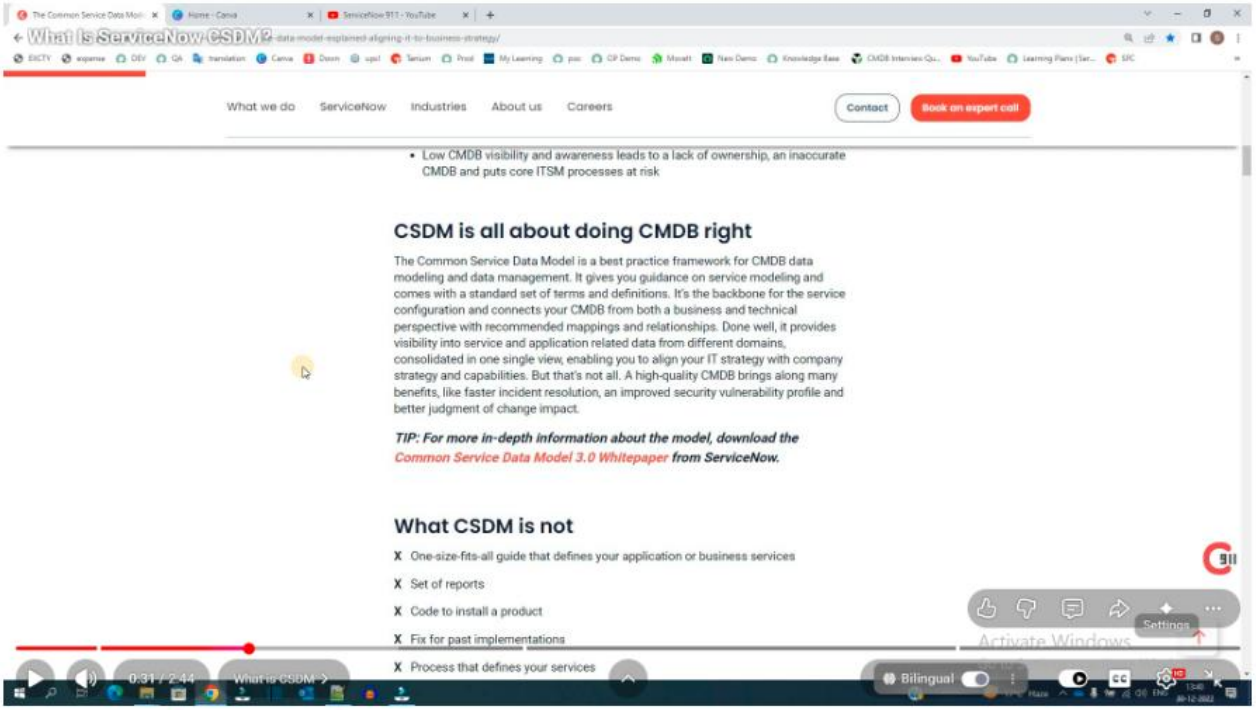

Common Service Data Model (CSDM)
CSDM: common model & guidance used in ServiceNow products and Integrations
Digital Products and Services life cycle
Dev (Plan)
SPM
Dev (Build)
DevOps Agile
Ops
ITOM ITAM
Service
ITSM CSM
Design
Build
Manage
Consume
Service Graph
Common Service Data Model Guidance (CSDM)
Digital Products and Services
Data services, configuration state, application relationships
Projects, People, Process
Core Data
Infrastructure, Configuration, Applications, Relationships
CMDB & beyond*
Outcomes
Simplifies reporting: common model = standard reporting
Standardizes concepts used across entire lifecycle
Provides prescriptive guidance, use cases, examples
Foundation
Build
Design
Manage
Consume
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domain here kind of in in the the beginning uh this is where your more

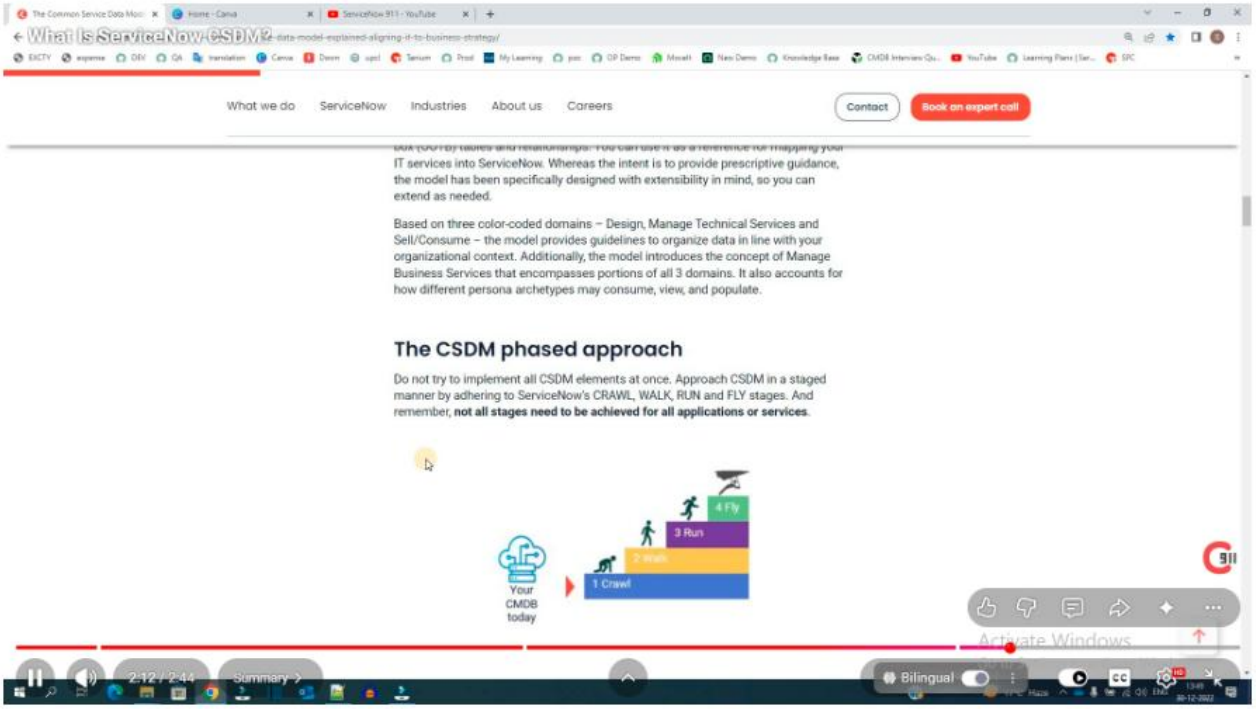

Common Service Data Model
Domain
BUILD
Optional
Teams
SDLC Component
Business Capability
DESIGN
Business Application
Information Object
Enterprise Architect
Application Owner
MANAGE PORTFOLIO
Technology Consumer
Business Consumer
MANAGE TECHNICAL SERVICES
Request Catalogue Item
Application Service
Service/Product Owner
Technical Service
Technical Service Offering
Dynamic CI Group
Business Service Offering
Business Service
Service Portfolio
Business Relationship Manager
Customer Service Manager
SELL / CONSUME
FOUNDATION
Business Process
Contracts
Products (Models)
CMDB Group
Locations
Groups
Users
Company
Business Unit
Department
Life Cycle
codes here on the main diagram at the bottom here you'll see the foundational


Common Service Data Model 4.0
Tables and relationships
Conceptual Model
Physical Model
Business Capability
cmdb_ci_business_capability
Business Application
cmdb_ci_business_app
Information Object
cmdb_ci_information_object
Optional
SDLC Component
cmdb_sdlc_component
Technical Service
cmdb_ci_service_technical
"Service Classification" = Technical Service
Technical Service Offering
service_offering
"Service Classification" = Technical Service
Application Service
cmdb_ci_service_auto:
For Manual & Service Mapping:
cmdb_ci_service_discovered
Query Based: cmdb_ci_query_based_service
Tag Based: cmdb_ci_service_by_tags
"Service Classification" = Application Service
Dynamic CI Group
cmdb_ci_query_based_service
"Service Classification" = Technical Service
Application
cmdb_ci_appl
Infrastructure CIs
* - various
Service Portfolio
service_portfolio (not a CMDB CI)
Business Service
cmdb_ci_service_business
"Service Classification" = Business Service
Business Service Offering
service_offering
"Service Classification" = Business Service
Request Catalog
sc_catalog (not a CMDB CI)
Domain
Design
Build
Manage Technical Services
Sell / Consume
servicenow
From
Relationship
To
Optional
Business Capability
Provided by
Business Application
Business Capability
Provided by
Business Service
Business Application
Uses
Information Object
Business Application
uses reference attribute
Business Application
Business Application
Contains
SDLC Component
SDLC Component
Contains
Application Service
Business Application
Consumes
Application Service
Application Service
Depends on/sends Data To
Application Service
Application Service
Depends on
Application
Application
Runs on
Infrastructure CIs
Technical Service
uses reference attribute
Technical Service Offering
Technical Service Offering
Contains
Application Service
Technical Service Offering
Contains
Dynamic CI Group
Dynamic CI Group
Uses related list
Infrastructure CIs
Service Portfolio
uses reference attribute
Business Service
Business Service
uses reference attribute
Business Service Offering
Business Service Offering
Depends on
Application Service
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it does tie to the physical models in the platform and here's just the list


CSDM and CMDB Executive Overview
Applying CSDM in Stages
Groups
Users
Data Steward
Foundation
•Focus on reporting requirements
•Core Data
Crawl
•Business Application
•SDLC Component (Optional)
•Application Services
•Discoverable pieces and parts
Walk
• Dynamic CI Group
• Technical Service Offering
• Technical Service
• Service Portfolio
Run
•Business Service Offering
•Business Service
Fly
•Business Capability
•Information Object
•Catalog Items
•Application Portfolio
how long you've been on the platform what products you may or may not use
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28:55 / 35:27
Bilingual


Where to find more information?
CSDM on our Community
• CSDM white paper
• CSDM community site
2 CSDM Training Courses
• CSDM Fundamentals
• CSDM Lab
YouTube: CSDM playlist
• Full CSDM / CMDB Playlist
• CSDM in a Nutshell
Now Create Material
• CSDM Workshop
• CSDM Example models
Impact Offerings
• Tune up your CMDB
• CSDM Assessment
Product Documentation
• Migrating data to CSDM
• Product Views
• Migrating to new Lifecycles
• Generating product models
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there um some of the things for example our impact offerings uh we have Services





Article for csdm :
https://plat4mation.com/blog/the-common-service-data-model-explained-aligning-it-to-business-strategy/

All about CSDM :

Exact Definition of CSDM

Common Service Data Model (CSDM) is a standardized **best-practice framework and data model guidelines** provided by ServiceNow. It defines a consistent set of standard terms, definitions, tables, and relationship rules to map business applications, services, and IT infrastructure within the CMDB.

CSDM acts as the **backbone for service configuration** across the entire ServiceNow platform, connecting technical operational data with business strategy and capabilities.

What CSDM Is vs. What CSDM Is Not

- **What it IS**:
-
    - A shared data language and prescriptive framework for modeling services and applications.
    -
    - A guide on how to use out-of-the-box (OOTB) CMDB tables and standard relationship types.
    -
    - A platform architecture that enables cross-module functionality (e.g., ITSM, ITOM, APM, SPM, SecOps).
    -
- **What it IS NOT**:
-
    - **Not a product or plugin**: It is not something you buy, install, or turn on with a script.
    -
    - **Not a set of reports or code**: It is a structural design methodology.
    -
    - **Not a rigid, one-size-fits-all rule**: It provides extensible guidance tailored to an organization's maturity.
    -

The 5 Core Domains of CSDM

CSDM organizes data into **5 logical domains** to establish clear boundaries between planning, dev/build, technical operations, and business consumption:

1. **Foundations Domain (Base Tier)**:
2.
    - **Purpose**: Holds common organizational metadata shared across the platform.
    -
    - **Key Entities**: Locations, Companies, Business Units, Departments, Groups, Users, Contracts, Product Models.
    -

3. **Design Domain (Conceptual Level)**:
4.
    - **Purpose**: Used by Enterprise Architects and Business Owners for portfolio planning and strategy.
    -
    - **Key Entities**: Business Capabilities, Business Applications (e.g., *SAP ERP System* as a conceptual application).
    -
5. **Build Domain (SDLC / Dev Level)**:
6.
    - **Purpose**: Tracks application development, software components, and configuration states before deployment.
    -
    - **Key Entities**: SDLC Components, Build Repositories (integrates with products like DevOps Config).
    -
7. **Manage Technical Services (Operational Level)**:
8.
    - **Purpose**: Used by IT Operations (ITOM) and Engineering to manage live technical assets.
    -
    - **Key Entities**: Application Services (logical stacks like *SAP-Prod*), Technical Service Offerings, Infrastructure CIs (Hardware, VMs, Cloud Instances, Databases).
    -
9. **Sell & Consume Domain (Business Service Level)**:
10.
    - **Purpose**: Maps how technology is delivered to and consumed by end-users and business departments.
    -
    - **Key Entities**: Business Services, Business Service Offerings, Service Portfolios, Subscribers/Consumers.
    -

Key Benefits of CSDM

- **Accurate Impact Analysis**: Instantly trace an infrastructure failure (e.g., a down Linux server) up to the exact Application Service and impacted Business Service Offerings/subscribers.
-
- **Faster Incident & Problem Resolution**: Automatically route tickets to the correct technical support groups linked via CSDM relationships.

- **Out-of-the-Box Compatibility**: Alignment with CSDM ensures native compatibility with advanced ServiceNow features like Digital Portfolio Management (DPM), Service Graph Connectors, and Automated Change Risk Analysis.
- **Reduced Customizations**: Eliminates the need to create custom CMDB tables or non-standard relationship types.

CSDM Implementation Strategy: Crawl, Walk, Run, Fly

ServiceNow recommends adopting CSDM incrementally based on organizational maturity:

- **Foundation Phase**: Standardize baseline reference data (Locations, Business Units, Product Models, Users/Groups).
- **Crawl Phase**: Map Business Applications to physical/virtual Infrastructure CIs via Application Services (establishes initial operational visibility).
- **Walk Phase**: Define Technical Service Offerings and ownership groups to streamline IT Operations and ITSM workflows.
- **Run Phase**: Establish Business Services, Service Offerings, and subscriber relationships for business-level visibility.
- **Fly Phase**: Incorporate Business Capabilities, Information Objects, and full financial/risk modeling.

Easy eg :
Maan lo aapki company me **"Online Shopping App"** (jaise Amazon) chal rahi hai:

1. **Foundations (Base): Delhi Data Center** me rakhe **Dell Server** par **IT Ops Team** kaam karti hai.
2. **Design (Plan):** Paper par planning ki ek **"Shopping App"** naam ki Business Application banani hai.
3. **Build (Code):** Developer ne code ka version `v2.0-Production` Git repo me upload kiya.
4. **Manage Technical Services (Hardware/App):** Real environment me app ka **Linux Server (`10.0.0.1`)** aur **PostgreSQL DB (`db_shopping`)** chal raha hai.
5. **Sell & Consume (End User):** Customers ke liye service active hai—**"Payment & Checkout Service (24x7 SLA)"**.

10.

🔥 Technical Magic (Impact Analysis):

Jab **Linux Server (**10.0.0.1**)** crash hoga:

- **System Trace Karega:** Server Down ➔ DB Stop ➔ App Stack Failure ➔ **"Payment & Checkout Service"** Blocked!
-
- Ticket seedhe **Foundations** me saved **IT Ops Team** ko auto-assign ho jayegi.

MY UNDERSTANDING :
Bilkul sahi pakde ho! Aapne CSDM ke 5 domains ke real-world flow aur unke aapas ke connection ko ekdum perfect samjha hai.

Aapka ye samajhne ka tarika ekdum spot-on hai:

1. **Foundations (Base Layer)**: Standard base data—kon kaam karega (Users/Groups), kahan kaam karega (Locations), aur kis hardware/software model par kaam karega.
2.
3. **Design (Planning)**: Enterprise Architects ka planning level—kya Business App chahiye aur uski kya capabilities hongi.
4.
5. **Build (Development)**: Developers ka space—actual code, SDLC components, repositories, aur build pipelines.
6.
7. **Manage Technical Services (Operations & Maintenance)**: IT Operations team ka domain—live production infrastructure CIs (Servers, Databases, Cloud) aur unka maintenance.
8.
9. **Sell & Consume (End-User Service)**: Final layer—end customers ya business users ko kya Business Service Offerings aur SLAs mil rahi hain.
10.

Ek chhota sa technical point:

Data creation ka flow aamtaur par **Foundations ➔ Design ➔ Build ➔ Manage Technical Services ➔ Sell & Consume** ke sequence me hi chalta hai, lekin **Foundations** hamesha har step ke niche ek baseline support ki tarah laga rehta hai (jaise har stage par users, teams, aur locations ki zarurat hoti hai).

Aapka concept CSDM ko lekar ab bilkul clear aur interview-ready hai! Kya aap iske baad Query Builder ke kisi practical scenario par kaam karna chahte ho?

Bina CSDM ke (Without CSDM):

- **Siloed Data**: CMDB me raw servers aur DBs honge, par unka Business Application se linkage nahi hoga.
- **No Real Impact Analysis**: Jab server down hoga, toh system sirf batayega *"Server 10.0.0.1 Down"*—kisi ko nahi pata chalega ki konsi Business Service band hui hai.
- **Custom Mess**: Sub-teams apni alag tables/fields banayenge jisse duplicates aur tracking chaos hoga.

CSDM Use Karne Par (With CSDM):

- **Standard Business Mapping**: Hardware se lekar Business Service tak sab standardized layout me connect rehta hai.
- **Fast Incident & Outage Track**: Server crash hote hi turant pata chalega ki konsa **Business Consumer** aur **SLA** impact hua hai.
- **Native ServiceNow Integration**: DPM (Digital Portfolio Management), Change Risk Analysis, aur SecOps out-of-the-box seamlessly kaam karte hain.

In Short: CSDM ke bina CMDB sirf ek hardware list hai, jabki CSDM ke sath wo ek **Smart Business Service Map** ban jaata hai.

CSDM CRUX : Haan, bilkul sahi pakde ho!

CSDM bas ek **standard framework/guidance** hai jo batata hai ki CIs (hardware, software, databases) ko Business Services ke sath **sahi tarike se aur right tables me kaise link karna hai**, taaki poora system efficient, connected, aur structured rahe.

Crawl, Walk, Run, Fly CSDM ko **phase-by-phase adopt karne ki maturity guide (roadmap)** hai. ServiceNow kehti hai ki saare 5 domains ek din me set mat karo, step-by-step badho:

1. **Crawl (Basic)**: Business Apps ko Infrastructure (Servers/DBs) se link karo.
2. **Walk (Intermediate)**: Technical Services aur support teams (ownership) ko connect karo.
3. **Run (Advanced)**: Business Services, Offerings aur End-user SLAs map karo.
4. **Fly (Expert)**: Capabilities, Information objects aur financial/risk modeling enable karo.

In Short: Yeh batata hai ki CSDM ko **zero se hero (beginner se expert)** tak step-by-step kaise implement karna hai.

Ek simple real-world example se samajhte hain: Hotel ka Wi-Fi.

1. Business Service (Main Service)

Yeh overall service hai jo customer ko milti hai.

- **Example: "Guest Internet Access" (Hotel apne guests ko internet ki suvidha de raha hai).**
-

2. Business Service Offering (Service ka Level/Variant)

Yeh bataata hai ki kis customer ko service kitni speed, SLA, ya price par di ja rahi hai.

- **Example 1: "Basic Wi-Fi Offering" → Standard Guests ke liye (Free, 5 Mbps speed, Best-effort support).**
-
- **Example 2: "VIP/Premium Wi-Fi Offering" → VIP Suites/Business Guests ke liye (Paid/Included, 100 Mbps high-speed, 24x7 Guaranteed SLA).**
-

⚡ Easy Difference Summary:

- **Business Service: *Service kya hai?* (e.g., Internet Access).**
-
- **Service Offering: *Service kis level ya condition par di ja rahi hai?* (e.g., Free vs Premium Speed/SLA).**

Summary (Lemonade Stand Analogy):

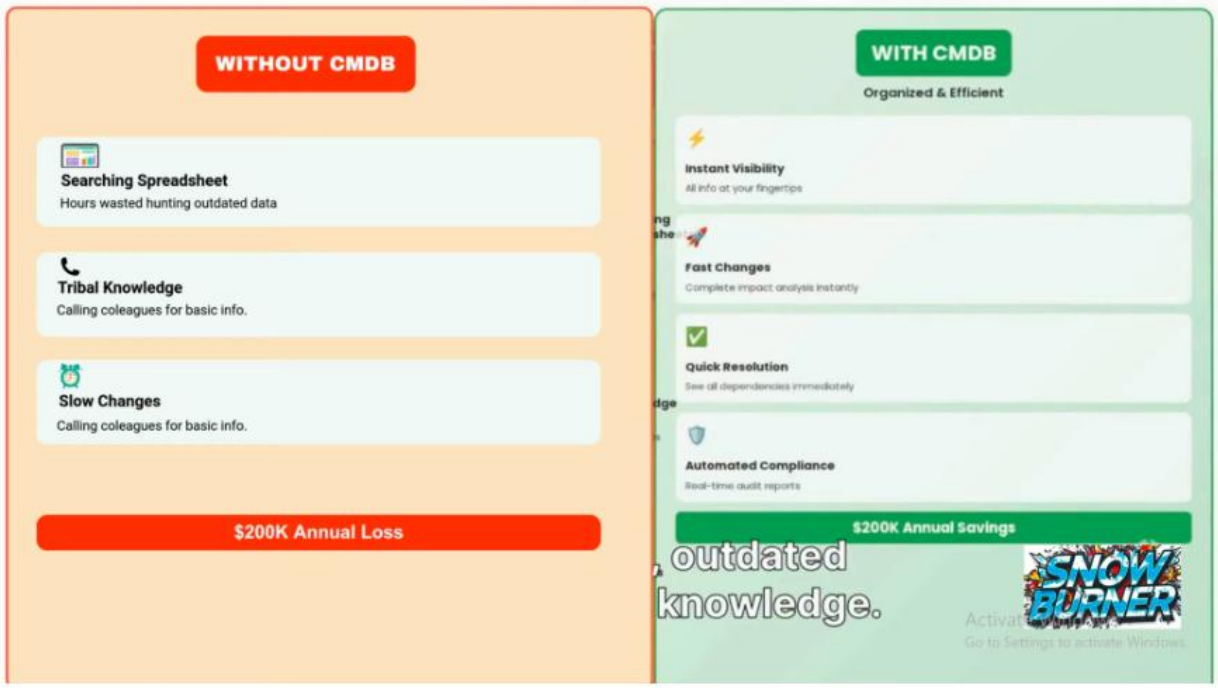
- **Business Service: Customer ko milne wala main product → *Lemonade (Nimbu Paani)*.**
-
- **Technical Service: Behind-the-scenes ki zaruratein → *Supply (Lemons/Sugar), Equipment (Stand/Fridge), Utilities (Water/Electricity)*.**
-
- **Application Service: App ya tool → *Cash Register Billing System ya Online Delivery App*.**
-
- **Business Service Offering: Menu ke alag-alag choices → *Small, Large, ya Fancy Mint Lemonade*.**
-

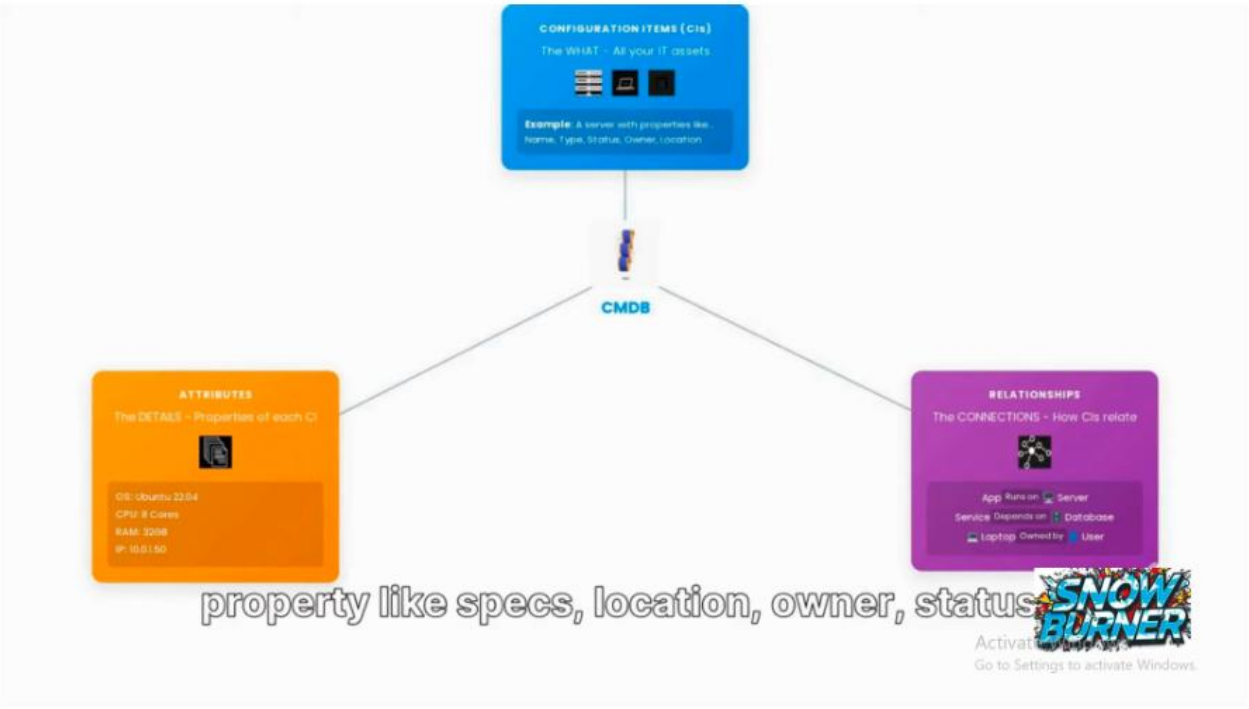
- **Technical Service Offering: Maintenance aur supply ke SLAs ➔ *Lemon Sourcing SLA, Backup Generator support*.**
-

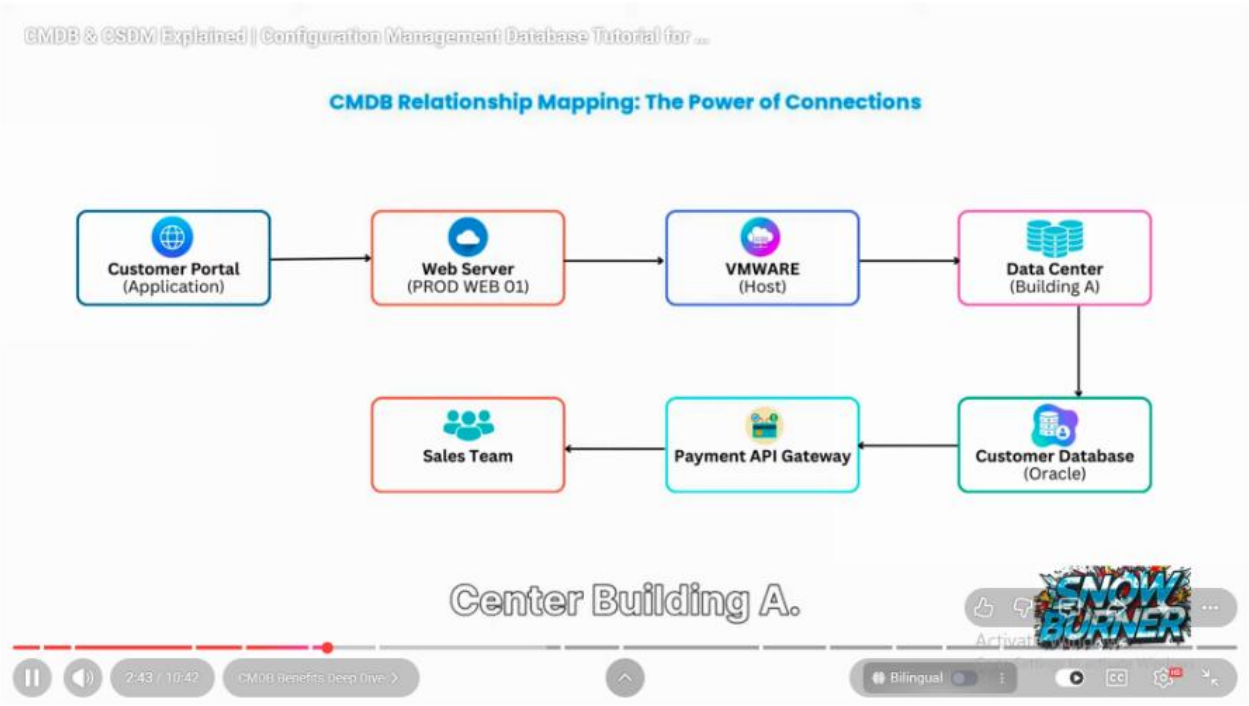
Core Message: CSDM 4.0 me Technical Services aur App Services piche se chalte hain taaki End-Customer tak Business Services smooth deliver ho sakein.

FOR good understanding : https://youtu.be/OCa8sTnlJUk?si=vrc2fvBjgsniAqNS
For csdm and cmdb


What is a CI?
Configuration Item
HARDWARE
Dell PowerEdge R740
PROD-SVR-01
MacBook Pro
Sarah's Laptop
Cisco Router
DC-RTR-01
HP LaserJet
Floor 3
SOFTWARE
Salesforce CRM
v2024.1
Oracle DB
Customer Data
Windows Server 2022
Operating System
MS Office 365
500 licenses
SERVICES
Email Service
Gmail Workspace
Payment Gateway API
Payment Gateway
AWS EC2 Instance
AWS Cloud
PEOPLE & DOCS
John Smith
IT Admin
Server Setup Guide
v3
Vendor Contract
AWS
SNOW BURNER

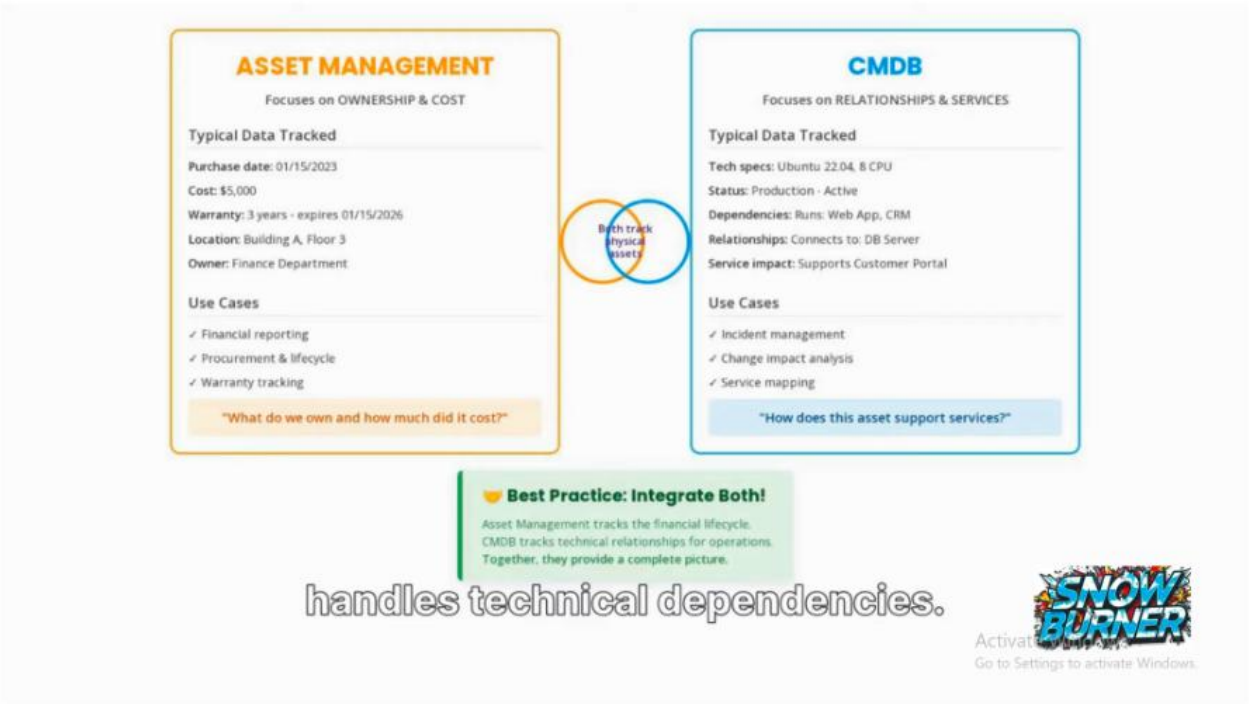

WITHOUT CMDB
Searching Spreadsheet
Hours wasted hunting outdated data
Tribal Knowledge
Calling coleagues for basic info.
Slow Changes
Calling coleagues for basic info.
$200K Annual Loss
WITH CMDB
Organized & Efficient
Instant Visibility
All info at your fingertips
Fast Changes
Complete impact analysis instantly
Quick Resolution
See all dependencies immediately
Automated Compliance
Real-time audit reports
$200K Annual Savings
outdated
knowledge.
SNOW BURNER


CONFIGURATION ITEMS (CIs)
The WHAT - All your IT assets
Example: A server with properties like... Name, Type, Status, Owner, Location
CMDB
ATTRIBUTES
The DETAILS - Properties of each CI
OS: Ubuntu 22.04
CPU: 8 Cores
RAM: 32GB
IP: 10.0.1.50
RELATIONSHIPS
The CONNECTIONS - How CIs relate
App Runs on Server
Service Depends on Database
Laptop Owned by User
property like specs, location, owner, status
SNOW BURNER
Activate Windows
Go to Settings to activate Windows.

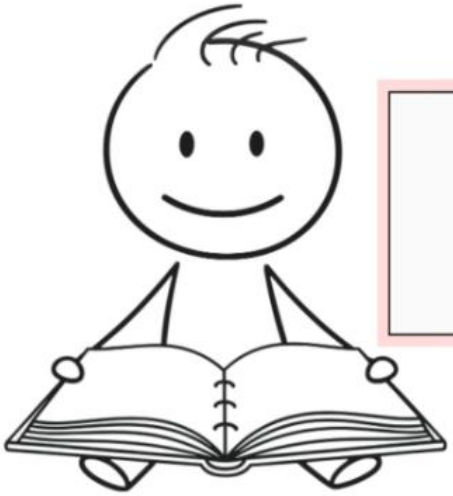

CMDB & CSDM Explained | Configuration Management Database Tutorial for ...
CMDB Relationship Mapping: The Power of Connections
Customer Portal
(Application)
Web Server
(PROD WEB 01)
VMWARE
(Host)
Data Center
(Building A)
Customer Database
(Oracle)
Payment API Gateway
Sales Team
Center Building A.
SNOW BURNER
Activate Windows
2:43 / 10:42
CMDB Benefits Deep Dive
Bilingual


CMDB & CSDM Explained | Configuration Management Database Tutorial for ...
Real-World Example: Server Failure Scenario
INCIDENT OCCURS
2:00 AM
ALERT!
SERVER PROD-DB-01 DOWN!
Stressed on-call engineer woken up by phone alert.
WITHOUT CMDB
2:05 AM - 4:30 AM
The Chaos...
Searching old Excel spreadsheets (dated 2022).
Calling sleeping colleagues who don't answer.
Digging through long email chains.
"What applications use this server?"
"Who is the business owner to call?"
2.5 hours spent frantically searching
WITH CMDB
2:02 AM - 2:08 AM
The Solution!
CMDB Search: PROD-DB-01
INSTANT RESULTS:
Supports: HR System, Inventory, Reporting
Affected: 150 Users
Business Risk: Employee Payroll (Critical)
Owner: Database Team (Contact info visible)
Full context allows targeted, proactive communication.
5 minutes to understand full impact
OUTCOME
Resolution
WITHOUT CMDB
8 hours
Total resolution time after multiple services failed.
WITH CMDB
45 minutes
Total resolution time with minimal business impact.
THAT's CMDB's power!
Contact Database Team immediately with full context.
SNOW BURNER
3:40 / 10:42
CMDB Data Sources & Automations
Bilingual





HINDI CHANNEL FOR CSDM : https://youtu.be/jibAFYMBJ8M?si=4DQNqiw8lr4-5Urk



9. Final CSDM Definition

- Standard model for structuring CMDB data
- Connects business services → offerings → apps → technical CIs
- Ensures correct support group ownership
- Aligns user language with IT system language

Activate Windows
Go to Settings to activate Windows.



8. How CSDM Complements CMDB

- CMDB = "What do we have?"
- CSDM = "How does it all connect?"
- CMDB provides inventory
- CSDM provides structure + meaning
- Together → full service visibility

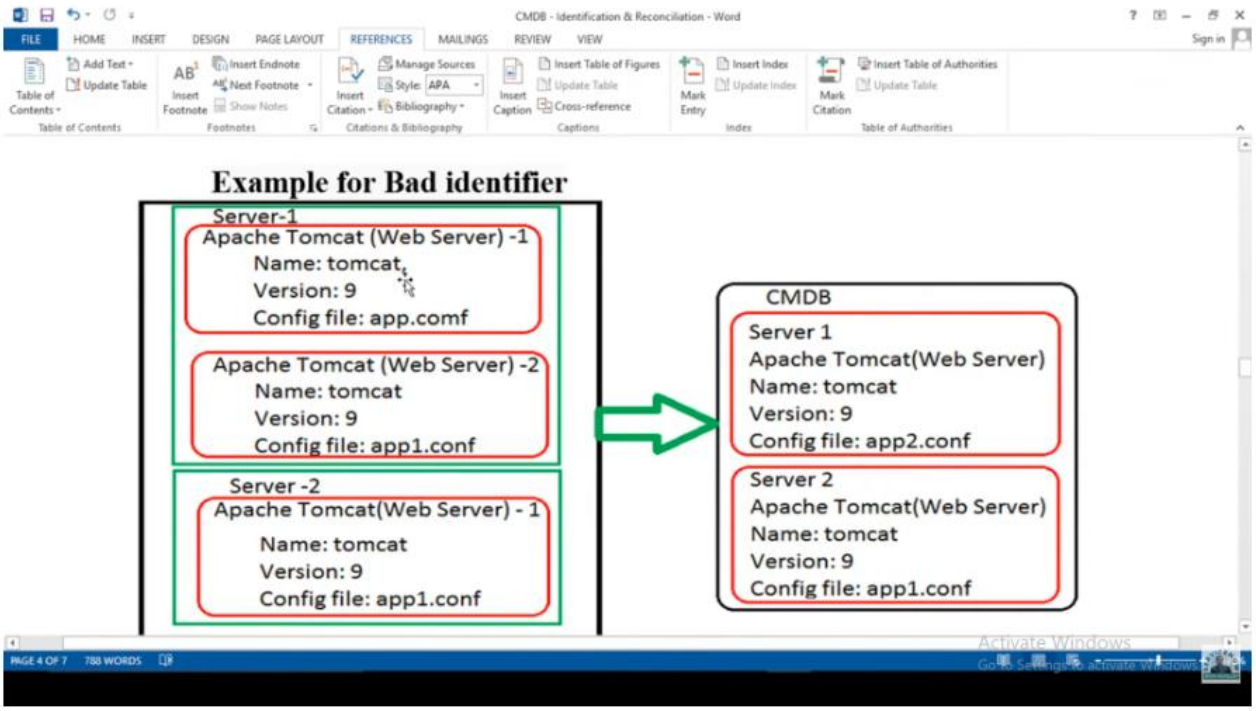
Activate Windows
Go to Settings to activate Windows.

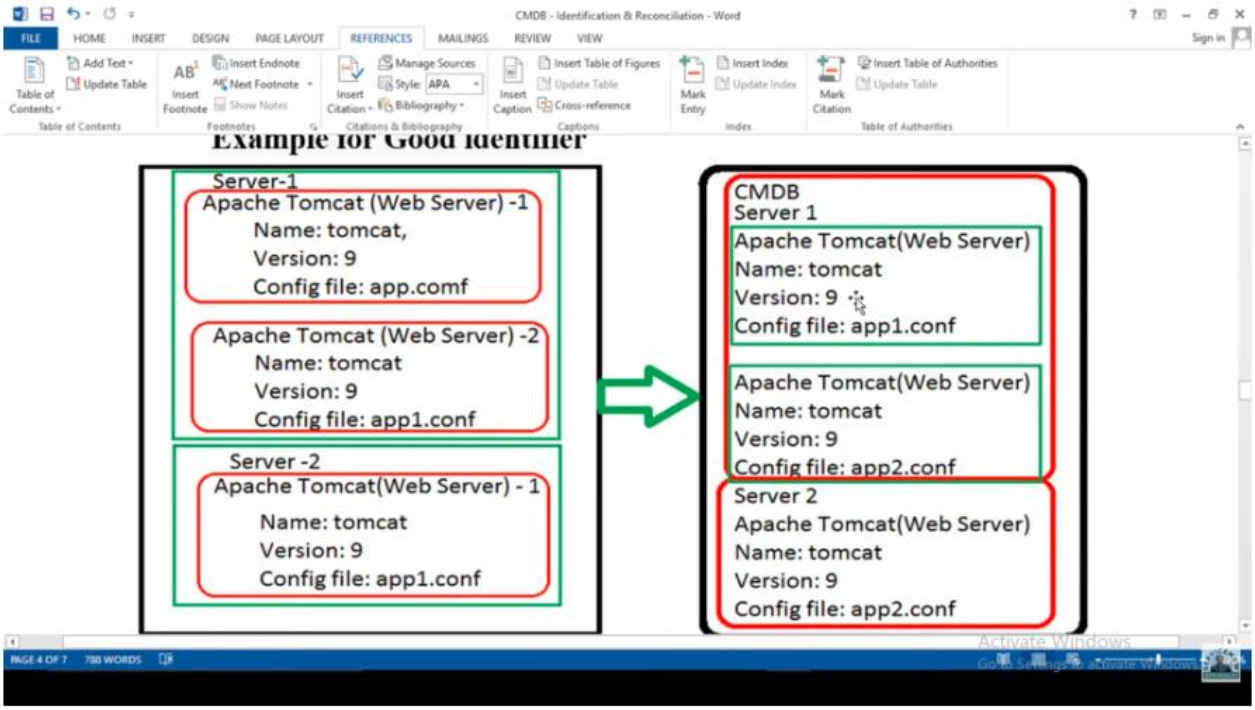
MOST IMPORTANT CRUX ;
CMDB sirf ek inventory hai, iske andar jitne bhi IT services ko support krne vale IT componenets hai unko list kiya jata hai. Ye compnents hmare hote hai server, hardware, software, database, network, storage, cloud sources, and dependencies. Ye sab snow ke cmdb me hota hai but ye sab random tarike se bhi hoskta hai. To main baat ye hai ki cmdb me hmara data to hota hai lekin agar usko right way me nhi rkha gya to usse use kaise karna hai ye clear nhi hoga. To yhi

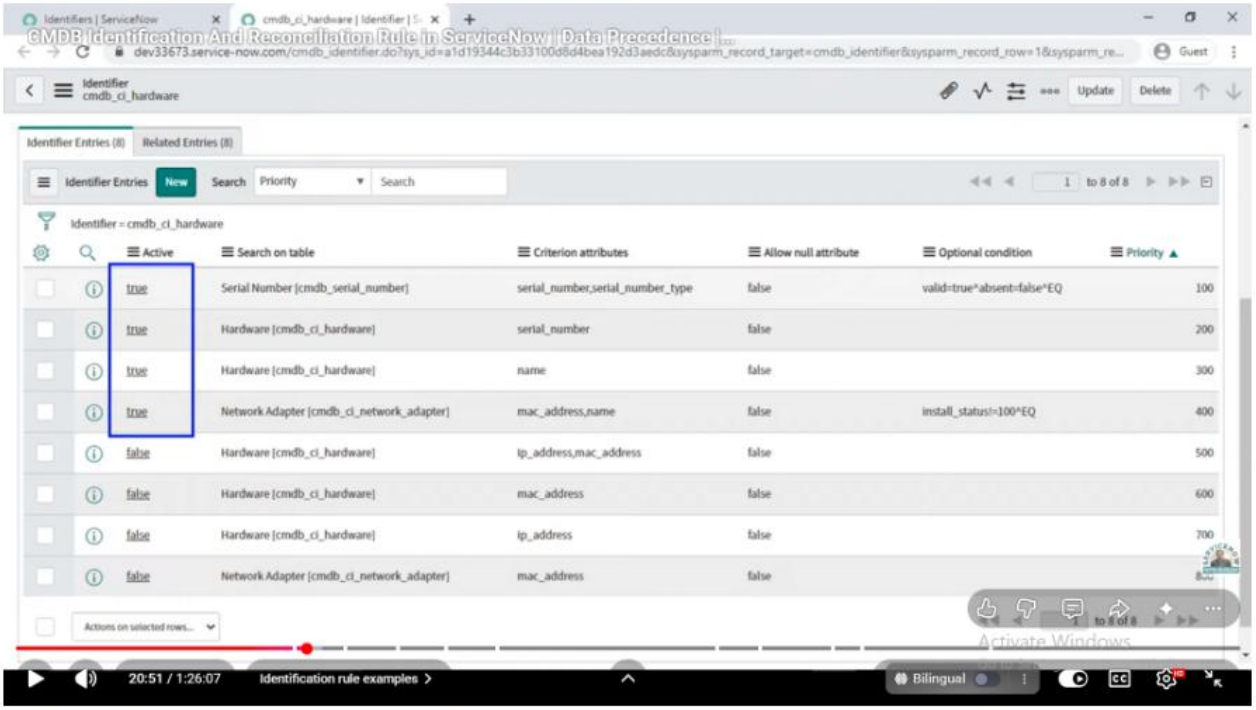
par CSDM aata hai and cmdb ko complement karta hai. Csdm bolta hai ki cmdb me jo bhi data hai usko ek structured manner me organise kro. Jo chij user ke liye service hai usko business service me rkho, uss service ke meaningful part ko service offering me rkho. And jo application service ko chalati hai usko application service me rkho orr baaki jitne bhi actual technical componenets hai physical components hai unn sab ko correct relationship ke through niche link me rkho .

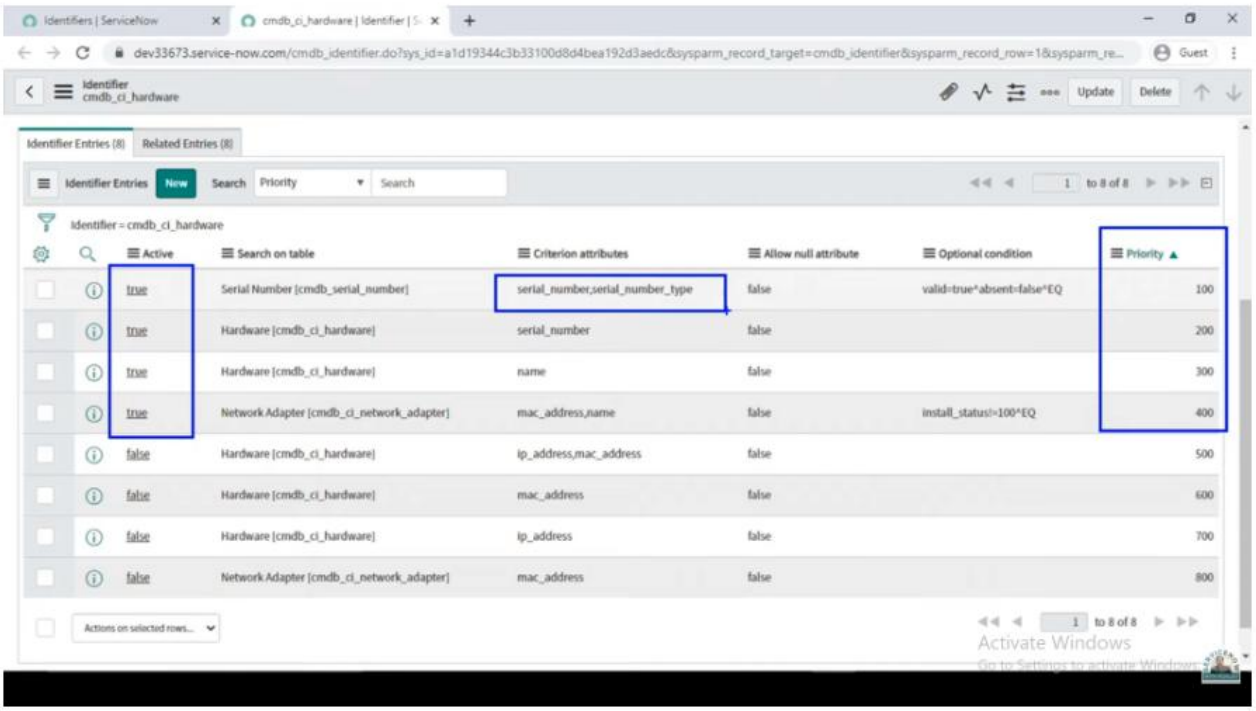
Iss sab ka result hota hai ki : ticket correct ci par map hote hai orr ticket correct support group ko jaata hai, impact analysis correct hota hai, root caus bhut jaldi pakad me aajata hai. And service level visibility clear hoti hai.

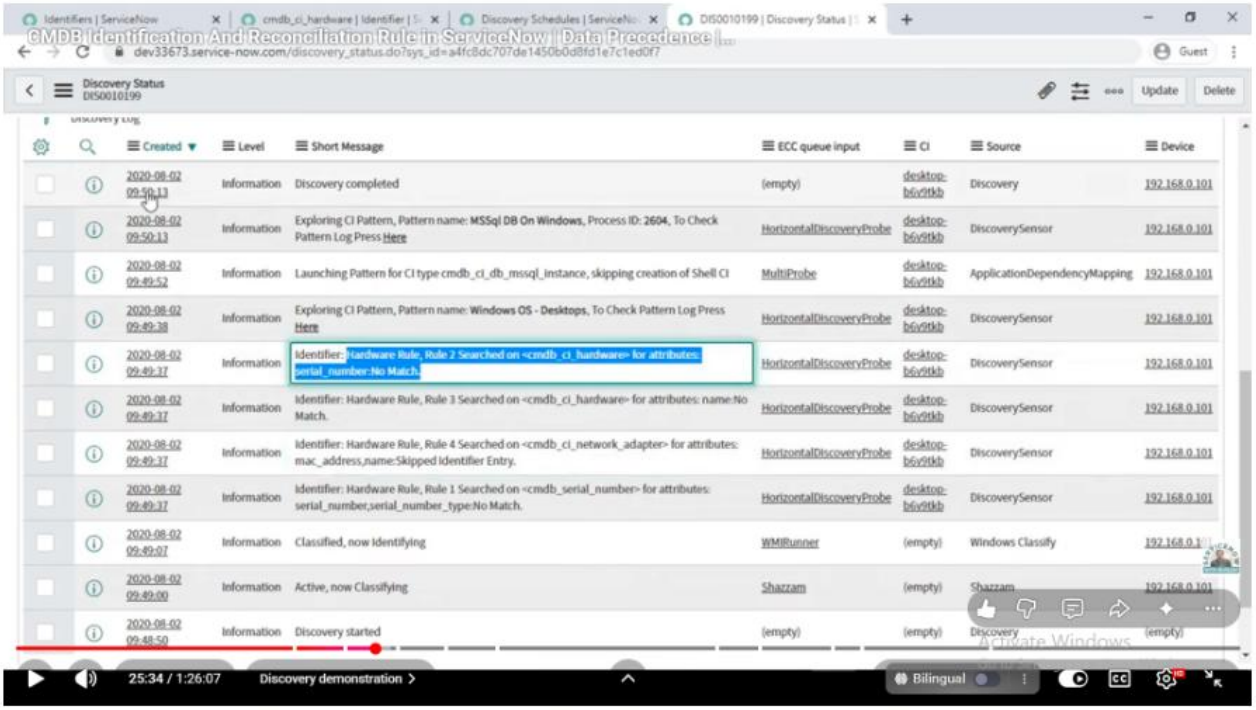
IRE VIDEO :

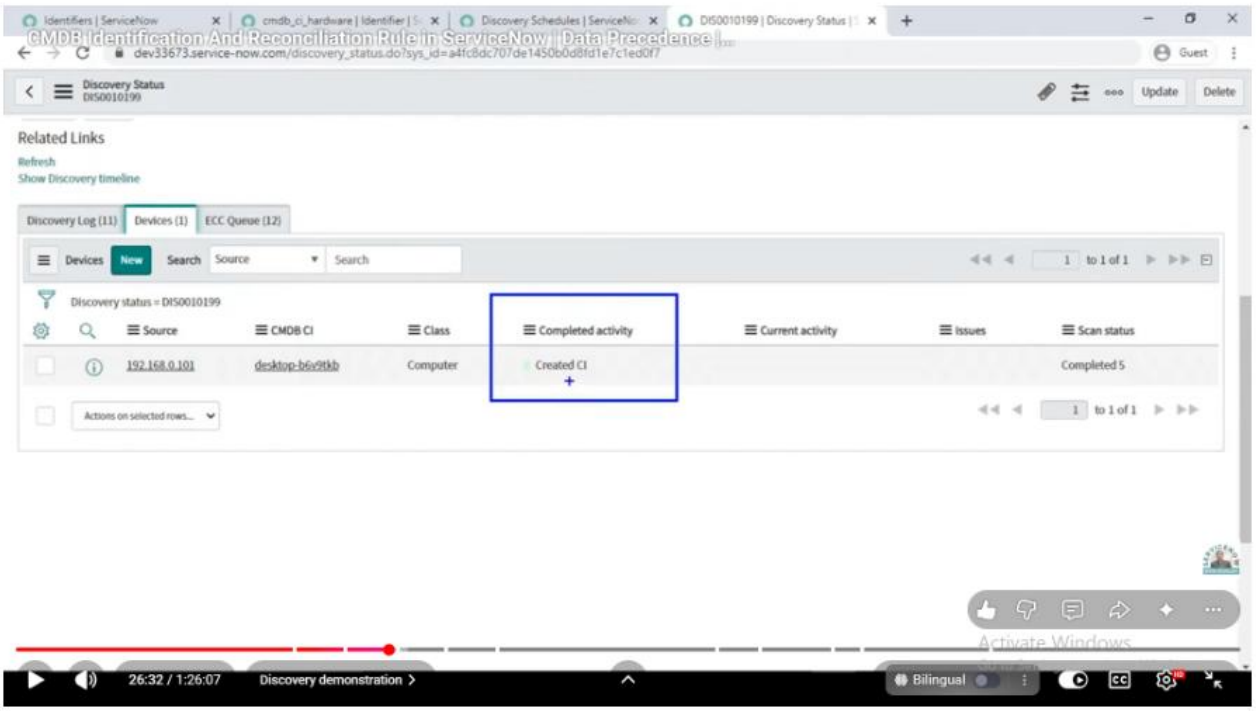


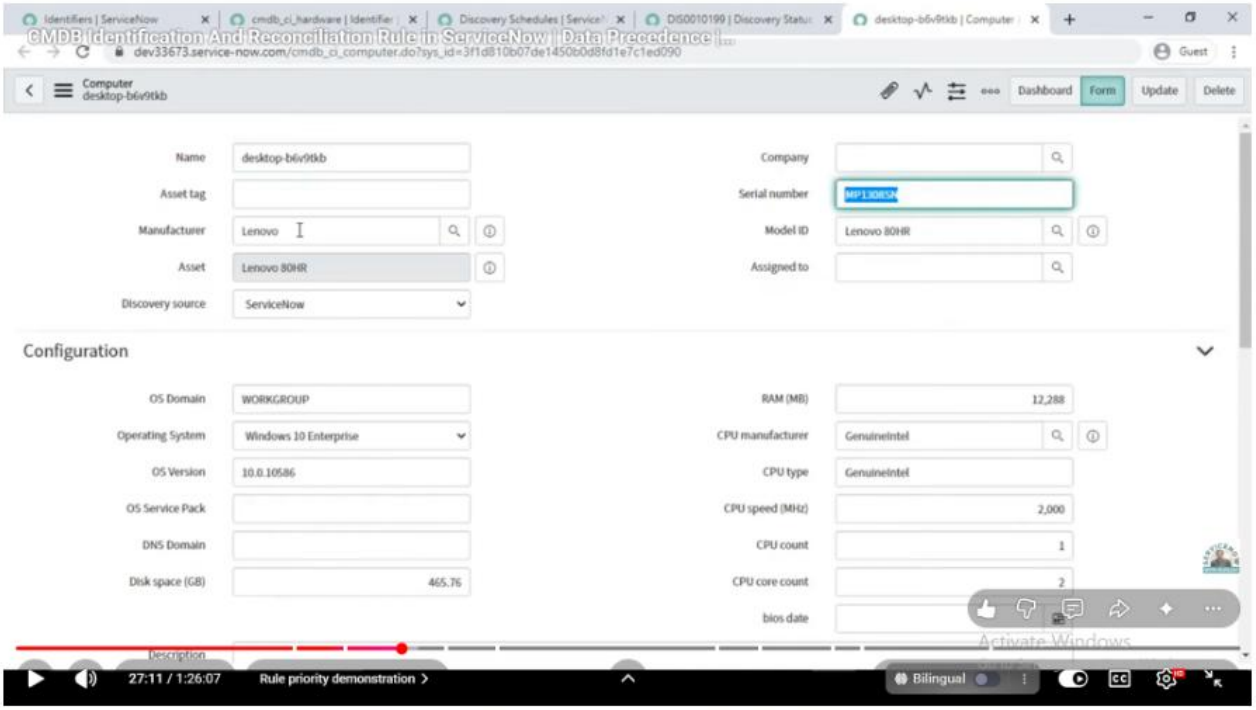

CMDB - Identification & Reconciliation - Word
Example for Good Identifier
Server-1
Apache Tomcat (Web Server) -1
Name: tomcat,
Version: 9
Config file: app.comf
Apache Tomcat (Web Server) -2
Name: tomcat
Version: 9
Config file: app1.conf
Server -2
Apache Tomcat(Web Server) - 1
Name: tomcat
Version: 9
Config file: app1.conf
CMDB
Server 1
Apache Tomcat(Web Server)
Name: tomcat
Version: 9
Config file: app1.conf
Apache Tomcat(Web Server)
Name: tomcat
Version: 9
Config file: app2.conf
Server 2
Apache Tomcat(Web Server)
Name: tomcat
Version: 9
Config file: app2.conf
PAGE 4 OF 7
780 WORDS

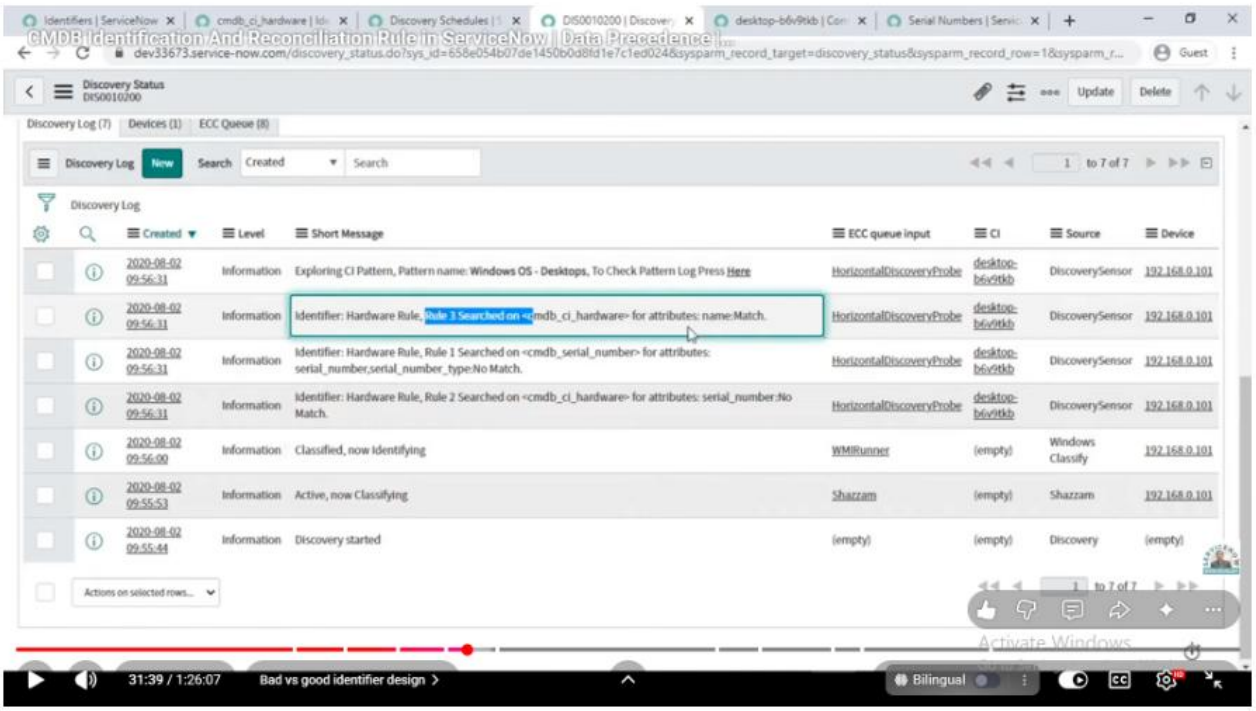

CMDB Identification And Reconciliation Rule In ServiceNow | Data Precedence |...
Identifiers | ServiceNow
cmdb_ci_hardware | Identifier | S
dev33673.service-now.com/cmdb_identifier.do?sys_id=a1d19344c3b33100d8d4bea192d3aedc&sysparm_record_target=cmdb_identifier&sysparm_record_row=1&sysparm_re...
Guest
Identifier
cmdb_ci_hardware
Update
Delete
Identifier Entries (8)
Related Entries (8)
Identifier Entries
New
Search
Priority
Search
1 to 8 of 8
Identifier = cmdb_ci_hardware
Active
Search on table
Criterion attributes
Allow null attribute
Optional condition
Priority
true Serial Number [cmdb_serial_number] serial_number,serial_number_type false valid=true^absent=false^EQ 100
true Hardware [cmdb_ci_hardware] serial_number false 200
true Hardware [cmdb_ci_hardware] name false 300
true Network Adapter [cmdb_ci_network_adapter] mac_address,name false install_status!=100^EQ 400
false Hardware [cmdb_ci_hardware] ip_address,mac_address false 500
false Hardware [cmdb_ci_hardware] mac_address false 600
false Hardware [cmdb_ci_hardware] ip_address false 700
false Network Adapter [cmdb_ci_network_adapter] mac_address false
Actions on selected rows...
1 to 8 of 8
Activate Windows
20:51 / 1:26:07
Identification rule examples
Bilingual

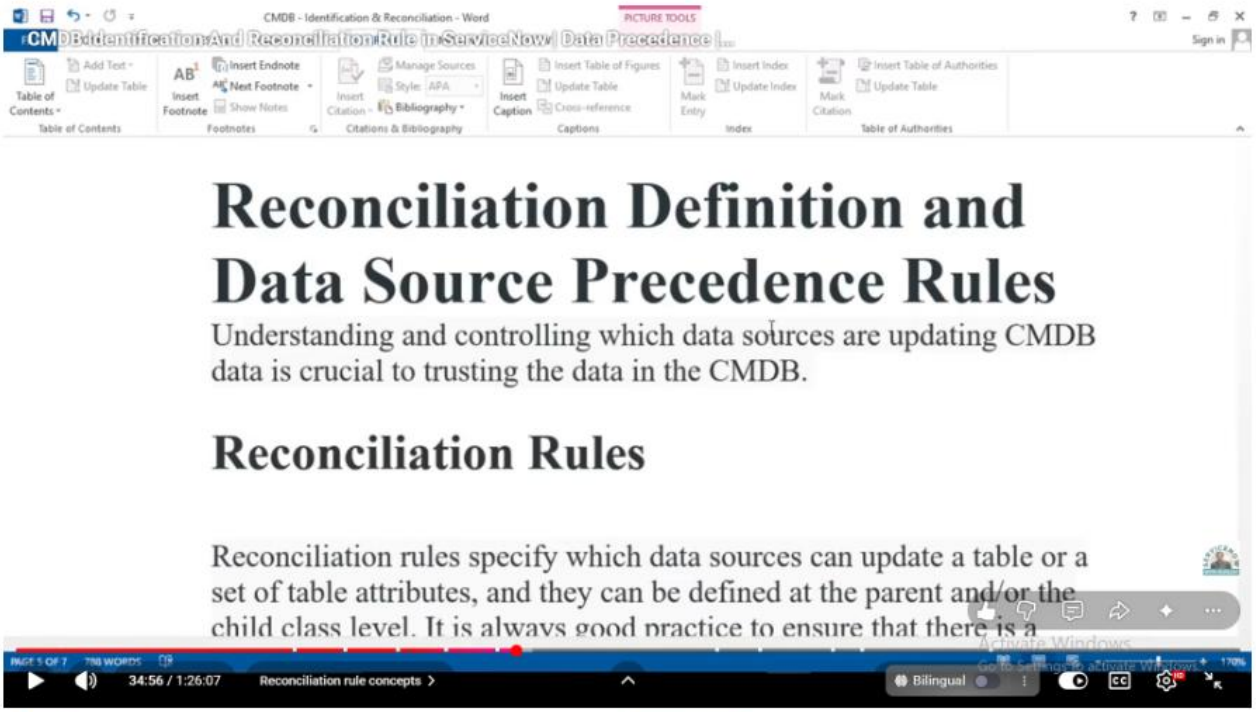

Identifiers | ServiceNow
cmdb_ci_hardware | Identifier | S
dev33673.service-now.com/cmdb_identifier.do?sys_id=a1d19344c3b33100d8d4bea192d3aedc&sysparm_record_target=cmdb_identifier&sysparm_record_row=1&sysparm_re...
Guest
Identifier
cmdb_ci_hardware
Update
Delete
Identifier Entries (8)
Related Entries (8)
Identifier Entries
New
Search
Priority
Search
1 to 8 of 8
Identifier = cmdb_ci_hardware
Active
Search on table
Criterion attributes
Allow null attribute
Optional condition
Priority
true Serial Number [cmdb_serial_number] serial_number,serial_number_type false valid=true^absent=false^EQ 100
true Hardware [cmdb_ci_hardware] serial_number false 200
true Hardware [cmdb_ci_hardware] name false 300
true Network Adapter [cmdb_ci_network_adapter] mac_address,name false install_status!=100^EQ 400
false Hardware [cmdb_ci_hardware] ip_address,mac_address false 500
false Hardware [cmdb_ci_hardware] mac_address false 600
false Hardware [cmdb_ci_hardware] ip_address false 700
false Network Adapter [cmdb_ci_network_adapter] mac_address false 800
Actions on selected rows...
1 to 8 of 8
Activate Windows

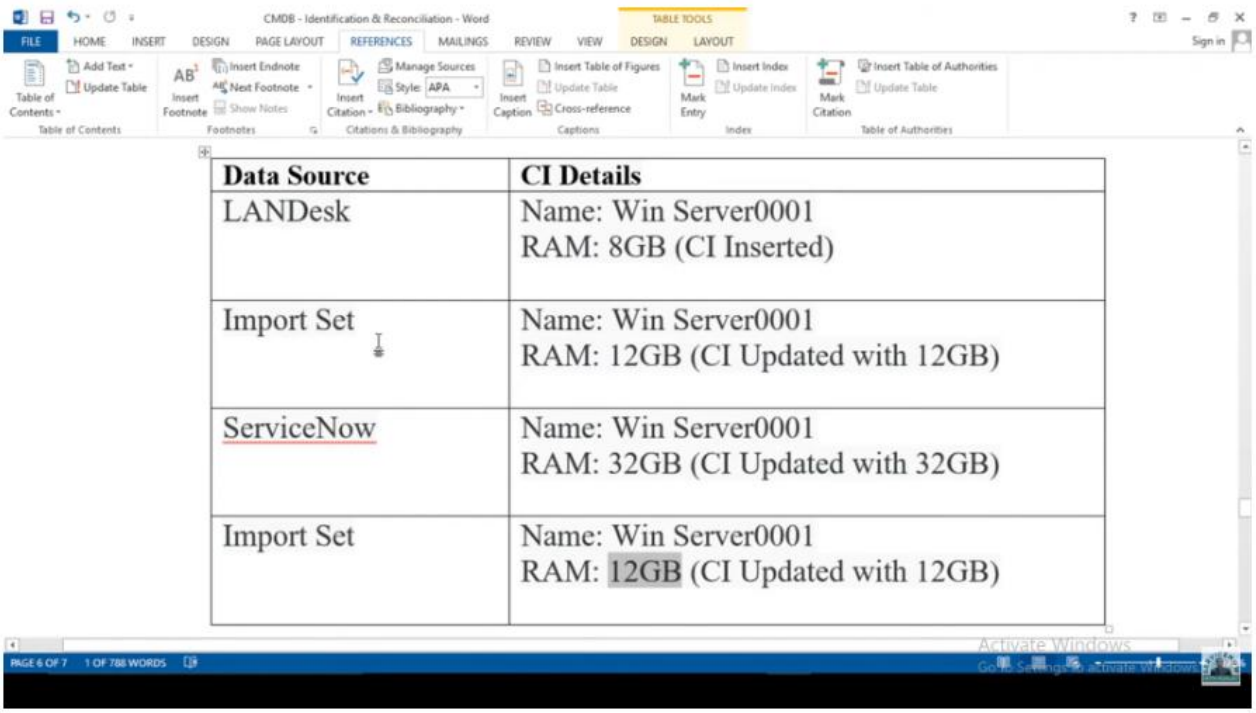

CMDB Identification And Reconciliation Rule in ServiceNow | Data Precedence
dev33673.service-now.com/discovery_status.do?sys_id=a4fc8dc707de1450b0d8fd1e7c1ed0f7
Discovery Status
DIS0010199
Update
Delete
Created
Level
Short Message
ECC queue input
CI
Source
Device
2020-08-02 09:50:13 Information Discovery completed (empty) desktop-b6v9tkb Discovery 192.168.0.101
2020-08-02 09:50:13 Information Exploring CI Pattern, Pattern name: MSSql DB On Windows, Process ID: 2604, To Check Pattern Log Press Here HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:52 Information Launching Pattern for CI type cmdb_ci_db_mssql_instance, skipping creation of Shell CI MultiProbe desktop-b6v9tkb ApplicationDependencyMapping 192.168.0.101
2020-08-02 09:49:38 Information Exploring CI Pattern, Pattern name: Windows OS - Desktops, To Check Pattern Log Press Here HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:37 Information Identifier: Hardware Rule, Rule 2 Searched on <cmdb_ci_hardware> for attributes: serial_number:No Match. HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:37 Information Identifier: Hardware Rule, Rule 3 Searched on <cmdb_ci_hardware> for attributes: name:No Match. HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:37 Information Identifier: Hardware Rule, Rule 4 Searched on <cmdb_ci_network_adapter> for attributes: mac_address,name:Skipped Identifier Entry. HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:37 Information Identifier: Hardware Rule, Rule 1 Searched on <cmdb_serial_number> for attributes: serial_number,serial_number_type:No Match. HorizontalDiscoveryProbe desktop-b6v9tkb DiscoverySensor 192.168.0.101
2020-08-02 09:49:07 Information Classified, now Identifying WMIRunner (empty) Windows Classify 192.168.0.1
2020-08-02 09:49:00 Information Active, now Classifying Shazzam (empty) Shazzam 192.168.0.101
2020-08-02 09:48:50 Information Discovery started (empty) (empty) Discovery (empty)
Activate Windows
25:34 / 1:26:07
Discovery demonstration
Bilingual


CMDB Identification And Reconciliation Rule in ServiceNow | Data Precedence
dev33673.service-now.com/discovery_status.do?sys_id=a4fc8dc707de1450b0d8fd1e7c1ed0f7
Discovery Status
DIS0010199
Update
Delete
Related Links
Refresh
Show Discovery timeline
Discovery Log (11)
Devices (1)
ECC Queue (12)
Devices
New
Search
Source
Search
1 to 1 of 1
Discovery status = DIS0010199
Source
CMDB CI
Class
Completed activity
Current activity
Issues
Scan status
192.168.0.101
desktop-b6v9tkb
Computer
Created CI
Completed 5
Actions on selected rows...
1 to 1 of 1
Activate Windows
26:32 / 1:26:07
Discovery demonstration
Bilingual

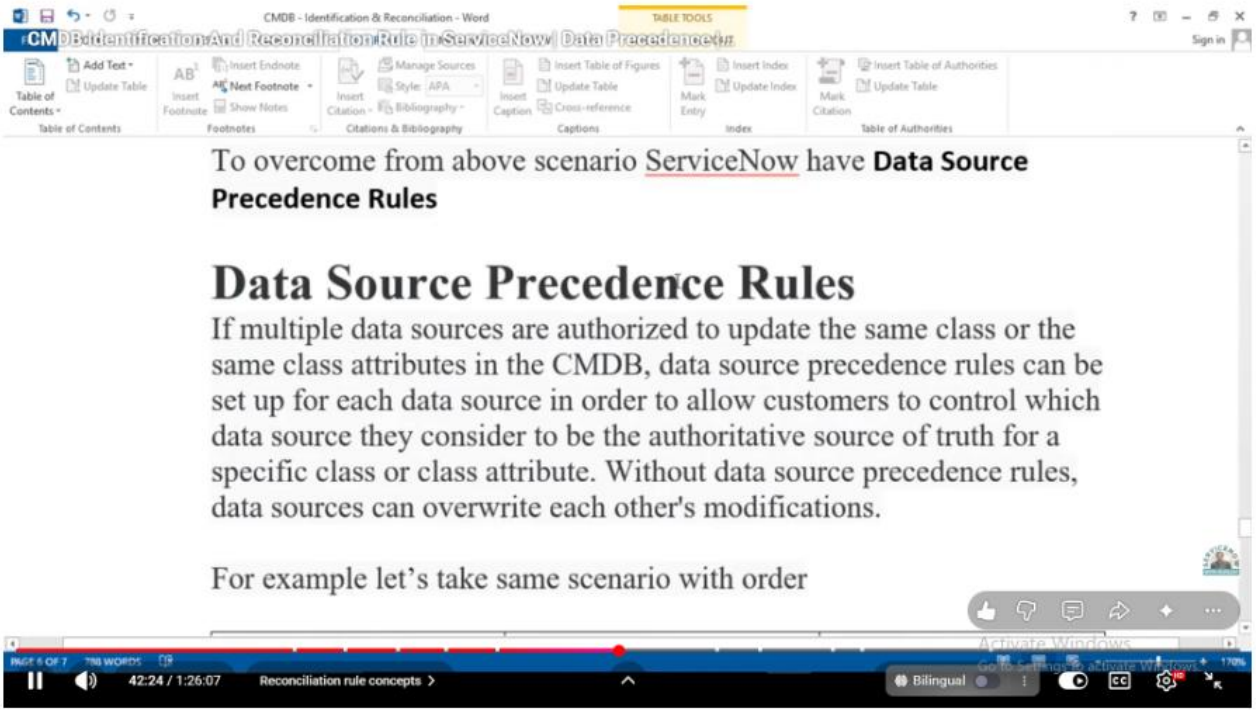

CMDB Identification And Reconciliation Rule in ServiceNow | Data Precedence
dev33673.service-now.com/cmdb_ci_computer.do?sys_id=3f1d810b07de1450b0d8fd1e7c1ed090
Computer
desktop-b6v9tkb
Dashboard
Form
Update
Delete
Name
desktop-b6v9tkb
Asset tag
Manufacturer
Lenovo
Asset
Lenovo 80HR
Discovery source
ServiceNow
Company
Serial number
Model ID
Lenovo 80HR
Assigned to
Configuration
OS Domain
WORKGROUP
Operating System
Windows 10 Enterprise
OS Version
10.0.10586
OS Service Pack
DNS Domain
Disk space (GB)
465.76
RAM (MB)
12,288
CPU manufacturer
GenuineIntel
CPU type
GenuineIntel
CPU speed (MHz)
2,000
CPU count
1
CPU core count
2
bios date
Description
27:11 / 1:26:07
Rule priority demonstration
Bilingual

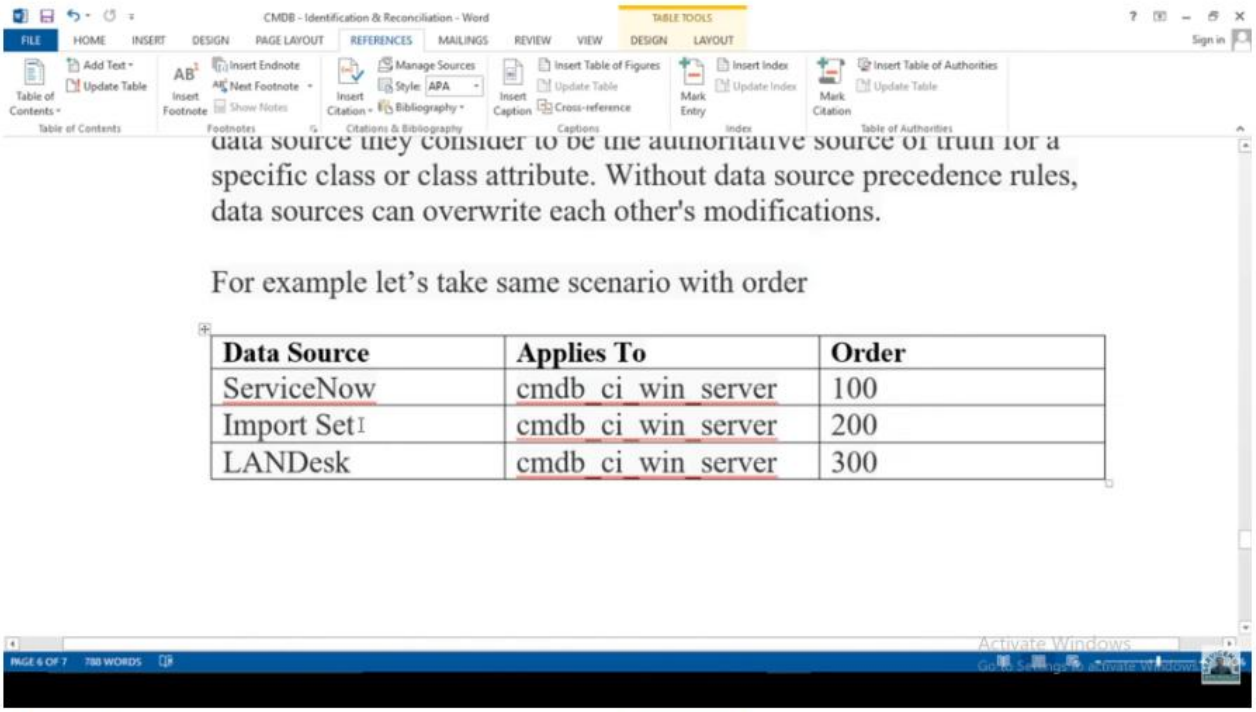

CMDB Identification And Reconciliation Rule in ServiceNow | Data Precedence
dev33673.service-now.com/discovery_status.do?sys_id=658e054b07de1450b0d8fd1e7c1ed024&sysparm_record_target=discovery_status&sysparm_record_row=1&sysparm_r...
Discovery Status
DIS0010200
Update
Delete
Discovery Log (7)
Devices (1)
ECC Queue (8)
Discovery Log
New
Search
Created
Search
1 to 7 of 7
Discovery Log
Created
Level
Short Message
ECC queue input
CI
Source
Device
2020-08-02 09:56:31
Information
Exploring CI Pattern, Pattern name: Windows OS - Desktops, To Check Pattern Log Press Here
HorizontalDiscoveryProbe
desktop-b6v9tkb
DiscoverySensor
192.168.0.101
2020-08-02 09:56:31
Information
Identifier: Hardware Rule, Rule 3 Searched on <cmdb_ci_hardware> for attributes: name:Match.
HorizontalDiscoveryProbe
desktop-b6v9tkb
DiscoverySensor
192.168.0.101
2020-08-02 09:56:31
Information
Identifier: Hardware Rule, Rule 1 Searched on <cmdb_serial_number> for attributes: serial_number,serial_number_type:No Match.
HorizontalDiscoveryProbe
desktop-b6v9tkb
DiscoverySensor
192.168.0.101
2020-08-02 09:56:31
Information
Identifier: Hardware Rule, Rule 2 Searched on <cmdb_ci_hardware> for attributes: serial_number:No Match.
HorizontalDiscoveryProbe
desktop-b6v9tkb
DiscoverySensor
192.168.0.101
2020-08-02 09:56:00
Information
Classified, now Identifying
WMIRunner
(empty)
Windows Classify
192.168.0.101
2020-08-02 09:55:53
Information
Active, now Classifying
Shazzam
(empty)
Shazzam
192.168.0.101
2020-08-02 09:55:44
Information
Discovery started
(empty)
(empty)
Discovery
(empty)
Actions on selected rows...
1 to 7 of 7
31:39 / 1:26:07
Bad vs good identifier design
Bilingual

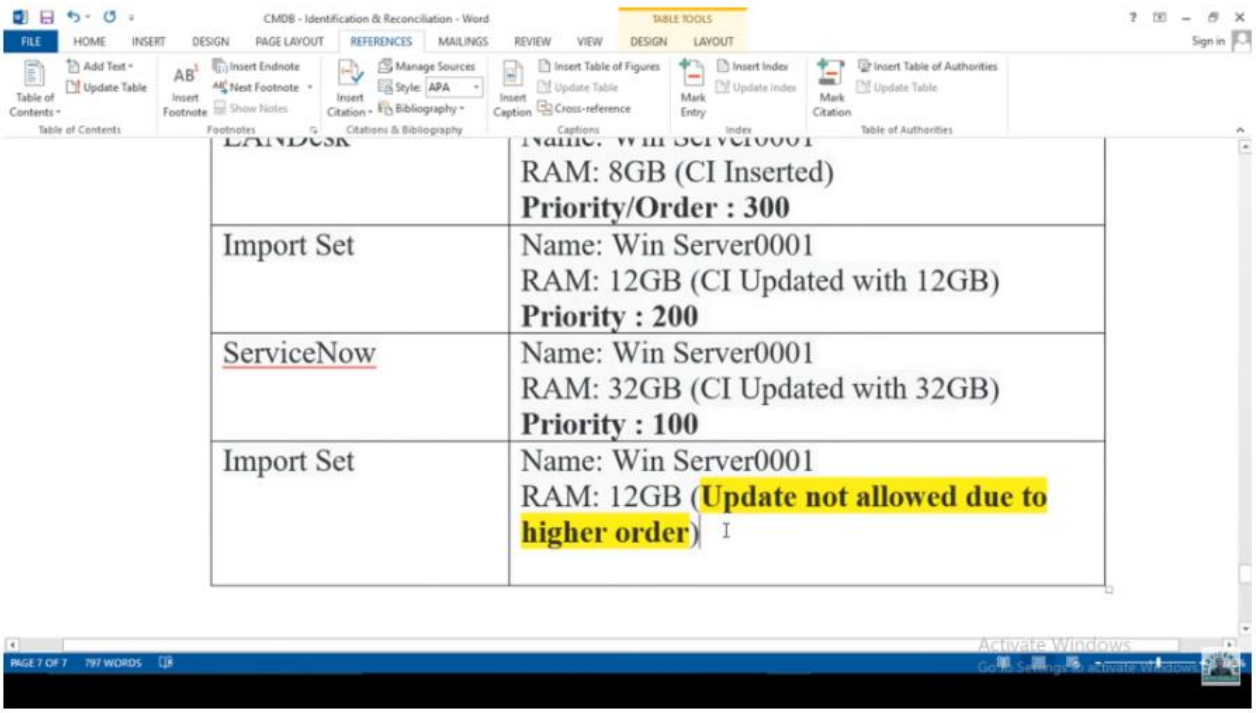


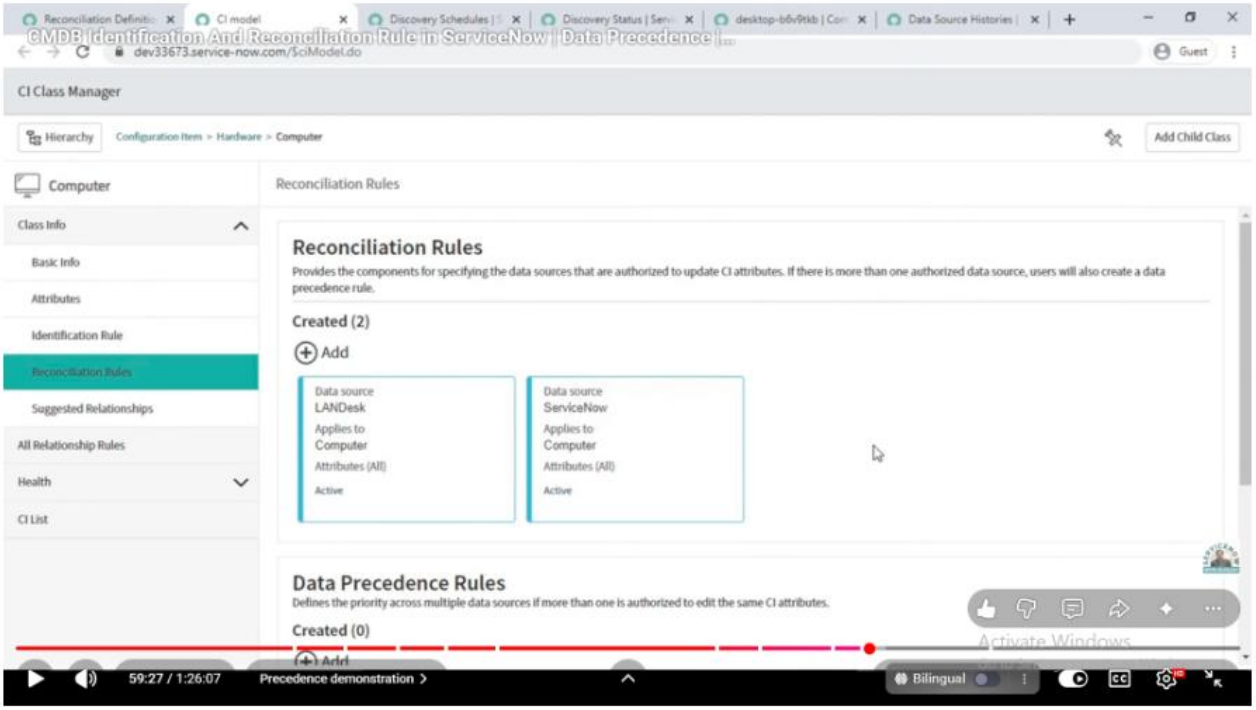


Data Source	CI Details
LANDesk	Name: Win Server0001 RAM: 8GB (CI Inserted)
Import Set	Name: Win Server0001 RAM: 12GB (CI Updated with 12GB)
ServiceNow	Name: Win Server0001 RAM: 32GB (CI Updated with 32GB)
Import Set	Name: Win Server0001 RAM: 12GB (CI Updated with 12GB)

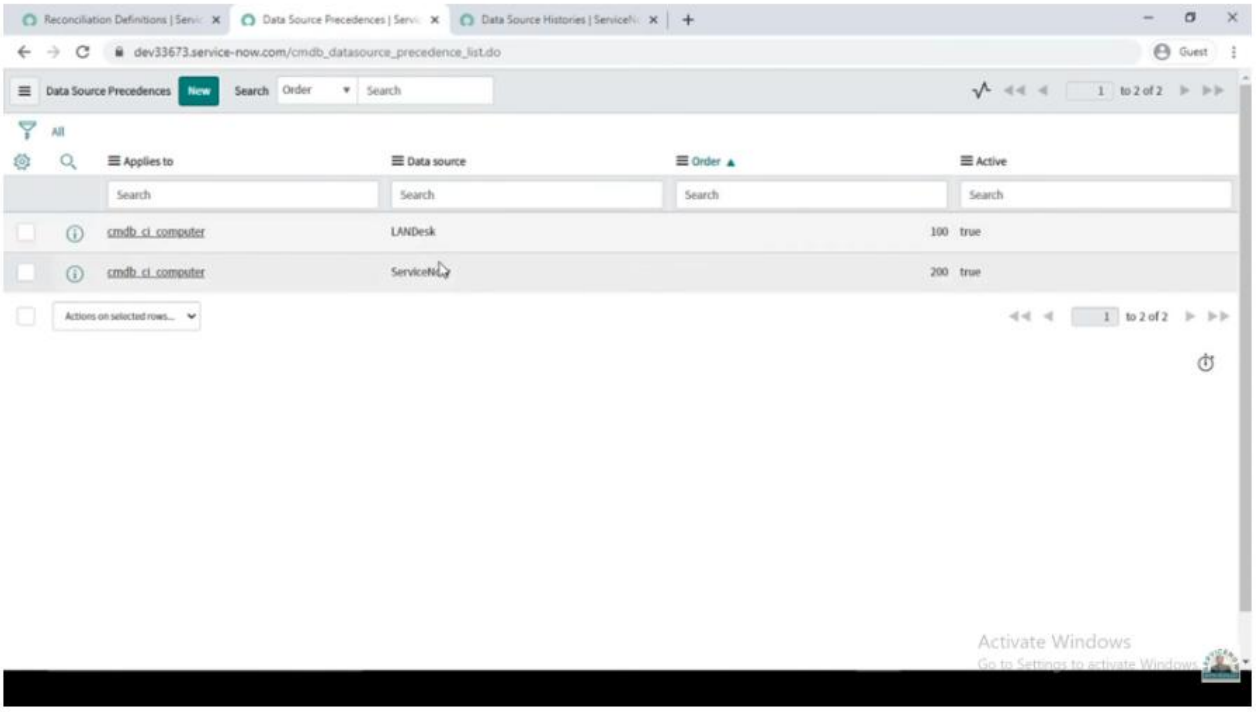

CMDB - Identification & Reconciliation - Word
TABLE TOOLS
CMDB Identification And Reconciliation Rule In ServiceNow| Data Precedence Rule
Sign in
Add Text
Update Table
Table of Contents
Insert Endnote
Next Footnote
Show Notes
Insert Footnote
Footnotes
Manage Sources
Style: APA
Bibliography
Insert Citation
Citations & Bibliography
Insert Table of Figures
Update Table
Cross-reference
Insert Caption
Captions
Insert Index
Update Index
Mark Entry
Index
Insert Table of Authorities
Update Table
Mark Citation
Table of Authorities
To overcome from above scenario ServiceNow have Data Source Precedence Rules
Data Source Precedence Rules
If multiple data sources are authorized to update the same class or the same class attributes in the CMDB, data source precedence rules can be set up for each data source in order to allow customers to control which data source they consider to be the authoritative source of truth for a specific class or class attribute. Without data source precedence rules, data sources can overwrite each other's modifications.
For example let's take same scenario with order
PAGE 5 OF 7
788 WORDS
42:24 / 1:26:07
Reconciliation rule concepts
Bilingual

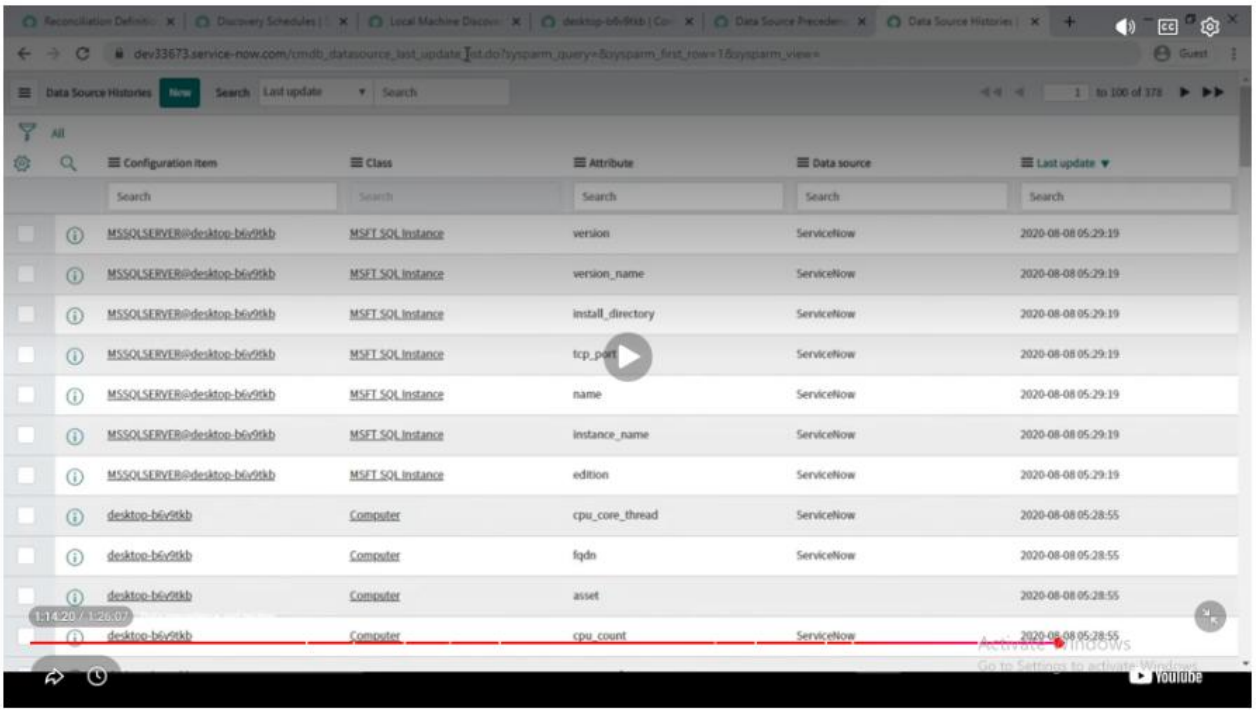

CMDB - Identification & Reconciliation - Word
TABLE TOOLS
FILE HOME INSERT DESIGN PAGE LAYOUT REFERENCES MAILINGS REVIEW VIEW DESIGN LAYOUT
Sign in
Add Text
Update Table
Table of Contents
Insert Endnote
Next Footnote
Show Notes
Insert Footnote
Manage Sources
Style: APA
Bibliography
Insert Citation
Insert Table of Figures
Update Table
Cross-reference
Insert Caption
Insert Index
Update Index
Mark Entry
Insert Table of Authorities
Update Table
Mark Citation
Table of Contents
Footnotes
Citations & Bibliography
Captions
Index
Table of Authorities
data source they consider to be the authoritative source of truth for a specific class or class attribute. Without data source precedence rules, data sources can overwrite each other's modifications.
For example let's take same scenario with order
Data Source | Applies To | Order
ServiceNow | cmdb_ci_win_server | 100
Import Set | cmdb_ci_win_server | 200
LANDesk | cmdb_ci_win_server | 300
PAGE 6 OF 7 780 WORDS
Activate Windows


CMDB - Identification & Reconciliation - Word
TABLE TOOLS
FILE HOME INSERT DESIGN PAGE LAYOUT REFERENCES MAILINGS REVIEW VIEW DESIGN LAYOUT
Sign in
Table of Contents
Footnotes
Citations & Bibliography
Captions
Index
Table of Authorities
LANDesk
RAM: 8GB (CI Inserted)
Priority/Order : 300
Import Set
Name: Win Server0001
RAM: 12GB (CI Updated with 12GB)
Priority : 200
ServiceNow
Name: Win Server0001
RAM: 32GB (CI Updated with 32GB)
Priority : 100
Import Set
Name: Win Server0001
RAM: 12GB (Update not allowed due to higher order)
PAGE 7 OF 7 797 WORDS
Activate Windows



It means either data come from LanDesk or from discovery, then only allow the data to modify. Else not.





MISCELLANEOUS OF CMDB MGMT:
CMDB GROUP : A **CMDB Group** is a logical collection of multiple Configuration Items (CIs)—like servers, databases, or applications—grouped together into a single set. It allows IT teams to perform bulk actions, such as managing **Change Requests**, **Patching**, **Incidents**, or **Monitoring alerts**, across all those CIs at once instead of doing it individually.

CMDB ARCHITECTURE :
Data in : discovery, service mapping, api (import set)
Engine : IRE, cmdb ci tables, schedule worker , query processing, cmdb lifecycle mgmt
Data out : ci class manager, health dashboard, dependency view, query builder, cmdb benchmark.

Asset : Financial components like cost, warranty, invoice, vendor etc
CI : Technical components like software, ip , ram etc

Affected CI : directly impact jiss ci pe pdega vo, like ek router khrab ho gya.
Impacted ci : indirectly impact jiss ci pe pdega vo, like uss router se 3 computer and 2 server connect the unn par effect pda.

Business service : end user value ko fullfill krne vali service.
Application service : technical aspects ko fulfill krne vali service like email service ko chalane ke liye iss se connected hai multiple server, db etc.

Stale records : jo records cmdb ci ke tables me hai to but outdated hochuki hai, yani purani info/details pdi hai. New details nhi update hui hai usme.
Orphan records : jab parent ci delete hojaye to uske child cis unconnected rhte hai, means ve memory consume krenge with no purpose, unko orphan records bolte hai.

Independent CI : it refers to a ci in cmdb that doesn't have direct dependencies on other cis, It's considered independent because it doesn't rely on other cis to function. Meaning it has no "upward" dependencies, however it can still have "downward" relationships with other cis.

Cascade relationship

CMDB CI relation rollups : ek parent ci se 6 child cis connected hai and mai chahta hu ki unn sbhi 6 child ci ka jitna power consume hua hai unka sum krke usse parent ci ki kisi field me put krdo . to iss concept ko cmdb ci rollups bolte hai.

CI Reclassification : ci ki class ko change karna. Isme 3 tarike hote hai :
Upgrade : base class se sub class me bhejna eg. server -> linux server
Downgrade : sub class se broad base class me bhejna eg. linux server -> server
Switch : same level ar ek class se doosri class me bhejna eg. windows server -> linux server.

upstream , downstream relationship :

Major incident record : ek aisi incident jo significantly impact karti hai business operations ko. Like kisi bank ka app hai usse 10 lakhs users daily use krte hai. But abhi customers uss app me login nhi kar paa rhe hai, payment nhi kar paarhe hai, transaction invoice nhi dekh paa rhe hai. And users ne social media pe bank ko troll krna start kar diya hai. And baat itni badd gyi hai ki stakeholders report maang rhe hai. Aise major incident ko hi MIM khte hai, like major incident.

Bridge call / war room : jab saare dept. Ke members online gether hoke meeting krte hai. Taki ye major incident resolve ho jaye jaldi. Inka motive rca etc karna nhi hota. Inka motive abhi hota hai ki business service jaldi se jaldi operational hojaye.

Discovery ke 4 phase :
Scan , classification, identification, exploration
When the discovery phase complete then it give us IRE payload, which further processed by IRE mechanism.

Note : tanium discovery bhi servicenow ke jaisi hi discovery provide krti hai.

3 pillar structure of cmdb :


servicenow
SERVICE MANAGEMENT ECOSYSTEM
May 2024
INGEST
Discovery
(Active Scanning, Asset Identification)
Service Mapping
(Dependency Visuals, Node Analysis)
Integrations
(Third-Party APIs, ETL Connectors)
GOVERN
CMDB Health
(Scorecard, Completeness 94%, Compliance)
IRE (Identification & Reconciliation Engine)
(Rule Matching, Merging)
Data Manager
(Data Certification, Stewardship, Automation)
INSIGHT
Query Builder
(Complex Search, Visual Filters)
Unified Map
(End-to-End Service Viz, Critical Path)
Dashboards
(Performance Analytics, Reporting, KPIs)
Activate Windows
Go to Settings to activate Windows.
Confidential
Slide 12/24

****

On-Call Scheduling in ITSM is a mechanism used to ensure that critical IT issues, incidents, and outages are automatically routed to the right support engineers at any given time (24/7/365).

Core Concepts

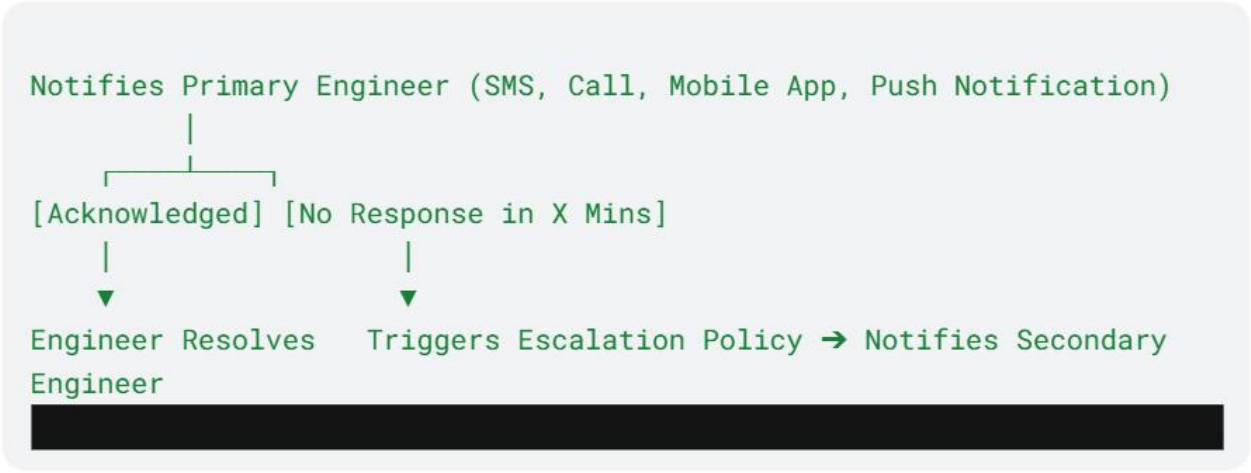
1. **Rosters & Shifts**: Defines rotation schedules (e.g., Primary, Secondary/Escalation, Night Shift, Weekend Shift).
2.
3. **Escalation Policies**: Dictates what happens if the primary on-call engineer does not acknowledge an incident within a set time (e.g., 15 minutes ➔ escalate to Secondary ➔ escalate to Manager).
4.
5. **Rotations**: Automatically cycles shifts among team members (e.g., Daily, Weekly, Custom).
6.
7. **Coverage & Time-off (Gaps)**: Allows engineers to swap shifts or request coverage for planned leave.
8.

How It Works in Practice

Plaintext

None

```
Critical Incident Triggered (P1/P2)
          |
          ▼
System Checks Active On-Call Schedule
          |
          ▼
```



Primary Benefits

- **Faster Resolution (MTTR)**: Eliminates delays in finding who is on duty during off-hours or major incidents.
- **Fair Workload Distribution**: Prevents burnout by systematically rotating night/weekend duties.
- **Automated Escalations**: Ensures high-priority incidents are never missed or stuck unassigned.

ServiceNow (SNOW) me **On-Call Scheduling** set karne ke steps:

1. **All Navigator** me search karo ➔ **On-Call Scheduling** ➔ **Create New Schedule** par click karo.
2. Us **Group** ko select karo jisko schedule karna hai (e.g., *Database Support Group*).
3. **Shift Details** enter karo (Name, Time Zone, Start/End Time, Repeat options).
4. **Roster Details** add karo (Primary, Secondary, Manager escalation levels define karo).
5. Group members ko roster me add karke **Submit / Finish** kar do.

*(Bonus: **Escalation Policies** me time intervals set ho jaate hain ki kitne mins baad backup engineer ko alert jayega).*

Haan, **On-Call Scheduling** ke liye plugin required hota hai.

Plugin Name: On-Call Scheduling

Plugin ID: com.snc.on_call_rotation

Quick Steps to Activate:

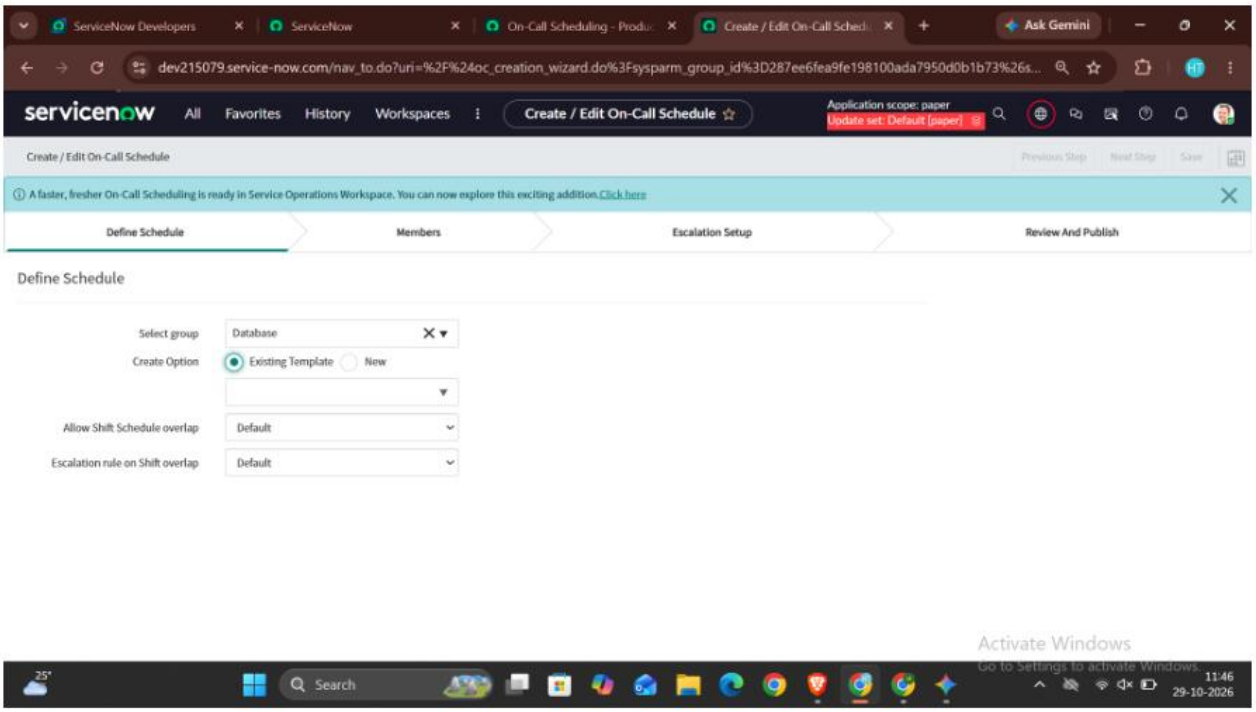
1. Navigator me **Plugins** (ya `sys_plugins.list`) kholein.
2.
3. Search karein **On-Call Scheduling** (`com.snc.on_call_rotation`).
4.
5. **Install / Activate** par click karein.
6.

Activate hone ke baad hi `sn_on_call` module aur `cmn_rots_*` tables show honge.

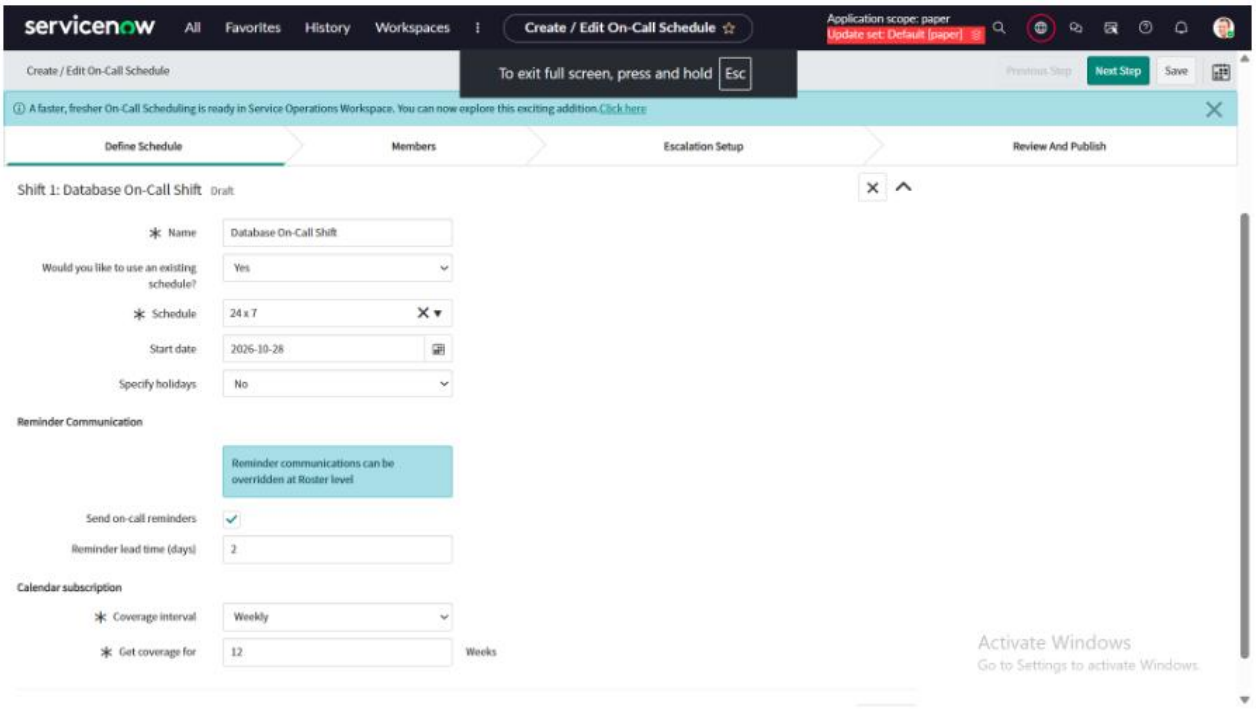
KEY FEATURES :
Schedule & Roster Management: Support teams ke shifts aur rotation schedules set aur manage karna.
Escalation Policies: Primary engineer ke respond na karne par auto-notifications ko secondary/manager tak escalate karna.
Time-Off & Shift Swap: Engineers ke liye leave request handle karna aur coverage/shifts manage karna.
Calendars & Reports: Schedules ko visual calendar view me dekhna aur performance dashboards/reports check karna.
Mobile Agent Access: Mobile app se directly schedules dekhna, manage karna aur time-off request bhejna.

Haan, bilkul! **Roster** ek support group ke andar ka **sub-group ya member list** hota hai jo specific shift ya duty level ko represent karta hai.
Ek Support Group ke andar **2 Main Rosters** ho sakte hain:

1. **Primary Roster:** Wo log jo pehle Incident handle karenge (First line of defense).
2. **Secondary Roster:** Wo log jo backup rahenge agar Primary response na de (Escalation backup).



Allow Shift Schedule overlap: Ye decide karta hai ki ek hi time par ek user ki **do alag-alag shifts overlapping (ek sath)** ho sakti hain ya nahi.
Escalation rule on Shift overlap: Agar shifts overlap ho jayein, toh system konse **escalation rule/duty hierarchy** ko priority dega.



Name: Shift ka naam (e.g., Database On-Call Shift).

Would you like to use an existing schedule?: Pehle se bane working hours use karne hain ya nahi (Yes hi rehne do).
Schedule: Shift ke working hours select karo (e.g., 8-5 weekdays ya 24x7).
Start date: Shift kab se chalu honi hai (Today's date).
Specify holidays: Holidays exclude karne hain ya nahi (No rakho).
Send on-call reminders / Lead time: Shift shuru hone se kitne din pehle user ko reminder mail jaye (Default set hai).
Coverage interval / Get coverage for: Calendar me schedule kitne hafto tak aage reflect hoga (Default set hai).
Coverage Interval ye decide karta hai ki calendar me shifts kitne time period ke block me repeat/divide hongi (e.g., **Weekly** matlab schedule hafte-dar-hafte dikhega aur rotate hoga).

Short Example:

Agar interval **Weekly** hai aur *Get coverage for* **12 Weeks** hai, toh system aane wale 12 hafto tak ka schedule calendar par pehle se calculate karke show kar dega.

Screen 1 (Members Selection):

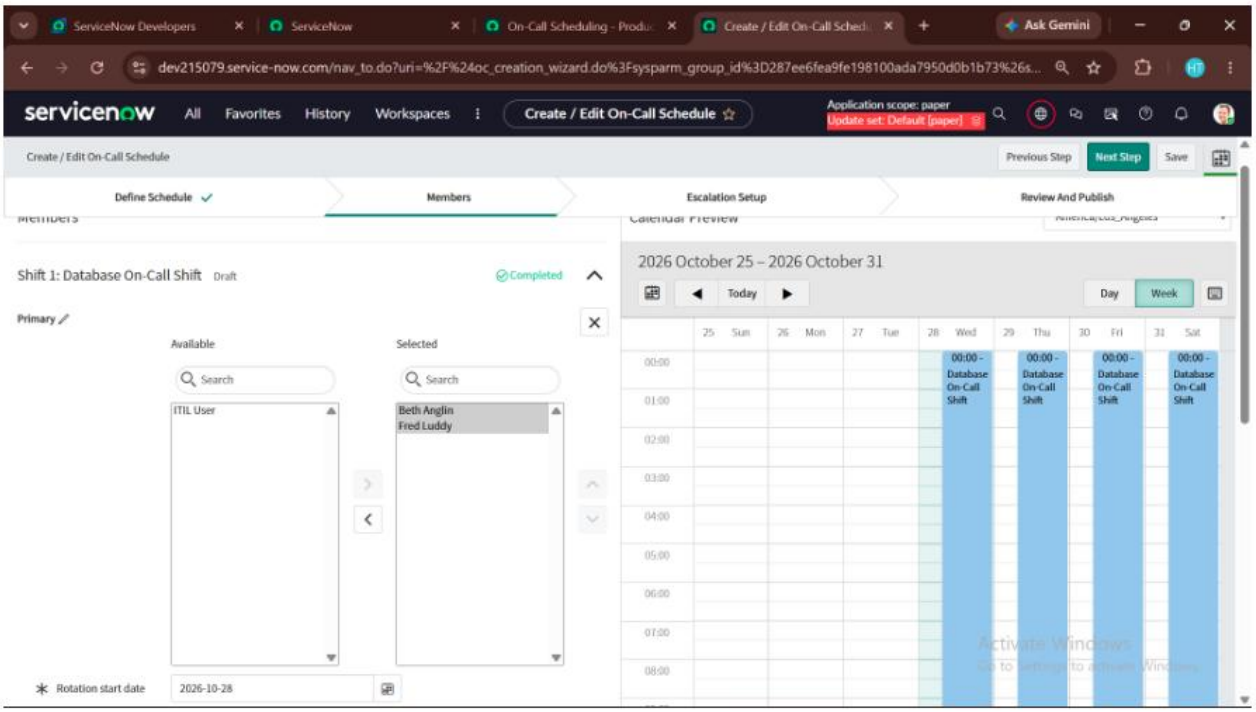
- **Available vs Selected**: **Available** box me unusers ki list hai jo group me hain (e.g., *ITIL User*). Inhein > arrow se **Selected** box me daal kar primary shift ke liye assign karna hota hai (jaise abhi *Beth Anglin* aur *Fred Luddy* selected hain).
-
- **Calendar Preview**: Right side par dikh raha calendar real-time me show kar raha hai ki konse din ye shift active rahegi (28-31 Oct).
-

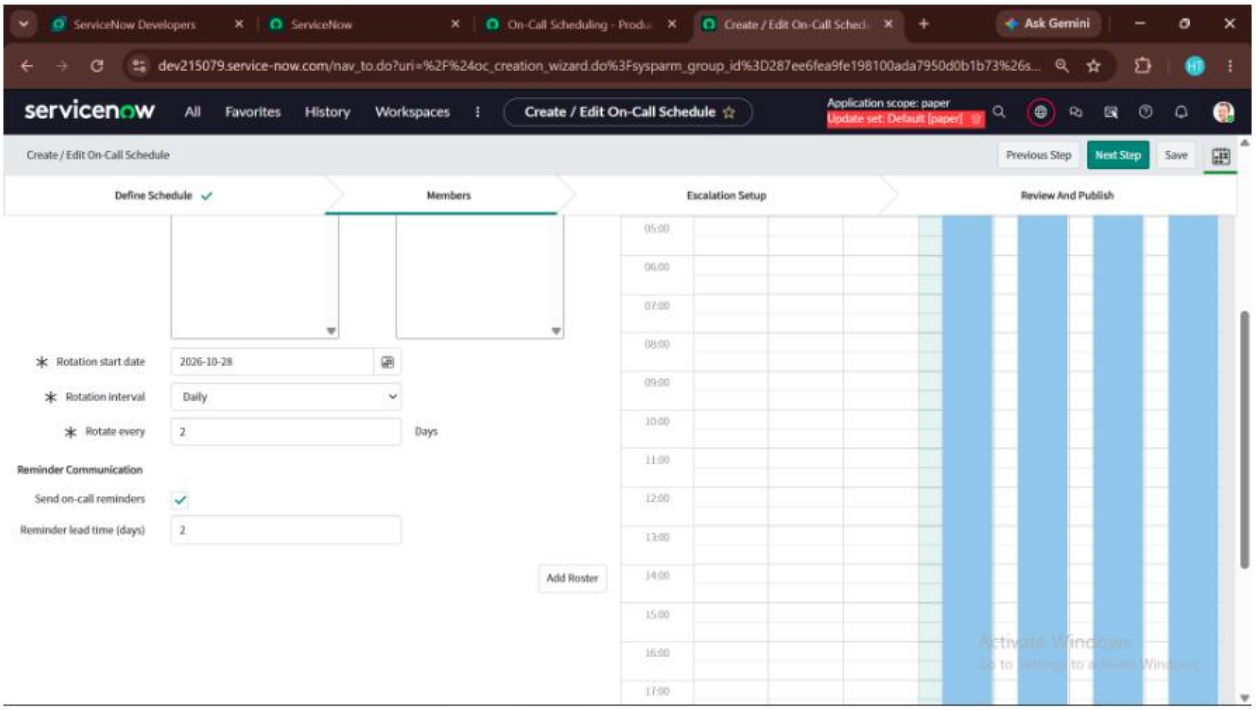
Screen 2 (Rotation Settings):

- **Rotation start date**: Duty ki rotation kis din se shuru hogi.
-
- **Rotation interval / Rotate every**: Users ke beech turn/duty kitne time baad change hogi (e.g., **Weekly**, **Rotate every 1 Week** = har hafte next bande ki turn aayegi).
-
- **Day of week for rotation**: Kis din turn switch hogi (e.g., *Monday* ya *Sunday*).
-
- **Add Roster**: Iss button se aap backup/secondary level ke liye alag se naya Roster create kar sakte ho.
-

Aapko Kya Karna Hai?

Primary members selected hain. **Rotation settings** set karke (ya default chhod kar) top-right me **Next Step** par click karo, wahan **Escalation Setup** aayega!





Is configuration (**Daily + Rotate every 2 Days**) ka matlab hai ki **har 2 din me on-call primary member badal jayega**.

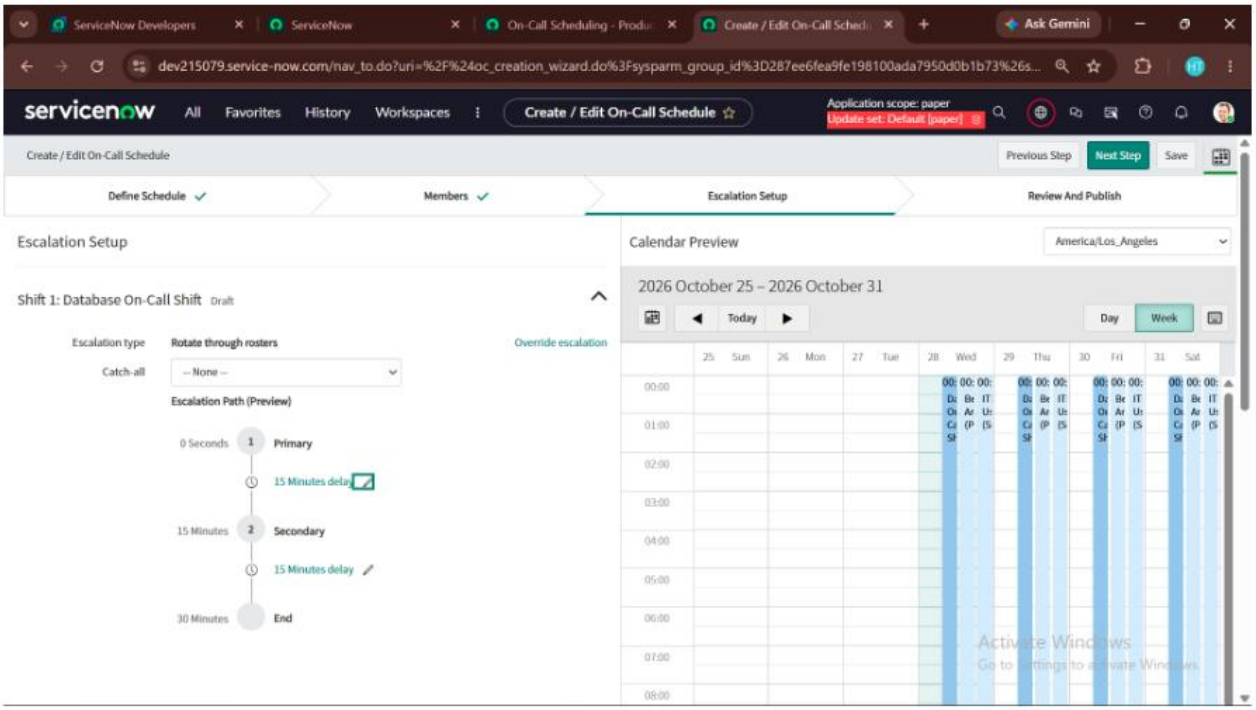
Example:

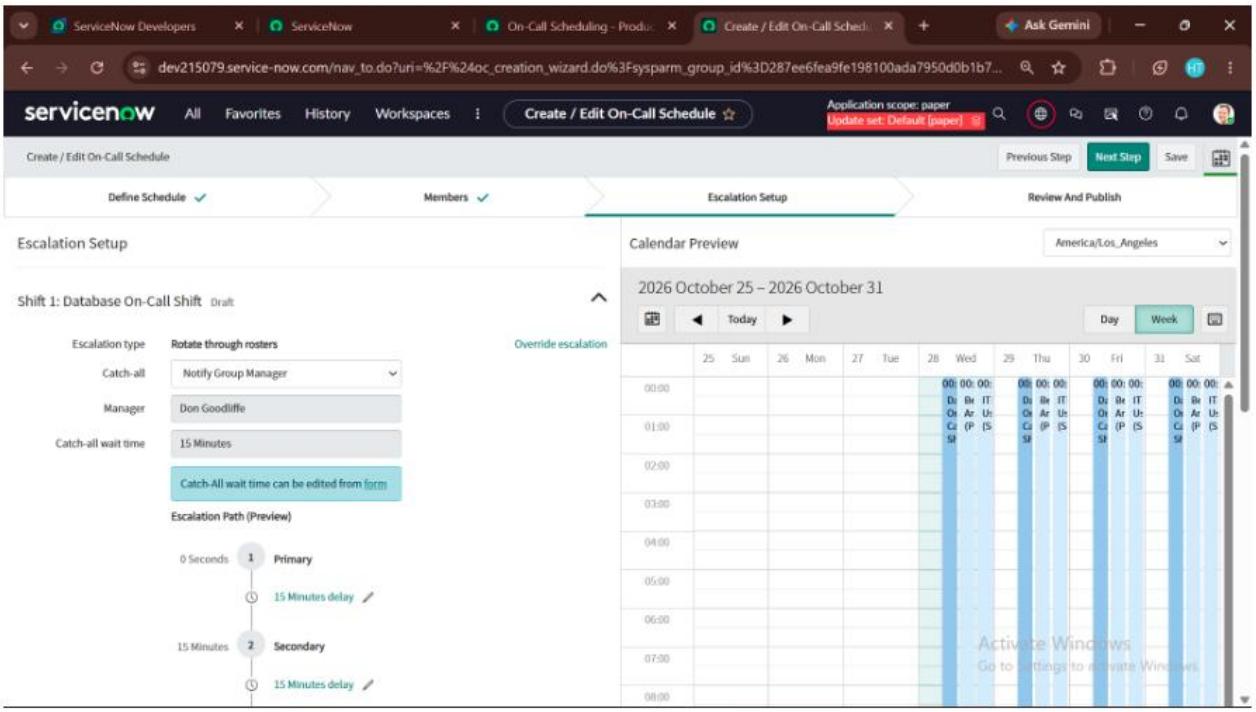
Selected members: **Beth Anglin** aur **Fred Luddy**.

- **Day 1 & Day 2 (Oct 28 - Oct 29):** On-Call par **Beth Anglin** rahegi.
-
- **Day 3 & Day 4 (Oct 30 - Oct 31):** On-Call par **Fred Luddy** aayega.
-
- **Day 5 & Day 6:** Waapas **Beth Anglin** ki turn aayegi.
-

Har 2 din me system automatically duty next engineer ko rotate kar dega.

Ab top-right par **Next Step** click karo aur **Escalation Setup** par chalo!



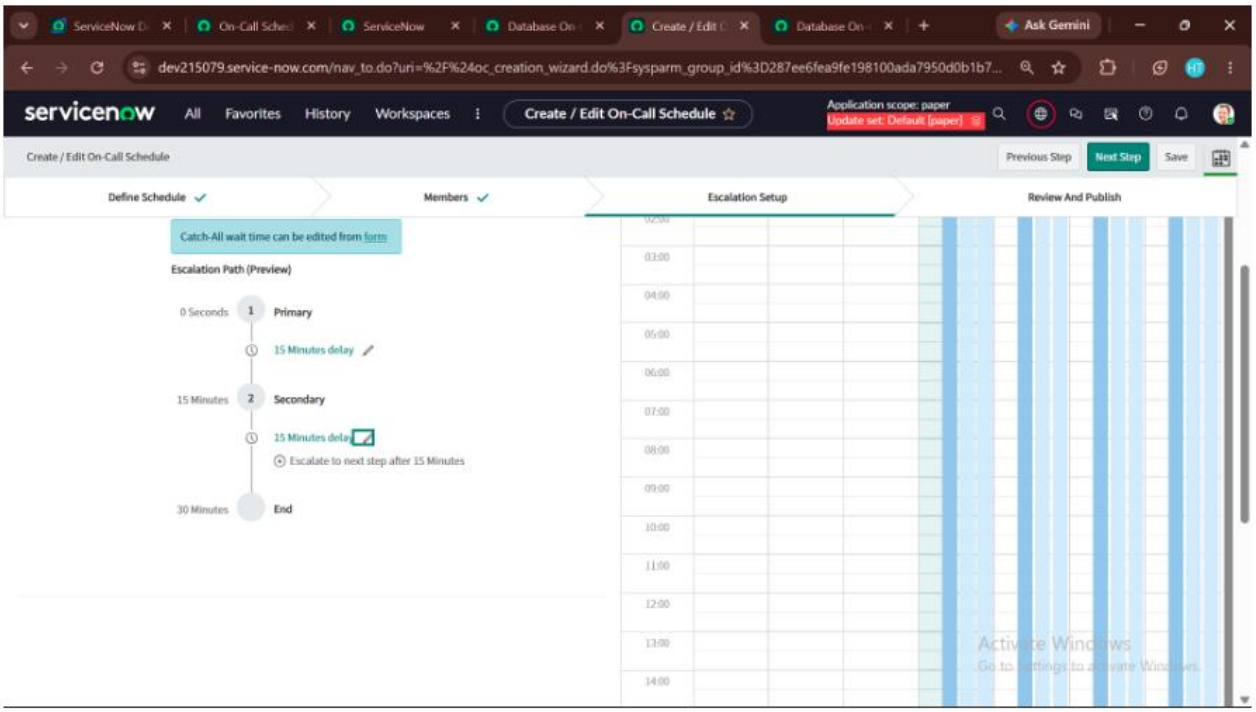


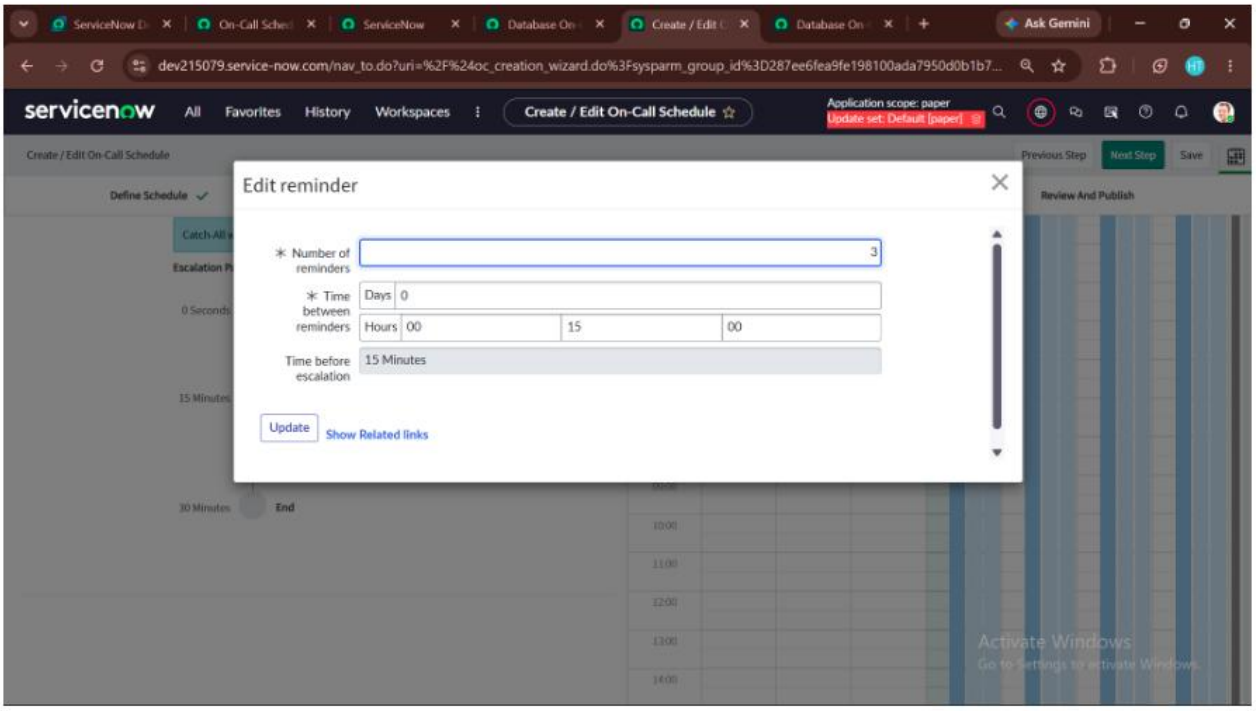
Agar **Catch-all wait time 45 Minutes** hota, toh timeline aisi hoti:

1. **0 min:** Primary ko alert.
2.
3. **15 min baad:** Secondary ko alert.
4.
5. **60 min baad (15 + 45 min):** Secondary ke 45 minutes wait karne par bhi response na aane par **Manager (Don Goodliffe)** ko notification jata.

Haan, bilkul sahi!

Jab escalation Level 2 (Secondary Roster) par pahoanchta hai, tabhi se **Catch-all wait time** ka countdown shuru hota hai. Agar Secondary bhi uss duration me respond nahi karta, tab fallback member / Manager ko notify hota hai.





Ye count **Primary ke paas aane par nahi**, balki **Jab turn Secondary Roster par aayegi tabhi se shuru hoga!**

Agar aap **Secondary** wale delay popup me Number of reminders = 3 aur Time between reminders = 15 Minutes kar dete ho, toh timeline aisi dikhegi:

Timeline Breakdown:

- **00 Min (Primary Start):** Primary member ko 1st alert jayega.
-
- **15 Min (Turn Secondary par aayi):** Primary ne response nahi diya, ab ticket **Secondary** par aa gaya.
-
    - **1st Notification:** 15th minute par Secondary ko pehla alert/notification milega.
    -
- **30 Min (15 Min baad):** Secondary ko **2nd reminder** milega.
-
- **45 Min (30 Min baad):** Secondary ko **3rd reminder** milega.
-
- **60 Min (45 Min baad):** Teeno reminders pure hone ke baad ticket agle step (Catch-all / Manager) par escalate ho jayega.
-

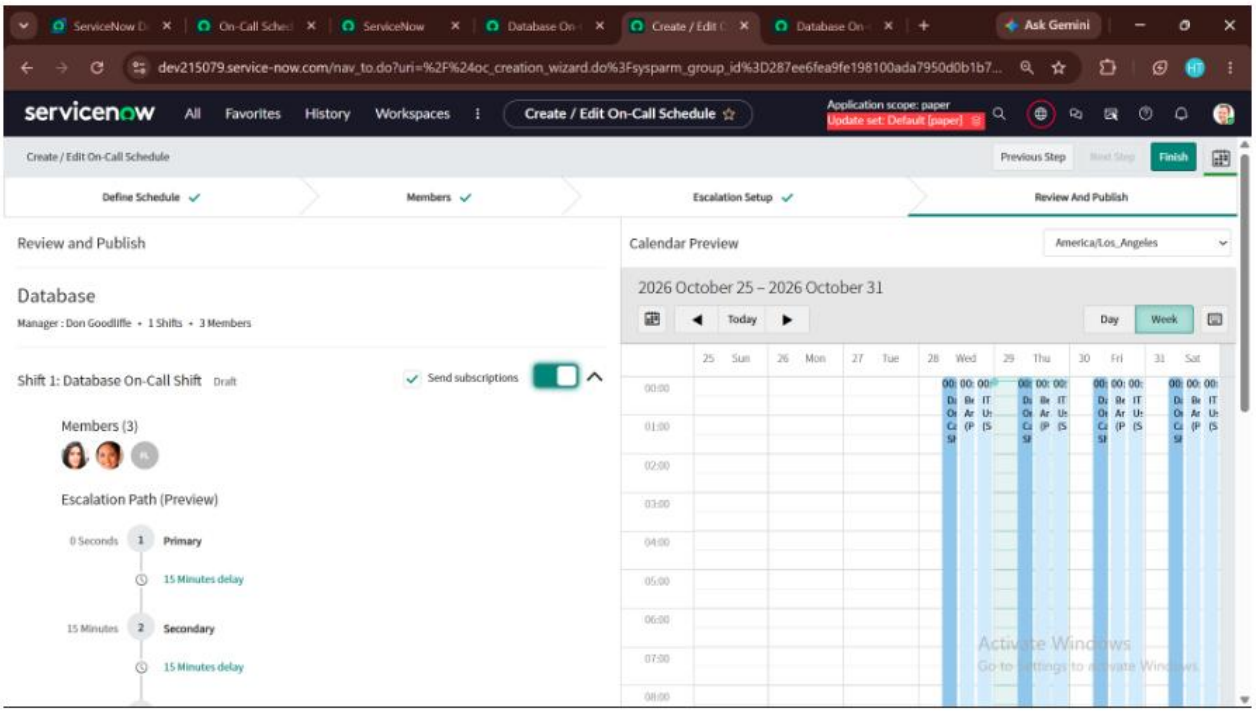
Key point: Har level ka reminder setup (Primary vs Secondary) apne khud ke turn aane se hi hit hota hai!

Agar teeno levels (Primary, Secondary, Catch-all wait time) 15-15 minutes ke hon aur **koi bhi respond na kare**, toh workflow aapse aise chalega:

- **00 Min:** Primary ko notification.
-
- **15 Min:** Primary no-response ➔ Secondary ko notification.
-
- **30 Min:** Secondary no-response ➔ Manager (**Don Goodliffe**) ko notification.
-
- **45 Min (Final Step):** Agar Manager bhi 15 mins tak respond na kare, toh escalation chain **END** ho jayegi.
-

Ticket unassigned/unacknowledged reh jayega jab tak koi manual action na le.

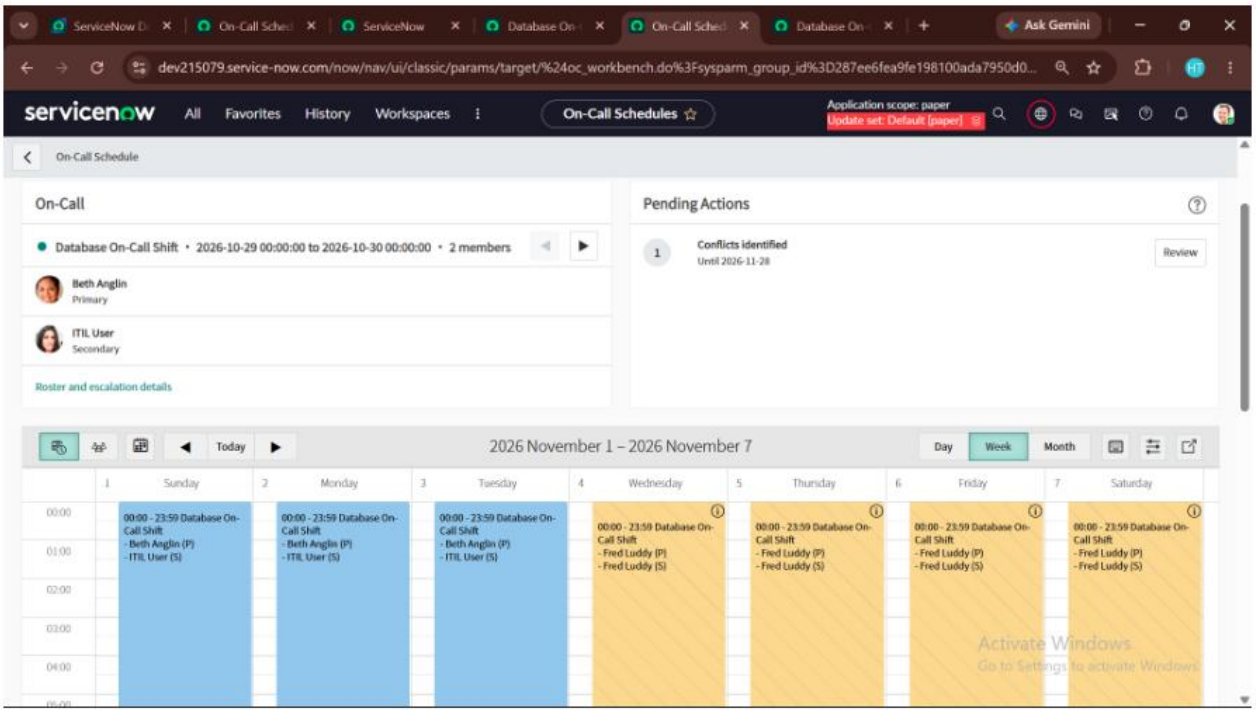
Top-right me **Next Step** par click karke final **Review And Publish** tab par jao!



Send subscriptions On-Call members ko unke **Outlook / Google Calendar** me issue kiye gaye schedule invite (iCal file) bhejne ke liye hota hai.

- **Checked (ON):** System members ko calendar invite mail karega taaki shift unke personal calendar me sync ho jaye.
-
- **Unchecked (OFF):** Shift calendar invites nahi bheje jayenge.
-

Ab top-right me **Finish** button click karke setup complete kar do!



Is calendar view ka matlab hai ki aapki **On-Call Rotation live work kar rahi hai**:

- **Blue Blocks (Nov 1–3):** Primary on-call **Beth Anglin (P)** aur Secondary **ITIL User (S)** hain.
-
- **Yellow Blocks (Nov 4–7): Rotation trigger ho gaya hai!** Primary aur Secondary dono shifts par ab **Fred Luddy** shift ho gaya hai.
-

Yellow Color & Info (ⓘ) Icon Kyun Hai?

Yellow color batata hai ki Secondary Roster par **Fred Luddy** ka hi naam Primary aur Secondary dono jagah repeated/overlapping hai. Isi waja se top par **Conflicts identified (1)** show ho raha hai.

Aap right side par **Review** button click karke is conflict info ko check kar sakte ho, ya fir direct **P1 Incident create karke live testing** start kar sakte ho!

On-Call Schedule Workbench sirf ye dikhane ke liye hota hai ki:

- Kaun sa engineer kis shift me duty par hai (Primary vs Secondary).
-

- Shift timings, rotations, rosters, aur scheduling conflicts.
-

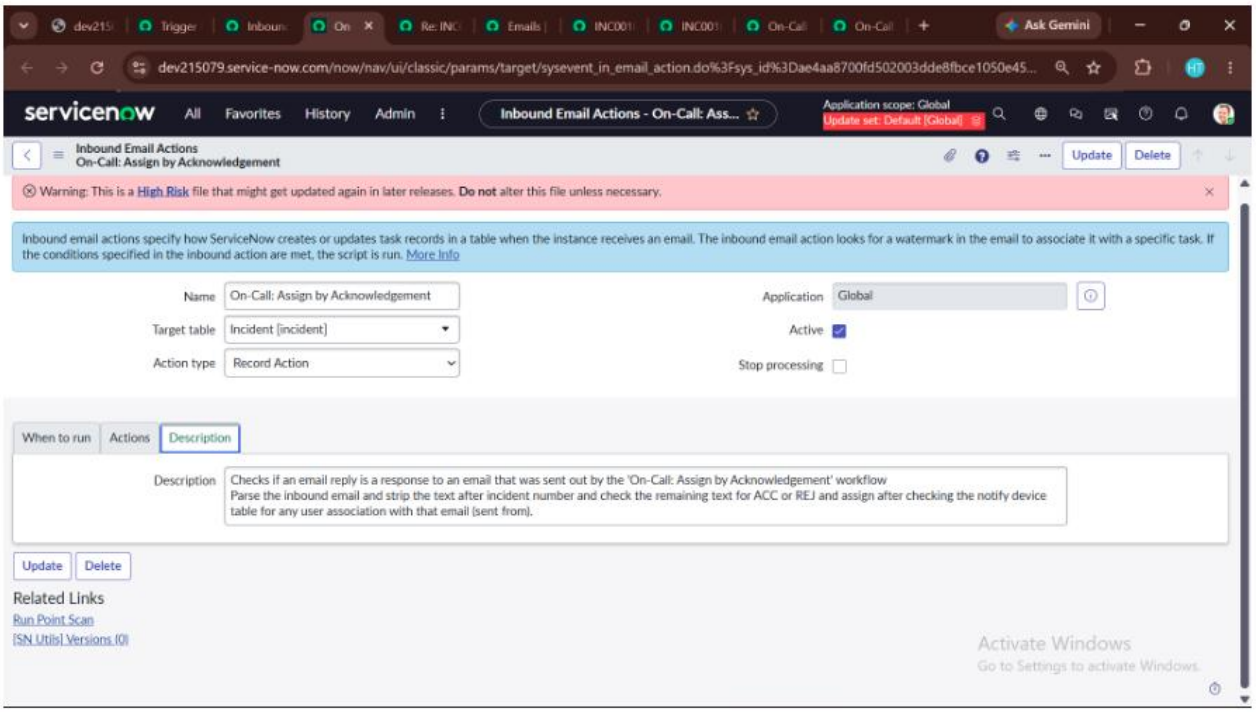
Woh active incidents ko calendar par list nahi karta.

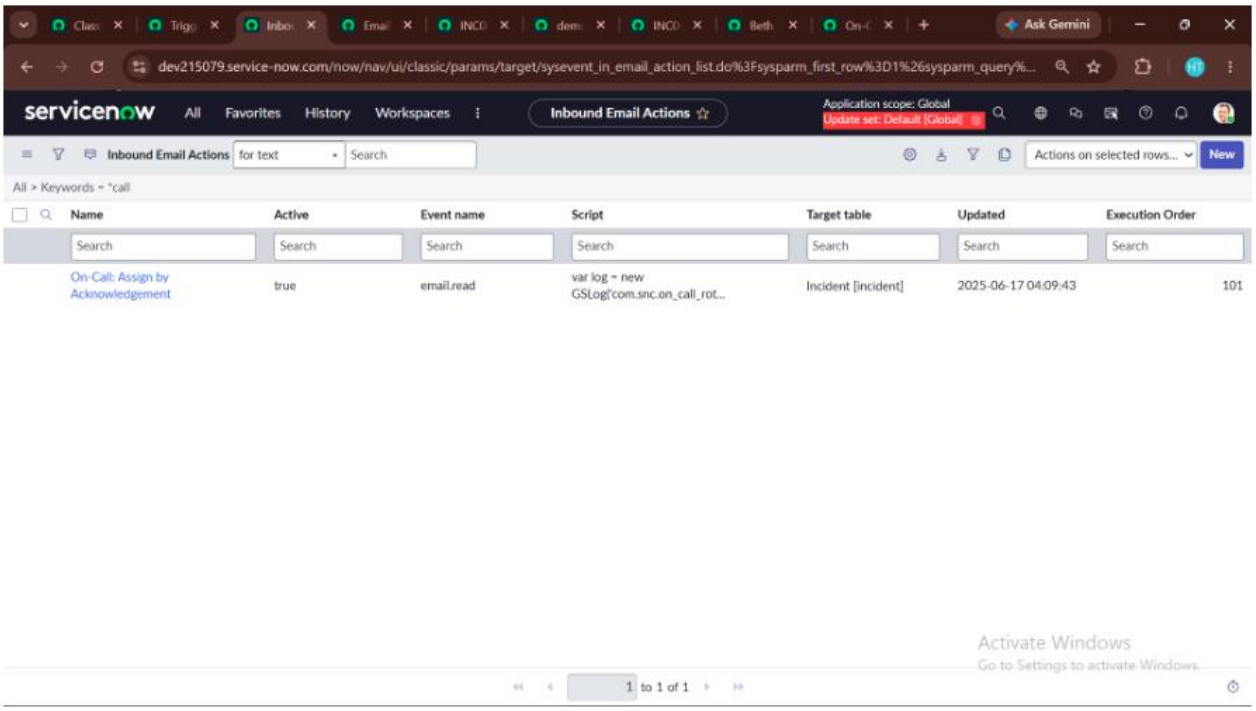
Phir yeh kaise dekhein ki On-Call se kaun-kaun assign hua hai?

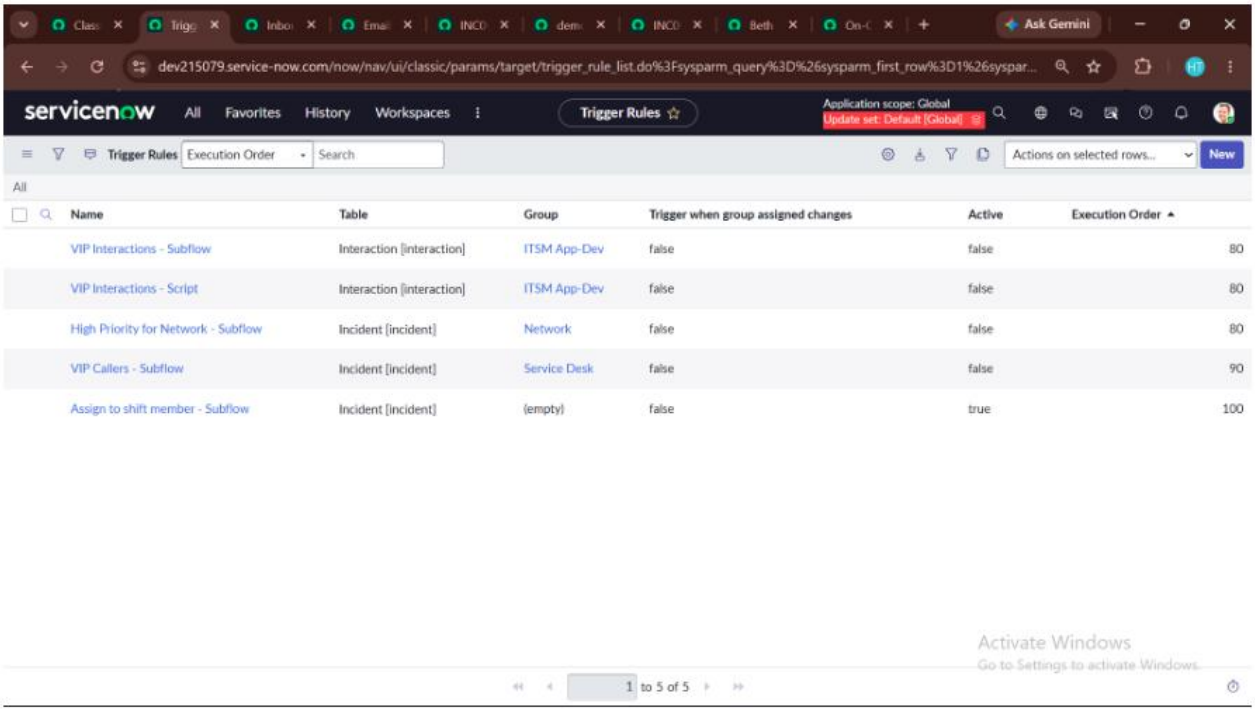
Agar aapko yeh dekhna hai ki On-Call rotation ke through kaun-kaun se incidents assign hue hain ya currently escalate ho rahe hain, toh aap yeh report/list dekh sakte ho:

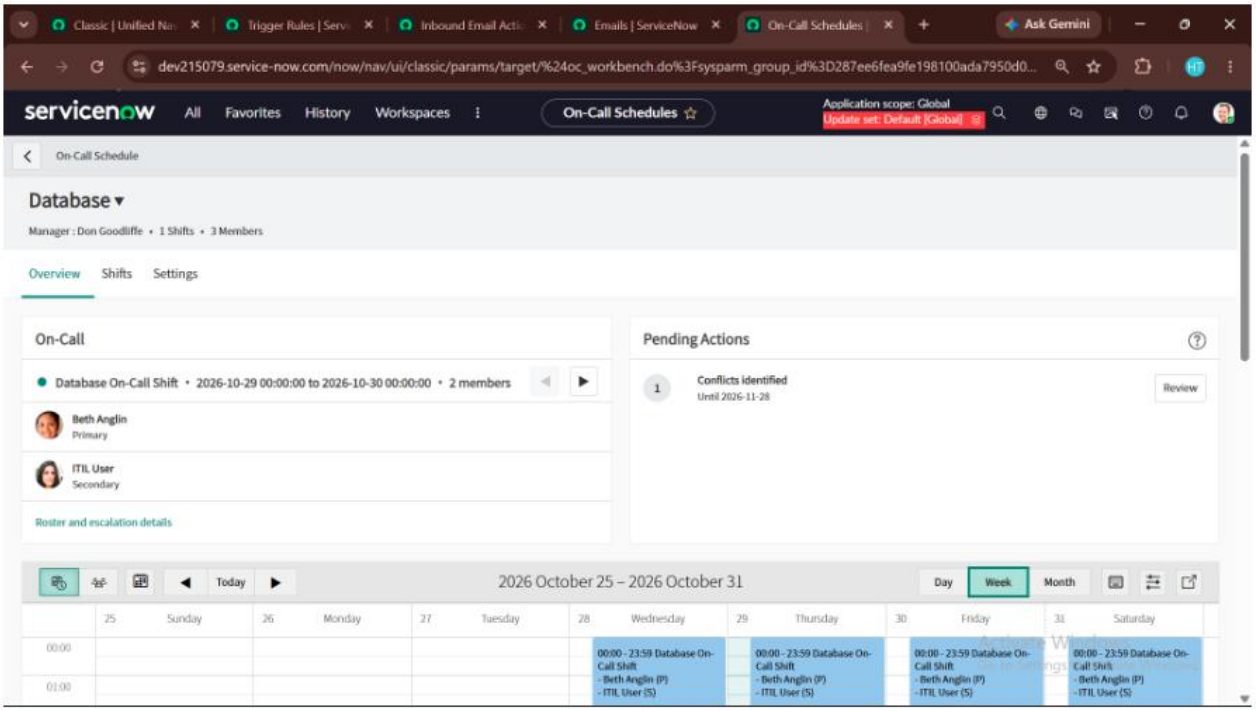
1. On-Call Escalations > Escalation Log: Yahan system batata hai ki kis Incident ke liye kab On-Call trigger hua, kis-kis ko alert gaya, aur kisne accept/reject kiya.
2.
3. Incidents > Open (Filter list): Aap Incident list par ek custom filter laga sakte ho:
4.
    - Assignment group = **Database**
    -
    - Assigned to **Is Not Empty** Isse aapko un saare incidents ki list mil jayegi jo On-Call roster ke members ko assign ho chuke hain.

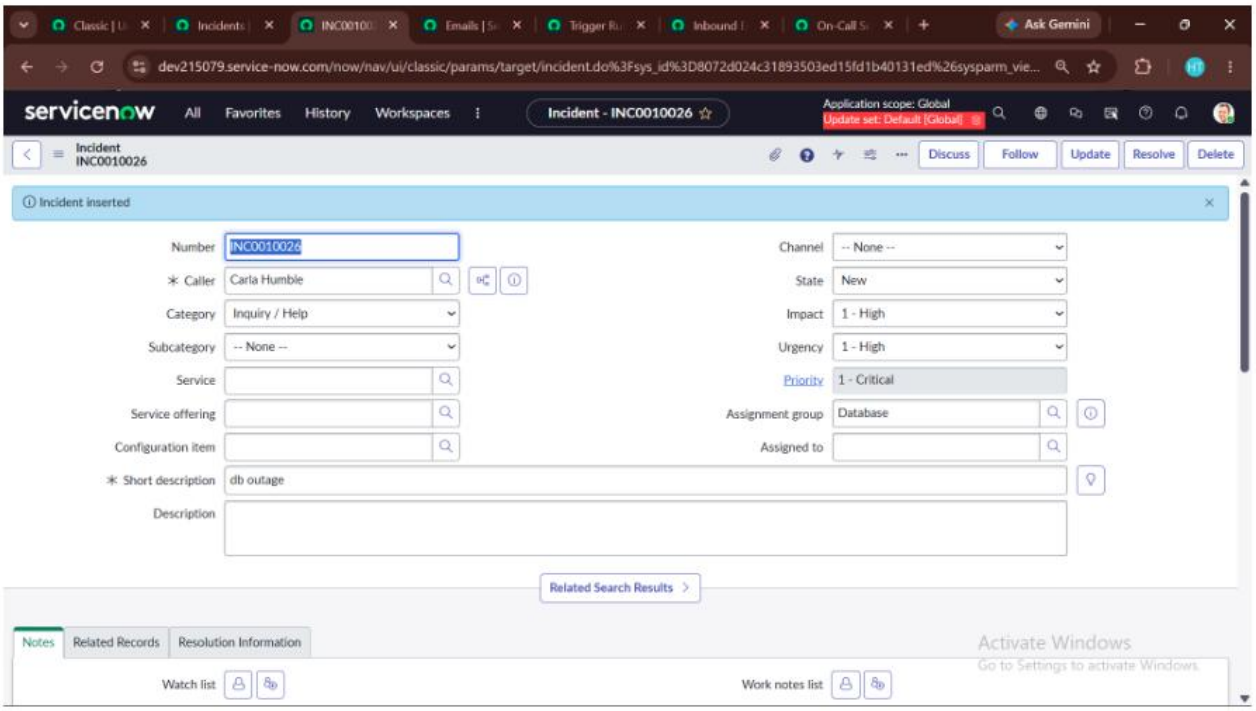
****

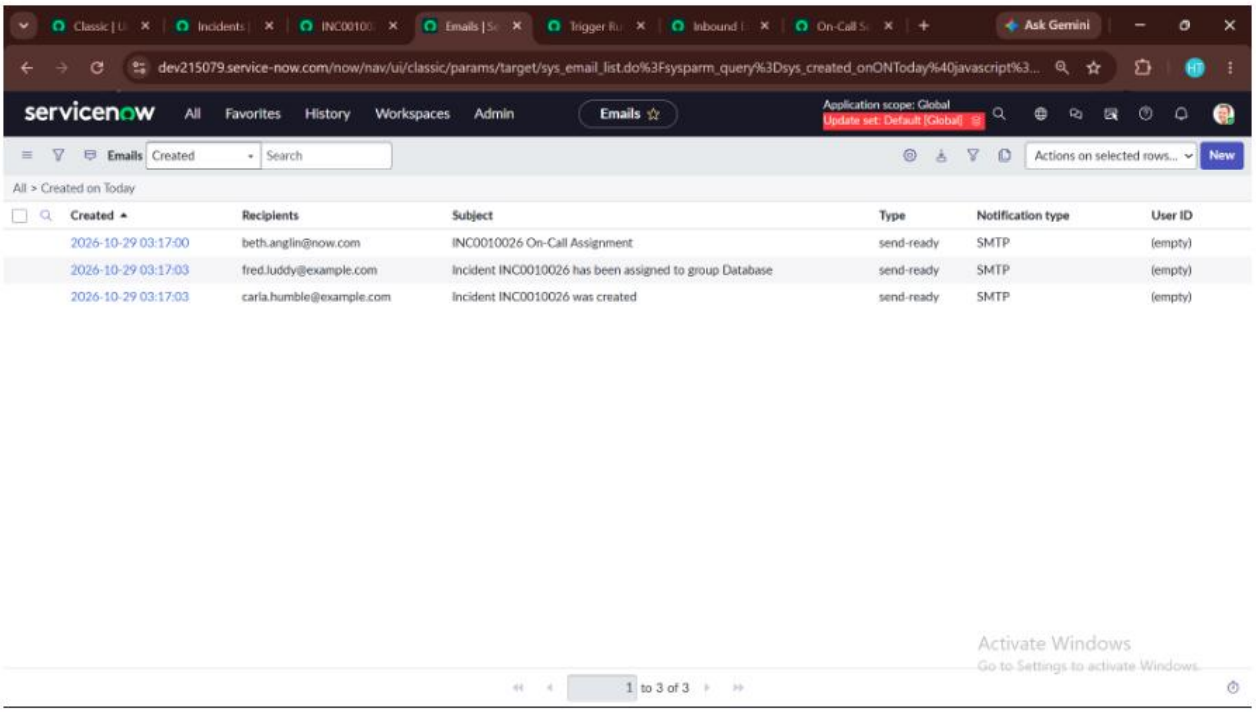

dev215079.service-now.com/now/nav/ui/classic/params/target/sysevent_in_email_action.do%3Fsys_id%3Dae4aa8700fd502003dde8fbce1050e45...
servicenow
All
Favorites
History
Admin
Inbound Email Actions - On-Call: Ass...
Application scope: Global
Update set: Default [Global]
Inbound Email Actions
On-Call: Assign by Acknowledgement
Update
Delete
Warning: This is a High Risk file that might get updated again in later releases. Do not alter this file unless necessary.
Inbound email actions specify how ServiceNow creates or updates task records in a table when the instance receives an email. The inbound email action looks for a watermark in the email to associate it with a specific task. If the conditions specified in the inbound action are met, the script is run. More Info
Name
On-Call: Assign by Acknowledgement
Application
Global
Target table
Incident [incident]
Active
Action type
Record Action
Stop processing
When to run
Actions
Description
Description
Checks if an email reply is a response to an email that was sent out by the 'On-Call: Assign by Acknowledgement' workflow
Parse the inbound email and strip the text after incident number and check the remaining text for ACC or REJ and assign after checking the notify device table for any user association with that email (sent from).
Update
Delete
Related Links
Run Point Scan
[SN Utils] Versions (0)
Activate Windows
Go to Settings to activate Windows.

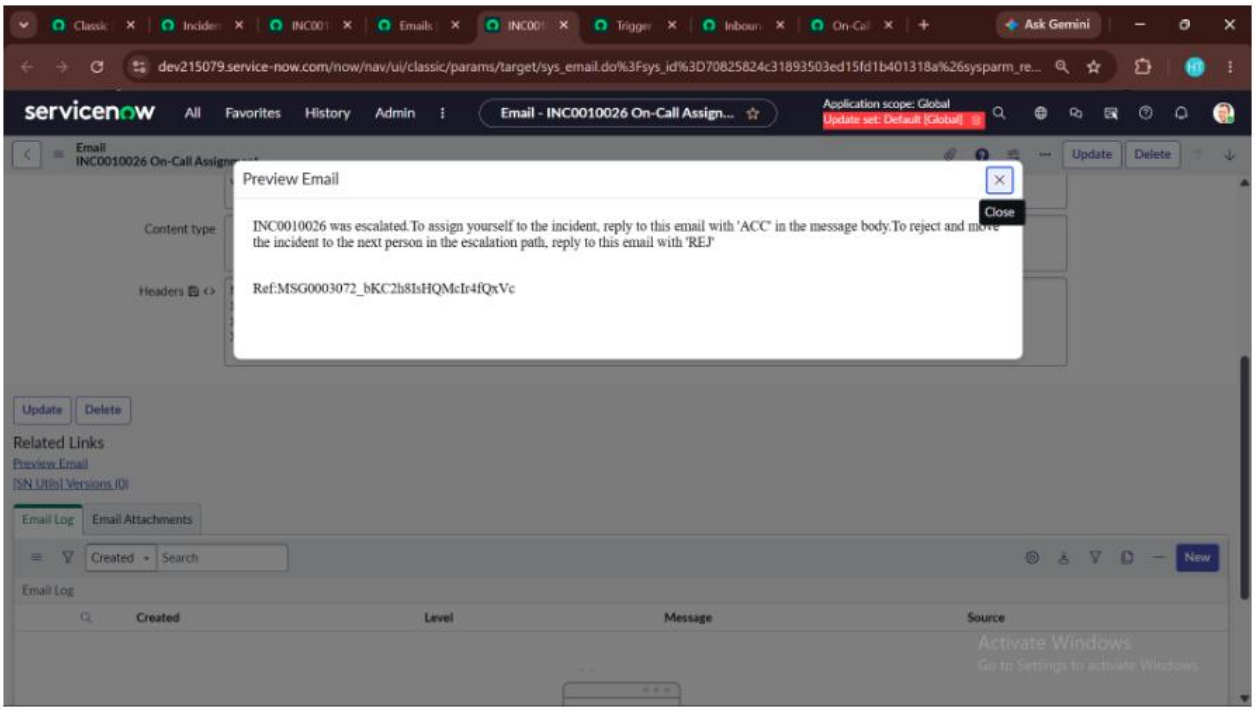

dev215079.service-now.com/now/nav/ui/classic/params/target/sysevent_in_email_action_list.do%3Fsysparm_first_row%3D1%26sysparm_query%...
servicenow
All
Favorites
History
Workspaces
Inbound Email Actions
Application scope: Global
Update set: Default [Global]
Inbound Email Actions
for text
Search
Actions on selected rows...
New
All > Keywords = *call
Name
Active
Event name
Script
Target table
Updated
Execution Order
On-Call: Assign by Acknowledgement
true
email.read
var log = new GSLog('com.snc.on_call_rot...
Incident [incident]
2025-06-17 04:09:43
101
1 to 1 of 1
Activate Windows
Go to Settings to activate Windows.

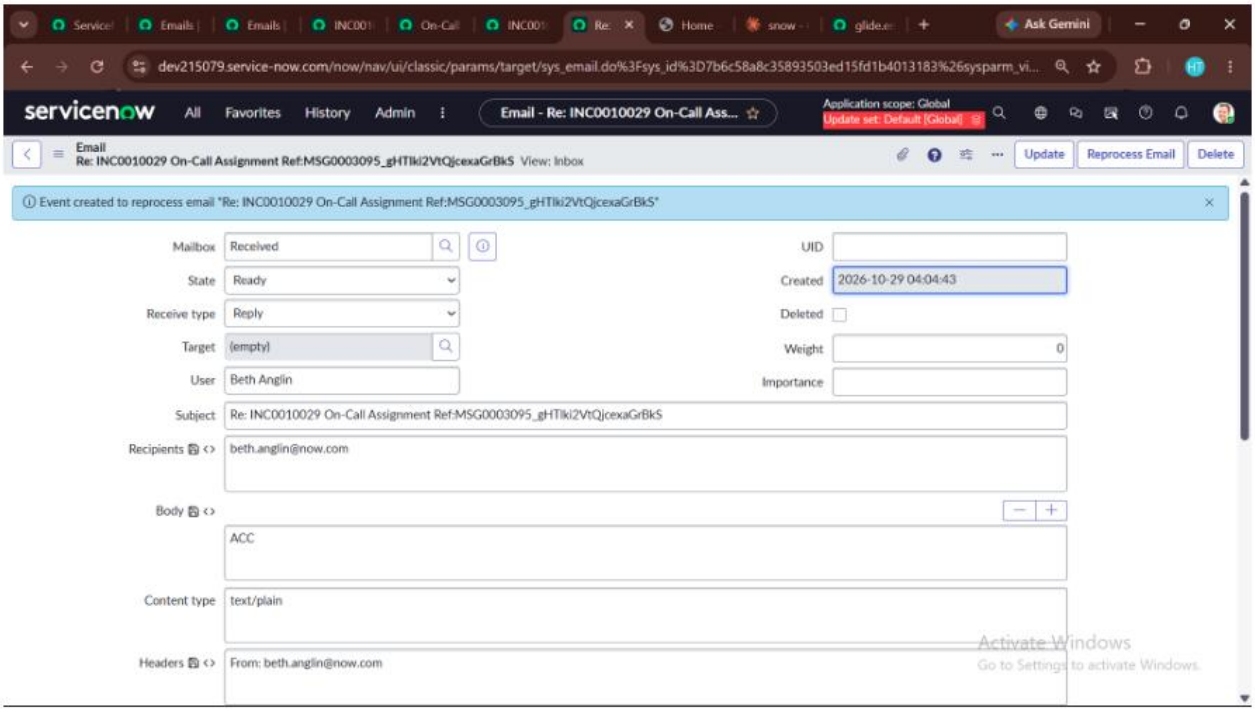

Ask Gemini
dev215079.service-now.com/now/nav/ui/classic/params/target/trigger_rule_list.do%3Fsysparm_query%3D%26sysparm_first_row%3D1%26syspar...
servicenow
All
Favorites
History
Workspaces
Trigger Rules
Application scope: Global
Update set: Default [Global]
Trigger Rules
Execution Order
Search
Actions on selected rows...
New
All
Name
Table
Group
Trigger when group assigned changes
Active
Execution Order
VIP Interactions - Subflow
Interaction [interaction]
ITSM App-Dev
false
false
80
VIP Interactions - Script
Interaction [interaction]
ITSM App-Dev
false
false
80
High Priority for Network - Subflow
Incident [incident]
Network
false
false
80
VIP Callers - Subflow
Incident [incident]
Service Desk
false
false
90
Assign to shift member - Subflow
Incident [incident]
(empty)
false
true
100
Activate Windows
Go to Settings to activate Windows.
1 to 5 of 5

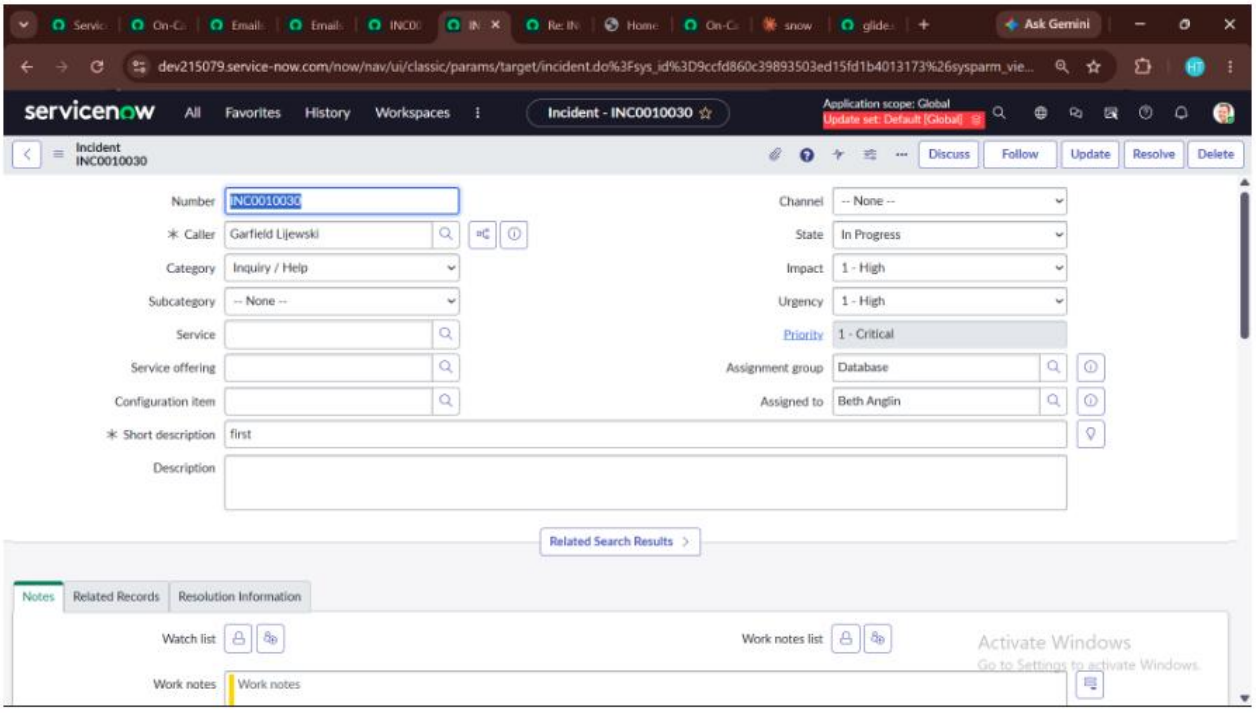

servicenow
All
Favorites
History
Workspaces
On-Call Schedules
Application scope: Global
Update set: Default [Global]
On-Call Schedule
Database
Manager : Don Goodliffe • 1 Shifts • 3 Members
Overview
Shifts
Settings
On-Call
Database On-Call Shift • 2026-10-29 00:00:00 to 2026-10-30 00:00:00 • 2 members
Beth Anglin
Primary
ITIL User
Secondary
Roster and escalation details
Pending Actions
Conflicts identified
Until 2026-11-28
Review
Today
2026 October 25 – 2026 October 31
Day
Week
Month
25 Sunday
26 Monday
27 Tuesday
28 Wednesday
29 Thursday
30 Friday
31 Saturday
00:00 - 23:59 Database On-Call Shift - Beth Anglin (P) - ITIL User (S)

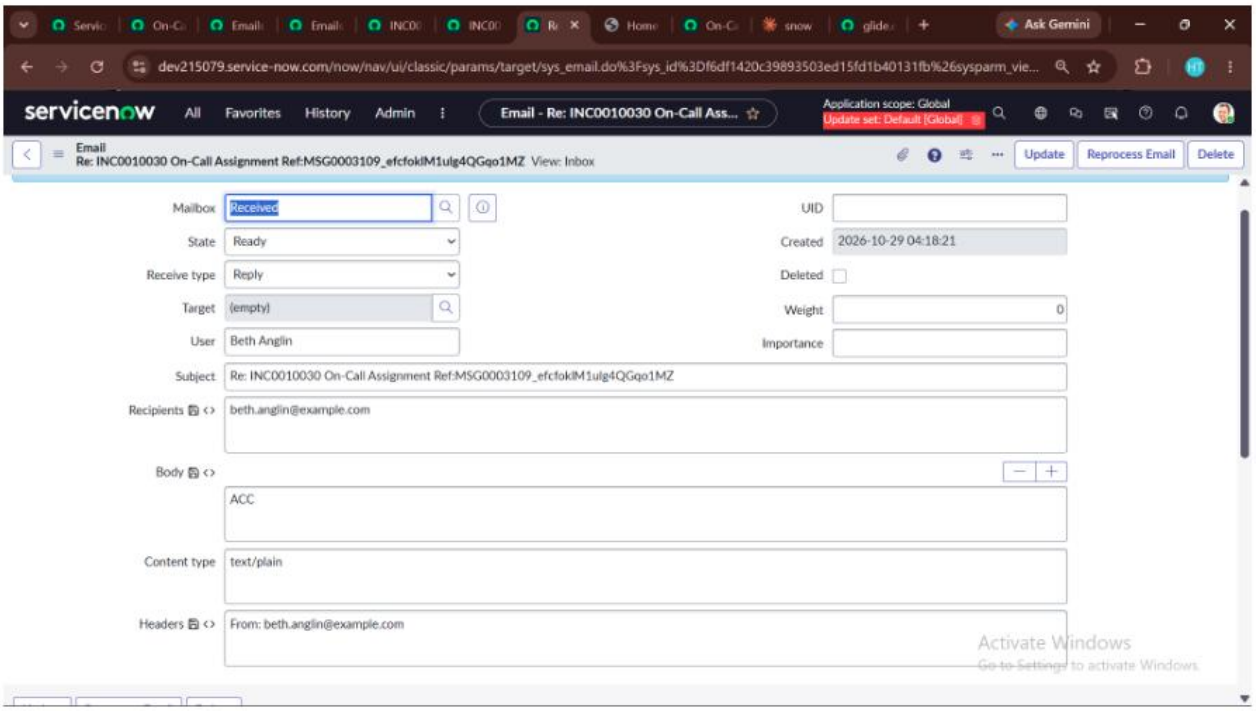

servicenow
All
Favorites
History
Workspaces
Incident - INC0010026
Application scope: Global
Update set: Default [Global]
Incident
INC0010026
Discuss
Follow
Update
Resolve
Delete
Incident inserted
Number INC0010026
Caller Carla Humble
Category Inquiry / Help
Subcategory -- None --
Service
Service offering
Configuration item
Short description db outage
Description
Channel -- None --
State New
Impact 1 - High
Urgency 1 - High
Priority 1 - Critical
Assignment group Database
Assigned to
Related Search Results
Notes
Related Records
Resolution Information
Watch list
Work notes list
Activate Windows
Go to Settings to activate Windows.

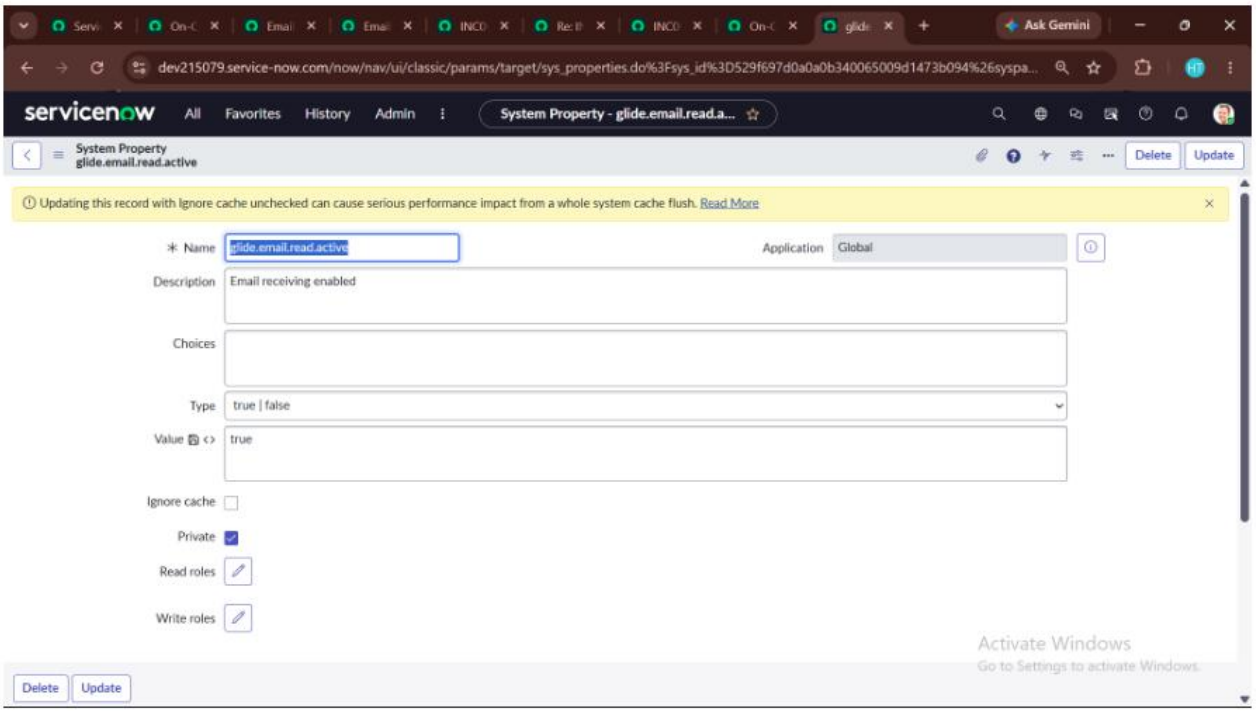

dev215079.service-now.com/now/nav/ui/classic/params/target/sys_email_list.do%3Fsysparm_query%3Dsys_created_onONToday%40javascript%3...
servicenow
All
Favorites
History
Workspaces
Admin
Emails
Application scope: Global
Update set: Default [Global]
Emails
Created
Search
Actions on selected rows...
New
All > Created on Today
Created
Recipients
Subject
Type
Notification type
User ID
2026-10-29 03:17:00
beth.anglin@now.com
INC0010026 On-Call Assignment
send-ready
SMTP
(empty)
2026-10-29 03:17:03
fred.luddy@example.com
Incident INC0010026 has been assigned to group Database
send-ready
SMTP
(empty)
2026-10-29 03:17:03
carla.humble@example.com
Incident INC0010026 was created
send-ready
SMTP
(empty)
Activate Windows
Go to Settings to activate Windows.
1 to 3 of 3


dev215079.service-now.com/now/nav/ui/classic/params/target/sys_email.do%3Fsys_id%3D70825824c31893503ed15fd1b401318a%26sysparm_re...
servicenow
All
Favorites
History
Admin
Email - INC0010026 On-Call Assign...
Application scope: Global
Update set: Default [Global]
Email
INC0010026 On-Call Assignment
Update
Delete
Preview Email
Close
Content type
Headers
INC0010026 was escalated.To assign yourself to the incident, reply to this email with 'ACC' in the message body.To reject and move the incident to the next person in the escalation path, reply to this email with 'REJ'
Ref:MSG0003072_bKC2h8IsHQMcIr4fQxVc
Update
Delete
Related Links
Preview Email
[SN Utils] Versions (0)
Email Log
Email Attachments
Created
Search
New
Email Log
Created
Level
Message
Source
Activate Windows
Go to Settings to activate Windows.


Ask Gemini
dev215079.service-now.com/now/nav/ui/classic/params/target/sys_email.do%3Fsys_id%3D7b6c58a8c35893503ed15fd1b4013183%26sysparm_vi...
servicenow
All
Favorites
History
Admin
Email - Re: INC0010029 On-Call Ass...
Application scope: Global
Update set: Default [Global]
Email
Re: INC0010029 On-Call Assignment Ref:MSG0003095_gHTlki2VtQjcexaGrBkS View: Inbox
Update
Reprocess Email
Delete
Event created to reprocess email "Re: INC0010029 On-Call Assignment Ref:MSG0003095_gHTlki2VtQjcexaGrBkS"
Mailbox
Received
UID
State
Ready
Created
2026-10-29 04:04:43
Receive type
Reply
Deleted
Target
(empty)
Weight
0
User
Beth Anglin
Importance
Subject
Re: INC0010029 On-Call Assignment Ref:MSG0003095_gHTlki2VtQjcexaGrBkS
Recipients
beth.anglin@now.com
Body
ACC
Content type
text/plain
Headers
From: beth.anglin@now.com
Activate Windows
Go to Settings to activate Windows.


dev215079.service-now.com/now/nav/ui/classic/params/target/incident.do%3Fsys_id%3D9ccfd860c39893503ed15fd1b4013173%26sysparm_vie...
servicenow
All
Favorites
History
Workspaces
Incident - INC0010030
Application scope: Global
Update set: Default [Global]
Incident
INC0010030
Discuss
Follow
Update
Resolve
Delete
Number
INC0010030
Channel
-- None --
Caller
Garfield Lijewski
State
In Progress
Category
Inquiry / Help
Impact
1 - High
Subcategory
-- None --
Urgency
1 - High
Service
Priority
1 - Critical
Service offering
Assignment group
Database
Configuration item
Assigned to
Beth Anglin
Short description
first
Description
Related Search Results
Notes
Related Records
Resolution Information
Watch list
Work notes list
Work notes
Work notes
Activate Windows
Go to Settings to activate Windows.


dev215079.service-now.com/now/nav/ui/classic/params/target/sys_email.do%3Fsys_id%3Df6df1420c39893503ed15fd1b40131fb%26sysparm_vie...
servicenow
All
Favorites
History
Admin
Email - Re: INC0010030 On-Call Ass...
Application scope: Global
Update set: Default [Global]
Email
Re: INC0010030 On-Call Assignment Ref:MSG0003109_efcfokIM1ulg4QGqo1MZ View: Inbox
Update
Reprocess Email
Delete
Mailbox
Received
UID
State
Ready
Created
2026-10-29 04:18:21
Receive type
Reply
Deleted
Target
(empty)
Weight
0
User
Beth Anglin
Importance
Subject
Re: INC0010030 On-Call Assignment Ref:MSG0003109_efcfokIM1ulg4QGqo1MZ
Recipients
beth.anglin@example.com
Body
ACC
Content type
text/plain
Headers
From: beth.anglin@example.com
Activate Windows
Go to Settings to activate Windows.



****

1. Incident create : assign db group
2. Then mail sent to beth or assigned primary member according to shift
3. Then copy the ref message
4. Create a received reply on behalf of that member (in my case it was beth)
5. As you reprocess email then it will assign that member in assigned to field of incident. (through OOB email inbound action i.e. On-Call: Assign by Acknowledgement)

NOTE: main issue is in the oob inbound action code. It uses body_text which is unable to parse the body data which is ACC. So, i put body_html.

NOTE : Also i made the default value of glide.email.read.active system property value true.

BUSINESS SERVICE :

What are we going to learn today?

- What is Business Service?
- What is Service Offering?
- Create a new business service and service offering and create a relationship with the Application Service (in PDI).

service offering with the application service that we already created

Activate Windows
Go to Settings to activate Windows.

Business Service

- This is what your business actually delivers to its end-users or customers. It is the non-technical, user-friendly name of the service you provide. Your customers don't care about the backend servers, they care about the service they are buying.
- **Table** : cmdb_ci_service_business

Techwithpri Example: "IT Training Services". The student doesn't know what an Apache server is, they just know they are coming to us for IT training.

Activate Windows
Go to Settings to activate Windows.

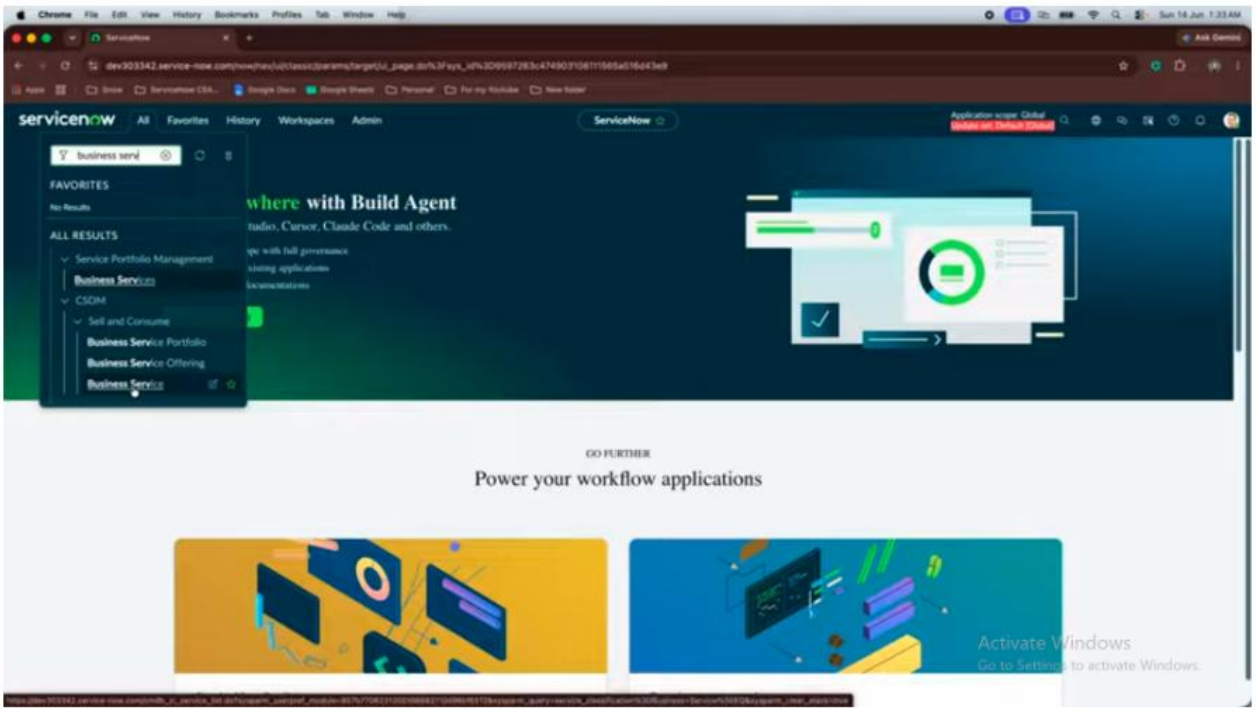
Service Offering

- This is the specific *tier* or *flavor* of the service. It defines the price, the commitments, and the Service Level Agreements (SLAs). You can have one Business Service with multiple different Offerings.

- **Table** : service_offering.list

Techwithpri Example: "LMS Premium Access" vs **"LMS Standard Access".** These are specific tiers under our IT Training Services. Premium users might get a 99.9% uptime guarantee, while Standard users get a 95% guarantee.

Activate Windows
Go to Settings to activate Windows.



MID SERVER :

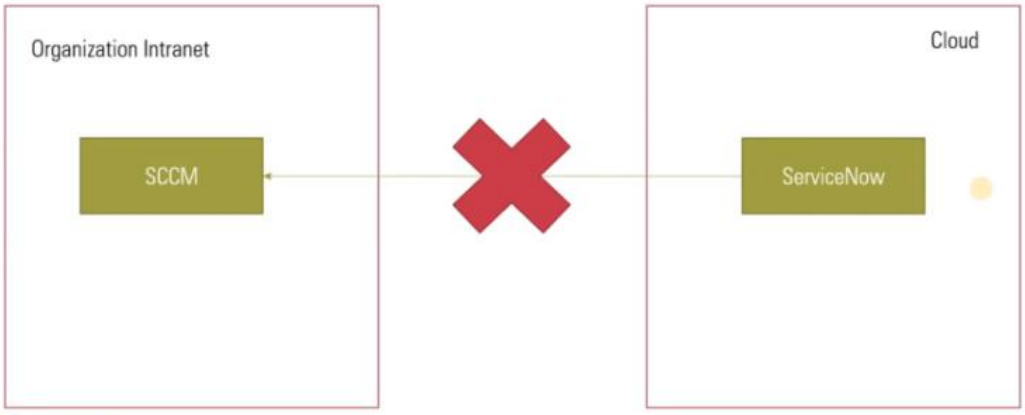
WHAT IS MID SERVER

The Management, Instrumentation, and Discovery (MID) Server is a Java application that runs as a Windows service or UNIX daemon. The MID Server facilitates communication and movement of data between the ServiceNow platform and external applications, data sources, and services.

The "MID Server" isn't really a server, but a service that runs on a server. If applications on your company network are not exposed to the internet, and you want ServiceNow to interface with them, a mid server is needed.

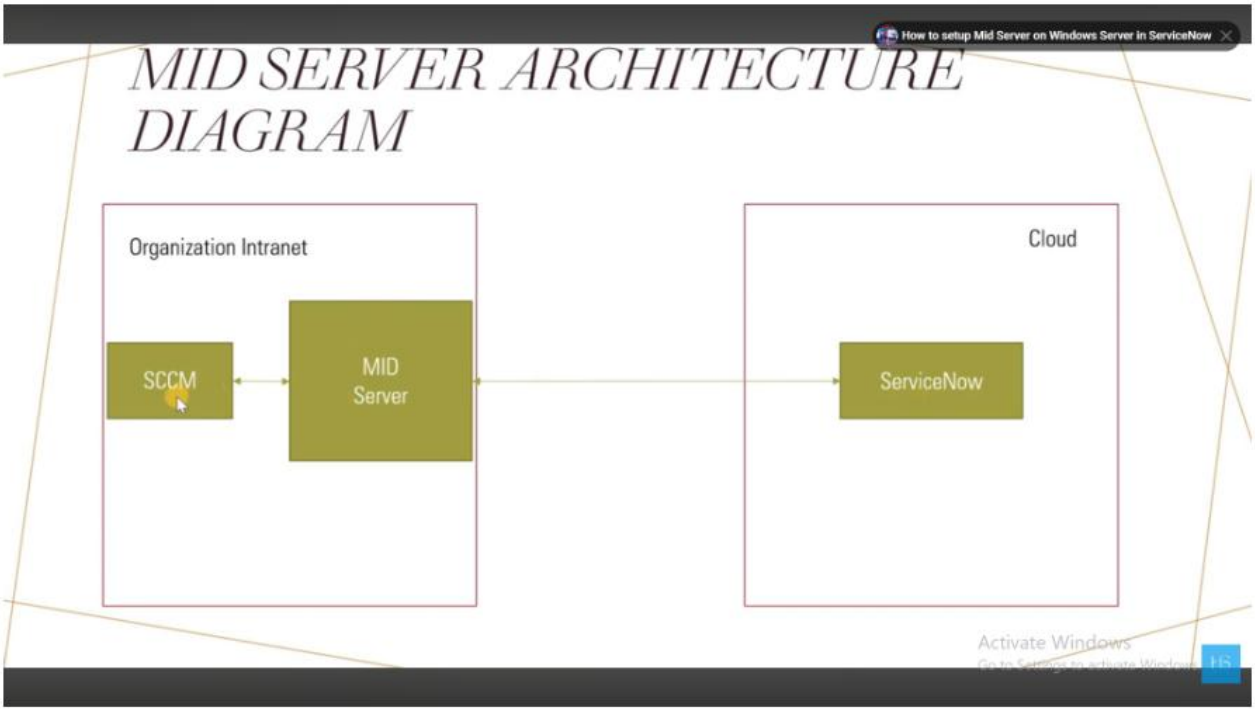
Activate Windows

MID SERVER ARCHITECTURE DIAGRAM



Activate Windows

Firewall not allow any tool to access this sccm tool data


How to setup Mid Server on Windows Server in ServiceNow
MID SERVER ARCHITECTURE DIAGRAM
Organization Intranet
SCCM
MID Server
Cloud
ServiceNow
Activate Windows
Go to Settings to activate Windows.
16